FUNDAMENTALS OF MATHEMATICS AND TRIGNOMETRY

Number System :

(i) **Natural numbers :** The counting numbers 1, 2, 3, 4, ... are called Natural Numbers. The set of natural numbers is denoted by N. Thus N = {1, 2, 3, 4, ...}.

(ii) **Whole numbers :** Natural numbers including zero are called whole numbers. The set of whole numbers is denoted by W. Thus W = {0, 1, 2, ...}

(iii) **Integers :** The numbers ... – 3, – 2, – 1, 0, 1, 2, 3 .... are called integers and the set is denoted by I or Z. Thus I (or Z) = {.. – 3, – 2, – 1, 0, 1, 2, 3...}

Note : (a) Positive integers I^+ = {1, 2, 3 ....} = N

(b) Negative integers I^- = {....., –3, –2, –1}.

(c) Non-negative integers (whole numbers) = {0, 1, 2, ...}.

(d) Non-positive integers = {....., –3, –2, –1, 0}.

(iv) **Even integers :** Integers which are divisible by 2 are called even integers.

e.g. 0, ± 2, ± 4,.....

(v) **Odd integers :** Integers which are not divisible by 2 are called odd integers.

e.g. ± 1, ± 3, ± 5, ± 7.....

(vi) **Prime numbers :** Natural numbers which are divisible by 1 and itself only are called prime numbers.

e.g. 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, ...

(vii) **Composite number :** Let 'a' be a natural number, 'a' is said to be composite if, it has atleast three distinct factors.

e.g. 4, 6, 8, 9, 10, 12, 14, 15 .....

Note : (i) 1 is neither a prime number nor a composite number.

(ii) Numbers which are not prime are composite numbers (except 1).

(iii) '4' is the smallest composite number.

(iv) '2' is the only even prime number.

(viii) **Co-prime numbers :** Two natural numbers (not necessarily prime) are called coprime, if there H.C.F (Highest common factor) is one.

e.g. (1, 2), (1, 3), (3, 4), (3, 10), (3, 8), (5, 6), (7, 8) (15, 16) etc.

These numbers are also called as **relatively prime** numbers.

Note : (a) Two prime number(s) are always co-prime but converse need not be true.

(b) Consecutive natural numbers are always co-prime numbers.

(ix) Twin prime numbers : If the difference between two prime numbers is two, then the numbers are called twin prime numbers.

e.g. $\{3, 5\}, \{5, 7\}, \{11, 13\}, \{17, 19\}, \{29, 31\}$

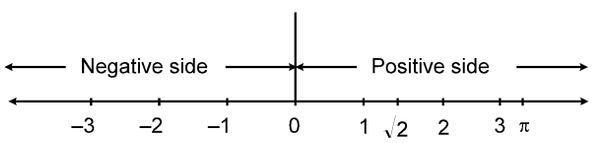
Note : Number between twin prime numbers is divisible by 6 (except (3, 5)).

(x) Rational numbers : All the numbers that can be represented in the form p/q, where p and q are integers and $q \neq 0$, are called rational numbers and their set is denoted by Q. Thus $Q = \{ \frac{p}{q} : p, q \in I \text{ and } q \neq 0\}$. It may be noted that every integer is a rational number since it can be written as p/1. It may be noted that all recurring decimals are rational numbers.

(xi) Irrational numbers : The numbers which can not be expressed in p/q form where $p, q \in I$ and $q \neq 0$ i.e. the numbers which are not rational are called irrational numbers and their set is denoted by Q^c. (i.e. complementary set of Q) e..g. $\sqrt{2}$, $1 + \sqrt{3}$ etc. Irrational numbers can not be expressed as recurring decimals.

Note : $e \approx 2.71$ is called Napier's constant and $\pi \approx 3.14$ are irrational numbers.

(xii) Real numbers : Numbers which can be expressed on number line are called real numbers. The complete set of rational and irrational numbers is the set of real numbers and is denoted by R. Thus $R = Q \cup Q^c$.



Real line

All real numbers follow the order property i.e. if there are two distinct real numbers a and b then either $a < b$ or $a > b$.

(xiii) Complex number : A number of the form $a + ib$ is called a complex number, where $a,b \in R$ and $i = \sqrt{-1}$. Complex number is usually denoted by Z and the set of complex number is represented by C. Thus $C = \{a + ib : a, b \in R \text{ and } i = \sqrt{-1}\}$

Note : It may be noted that $N \subset W \subset I \subset Q \subset R \subset C$.

Divisibility test :

S.No.	Divisibility of	Test
1	2	The digit at the unit place of the number is divisible by 2.
2	3	The sum of digits of the number is divisible by 3.
3	4	The last two digits of the number together are divisible by 4.
4	5	The digit of the number at the unit place is either 0 or 5.
5	6	The digit at the unit place of the number is divisible by 2 & the sum of all digits of the number is divisible by 3.
6	8	The last 3 digits of the number all together are divisible by 8.
7	9	The sum of all it's digits is divisible by 9.
8	10	The digit at unit place is 0.
9	11	The difference between the sum of the digits at even places and the sum of digits at odd places is 0 or multiple of 11. e.g.1298, 1221, 123321, 12344321, 1234554321, 123456654321

Remainder theorem : Let p(x) be any polynomial of degree greater than or equal to one and 'a' be any real number. If p(x) is divided by (x – a), then the remainder is equal to p(a).

Factor theorem : Let p(x) be a polynomial of degree greater than or equal to 1 and 'a' be a real number such that p(a) = 0, then (x – a) is a factor of p(x). Conversely, if (x – a) is a factor of p(x), then p(a) = 0.

Some important identities:

(1) $(a + b)^2 = a^2 + 2ab + b^2 = (a - b)^2 + 4ab$

(2) $(a - b)^2 = a^2 - 2ab + b^2 = (a + b)^2 - 4ab$

(3) $a^2 - b^2 = (a + b)(a - b)$

(4) $(a + b)^3 = a^3 + b^3 + 3ab(a + b)$

(5) $(a - b)^3 = a^3 - b^3 - 3ab(a - b)$

(6) $a^3 + b^3 = (a + b)^3 - 3ab(a + b) = (a + b)(a^2 + b^2 - ab)$

(7) $a^3 - b^3 = (a - b)^3 + 3ab(a - b) = (a - b)(a^2 + b^2 + ab)$

(8) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$

$$= a^2 + b^2 + c^2 + 2abc\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$$

(9) $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a - b)^2 + (b - c)^2 + (c - a)^2]$

(10) $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

$$= \frac{1}{2}(a + b + c)[(a - b)^2 + (b - c)^2 + (c - a)^2]$$

If $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$

(11) $a^4 - b^4 = (a + b)(a - b)(a^2 + b^2)$

(12) $a^4 + a^2 + 1 = (a^2 + 1)^2 - a^2 = (1 + a + a^2)(1 - a + a^2)$

Definition of indices :

If 'a' is any non zero real or imaginary number and 'm' is the positive integer, then a^m = a. a. a. ...a (m times). Here a is called the base and m is called the index, power or exponent.

Law of indices :

(1) $a^0 = 1, \quad (a \neq 0)$

(2) $a^{-m} = \frac{1}{a^m}, \quad (a \neq 0)$

(3) $a^{m+n} = a^m \cdot a^n$, where m and n are rational numbers

(4) $a^{m-n} = \frac{a^m}{a^n}$, where m and n are rational numbers, $a \neq 0$

(5) $(a^m)^n = a^{mn}$

(6) $a^{p/q} = \sqrt[q]{a^p}$

Cross Multiplication :

If two equations containing three unknowns are

$a_1x + b_1y + c_1z = 0$ .....(i)

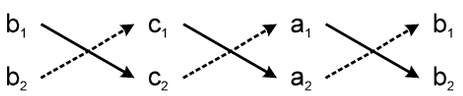
$a_2x + b_2y + c_2z = 0$ .....(ii)

Then by the rule of cross multiplication

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{c_1a_2 - c_2a_1} = \frac{z}{a_1b_2 - a_2b_1} \quad \text{.....(iii)}$$

In order to write down the denominators of x, y and z in (iii) apply the following rule,

"write down the coefficients of x, y and z in order beginning with the coefficients of y and repeat them as in the diagram"



Multiply the coefficients across in the way indicated by the arrows; remembering that informing the products any one obtained by descending is positive and any one obtained by ascending is negative.

Intervals :

Intervals are basically subsets of R and are commonly used in solving inequalities or in finding domains. If there are two numbers a, b ∈ R such that a < b, we can define four types of intervals as follows :

Name	Representation	Discription
Open Interval	(a, b)	{x : a < x < b} i.e. end points are not included.
Close Interval	[a, b]	{x : a ≤ x ≤ b} i.e. end points are also included. This is possible only when both a and b are finite.
Open - Closed Interval	(a, b]	{x : a < x ≤ b} i.e. a is excluded and b is included.
Close - Open Interval	[a, b)	{x : a ≤ x < b} i.e. a is included and b is excluded.

Note : (1) The infinite intervals are defined as follows :

(i) $(a, \infty) = \{x : x > a\}$ (ii) $[a, \infty) = \{x : x \geq a\}$

(iii) $(-\infty, b) = \{x : x < b\}$ (iv) $(\infty, b] = \{x : x \leq b\}$

(v) $(-\infty, \infty) = \{x : x \in R\}$

(2) $x \in \{1, 2\}$ denotes some particular values of x, i.e. x = 1, 2

(3) If there is no value of x, then we say $x \in \phi$ (null set)

General Method to solve Inequalities :

(Method of intervals (Wavy curve method)

Let $g(x) = \left(\frac{(x-b_1)^{k_1}(x-b_2)^{k_2} \cdots (x-b_n)^{k_n}}{(x-a_1)^{r_1}(x-a_2)^{r_2} \cdots (x-a_n)^{r_n}}\right)$ ... (i)

Where $k_1, k_2 \ldots k_n$ and $r_1, r_2 \ldots r_n \in N$ and $b_1, b_2 \ldots b_n$ and $a_1, a_2 \ldots a_n$ are real numbers.

Then to solve the inequality following steps are taken.

Steps : - Points where numerator becomes zero are called zeros or roots of the function and where

denominator becomes zero are called poles of the function.

(i) First we find the zeros and poles of the function.

(ii) Then we mark all the zeros and poles on the real line and put a vertical bar there dividing the real line in many intervals.

(iii) Determine sign of the function in any of the interval and then alternates the sign in the neghbouring interval if the poles or zeros dividing the two interval has appeared odd number of times otherwise retain the sign.

(iv) Thus we consider all the intervals. The solution of the $g(x) > 0$ is the union of the intervals in which we have put the plus sign and the solution of $g(x) < 0$ is the union of all intervals in which we have put the minus sign.

Various types of functions :

(i) Polynomial Function :

If a function f is defined by $f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + ... + a_{n-1} x + a_n$ where n is a **non negative integer** and a_0, a_1, a_2,....., a_n are real numbers and $a_0 \neq 0$, then f is called a polynomial function of degree n.

Note : ☞ There are two polynomial functions, satisfying the relation; $f(x).f(1/x) = f(x) + f(1/x)$, which are
$f(x) = 1 \pm x^n$

(ii) Constant function :

A function $f : A \to B$ is said to be a constant function, if every element of A has the same f image in B. Thus $f : A \to B$; $f(x) = c$, $\forall\ x \in A$, $c \in B$ is a constant function.

(iii) Identity function :

The function $f : A \to A$ defined by $f(x) = x\ \forall\ x \in A$ is called the identity function on A and is denoted by I_A. It can be observed that identity function is a bijection.

(iv) Algebraic Function :

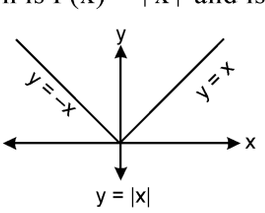
y is an algebraic function of x, if it is a function that satisfies an algebraic equation of the form, $P_0(x)\, y^n + P_1(x)\, y^{n-1} + ... + P_{n-1}(x)\, y + P_n(x) = 0$ where n is a positive integer and $P_0(x)$, $P_1(x)$..... are polynomials in x. e.g. $y = |x|$ is an algebraic function, since it satisfies the equation $y^2 - x^2 = 0$.

Note : ☞ All polynomial functions are algebraic but not the converse.

☞ A function that is not algebraic is called **Transcendental Function**.

(v) Absolute value function / modulus function :

The symbol of modulus function is $f(x) = |x|$ and is defined as: $y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$.



Properties of modulus : For any $a, b \in R$

(i) $|a| \geq 0$ (ii) $|a| = |-a|$

(iii) $|a| \geq a, |a| \geq -a$ (iv) $|ab| = |a|\,|b|$

(v) $\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$ (vi) $|a + b| \leq |a| + |b|$

(vii) $|a - b| \geq ||a| - |b||$

(vi) Rational function :

A rational function is a function of the form, $y = f(x) = \frac{g(x)}{h(x)}$, where g (x) & h (x) are polynomial functions.

(vii) Irrational function :

An irrational function is a function y = f(x) in which the operations of addition, substraction, multiplication, division and raising to a fractional power are used.

For example $y = \frac{x^3 + x^{1/3}}{2x + \sqrt{x}}$ is an irrational function

(a) The equation $\sqrt{f(x)} = g(x)$, is equivalent to the following system

$f(x) = g^2(x)$ & $g(x) \geq 0$

(b) The inequation $\sqrt{f(x)} < g(x)$, is equivalent to the following system

$f(x) < g^2(x)$ & $f(x) \geq 0$ & $g(x) \geq 0$

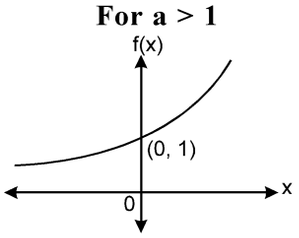
(c) The inequation $\sqrt{f(x)} > g(x)$, is equivalent to the following system

$g(x) \leq 0$ & $f(x) \geq 0$ or $g(x) \geq 0$ & $f(x) > g^2(x)$

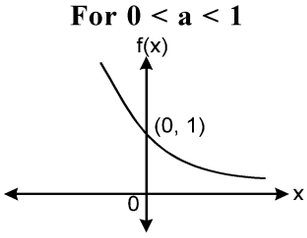
(viii) Exponential function :

A function $f(x) = a^x = e^{x \ln a}$ ($a > 0, a \neq 1, x \in R$) is called an exponential function. Graph of exponential function can be as follows :

Case - I
For a > 1

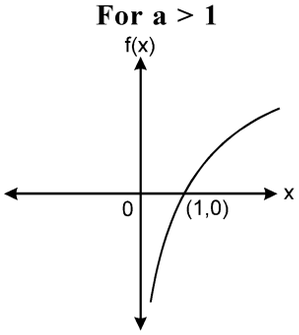


Case - II
For 0 < a < 1

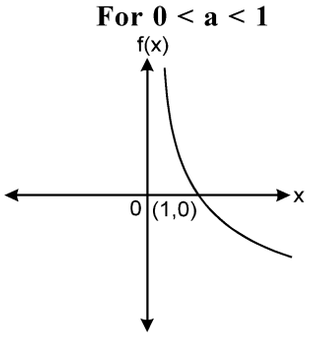


(ix) Logarithmic function : $f(x) = \log_a x$ is called logarithmic function where $a > 0$ and $a \neq 1$ and $x > 0$. Its graph can be as follows :

Case- I
For a > 1



Case- II
For 0 < a < 1



Logarithm of A Number :

The logarithm of the number N to the base 'a' is the exponent indicating the power to which the base 'a' must be raised to obtain the number N. This number is designated as $\log_a N$. Hence:

$$\log_a N = x \Leftrightarrow a^x = N \text{ , } a > 0, a \neq 1 \text{ \& } N > 0$$

If a = 10, then we write log b rather than $\log_{10} b$.

If a = e, we write *l*n b rather than $\log_e b$. Here 'e' is called as Napier's base & has numerical value equal to 2.7182.

Remember

$\log_{10} 2 \approx 0.3010$; $\log_{10} 3 \approx 0.4771$

$\ell n\ 2 \approx 0.693$; $\ell n\ 10 \approx 2.303$

Domain of Definition :

The existence and uniqueness of the number $\log_a N$ can be determined with the help of set of conditions, $a > 0$ & $a \neq 1$ & $N > 0$.

The base of the logarithm 'a' must not equal unity otherwise numbers not equal to unity will not have a logarithm and any number will be the logarithm of unity.

Fundamental Logarithmic Identity :

$$a^{\log_a N} = N,\ a > 0,\ a \neq 1 \text{ \& } N > 0$$

The Principal Properties of Logarithm:

Let M & N are arbitrary positive numbers, $a > 0$, $a \neq 1$, $b > 0$, $b \neq 1$ and α, β are any real numbers, then :

(i) $\log_a (M.N) = \log_a M + \log_a N$; in general $\log_a (x_1\ x_2\ .....x_n) = \log_a x_1 + \log_a x_2 + ...+ \log_a x_n$

(ii) $\log_a (M/N) = \log_a M - \log_a N$

(iii) $\log_a M^\alpha = \alpha . \log_a M$

(iv) $\log_{a^\beta} M = \frac{1}{\beta} \log_a M$

(v) $\log_b M = \frac{\log_a M}{\log_a b}$ (base changing theorem)

NOTE :

- $\log_a 1 = 0$
- $\log_a a = 1$
- $\log_{1/a} a = -1$
- $\log_b a = \frac{1}{\log_a b}$
- $a^x = e^{x\,\ell na}$
- $a^{\log_c b} = b^{\log_c a}$

Logarithmic Equation :

The equality $\log_a x = \log_a y$ is possible if and only if x = y i.e.

$\log_a x = \log_a y \Leftrightarrow x = y$

Always check validity of given equation, $(x > 0, y > 0, a > 0, a \neq 1)$

Logarithmic Inequality :

Let 'a' is a real number such that

(i) If $a > 1$, then $\log_a x > \log_a y \Rightarrow x > y$

(ii) If $a > 1$, then $\log_a x < \alpha \Rightarrow 0 < x < a^\alpha$

(iii) If $a > 1$, then $\log_a x > \alpha \Rightarrow x > a^\alpha$

(iv) If $0 < a < 1$, then $\log_a x > \log_a y \Rightarrow 0 < x < y$

(v) If $0 < a < 1$, then $\log_a x < \alpha \Rightarrow x > a^\alpha$

Form - I : $f(x) > 0, g(x) > 0, g(x) \neq 1$

	Form		Collection of system
(a)	$\log_{g(x)} f(x) \geq 0$	$\Leftrightarrow$	$\begin{cases} f(x) \geq 1 & , \quad g(x) > 1 \\ 0 < f(x) \leq 1 & , \quad 0 < g(x) < 1 \end{cases}$
(b)	$\log_{g(x)} f(x) \leq 0$	$\Leftrightarrow$	$\begin{cases} f(x) \geq 1 & , \quad 0 < g(x) < 1 \\ 0 < f(x) \leq 1 & , \quad g(x) > 1 \end{cases}$
(c)	$\log_{g(x)} f(x) \geq a$	$\Leftrightarrow$	$\begin{cases} f(x) \geq (g(x))^a & , \quad g(x) > 1 \\ 0 < f(x) \leq (g(x))^a & , \quad 0 < g(x) < 1 \end{cases}$
(d)	$\log_{g(x)} f(x) \leq a$	$\Leftrightarrow$	$\begin{cases} 0 < f(x) \leq (g(x))^a & , \quad g(x) > 1 \\ f(x) \geq (g(x))^a & , \quad 0 < g(x) < 1 \end{cases}$

From - II : When the inequality of the form

	Form		Collection of system
(a)	$\log_{\phi(x)} f(x) \geq \log_{\phi(x)} g(x)$	$\Leftrightarrow$	$\begin{cases} f(x) \geq g(x), \phi(x) > 1, \\ 0 < f(x) \leq g(x); 0 < \phi(x) < 1 \end{cases}$
(b)	$\log_{\phi(x)} f(x) \leq \log_{\phi(x)} g(x)$	$\Leftrightarrow$	$\begin{cases} 0 < f(x) \leq g(x), \phi(x) > 1, \\ f(x) \geq g(x) > 0, 0 < \phi(x) < 1 \end{cases}$

Trigonometric Functions of Sum or Difference of Two Angles:

(a) $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

(b) $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$

(c) $\sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A = \sin(A+B).\sin(A-B)$

(d) $\cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A = \cos(A+B).\cos(A-B)$

(e) $\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

(f) $\cot(A \pm B) = \dfrac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$

(g) $\tan(A+B+C) = \dfrac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$.

Factorisation of the sum or difference of two sines or cosines :

(a) $\sin C + \sin D = 2 \sin \dfrac{C+D}{2} \cos \dfrac{C-D}{2}$ (b) $\sin C - \sin D = 2 \cos \dfrac{C+D}{2} \sin \dfrac{C-D}{2}$

(c) $\cos C + \cos D = 2 \cos \dfrac{C+D}{2} \cos \dfrac{C-D}{2}$ (d) $\cos C - \cos D = -2 \sin \dfrac{C+D}{2} \sin \dfrac{C-D}{2}$

Transformation of products into sum or difference of sines & cosines:

(a) $2 \sin A \cos B = \sin(A+B) + \sin(A-B)$ (b) $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$

(c) $2 \cos A \cos B = \cos(A+B) + \cos(A-B)$ (d) $2 \sin A \sin B = \cos(A-B) - \cos(A+B)$

Multiple and sub-multiple angles :

(a) $\sin 2A = 2 \sin A \cos A$; $\sin\theta = 2 \sin \dfrac{\theta}{2} \cos \dfrac{\theta}{2}$

(b) $\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$; $2\cos^2 \dfrac{\theta}{2} = 1 + \cos\theta$, $2\sin^2 \dfrac{\theta}{2} = 1 - \cos\theta$.

(c) $\tan 2A = \dfrac{2\tan A}{1 - \tan^2 A}$; $\tan\theta = \dfrac{2\tan\frac{\theta}{2}}{1 - \tan^2 \frac{\theta}{2}}$

(d) $\sin 2A = \dfrac{2\tan A}{1 + \tan^2 A}$, $\cos 2A = \dfrac{1 - \tan^2 A}{1 + \tan^2 A}$

(e) $\sin 3A = 3 \sin A - 4 \sin^3 A$

(f) $\cos 3A = 4 \cos^3 A - 3 \cos A$

(g) $\tan 3A = \dfrac{3\tan A - \tan^3 A}{1 - 3\tan^2 A}$

Important trigonometric ratios :

(a) $\sin n\pi = 0$; $\cos n\pi = (-1)^n$; $\tan n\pi = 0$, where $n \in I$

(b) $\sin 15^\circ$ or $\sin \dfrac{\pi}{12} = \dfrac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75^\circ$ or $\cos \dfrac{5\pi}{12}$;

$\cos 15^\circ$ or $\cos \dfrac{\pi}{12} = \dfrac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75^\circ$ or $\sin \dfrac{5\pi}{12}$;

$\tan 15^\circ = \dfrac{\sqrt{3}-1}{\sqrt{3}+1} = 2-\sqrt{3} = \cot 75^\circ$; $\tan 75^\circ = \dfrac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3} = \cot 15^\circ$

(c) $\sin \dfrac{\pi}{10}$ or $\sin 18^\circ = \dfrac{\sqrt{5}-1}{4}$ & $\cos 36^\circ$ or $\cos \dfrac{\pi}{5} = \dfrac{\sqrt{5}+1}{4}$

Conditional identities:

If $A + B + C = \pi$ then :

(i) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

(ii) $\sin A + \sin B + \sin C = 4 \cos\frac{A}{2} \cos\frac{B}{2} \cos\frac{C}{2}$

(iii) $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$

(iv) $\cos A + \cos B + \cos C = 1 + 4 \sin\frac{A}{2} \sin\frac{B}{2} \sin\frac{C}{2}$

(v) $\tan A + \tan B + \tan C = \tan A \tan B \tan C$

(vi) $\tan\frac{A}{2}\tan\frac{B}{2} + \tan\frac{B}{2}\tan\frac{C}{2} + \tan\frac{C}{2}\tan\frac{A}{2} = 1$

(vii) $\cot\frac{A}{2} + \cot\frac{B}{2} + \cot\frac{C}{2} = \cot\frac{A}{2} \cdot \cot\frac{B}{2} \cdot \cot\frac{C}{2}$

(viii) $\cot A \cot B + \cot B \cot C + \cot C \cot A = 1$

(ix) If $A + B + C = \frac{\pi}{2}$, then $\tan A \tan B + \tan B \tan C + \tan C \tan A = 1$

Range of trigonometric expression :

$E = a \sin\theta + b \cos\theta$

$E = \sqrt{a^2+b^2} \sin(\theta + \alpha)$, where $\tan\alpha = \frac{b}{a}$ $= \sqrt{a^2+b^2} \cos(\theta - \beta)$, where $\tan\beta = \frac{a}{b}$

Hence for any real value of θ, $-\sqrt{a^2+b^2} \le E \le \sqrt{a^2+b^2}$

Sine and cosine series:

$$\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + \ldots\ldots + \sin\left(\alpha + \overline{n-1}\beta\right) = \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}} \sin\left(\alpha + \frac{n-1}{2}\beta\right)$$

$$\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + \ldots\ldots + \cos\left(\alpha + \overline{n-1}\beta\right) = \frac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}} \cos\left(\alpha + \frac{n-1}{2}\beta\right)$$

Trigonometric equations

Principal Solutions:

The solutions of a trigonometric equation which lie in the interval $[0, 2\pi)$ are called **Principal solutions.**

General solution :

The expression involving an integer 'n' which gives all solutions of a trigonometric equation is called general solution.

General solution of some standard trigonometric equations are given below :

(i) If $\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n \alpha$ where $\alpha \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $n \in I$.

(ii) If $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha$ where $\alpha \in [0, \pi]$, $n \in I$.

(iii) If $\tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha$ where $\alpha \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $n \in I$.

(iv) If $\sin^2\theta = \sin^2\alpha \Rightarrow \theta = n\pi \pm \alpha$; $n \in I$ **(v)** $\cos^2\theta = \cos^2\alpha \Rightarrow \theta = n\pi \pm \alpha$; $n \in I$

(vi) $\tan^2\theta = \tan^2\alpha \Rightarrow \theta = n\pi \pm \alpha$; $n \in I$ [Note: α is called the principal angle]

Trigonometric Inequalities :

Solutions of elementary trigonometric inequalities are obtained from graphs.

Inequality	**Set of solutions of inequality ($n \in z$)**
$\sin x > a$ ($\|a\| < 1$)	$x \in (\sin^{-1}a + 2\pi n, \pi - \sin^{-1}a + 2\pi n)$
$\sin x < a$ ($\|a\| < 1$)	$x \in (-\pi - \sin^{-1}a + 2\pi n, \sin^{-1}a + 2\pi n)$
$\cos x > a$ ($\|a\| < 1$)	$x \in (-\cos^{-1} a + 2\pi n, \cos^{-1} a + 2\pi n)$
$\cos x < a$ ($\|a\| < 1$)	$x \in (\cos^{-1} a + 2\pi n, 2\pi - \cos^{-1} a + 2\pi n)$
$\tan x > a$	$x \in (\tan^{-1} a + \pi n, \pi/2 + \pi n)$
$\tan x < a$	$x \in \left(-\frac{\pi}{2} + \pi n, \tan^{-1} a + \pi n\right)$

Inequalities of the form $R(y) > 0$, $R(y) < 0$, where R is a certain rational function and y is a trigonometric function (sine, cosine or tangent), are usually solved in two stages: first the rational inequality is solved for the unknown y and then follows the solution of an elementary trigonometric inequality.

* * * * *

QUADRATIC EQUATION

1. Polynomial :

A function f defined by $f(x) = a_n x^n + a_{n-1}x^{n-1} + ...+ a_1 x + a_0$
where $a_0, a_1, a_2, ..., a_n \in R$ is called a polynomial of degree n with real coefficients ($a_n \neq 0$, $n \in W$). If $a_0, a_1, a_2, ..., a_n \in C$, it is called a polynomial with complex coefficients.

2. Quadratic polynomial & Quadratic equation :

A polynomial of degree 2 is known as quadratic polynomial. Any equation f(x) = 0, where f is a quadratic polynomial, is called a quadratic equation. The general form of a quadratic equation is

$ax^2 + bx + c = 0$ .....(i)

Where a, b, c are real numbers, $a \neq 0$.

If a = 0, then equation (i) becomes linear equation.

3. Difference between equation & identity :

If a statement is true for all the values of the variable, such statements are called as identities. If the statement is true for some or no values of the variable, such statements are called as equations.

Example : (i) $(x + 3)^2 = x^2 + 6x + 9$ is an identity
(ii) $(x + 3)^2 = x^2 + 6x + 8$, is an equation having no root.
(iii) $(x + 3)^2 = x^2 + 5x + 8$, is an equation having – 1 as its root.

A quadratic equation has exactly two roots which may be real (equal or unequal) or imaginary.

$ax^2 + bx + c = 0$ is:

★	a quadratic equation if	$a \neq 0$	Two Roots
★	a linear equation if	$a = 0, b \neq 0$	One Root
★	a contradiction if	$a = b = 0, c \neq 0$	No Root
★	an identity if	$a = b = c = 0$	Infinite Roots

If $ax^2 + bx + c = 0$ is satisfied by three distinct values of 'x', then it is an identity.

4. Relation Between Roots & Co-efficients:

(i) The solutions of quadratic equation, $ax^2 + bx + c = 0$, $(a \neq 0)$ is given by

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The expression, $b^2 - 4ac \equiv D$ is called discriminant of quadratic equation.

(ii) If α, β are the roots of quadratic equation, $ax^2 + bx + c = 0$ .....(i)
then equation (i) can be written as $a(x - \alpha)(x - \beta) = 0$
or $ax^2 - a(\alpha + \beta)x + a\alpha\beta = 0$ .....(ii)
equations (i) and (ii) are identical,

$\therefore$ by comparing the coefficients sum of the roots, $\alpha + \beta = -\frac{b}{a} = -\frac{\text{coefficient of } x}{\text{coefficient of } x^2}$

and product of the roots, $\alpha\beta = \frac{c}{a} = \frac{\text{constant term}}{\text{coefficient of } x^2}$

(iii) Dividing the equation (i) by a, $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$

$\Rightarrow$ $x^2 - \left(\frac{-b}{a}\right)x + \frac{c}{a} = 0$ $\Rightarrow$ $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

$\Rightarrow$ x^2 – (sum of the roots) x + (product of the roots) = 0

Hence we conclude that the quadratic equation whose roots are α & β is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

5. Theory Of Equations:

If $\alpha_1, \alpha_2, \alpha_3, \ldots\ldots \alpha_n$ are the roots of the equation;

$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \ldots + a_{n-1}x + a_n = 0$ where $a_0, a_1, \ldots, a_n$ are all real & $a_0 \neq 0$ then,

$$\sum \alpha_1 = -\frac{a_1}{a_0}, \sum \alpha_1\alpha_2 = +\frac{a_2}{a_0}, \sum \alpha_1\alpha_2\alpha_3 = -\frac{a_3}{a_0}, \ldots\ldots, \alpha_1\alpha_2\alpha_3 \ldots\ldots \alpha_n = (-1)^n\frac{a_n}{a_0}$$

Note : (i) If α is a root of the equation f(x) = 0, then the polynomial f(x) is exactly divisible by $(x - \alpha)$ or $(x - \alpha)$ is a factor of f(x) and conversely.

(ii) Every equation of n^{th} degree $(n \geq 1)$ has exactly n roots & if the equation has more than n roots, it is an identity.

(iii) If the coefficients of the equation f(x) = 0 are all real and $\alpha + i\beta$ is its root, then $\alpha - i\beta$ is also a root. i.e. imaginary roots occur in conjugate pairs.

(iv) An equation of odd degree will have odd number of real roots and an equation of even degree will have even numbers of real roots.

(v) If the coefficients in the equation are all rational & $\alpha + \sqrt{\beta}$ is one of its roots, then $\alpha - \sqrt{\beta}$ is also a root where $\alpha, \beta \in Q$ & β is not square of a rational number.

(vi) If there be any two real numbers 'a' & 'b' such that f(a) & f(b) are of opposite signs, then f(x) = 0 must have odd number of real roots (also atleast one real root) between 'a' and 'b'.

(vii) Every equation f(x) = 0 of degree odd has atleast one real root of a sign opposite to that of its last term. (If coefficient of highest degree term is positive).

6. Nature of Roots:

Consider the quadratic equation, $ax^2 + bx + c = 0$ having α, β as its roots;

$D \equiv b^2 - 4ac$

$D = 0$
Roots are equal i.e. $\alpha = \beta = -b/2a$
& the quadratic expression can be expressed as a perfect square of a linear polynomial

$D \neq 0$
Roots are unequal

$a, b, c \in R$ & $D > 0$
Roots are real

$a, b, c \in R$ & $D < 0$
Roots are imaginary $\alpha = p + iq$, $\beta = p - iq$

$a, b, c \in Q$ &
D is square of a rational number
$\Rightarrow$ Roots are rational

$a, b, c \in Q$ &
D is not square of a rational number
$\Rightarrow$ Roots are irrational
i.e. $\alpha = p + \sqrt{q}$, $\beta = p - \sqrt{q}$

$a = 1, b, c \in I$ & D is square of an integer

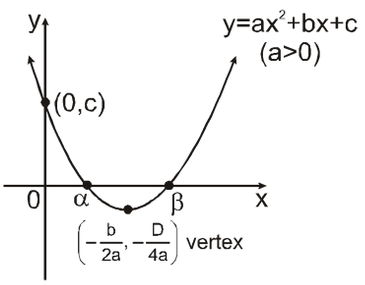
$\Rightarrow$ Roots are integral.

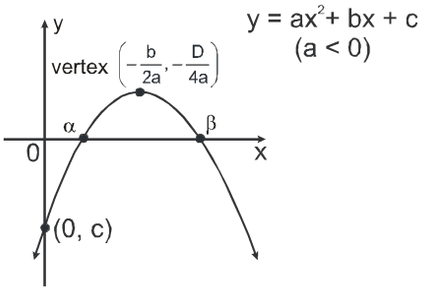
7. Graph of Quadratic Expression:

★ the graph between x, y is always a parabola.

★ the co–ordinate of vertex are $\left(-\frac{b}{2a}, -\frac{D}{4a}\right)$

★ If $a > 0$ then the shape of the parabola is concave upwards & if $a < 0$ then the shape of the parabola is concave downwards.





★ the parabola intersect the y–axis at point (0, c).

★ the x–co–ordinate of point of intersection of parabola with x–axis are the real roots of the quadratic equation $f(x) = 0$. Hence the parabola may or may not intersect the x–axis.

8. Range of Quadratic Expression $f(x) = ax^2 + bx + c$.

(i) Range :

If $a > 0 \quad \Rightarrow \quad f(x) \in \left[-\frac{D}{4a}, \infty\right)$

If $a < 0 \quad \Rightarrow \quad f(x) \in \left(-\infty, -\frac{D}{4a}\right]$

Hence maximum and minimum values of the expression $f(x)$ is $-\frac{D}{4a}$ in respective cases and it occurs at $x = -\frac{b}{2a}$ (at vertex).

(ii) Range in restricted domain:

Given $x \in [x_1, x_2]$

(a) If $-\frac{b}{2a} \notin [x_1, x_2]$ then,

$$f(x) \in [\min\{f(x_1), f(x_2)\}, \ \max\{f(x_1), f(x_2)\}]$$

(b) If $-\frac{b}{2a} \in [x_1, x_2]$ then,

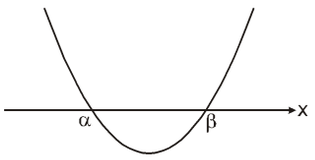
$$f(x) \in \left[\min\left\{f(x_1), f(x_2), -\frac{D}{4a}\right\}, \ \max\left\{f(x_1), f(x_2), -\frac{D}{4a}\right\}\right]$$

9. Sign of Quadratic Expressions :

The value of expression $f(x) = ax^2 + bx + c$ at $x = x_0$ is equal to y–co–ordinate of the point on parabola $y = ax^2 + bx + c$ whose x–co–ordinate is x_0. Hence if the point lies above the x–axis for some $x = x_0$, then $f(x_0) > 0$ and vice–versa.

We get six different positions of the graph with respect to x–axis as shown.

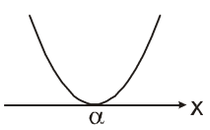
(i)



Conclusions :

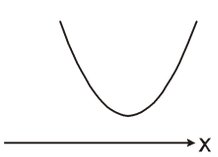
(a) a > 0
(b) D > 0
(c) Roots are real & distinct.
(d) $f(x) > 0$ in $x \in (-\infty, \alpha) \cup (\beta, \infty)$
(e) $f(x) < 0$ in $x \in (\alpha, \beta)$

(ii)



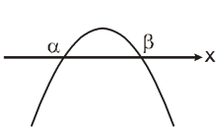
(a) a > 0
(b) D = 0
(c) Roots are real & equal.
(d) $f(x) > 0$ in $x \in R - \{\alpha\}$

(iii)



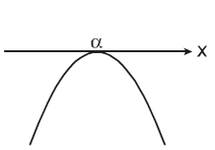
(a) a > 0
(b) D < 0
(c) Roots are imaginary.
(d) $f(x) > 0 \ \forall \ x \in R$.

(iv)



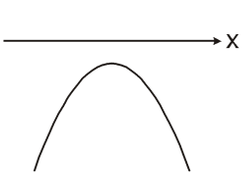
(a) a < 0
(b) D > 0
(c) Roots are real & distinct.
(d) $f(x) < 0$ in $x \in (-\infty, \alpha) \cup (\beta, \infty)$
(e) $f(x) > 0$ in $x \in (\alpha, \beta)$

(v)



(a) a < 0
(b) D = 0
(c) Roots are real & equal.
(d) $f(x) < 0$ in $x \in R - \{\alpha\}$

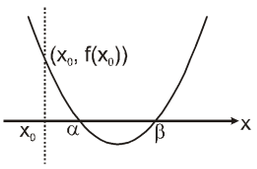
(vi)

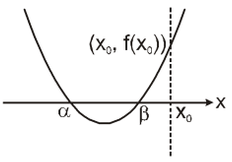


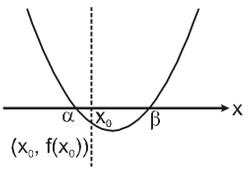
(a) a < 0
(b) D < 0
(c) Roots are imaginary.
(d) $f(x) < 0 \ \forall \ x \in R$.

10. Location of Roots :

Let $f(x) = ax^2 + bx + c$, where $a > 0$ & $a, b, c \in R$.

(i) 

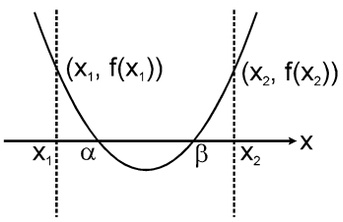
(ii) 

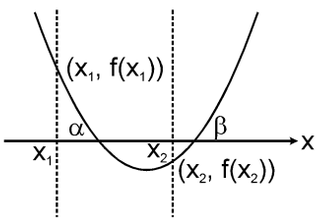
(iii) 

(i) Conditions for both the roots of $f(x) = 0$ to be greater than a specified number 'x_0' are $b^2 - 4ac \geq 0$ & $f(x_0) > 0$ & $(-b/2a) > x_0$.

(ii) Conditions for both the roots of $f(x) = 0$ to be smaller than a specified number 'x_0' are $b^2 - 4ac \geq 0$ & $f(x_0) > 0$ & $(-b/2a) < x_0$.

(iii) Conditions for a number 'x_0' to lie between the roots of $f(x) = 0$ is $f(x_0) < 0$.

(iv) 

(v) 

(iv) Conditions that both roots of $f(x) = 0$ to be confined between the numbers x_1 and x_2, $(x_1 < x_2)$ are $b^2 - 4ac \geq 0$ & $f(x_1) > 0$ & $f(x_2) > 0$ & $x_1 < (-b/2a) < x_2$.

(v) Conditions for exactly one root of $f(x) = 0$ to lie in the interval (x_1, x_2) i.e. $x_1 < x < x_2$ is $f(x_1).f(x_2) < 0$.

11. Common Roots:

Consider two quadratic equations, $a_1x^2 + b_1x + c_1 = 0$ & $a_2x^2 + b_2x + c_2 = 0$.

(i) If two quadratic equations have both roots common, then the equations are identical and their co-efficient are in proportion.

i.e. $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$.

(ii) If only one root is common, then the common root 'α' will be :

$$\alpha = \frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1} = \frac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1}$$

Hence the condition for one common root is :

$$\Rightarrow \quad (c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$$

Note : If $f(x) = 0$ & $g(x) = 0$ are two polynomial equation having some common root(s) then those common root(s) is/are also the root(s) of $h(x) \equiv a f(x) + bg(x) = 0$.

* * * * *

FUNCTIONS

Definition :

Function is a rule (or correspondence), from a non empty set A to a non empty set B, that associates each member of A to a unique member of B. Symbolically, we write f: A $\rightarrow$ B. We read it as "f is a function from A to B".

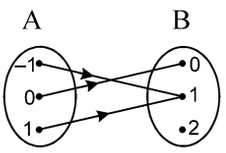
For example, let A $\equiv$ {–1, 0, 1} and B $\equiv$ {0, 1, 2}.

Then A × B $\equiv$ {(–1, 0), (–1, 1), (–1, 2), (0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2)}

Now, " f : A $\rightarrow$ B defined by $f(x) = x^2$ " is the function such that

f $\equiv$ {(–1, 1), (0, 0), (1, 1)}

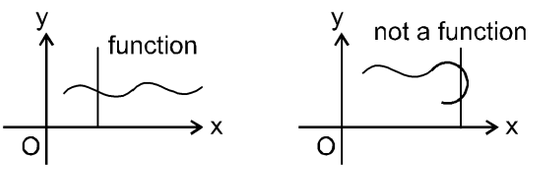
f can also be shown diagramatically by following mapping.



Note : Every function say y = f(x) : A $\rightarrow$ B. Here x is independent variable which takes its values from A while 'y' takes its value from B. A relation will be a function if and only if

(i) x must be able to take each and every value of A and

and (ii) one value of x must be related to one and only one value of y in set B.

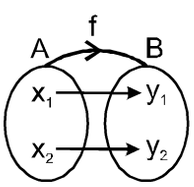


(a) (b)

Graphically : If any vertical line cuts the graph at more than one point, then the graph does not represent a function.

Domain, Co-domain and Range of a Function :

Let y = f(x) : A $\rightarrow$ B, then the set A is known as the domain of f and the set B is known as co-domain of f.



If x_1 is mapped to y_1, then y_1 is called as image of x_1 under f. Further x_1 is a pre-image of y_1 under f.

If only expression of f(x) is given (domain and co-domain are not mentioned), then domain is **complete** set of those values of x for which f(x) is real, while codomain is considered to be $(-\infty, \infty)$ (except in inverse trigonometric functions).

Range is the complete set of values that y takes. Clearly range is a subset of Co-domain.

A function whose domain and range are both subsets of real numbers is called **a real function**.

Algebraic Operations on Functions :

If f and g are real valued functions of x with domain set A and B respectively, then both f and g are defined in $A \cap B$. Now we define $f+g$, $f-g$, $(f \cdot g)$ and (f/g) as follows:

(i) $(f \pm g)(x) = f(x) \pm g(x)$

(ii) $(f.g)(x) = f(x).\ g(x)$

domain in each case is $A \cap B$

(iii) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$ domain is $\{x \mid x \in A \cap B \text{ such that } g(x) \neq 0\}$.

Note : ☞ For domain of $\phi(x) = \{f(x)\}^{g(x)}$, conventionally, the conditions are $f(x) > 0$ and $g(x)$ must be real.

☞ For domain of $\phi(x) = {}^{f(x)}C_{g(x)}$ or $\phi(x) = {}^{f(x)}P_{g(x)}$ conventional conditions of domain are $f(x) \geq g(x)$ and $f(x) \in N$ and $g(x) \in W$

Methods of determining range :

(i) **Representing x in terms of y**

If $y = f(x)$, try to express as $x = g(y)$, then domain of $g(y)$ represents possible values of y, which is range of f(x).

(ii) **Graphical Method :**

The set of y– coordinates of the graph of a function is the range.

(iii) Using monotonocity : Many of the functions are monotonic increasing or monotonic decreasing. In case of monotonic continuous functions the minimum and maximum values lie at end points of domain. Some of the common function which are increasing or decreasing in the interval where they are **continuous** is as under.

Monotonic increasing	Monotonic decreasing
$\log_a x,\ a > 1$	$\log_a x,\ 0 < a < 1$
e^x	e^{-x}
$\sin^{-1} x$	$\cos^{-1} x$
$\tan^{-1} x$	$\cot^{-1} x$
$\sec^{-1} x$	$\text{cosec}^{-1} x$

For monotonic increasing functions in [a, b]

(i) $f'(x) \geq 0$

(ii) range is [f(a), f(b)]

for monotonic decreasing functions in [a, b]

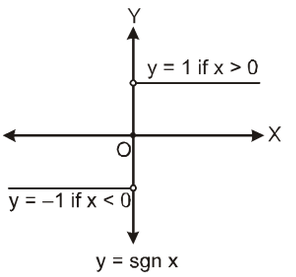
(i) $f'(x) \leq 0$

(ii) range is [f(b), f(a)]

Various Types of Functions :

(i) Signum Function : (Also known as sgn(x))

A function f (x) = sgn (x) is defined as follows :



$$f(x) = \text{sgn}(x) = \begin{cases} 1 & \text{for } x > 0 \\ 0 & \text{for } x = 0 \\ -1 & \text{for } x < 0 \end{cases}$$

It is also written as $\text{sgn } x = \begin{cases} \frac{|x|}{x}; & x \neq 0 \\ 0; & x = 0 \end{cases}$

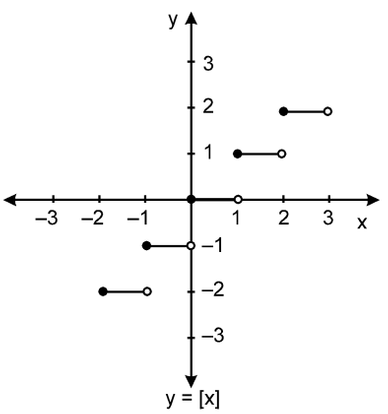
Note : $\text{sgn } f(x) = \begin{cases} \frac{|f(x)|}{f(x)}; & f(x) \neq 0 \\ 0; & f(x) = 0 \end{cases}$

(ii) Greatest Integer Function or Step Function :

The function y = f (x) = [x] is called the greatest integer function, where [x] equals to the greatest integer less than or equal to x. For example :

for $-1 \le x < 0$; $[x] = -1$; for $0 \le x < 1$; $[x] = 0$

for $1 \le x < 2$; $[x] = 1$; for $2 \le x < 3$; $[x] = 2$ and so on.



Properties of greatest integer function :

(a) $x - 1 < [x] \le x$

(b) If m is an integer, then $[x \pm m] = [x] \pm m$.

(c) $[x] + [y] \le [x + y] \le [x] + [y] + 1$

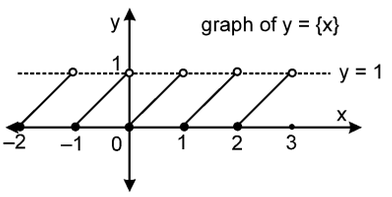
(d) $[x] + [-x] = \begin{cases} 0, & \text{if } x \text{ is an integer} \\ -1, & \text{if } x \text{ is not an integer} \end{cases}$

(iii) Fractional Part Function:

It is defined as, $y = \{x\} = x - [x]$, where [.] denotes greatest integer function.

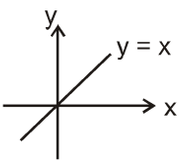
e.g. the fractional part of the number 2.1 is 2.1 – 2 = 0.1 and {–3.7} = 0.3.

The period of this function is 1 and graph of this function is as shown.



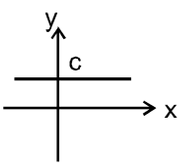
(iv) Identity function :

The function $f : A \to A$ defined by, $f(x) = x, \forall x \in A$ is called the identity function on A and is denoted by I_A. It is easy to observe
that identity function is a bijection.



(v) Constant function :

A function $f : A \to B$ is said to be a constant function, if every element of A has the same f image in B. Thus $f : A \to B$; $f(x) = c, \forall x \in A, c \in B$ is a constant function.



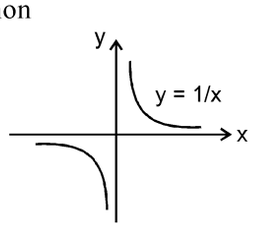
Equal or Identical Functions :

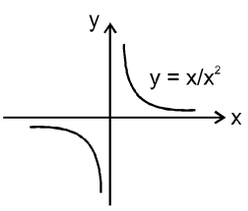
Two functions f and g are said to be identical (or equal) iff :

(i) The domain of f ≡ the domain of g.

(ii) $f(x) = g(x)$, for every x belonging to their common domain.

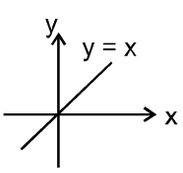
e.g. $f(x) = \frac{1}{x}$ and $g(x) = \frac{x}{x^2}$ are identical functions.

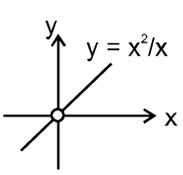




Clearly the graphs of f(x) and g(x) are exactly same

But $f(x) = x$ and $g(x) = \frac{x^2}{x}$ are not identical functions.





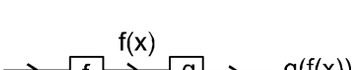
Clearly the graphs of f(x) and g(x) are different at x = 0.

Composite Function :

Let $f: X \to Y_1$ and $g: Y_2 \to Z$ be two functions and D is the set of values of x such that if $x \in X$, then $f(x) \in Y_2$. If $D \neq \phi$, then the function h defined on D by $h(x) = g\{f(x)\}$ is called composite function of g and f and is denoted by gof. It is also called function of a function.

Note : ☞ Domain of gof is D which is a subset of X (the domain of f). Range of gof is a subset of the range of g. If D = X, then $f(X) \subseteq Y_2$.

Pictorially gof(x) can be viewed as under



Note that gof(x) exists only for those x when range of f(x) is a subset of domain of g(x).

Properties of Composite Functions :

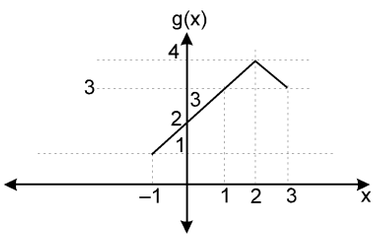
(a) In general gof ≠ fog (i.e. not commutative)

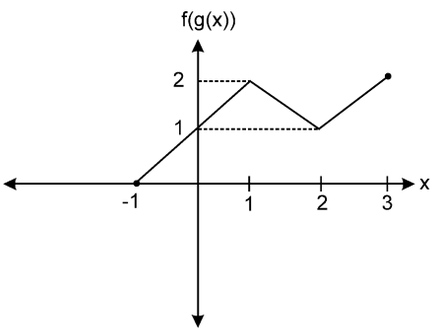
(b) The composition of functions are associative i.e. if three functions f, g, h are such that fo (goh) and (fog) oh are defined, then fo (goh) = (fog) oh.

Alternate method for finding fog

$$g(x) = \begin{cases} 2+x & -1 \le x < 2 \\ 6-x & 2 \le x \le 3 \end{cases}$$

graph of g(x) is





$$\therefore \quad fog(x) = \begin{cases} 1-g(x) & 0 \le g(x) < 1 \\ g(x)-1 & 1 \le g(x) < 3 \\ 5-g(x) & 3 \le g(x) \le 4 \end{cases}$$

$$= \begin{cases} 1-g(x) & \text{for no value} \\ g(x)-1 & -1 \le x < 1 \\ 5-g(x) & 1 \le x \le 3 \end{cases} = \begin{cases} 2+x-1 & -1 \le x < 1 \\ 5-(2+x) & 1 \le x < 2 \\ 5-(6-x) & 2 \le x \le 3 \end{cases} = \begin{cases} x+1 & -1 \le x < 1 \\ 3-x & 1 \le x < 2 \\ x-1 & 2 \le x \le 3 \end{cases}$$

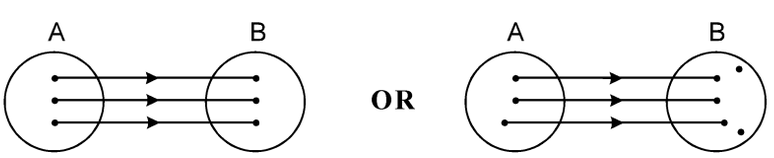
Classification of Functions :

Functions can be classified as "One – One Function (Injective Mapping)" and "Many – One Function":

One - One Function :

A function f : A → B is said to be a one-one function or injective mapping if different elements of A have different f images in B.

Thus for $x_1, x_2 \in A$ and $f(x_1), f(x_2) \in B$, $f(x_1) = f(x_2) \Leftrightarrow x_1 = x_2$ or $x_1 \ne x_2 \Leftrightarrow f(x_1) \ne f(x_2)$.
Diagrammatically an injective mapping can be shown as



Many - One function :

A function f : A → B is said to be a many one function if there exist at least two or more elements of A having the same f image in B.

Thus f : A → B is many one iff there exist atleast two elements $x_1, x_2 \in A$, such that $f(x_1) = f(x_2)$ but $x_1 \neq x_2$.

Diagrammatically a many one mapping can be shown as

Note : ☞ If a function is one–one, it cannot be many–one and vice versa.

Methods of determining whether a given function is ONE-ONE or MANY-ONE :

(a) If $x_1, x_2 \in A$ and $f(x_1), f(x_2) \in B$, equate $f(x_1)$ and $f(x_2)$ and if it implies that $x_1 = x_2$, then and only then function is ONE-ONE otherwise MANY-ONE.

(b) If there exists a straight line parallel to x-axis, which cuts the graph of the function atleast at two points, then the function is MANY-ONE, otherwise ONE-ONE.

(c) If either $f'(x) \geq 0$, $\forall$ x $\in$ domain or $f'(x) \leq 0$ $\forall$ x $\in$ domain, where equality can hold at discrete point(s) only i.e. strictly monotonic, then function is ONE-ONE, otherwise MANY-ONE.

Note : If f and g both are one-one, then gof and fog would also be one-one (if they exist). Functions can also be classified as "Onto function (Surjective mapping)" and "Into function":

Onto function :

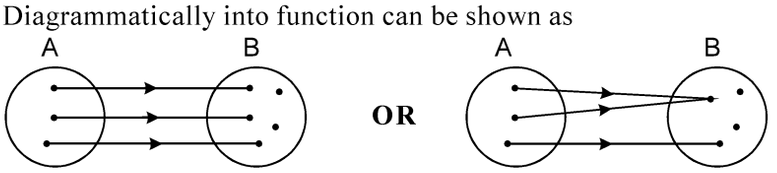
If the function f : A → B is such that each element in B (co–domain) must have atleast one pre–image in A, then we say that f is a function of A 'onto' B. Thus f : A → B is surjective iff $\forall$ b $\in$ B, there exists some a $\in$ A such that f (a) = b.

Diagrammatically surjective mapping can be shown as

Into function :

If f : A → B is such that there exists atleast one element in co–domain which is not the image of any element in domain, then f(x) is into.

Diagrammatically into function can be shown as

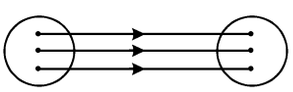


Note : (i) If range ≡ co–domain, then f(x) is onto, otherwise into

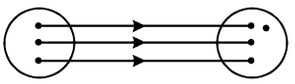
(ii) If a function is onto, it cannot be into and vice versa.

A function can be one of these four types:

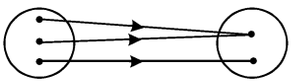
(a) one–one onto (injective and surjective)



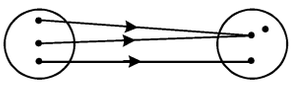
(b) one–one into (injective but not surjective)



(c) many–one onto (surjective but not injective)



(d) many–one into (neither surjective nor injective)



Note : (i) If f is both injective and surjective, then it is called a **bijective** mapping. The bijective functions are also named as invertible, non singular or biuniform functions.

(ii) If a set A contains 'n' distinct elements, then the number of different functions defined from $A \to A$ is n^n and out of which n! are one one.

(iii) If f and g both are onto, then gof or fog may or may not be onto.

(iv) The composite of two bijections is a bijection iff f and g are two bijections such that gof is defined, then gof is also a bijection only **when co-domain of f is equal to the domain of g**.

Odd and Even Functions :

(i) If $f(-x) = f(x)$ for all x in the domain of 'f', then f is said to be an even function.
e.g. $f(x) = \cos x$; $g(x) = x^2 + 3$.

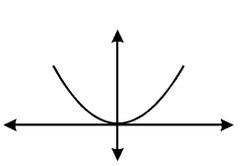
(ii) If $f(-x) = -f(x)$ for all x in the domain of 'f', then f is said to be an odd function.
e.g. $f(x) = \sin x$; $g(x) = x^3 + x$.

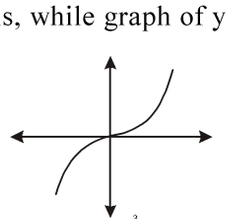
Note : (i) A function may neither be odd nor even. (e.g. $f(x) = e^x$, $\cos^{-1}x$)

(ii) If an odd function is defined at x = 0, then f(0) = 0

Properties of Even/Odd Function

(a) The graph of every even function is symmetric about the y–axis and that of every odd function is symmetric about the origin.
For example graph of $y = x^2$ is symmetric about y-axis, while graph of $y = x^3$ is symmetric about origin





(b) All functions (whose domain is symmetrical about origin) can be expressed as the sum of an even and an odd function, as follows

$$f(x) = \underbrace{\frac{f(x)+f(-x)}{2}}_{\text{even}} + \underbrace{\frac{f(x)-f(-x)}{2}}_{\text{odd}}$$

(c) The only function which is defined on the entire number line and is even and odd at the same time is f(x) = 0.

(d) If f and g both are even or both are odd, then the function f.g will be even but if any one of them is odd and the other even then f.g will be odd.

(e) If f(x) is even then f'(x) is odd while derivative of odd function is even. Note that same cannot be said for integral of functions.

Even extension / Odd extension :

Let f(x) be defined in [a, b] where $ab \geq 0$. Even extension of this function implies to define the function in [–b, –a] to make it even. In order to get even extension replace x by –x in the given definition.
Similarly, odd extension implies to define the function in [–b, –a] to make it odd. In order to get odd extension, multiply the definition of even extension by –1

Periodic Functions :

A function f(x) is called periodic with a period T if there exists a real number $T > 0$ such that for each x in the domain of f the numbers $x - T$ and $x + T$ are also in the domain of f and $f(x) = f(x + T)$ for all x in the domain of f(x). Graph of a periodic function with period T is repeated after every interval of 'T'.
e.g. The function sin x and cos x both are periodic over 2π and tan x is periodic over π.
The least positive period is called the principal or fundamental period of f(x) or simply the period of the function.

Note : Inverse of a periodic function does not exist.

Properties of Periodic Functions :

(a) If f(x) has a period T, then $\frac{1}{f(x)}$ and $\sqrt{f(x)}$ also have a period T.

(b) If f(x) has a period T, then f(ax + b) has a period $\frac{T}{|a|}$.

(c) Every constant function defined for all real x, is always periodic, with no fundamental period.

(d) If f(x) has a period T_1 and g(x) also has a period T_2 then period of $f(x) \pm g(x)$ or f(x) . g(x) or $\frac{f(x)}{g(x)}$ is L.C.M. of T_1 and T_2 provided their L.C.M. exists. However that L.C.M. (if exists) need not to be fundamental period. If L.C.M. does not exists then $f(x) \pm g(x)$ or f(x) . g(x) or $\frac{f(x)}{g(x)}$ is nonperiodic.

L.C.M. of $\left(\frac{a}{b}, \frac{p}{q}, \frac{\ell}{m}\right) = \frac{\text{L.C.M. } (a, p, \ell)}{\text{H.C.F. } (b, q, m)}$

e.g. $|\sin x|$ has the period π, $|\cos x|$ also has the period π

$\therefore$ $|\sin x| + |\cos x|$ also has a period π. But the fundamental period of $|\sin x| + |\cos x|$ is $\frac{\pi}{2}$.

(e) If g is a function such that gof is defined on the domain of f and f is periodic with T, then gof is also periodic with T as one of its periods.

Inverse of a Function :

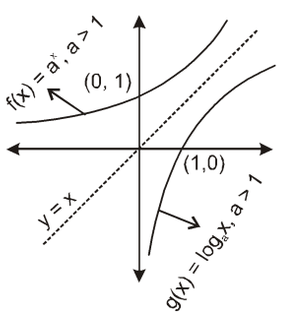
Let $y = f(x) : A \to B$ be a one-one and onto function. i.e. bijection, then there will always exist bijective function $x = g(y) : B \to A$ such that if (p, q) is an element of f, (q, p) will be an element of g and the functions f(x) and g(x) are said to be inverse of each other. g(x) is also denoted by $f^{-1}(x)$ and f(x) is denoted by $g^{-1}(x)$

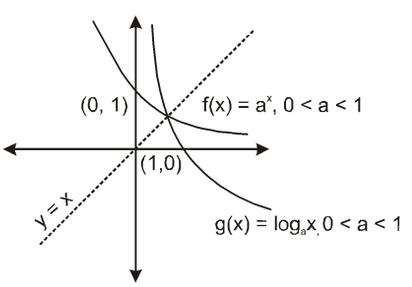
Note : (i) The inverse of a bijection is unique.

(ii) Inverse of an even function is not defined.

Properties of Inverse Function :

(a) The graphs of f and g are the mirror images of each other in the line y = x. For example $f(x) = a^x$ and $g(x) = \log_a x$ are inverse of each other, and their graphs are mirror images of each other on the line y = x as shown below.





(b) Normally points of intersection of f and f^{-1} lie on the straight line y = x. However it must be noted that f(x) and $f^{-1}(x)$ may intersect otherwise also. e.g f(x) = 1/x

(c) In general fog(x) and gof(x) are not equal. But if f and g are inverse of each other, then gof = fog. fog(x) and gof(x) can be equal even if f and g are not inverse of each other. e.g. f(x) = x + 1, g(x) = x + 2. However if fog(x) = gof(x) = x, then $g(x) = f^{-1}(x)$

(d) If f and g are two bijections $f : A \to B$, $g : B \to C$, then the inverse of gof exists and $(gof)^{-1} = f^{-1} \circ g^{-1}$.

(e) If f(x) and g(x) are inverse function of each other, then $f'(g(x)) = \dfrac{1}{g'(x)}$

General tips :

If x, y are independent variables, then:

(i) $f(xy) = f(x) + f(y) \Rightarrow f(x) = k \ln x$ or $f(x) = 0$.

(ii) $f(xy) = f(x).\ f(y) \Rightarrow f(x) = x^n$, $n \in R$

(iii) $f(x + y) = f(x).\ f(y) \Rightarrow f(x) = a^{kx}$.

(iv) $f(x + y) = f(x) + f(y) \Rightarrow f(x) = kx$, where k is a constant.

* * * * *

SETs And Relation

SET

A set is a collection of well defined objects which are distinct from each other. Set are generally denoted by capital letters A, B, C, ... etc. and the elements of the set by small letters a, b, c ..... etc.
If a is an element of a set A, then we write $a \in A$ and say a belongs to A.
If a does not belong to A then we write $a \notin A$,
e.g. the collection of first five prime natural numbers is a set containing the elements 2, 3, 5, 7, 11.

METHODS TO WRITE A SET :

(i) Roster Method or Tabular Method : In this method a set is described by listing elements, separated by commas and enclose then by curly brackets. Note that while writing the set in roster form, an element is not generally repeated e.g. the set of letters of word SCHOOL may be written as {S, C, H, O, L}.

(ii) Set builder form (Property Method) : In this we write down a property or rule which gives us all the element of the set.
$A = \{x : P(x)\}$ where P(x) is the property by which $x \in A$ and colon (:) stands for 'such that'

TYPES OF SETS

Null set or empty set : A set having no element in it is called an empty set or a null set or void set, it is denoted by ϕ or { }. A set consisting of at least one element is called a non-empty set or a non-void set.

Singleton set : A set consisting of a single element is called a singleton set.

Finite set : A set which has only finite number of elements is called a finite set.

Order of a finite set : The number of elements in a finite set A is called the order of this set and denoted by O(A) or n(A). It is also called cardinal number of the set.
e.g. $A = \{a, b, c, d\} \Rightarrow n(A) = 4$

Infinite set : A set which has an infinite number of elements is called an infinite set.

Equal sets : Two sets A and B are said to be equal if every element of A is member of B, and every element of B is a member of A. If sets A and B are equal, we write A = B and if A and B are not equal then $A \neq B$

Equivalent sets : Two finite sets A and B are equivalent if their number of elements are same
i.e. n(A) = n(B)
e.g. $A = \{1, 3, 5, 7\}, B = \{a, b, c, d\}$
$\Rightarrow$ n(A) = 4 and n(B) = 4
$\Rightarrow$ A and B are equivalent sets

Note - Equal sets are always equivalent but equivalent sets may not be equal

SUBSET AND SUPERSET :

Let A and B be two sets. If every element of A is an element B then A is called a subset of B and B is called superset of A. We write it as $A \subseteq B$.

e.g. $A = \{1, 2, 3, 4\}$ and $B = \{1, 2, 3, 4, 5, 6, 7\}$ $\Rightarrow$ $A \subseteq B$

If A is not a subset of B then we write $A \not\subseteq B$

PROPER SUBSET :

If A is a subset of B but $A \neq B$ then A is a proper subset of B and we write $A \subset B$. Set A is not proper subset of A so this is improper subset of A

Note : (i) Every set is a subset of itself

(ii) Empty set ϕ is a subset of every set

(iii) $A \subseteq B$ and $B \subseteq A \Leftrightarrow A = B$

(iv) The total number of subsets of a finite set containing n elements is 2^n.

(v) Number of proper subsets of a set having n elements is $2^n - 1$.

(vi) Empty set ϕ is proper subset of every set except itself.

POWER SET :

Let A be any set. The set of all subsets of A is called power set of A and is denoted by P(A)

UNIVERSAL SET :

A set consisting of all possible elements which occur in the discussion is called a universal set and is denoted by U.

e.g. if $A = \{1, 2, 3\}$, $B = \{2, 4, 5, 6\}$, $C = \{1, 3, 5, 7\}$ then $U = \{1, 2, 3, 4, 5, 6, 7\}$ can be taken as the universal set.

SOME OPERATION ON SETS :

(i) Union of two sets : $A \cup B = \{x : x \in A \text{ or } x \in B\}$

e.g. $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ then $A \cup B = \{1, 2, 3, 4\}$

(ii) Intersection of two sets : $A \cap B = \{x : x \in A \text{ and } x \in B\}$

e.g. $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ then $A \cap B = \{2, 3\}$

(iii) Difference of two sets : $A - B = \{x : x \in A \text{ and } x \notin B\}$. It is also written as $A \cap B'$.

Similarly $B - A = B \cap A'$ e.g. $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$; $A - B = \{1\}$

(iv) Symmetric difference of sets : It is denoted by $A \Delta B$ and $A \Delta B = (A - B) \cup (B - A)$

(v) Complement of a set : $A' = \{x : x \notin A \text{ but } x \in U\} = U - A$

e.g. $U = \{1, 2, ..., 10\}$, $A = \{1, 2, 3, 4, 5\}$ then $A' = \{6, 7, 8, 9, 10\}$

(vi) Disjoint sets : If $A \cap B = \phi$, then A, B are disjoint

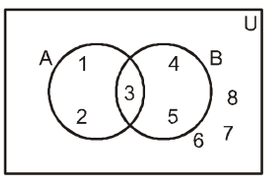
e.g. If $A = \{1, 2, 3\}$, $B = \{7, 8, 9\}$ then $A \cap B = \phi$

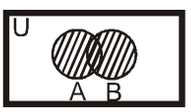
VENN DIAGRAM :

Most of the relationships between sets can be represented by means of diagrams which are known as venn

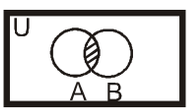
diagrams.These diagrams consist of a rectangle for universal set and circles in the rectangle for subsets of universal set. The elements of the sets are written in respective circles.

For example If A = {1, 2, 3}, B = {3, 4, 5}, U = {1, 2, 3, 4, 5, 6, 7, 8} then their venn diagram is



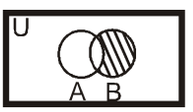


$A \cup B$

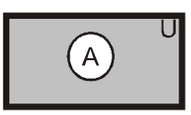


$A \cap B$

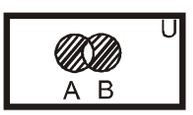
$A - B$



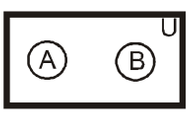
$B - A$



A'



$(A \Delta B) = (A - B) \cup (B - A)$



Disjoint

LAWS OF ALGEBRA OF SETS (PROPERTIES OF SETS):

(i) Commutative law : $(A \cup B) = B \cup A$; $A \cap B = B \cap A$

(ii) Associative law : $(A \cup B) \cup C = A \cup (B \cup C)$; $(A \cap B) \cap C = A \cap (B \cap C)$

(iii) Distributive law : $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$; $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(iv) De-morgan law : $(A \cup B)' = A' \cap B'$; $(A \cap B)' = A' \cup B'$

(v) Identity law : $A \cap U = A$; $A \cup \phi = A$

(vi) Complement law : $A \cup A' = U$, $A \cap A' = \phi$, $(A')' = A$

(vii) Idempotent law : $A \cap A = A$, $A \cup A = A$

NOTE :

(i) $A - (B \cup C) = (A - B) \cap (A - C)$; $A - (B \cap C) = (A - B) \cup (A - C)$

(ii) $A \cap \phi = \phi$, $A \cup U = U$

SOME IMPORTANT RESULTS ON NUMBER OF ELEMENTS IN SETS :

If A, B, C are finite sets and U be the finite universal set then

(i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

(iii) $n(A - B) = n(A) - n(A \cap B)$

(v) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(A \cap C) + n(A \cap B \cap C)$

(vi) Number of elements in exactly two of the sets A, B, C

$= n(A \cap B) + n(B \cap C) + n(C \cap A) - 3n(A \cap B \cap C)$

(vii) Number of elements in exactly one of the sets A, B, C

$= n(A) + n(B) + n(C) - 2n(A \cap B) - 2n(A \cap B) - 2n(B \cap C) - 2n(A \cap C) + 3n(A \cap B \cap C)$

RELATION

TYPES OF RELATIONS :

In this section we intend to define various types of relations on a given set A.

(i) **Void relation :** Let A be a set. Then $\phi \subseteq A \times A$ and so it is a relation on A. This relation is called the void or empty relation on A.

(ii) **Universal relation :** Let A be a set. Then $A \times A \subseteq A \times A$ and so it is a relation on A. This relation is called the universal relation on A.

(iii) **Identity relation :** Let A be a set. Then the relation $I_A = \{(a, a) : a \in A\}$ on A is called the identity relation on A. In other words, a relation I_A on A is called the identity relation if every element of A is related to itself only.

(iv) **Reflexive relation :** A relation R on a set A is said to be reflexive if every element of A is related to itself. Thus, R on a set A is not reflexive if there exists an element $a \in A$ such that $(a, a) \notin R$.

Note : Every identity relation is reflexive but every reflexive relation in not identity.

(v) **Symmetric relation :** A relation R on a set A is said to be a symmetric relation iff $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$. i.e. $a R b \Rightarrow b R a$ for all $a, b \in A$.

(vi) **Transitive relation :** Let A be any set. A relation R on A is said to be a transitive relation iff $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$ i.e. $a R b$ and $b R c \Rightarrow a R c$ for all $a, b, c \in A$

(vii) **Equivalence relation :** A relation R on a set A is said to be an equivalence relation on A iff

(i) it is reflexive i.e. $(a, a) \in R$ for all $a \in A$

(ii) it is symmetric i.e. $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$

(iii) it is transitive i.e. $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b \in A$

* * * * * *

INVERSE TRIGONOMETRIC FUNCTIONS

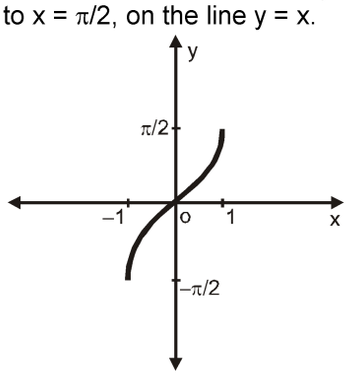
Introduction : The student may be familiar about trigonometric functions viz sin x, cos x, tan x, cosec x, sec x, cot x with respective domains R, R, R – {(2n + 1) π/2}, R – {nπ}, R – {(2n + 1) π/2}, R – {nπ} and respective ranges [–1, 1], [–1, 1], R, R – (–1, 1), R – (–1, 1), R.

Correspondingly, six inverse trigonometric functions (also called inverse circular functions) are defined.

$\sin^{-1}x$:

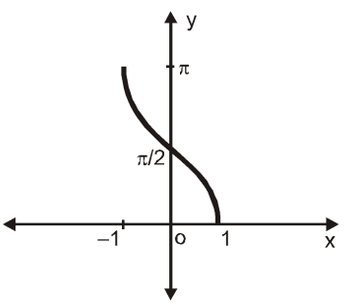
Thus $\sin^{-1}x$ is considered as a function with domain [–1, 1] and range [–π/2, π/2].

The graph of $y = \sin^{-1}x$ is as shown below, which is obtained by taking the mirror image, of the portion of the graph of y = sin x, from x = –π/2 to x = π/2, on the line y = x.

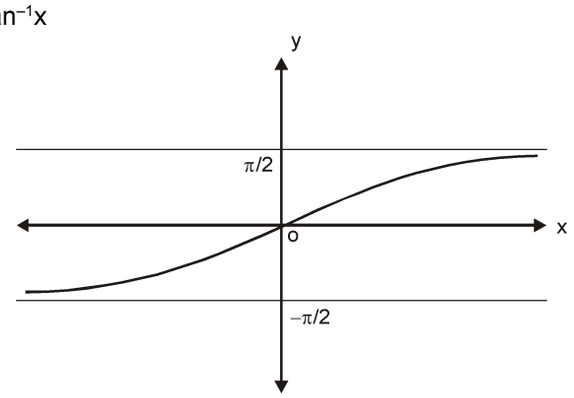


$\cos^{-1}x$: By following the discussions, similar to above, we have $\cos^{-1}x$ or arccos x as a function with domain [–1, 1] and range [0, π].

The graph of $y = \cos^{-1}x$ is similarly obtained as the mirror image of the portion of the graph of y = cos x from x = 0 to x = π.

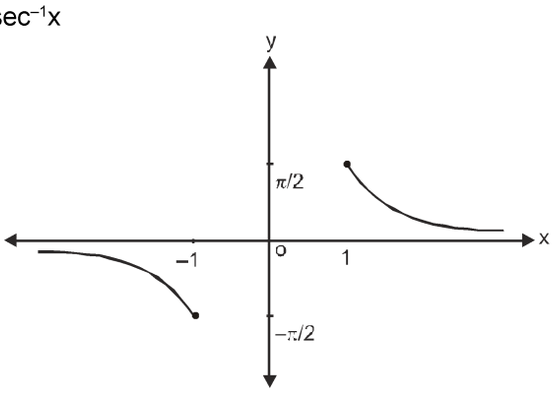


$\tan^{-1}x$: We get $\tan^{-1}x$ or arctanx as a function with domain R and range (–π/2, π/2).
Graph of $y = \tan^{-1}x$



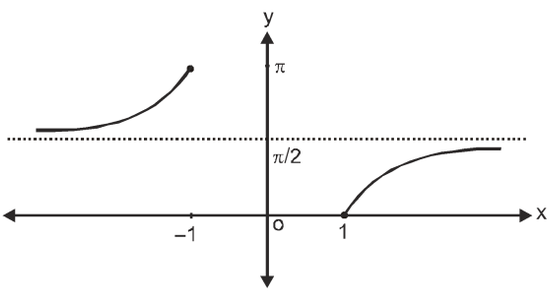
$\text{cosec}^{-1}x$: $\text{cosec}^{-1}x$ or arccosec x is a function with domain R – (–1, 1) and range [–π/2, π/2] – {0}.

Graph of $y = \text{cosec}^{-1}x$



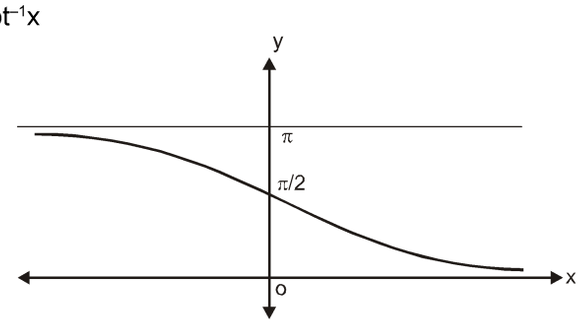
$\sec^{-1}x$: $\sec^{-1}x$ or arcsec x is a function with domain R – (–1, 1) and range $[0, \pi] - \{\pi/2\}$.

Graph of $y = \sec^{-1}x$



$\cot^{-1}$ x : $\cot^{-1}x$ or arccot x is a function with domain R and range $(0, \pi)$

Graph of $y = \cot^{-1}x$



Property 1 : "–x"

The graphs of $\sin^{-1}x$, $\tan^{-1}x$, $\text{cosec}^{-1}x$ are symmetric about origin.

Hence we get $\sin^{-1}(-x) = -\sin^{-1}x$

$\tan^{-1}(-x) = -\tan^{-1}x$

$\text{cosec}^{-1}(-x) = -\text{cosec}^{-1}x$.

Also the graphs of $\cos^{-1}x$, $\sec^{-1}x$, $\cot^{-1}x$ are symmetric about the point $(0, \pi/2)$. From this, we get

$\cos^{-1}(-x) = \pi - \cos^{-1}x$

$\sec^{-1}(-x) = \pi - \sec^{-1}x$

$\cot^{-1}(-x) = \pi - \cot^{-1}x$.

Property 2 : $T(T^{-1})$

(i) $\sin(\sin^{-1}x) = x$, $-1 \le x \le 1$

(ii) $\cos(\cos^{-1}x) = x$, $-1 \le x \le 1$

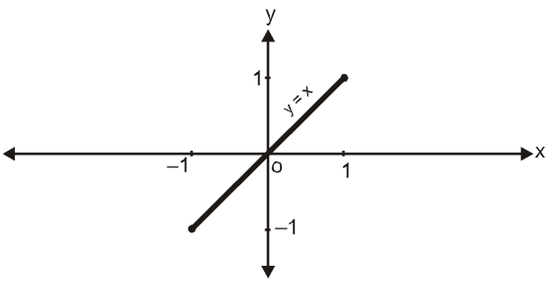
(iii) $\tan(\tan^{-1}x) = x$, $x \in R$

(iv) $\cot(\cot^{-1}x) = x$, $x \in R$

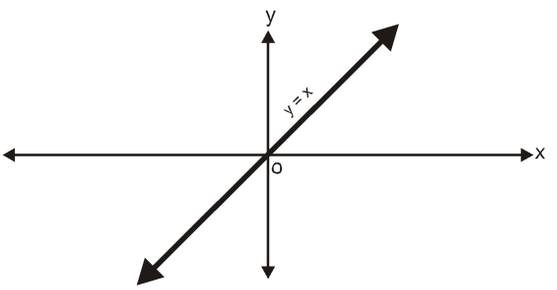
(v) $\sec(\sec^{-1}x) = x$, $x \le -1, x \ge 1$

(vi) $\text{cosec}(\text{cosec}^{-1}x) = x$, $|x| \ge 1$

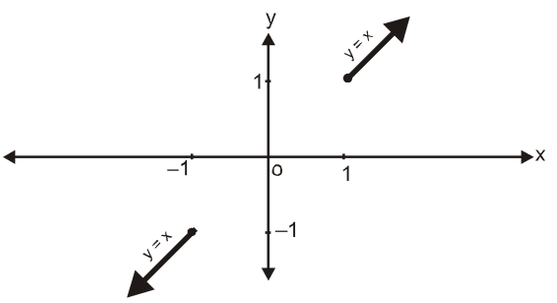
The graph of $y = \sin(\sin^{-1}x) \equiv \cos(\cos^{-1}x)$



The graph of $y = \tan(\tan^{-1}x) \equiv \cot(\cot^{-1}x)$

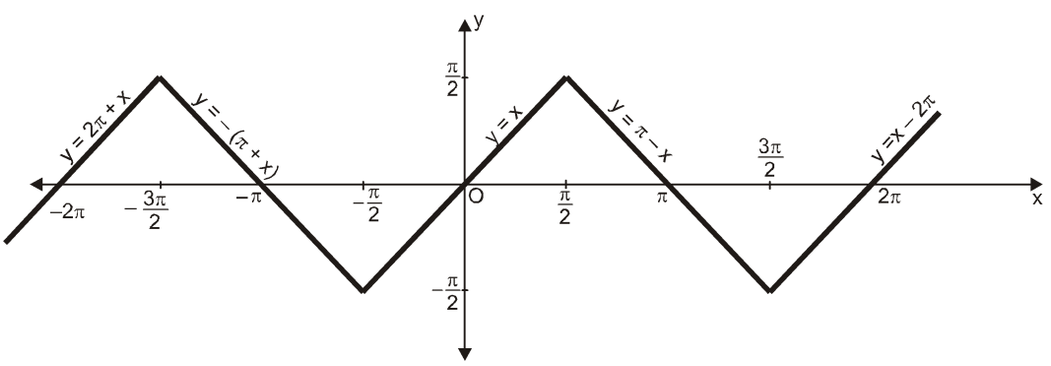


The graph of $y = \text{cosec}(\text{cosec}^{-1}x) \equiv \sec(\sec^{-1}x)$



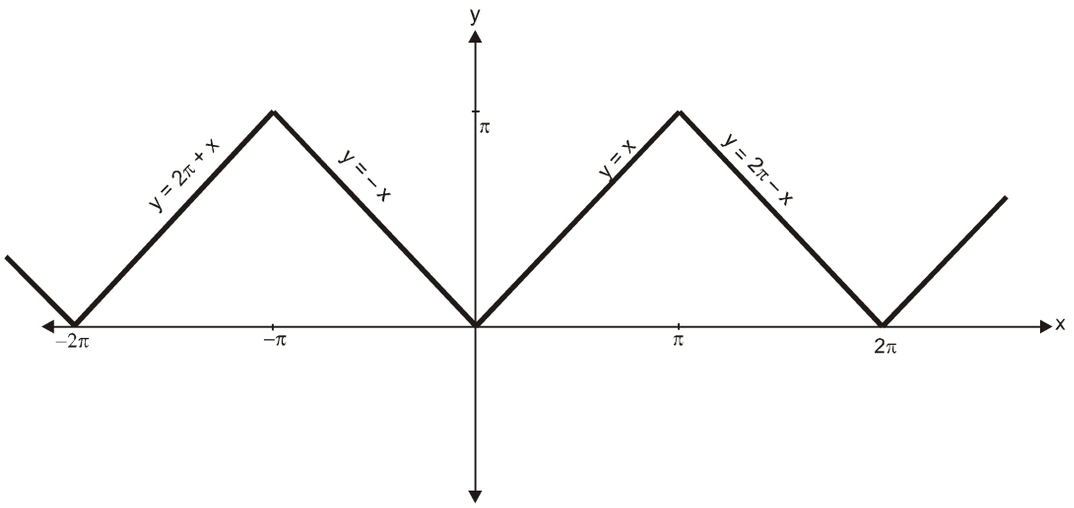
Property 3 : $T^{-1}(T)$

(i) $\sin^{-1}(\sin x) = \begin{cases} -2n\pi + x, & x \in [2n\pi - \pi/2, 2n\pi + \pi/2] \\ (2n+1)\pi - x, & x \in [(2n+1)\pi - \pi/2, (2n+1)\pi + \pi/2], n \in \not{Z} \end{cases}$



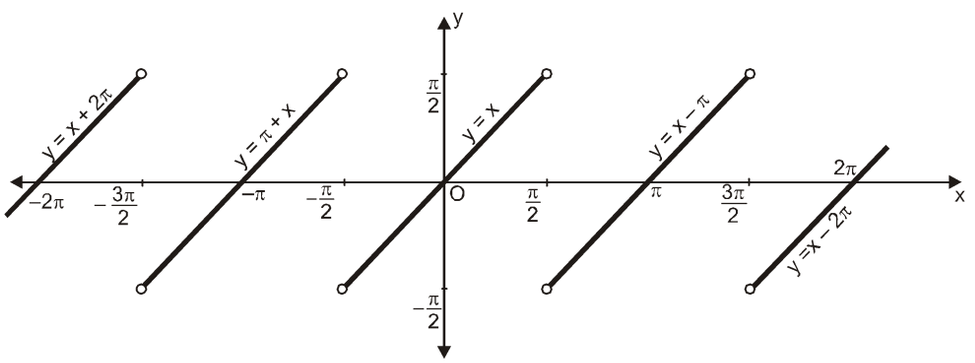
(ii) $\cos^{-1}(\cos x) = \begin{cases} -2n\pi + x, & x \in [2n\pi, (2n+1)\pi] \\ 2n\pi - x, & x \in [(2n-1)\pi, 2n\pi], n \in I \end{cases}$

Graph of $y = \cos^{-1}(\cos x)$



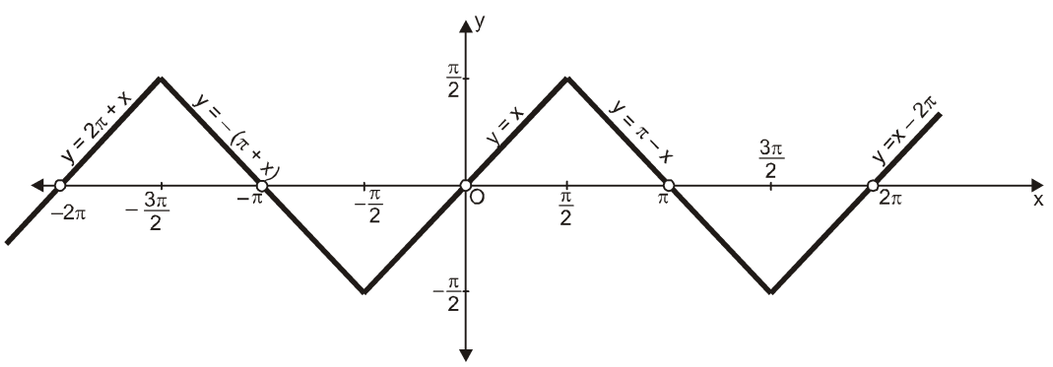
(iii) $\tan^{-1}(\tan x) = -n\pi + x,\ n\pi - \pi/2 < x < n\pi + \pi/2,\ n \in \mathbb{Z}$

Graph of $y = \tan^{-1}(\tan x)$



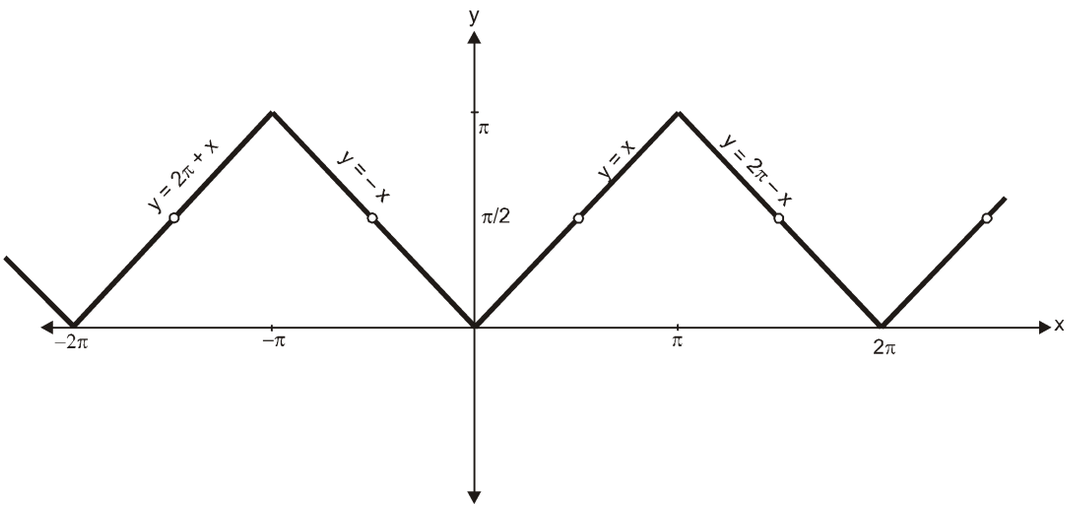
(iv) $\text{cosec}^{-1}(\text{cosec } x)$ is similar to $\sin^{-1}(\sin x)$

Graph of $y = \text{cosec}^{-1}(\text{cosec } x)$



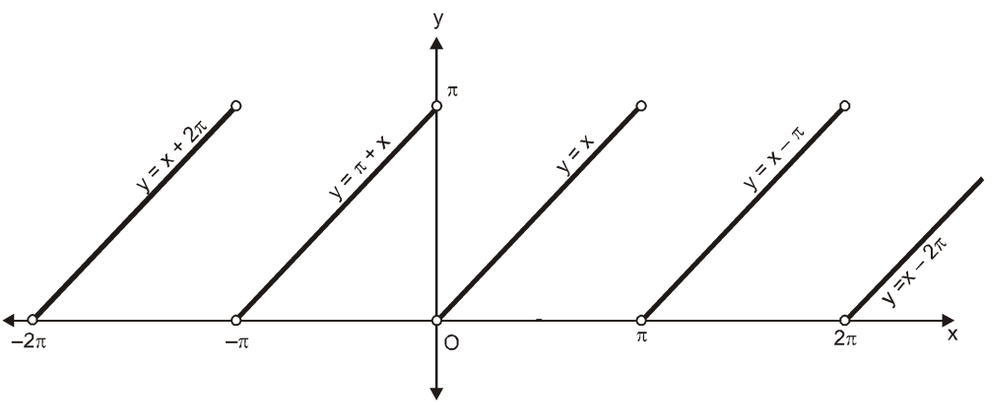
(v) $\sec^{-1}(\sec x)$ is similar to $\cos^{-1}(\cos x)$

Graph of $y = \sec^{-1}(\sec x)$



(vii) $\cot^{-1}(\cot x) = -n\pi + x,\ x \in (n\pi, (n+1)\pi),\ n \in Z$

Graph of $y = \cot^{-1}(\cot x)$



Remark : $\sin(\sin^{-1}x)$, $\cos(\cos^{-1}x)$, ... $\cot(\cot^{-1}x)$ are aperiodic (non periodic) functions where as $\sin^{-1}(\sin x)$, ..., $\cot^{-1}(\cot x)$ are periodic functions.

Property 4 : "1/x"

(i) $\operatorname{cosec}^{-1}(x) = \sin^{-1}(1/x),\ |x| \geq 1$

(ii) $\sec^{-1} x = \cos^{-1}(1/x),\ |x| \geq 1$

(iii) $\cot^{-1}x = \begin{cases} \tan^{-1}(1/x), & x > 0 \\ \pi + \tan^{-1}(1/x), & x < 0 \end{cases}$

Property 5 : "π/2"

(i) $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2},\ -1 \leq x \leq 1$

(ii) $\tan^{-1}x + \cot^{-1}x = \frac{\pi}{2}, \quad x \in R$

(iii) $\operatorname{cosec}^{-1}x + \sec^{-1}x = \frac{\pi}{2}, \quad |x| \geq 1$

Property 6 : Identities on addition and subtraction:

(i) $\sin^{-1}x + \sin^{-1}y \begin{cases} \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right), x \geq 0, y \geq 0 & \& \quad (x^2+y^2) \leq 1 \\ \pi - \sin^{-1}\left(x\sqrt{1-y^2} + y\sqrt{1-x^2}\right), x \geq 0, y \geq 0 & \& \quad x^2+y^2 \geq 1 \end{cases}$

(ii) $\sin^{-1}x - \sin^{-1}y = \sin^{-1}\left(x\sqrt{1-y^2} - y\sqrt{1-x^2}\right); \; x, y \in [0, 1]$

(iii) $\cos^{-1}x + \cos^{-1}y = \cos^{-1}\left(xy - \sqrt{1-x^2}\sqrt{1-y^2}\right); \; x, y \in [0, 1]$

(iv) $\cos^{-1}x - \cos^{-1}y = \begin{cases} \cos^{-1}\left(xy + \sqrt{1-x^2}\sqrt{1-y^2}\right); 0 \leq x < y \leq 1 \\ -\cos^{-1}\left(xy + \sqrt{1-x^2}\sqrt{1-y^2}\right); 0 \leq y < x \leq 1 \end{cases}$

(v) $\tan^{-1}x + \tan^{-1}y = \begin{cases} \pi/2 & \text{if} \quad x, y > 0 \ \& \ xy = 1 \\ -\pi/2 & \text{if} \quad x, y < 0 \ \& \ xy = 1 \\ \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if} \quad x, y \geq 0 \ \& \ xy < 1 \\ \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right) & \text{if} \quad x, y \geq 0 \ \& \ xy > 1 \end{cases}$

(vi) $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right), \; x \geq 0, y \geq 0$

Notes : (i) $x^2 + y^2 \leq 1$ & $x, y \geq 0 \quad \Rightarrow \quad 0 \leq \sin^{-1}x + \sin^{-1}y \leq \frac{\pi}{2}$

and $x^2 + y^2 \geq 1$ & $x, y \geq 0 \quad \Rightarrow \quad \frac{\pi}{2} \leq \sin^{-1}x + \sin^{-1}y \leq \pi$

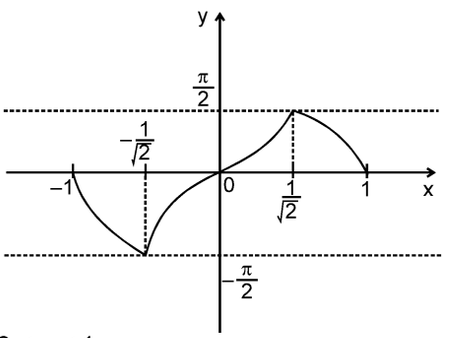
(ii) $xy < 1$ and $x, y \geq 0 \Rightarrow 0 \leq \tan^{-1}x + \tan^{-1}y < \frac{\pi}{2}$; $xy > 1$ and $x, y \geq 0 \Rightarrow \frac{\pi}{2} < \tan^{-1}x + \tan^{-1}y < \pi$

(iii) For $x < 0$ or $y < 0$ these identities can be used with the help of property "– x" i.e. change x or y to – x or – y which are positive .

Property 7 : Miscellaneous identities

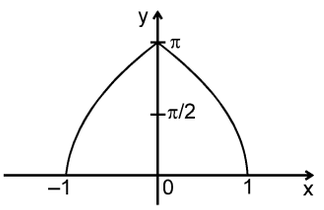
(i) $\sin^{-1}\left(2x\sqrt{1-x^2}\right) = \begin{cases} 2\sin^{-1}x & \text{if} \quad |x| \le \frac{1}{\sqrt{2}} \\ \pi - 2\sin^{-1}x & \text{if} \quad \frac{1}{\sqrt{2}} < x \le 1 \\ -\left(\pi + 2\sin^{-1}x\right) & \text{if} \quad -1 \le x < -\frac{1}{\sqrt{2}} \end{cases}$

graph of $y = \sin^{-1}\left(2x\sqrt{1-x^2}\right)$



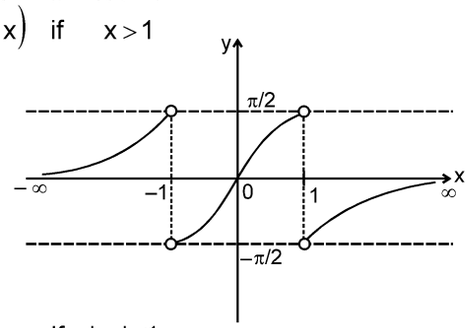
(ii) $\cos^{-1}(2x^2-1) = \begin{cases} 2\cos^{-1}x & \text{if} \quad 0 \le x \le 1 \\ 2\pi - 2\cos^{-1}x & \text{if} \quad -1 \le x < 0 \end{cases}$

graph of $y = \cos^{-1}(2x^2-1)$



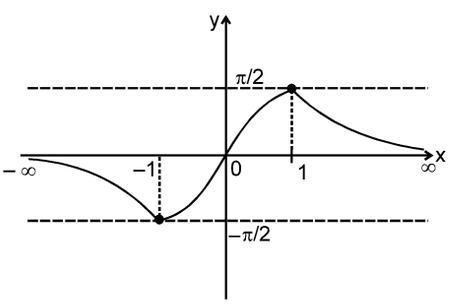
(iii) $\tan^{-1}\frac{2x}{1-x^2} = \begin{cases} 2\tan^{-1}x & \text{if} \quad |x| < 1 \\ \pi + 2\tan^{-1}x & \text{if} \quad x < -1 \\ -\left(\pi - 2\tan^{-1}x\right) & \text{if} \quad x > 1 \end{cases}$

graph of $y = \tan^{-1}\frac{2x}{1-x^2}$



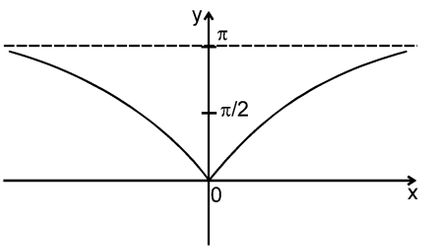
(iv) $\sin^{-1}\frac{2x}{1+x^2} = \begin{cases} 2\tan^{-1}x & \text{if} \quad |x| \le 1 \\ \pi - 2\tan^{-1}x & \text{if} \quad x > 1 \\ -\left(\pi + 2\tan^{-1}x\right) & \text{if} \quad x < -1 \end{cases}$

graph of $y = \sin^{-1}\frac{2x}{1+x^2}$



(v) $\cos^{-1}\frac{1-x^2}{1+x^2} = \begin{cases} 2\tan^{-1}x & \text{if } x \geq 0 \\ -2\tan^{-1}x & \text{if } x < 0 \end{cases}$

graph of $y = \cos^{-1}\frac{1-x^2}{1+x^2}$



(vi) If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \pi$, then $x + y + z = xyz$

(vii) If $\tan^{-1}x + \tan^{-1}y + \tan^{-1}z = \frac{\pi}{2}$, then $xy + yz + zx = 1$

(viii) $\tan^{-1}1 + \tan^{-1}2 + \tan^{-1}3 = \pi$

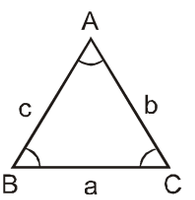
(ix) $\tan^{-1}1 + \tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \frac{\pi}{2}$

SOLUTION OF TRIANGLE

Sine Rule :

In any triangle ABC, the sines of the angles are proportional to the opposite sides

i.e. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$.



Cosine Formula :

In any ΔABC

(i) $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$ or $a^2 = b^2 + c^2 - 2bc\cos A = b^2 + c^2 + 2bc\cos(B + C)$

(ii) $\cos B = \frac{c^2 + a^2 - b^2}{2ca}$ **(iii)** $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

Projection Formula :

In any ΔABC

(i) $a = b\cos C + c\cos B$

(ii) $b = c\cos A + a\cos C$

(iii) $c = a\cos B + b\cos A$

Napier's Analogy - tangent rule :

In any ΔABC

(i) $\tan\frac{B-C}{2} = \frac{b-c}{b+c}\cot\frac{A}{2}$ **(ii)** $\tan\frac{C-A}{2} = \frac{c-a}{c+a}\cot\frac{B}{2}$

(iii) $\tan\frac{A-B}{2} = \frac{a-b}{a+b}\cot\frac{C}{2}$

Trigonometric Functions of Half Angles :

(i) $\sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}}$, $\sin\frac{B}{2} = \sqrt{\frac{(s-c)(s-a)}{ca}}$, $\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

(ii) $\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$, $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ca}}$, $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$

(iii) $\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}} = \frac{\Delta}{s(s-a)} = \frac{(s-b)(s-c)}{\Delta}$, where $s = \frac{a+b+c}{2}$ is semi perimeter and Δ is the area of triangle.

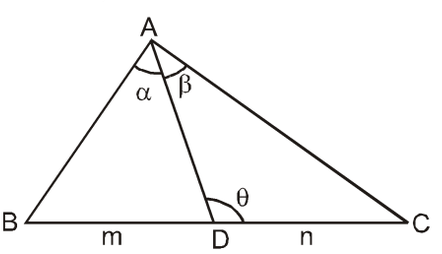
(iv) $\sin A = \frac{2}{bc}\sqrt{s(s-a)(s-b)(s-c)} = \frac{2\Delta}{bc}$

Area of Triangle (Δ)

$$\Delta = \frac{1}{2}ab\sin C = \frac{1}{2}bc\sin A = \frac{1}{2}ca\sin B = \sqrt{s(s-a)(s-b)(s-c)}$$

m - n Rule : In any triangle ABC if D be any point on the base BC, such that BD : DC :: m : n and if $\angle BAD = \alpha$, $\angle DAC = \beta$, $\angle CDA = \theta$, then

$$(m+n)\cot\theta = m\cot\alpha - n\cot\beta$$
$$= n\cot B - m\cot C$$



Radius of Circumcirlce :

If R be the circumradius of ΔABC, then $R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C} = \frac{abc}{4\Delta}$

Radius of The Incircle :

If 'r' be the inradius of ΔABC, then

(i) $r = \frac{\Delta}{s}$

(ii) $r = (s-a)\tan\frac{A}{2} = (s-b)\tan\frac{B}{2} = (s-c)\tan\frac{C}{2}$

(iii) $r = \frac{a\sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{A}{2}}$ and so on

(iv) $r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

Radius of The Ex- Circles :

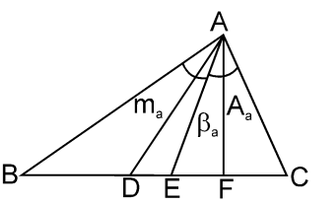
If r_1, r_2 , r_3 are the radii of the ex-circles of ΔABC opposite to the vertex A, B, C respectively, then

(i) $r_1 = \frac{\Delta}{s-a}$; $r_2 = \frac{\Delta}{s-b}$; $r_3 = \frac{\Delta}{s-c}$ **(ii)** $r_1 = s \tan\frac{A}{2}$; $r_2 = s \tan\frac{B}{2}$; $r_3 = s \tan\frac{C}{2}$

(iii) $r_1 = \frac{a\cos\frac{B}{2}\cos\frac{C}{2}}{\cos\frac{A}{2}}$ and so on **(iv)** $r_1 = 4R \sin\frac{A}{2} . \cos\frac{B}{2} . \cos\frac{C}{2}$

Length of Angle Bisectors, Medians & Altitudes :

(i) Length of an angle bisector from the angle A = $\beta_a = \frac{2bc\cos\frac{A}{2}}{b+c}$;



(ii) Length of median from the angle A = $m_a = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$

(iii) Length of altitude from the angle A = $A_a = \frac{2\Delta}{a}$

NOTE : $m_a^2 + m_b^2 + m_c^2 = \frac{3}{4}(a^2 + b^2 + c^2)$

The Distances of The Special Points from Vertices and Sides of Triangle :

(i) Circumcentre (O) : OA = R and O_a = R cos A

(ii) Incentre (I) : IA = r cosec $\frac{A}{2}$ and I_a = r

(iii) Excentre (I_1) : $I_1A = r_1$ cosec $\frac{A}{2}$ and $I_{1a} = r_1$

(iv) Orthocentre (H) : HA = 2R cos A and H_a = 2R cos B cos C

(v) Centroid (G) : GA = $\frac{1}{3}\sqrt{2b^2 + 2c^2 - a^2}$ and $G_a = \frac{2\Delta}{3a}$

Excentral Triangle :

The triangle formed by joining the three excentres I_1, I_2 and I_3 of Δ ABC is called the excentral or excentric triangle.

(i) Δ ABC is the pedal triangle of the $\Delta I_1I_2I_3$.

(ii) Its angles are $\frac{\pi}{2} - \frac{A}{2}$, $\frac{\pi}{2} - \frac{B}{2}$ and $\frac{\pi}{2} - \frac{C}{2}$.

(iii) Its sides are 4R cos $\frac{A}{2}$, 4R cos $\frac{B}{2}$ and 4R cos $\frac{C}{2}$.

(iv) $II_1 = 4R \sin\frac{A}{2}$; $II_2 = 4R \sin\frac{B}{2}$; $II_3 = 4R \sin\frac{C}{2}$.

(v) Incentre I of Δ ABC is the orthocentre of the excentral $\Delta I_1 I_2 I_3$.

Distance Between Special Points :

(i) Distance between circumcentre and orthocentre $OH^2 = R^2 (1 - 8 \cos A \cos B \cos C)$

(ii) Distance between circumcentre and incentre $OI^2 = R^2 (1 - 8 \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2}) = R^2 - 2Rr$

(iii) Distance between circumcentre and centroid $OG^2 = R^2 - \frac{1}{9}(a^2 + b^2 + c^2)$

* * * * *

SEQUENCE & SERIES

Arithmetic progression (A.P.) :

A.P. is a sequence whose successive terms are obtained by adding a fixed number 'd' to the preceding terms. This fixed number 'd' is called the common difference. If a is the first term & d the common difference, then A.P. can be written as a, a + d, a + 2 d,..... ,a + (n – 1) d,.....

e.g. – 4, – 1, 2, 5 .....

n^{th} term of an A.P. :

Let 'a' be the first term and 'd' be the common difference of an A.P., then

$t_n = a + (n - 1)\, d$, where $d = t_n - t_{n-1}$

The sum of first n terms of an A.P. :

If a is first term and d is common difference, then sum of the first n terms of AP is

$$S_n = \frac{n}{2}[2a + (n-1)\, d]$$

$= \frac{n}{2}[a + \ell\,] \equiv n t_{\left(\frac{n+1}{2}\right)}$, for n is odd. (Where ℓ is the last term and $t_{\left(\frac{n+1}{2}\right)}$ is the middle term.)

Note : For any sequence $\{t_n\}$, whose sum of first r terms is S_r, r^{th} term, $t_r = S_r - S_{r-1}$.

Remarks :

(i) The first term and common difference can be zero, positive or negative (or any complex number.)

(ii) If a, b, c are in A.P. $\Rightarrow 2b = a + c$ & if a, b, c, d are in A.P. $\Rightarrow a + d = b + c$.

(iii) Three numbers in A.P. can be taken as a – d, a, a + d ;
four numbers in A.P. can be taken as a – 3d, a – d, a + d, a + 3d;
five numbers in A.P. are a – 2d, a – d, a, a + d, a + 2d ;
six terms in A.P. are a – 5d, a – 3d, a – d, a + d, a + 3d, a + 5d etc.

(iv) The sum of the terms of an A.P. equidistant from the beginning & end is constant and equal to the sum of first & last terms.

(v) Any term of an A.P. (except the first) is equal to half the sum of terms which are equidistant from it. $a_n = 1/2\,(a_{n-k} + a_{n+k})$, $k < n$. For $k = 1$, $a_n = (1/2)\,(a_{n-1} + a_{n+1})$;
For $k = 2$, $a_n = (1/2)\,(a_{n-2} + a_{n+2})$ and so on.

(vi) If each term of an A.P. is increased, decreased, multiplied or divided by the same non-zero number, then the resulting sequence is also an AP.

(vii) The sum and difference of two AP's is an AP.

Arithmetic mean (mean or average) (A.M.) :

If three terms are in A.P. then the middle term is called the A.M. between the other two, so if a, b, c are in A.P., b is A.M. of a & c.

A.M. for any n numbers $a_1, a_2, ..., a_n$ is; $A = \frac{a_1 + a_2 + a_3 + ... + a_n}{n}$.

n - Arithmetic means between two numbers :

If a, b are any two given numbers & $a, A_1, A_2, ..., A_n, b$ are in A.P., then $A_1, A_2, ... A_n$ are the n A.M.'s between a & b.

$$A_1 = a + \frac{b-a}{n+1}, A_2 = a + \frac{2(b-a)}{n+1}, ..., A_n = a + \frac{n(b-a)}{n+1}$$

Note : Sum of n A.M.'s inserted between a & b is equal to n times the single A.M. between a & b

i.e. $\sum_{r=1}^{n} A_r = nA$, where A is the single A.M. between a & b i.e. $A = \frac{a+b}{2}$

Geometric progression (G.P.) :

G.P. is a sequence of numbers whose first term is non zero & each of the succeeding terms is equal to the preceeding terms multiplied by a constant. Thus in a G.P. the ratio of successive terms is constant. This constant factor is called the **common ratio** of the series & is obtained by dividing any term by that which immediately preceeds it. Therefore a, ar, ar^2, ar^3, ar^4,..... is a G.P. with 'a' as the first term & 'r' as common ratio.

e.g. (i) 2, 4, 8, 16, ... (ii) $\frac{1}{3}, \frac{1}{9}, \frac{1}{27}, \frac{1}{81}, ...$

Results : (i) n^{th} term of GP = $a r^{n-1}$

(ii) Sum of the first n terms of GP

$$S_n = \begin{cases} \frac{a(r^n - 1)}{r-1}, & r \neq 1 \\ na, & r = 1 \end{cases}$$

(iii) Sum of an infinite terms of GP when $|r| < 1$. When $n \to \infty$, $r^n \to 0$ if $|r| < 1$ therefore,

$S_\infty = \frac{a}{1-r} (|r| < 1)$.

Remarks :

(i) If a, b, c are in G.P. $\Rightarrow b^2 = ac$, in general if $a_1, a_2, a_3, a_4, ... a_{n-1}, a_n$ are in G.P., then $a_1 a_n = a_2 a_{n-1} = a_3 a_{n-2} = ...$

(ii) Any three consecutive terms of a G.P. can be taken as $\frac{a}{r}$, a, ar.

(iii) Any four consecutive terms of a G.P. can be taken as $\frac{a}{r^3}, \frac{a}{r}$, ar, ar^3.

(iv) If each term of a G.P. be multiplied or divided or raised to power by the same non–zero quantity, the resulting sequence is also a G.P..

(v) If $a_1, a_2, a_3, ...$ and $b_1, b_2, b_3, ...$ are two G.P's with common ratio r_1 and r_2 respectively, then the sequence $a_1b_1, a_2b_2, a_3b_3, ...$ is also a G.P. with common ratio $r_1 r_2$.

(vi) If $a_1, a_2, a_3, ...$are in G.P. where each $a_i > 0$, then log a_1, loga_2, loga_3,.....are in A.P. and its converse is also true.

Geometric means (mean proportional) (G.M.):

If a, b, c are in G.P., b is called as the G.M. of a & c.

If a and c are both positive, then $b = \sqrt{ac}$ and if a and c are both negative, then $b = -\sqrt{ac}$.

$b^2 = ac$, therefore $b = \sqrt{ac}$; $a > 0$, $c > 0$.

n-Geometric means between a, b :

If a, b are two given numbers & a, G_1, G_2,....., G_n, b are in G.P.. Then
G_1, G_2, G_3,...., G_n are n G.M.s between a & b.

$G_1 = a(b/a)^{1/n+1}$, $G_2 = a(b/a)^{2/n+1}$,....., $G_n = a(b/a)^{n/n+1}$

Note : The product of n G.M.s between a & b is equal to the nth power of the single G.M. between a & b

i.e. $\prod_{r=1}^{n} G_r = \left(\sqrt{ab}\right)^n = G^n$, where G is the single G.M. between a & b.

Harmonic progression (H.P.)

A sequence is said to be in H.P if the reciprocals of its terms are in A.P.. If the sequence a_1, a_2, a_3,...., a_n is in H.P. then $1/a_1$, $1/a_2$,...., $1/a_n$ is in A.P.

Note : (i) Here we do not have the formula for the sum of the n terms of an H.P.. For H.P. whose first term is a and second term is b, the n^{th} term is $t_n = \frac{ab}{b + (n-1)(a-b)}$.

(ii) If a, b, c are in H.P. $\Rightarrow b = \frac{2ac}{a+c}$ or $\frac{a}{c} = \frac{a-b}{b-c}$.

(iii) If a, b, c are in A.P. $\Rightarrow \frac{a-b}{b-c} = \frac{a}{a}$

(iv) If a, b, c are in G.P. $\Rightarrow \frac{a-b}{b-c} = \frac{a}{b}$

Harmonic mean (H.M.):

If a, b, c are in H.P., b is called as the H.M. between a & c, then $b = \frac{2ac}{a+c}$

If a_1, a_2 , ... a_n are 'n' non-zero numbers then H.M. 'H' of these numbers is given by

$$\frac{1}{H} = \frac{1}{n}\left[\frac{1}{a_1} + \frac{1}{a_2} + ... + \frac{1}{a_n}\right]$$

Relation between means :

(i) If A, G, H are respectively A.M., G.M., H.M. between a & b both being positive, then $G^2 = AH$ (i.e. A, G, H are in G.P.) and $A \geq G \geq H$.

A.M. ≥ G.M. ≥ H.M.

Let a_1, a_2, a_3, ...a_n be n positive real numbers, then we define their

A.M. = $\frac{a_1 + a_2 + a_3 + ... + a_n}{n}$, their

G.M. = $(a_1 a_2 a_3 .....a_n)^{1/n}$ and their

H.M. = $\frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + ... + \frac{1}{a_n}}$.

It can be shown that
A.M. ≥ G.M. ≥ H.M. and equality holds at either places iff
$a_1 = a_2 = a_3 = ...= a_n$

Arithmetico-geometric series :

A series, each term of which is formed by multiplying the corresponding terms of an A.P. & G.P. is called the Arithmetico-Geometric Series. e.g. $1 + 3x + 5x^2 + 7x^3 + ...$
Here 1, 3, 5,.... are in A.P. & 1, x, x^2, x^3.... are in G.P..

Sum of n terms of an arithmetico-geometric series:
Let $S_n = a + (a + d)\,r + (a + 2d)\,r^2 + ... + [a + (n-1)d]\,r^{n-1}$, then

$$S_n = \frac{a}{1-r} + \frac{dr\left(1-r^{n-1}\right)}{(1-r)^2} - \frac{[a+(n-1)d]r^n}{1-r}, \; r \neq 1.$$

Sum to infinity: If $|r| < 1$ & $n \to \infty$, then $\underset{n\to\infty}{\text{Limit}}\; r^n = 0$ and $\underset{n\to\infty}{\text{Limit}}\; n.r^n = 0$

$$\therefore \quad S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2} .$$

Sum of Miscellaneous Sequences :

(i) $\sum_{r=1}^{n} (a_r \pm b_r) = \sum_{r=1}^{n} a_r \pm \sum_{r=1}^{n} b_r.$

(ii) $\sum_{r=1}^{n} k\,a_r = k \sum_{r=1}^{n} a_r.$

(iii) $\sum_{r=1}^{n} k = k + k + k + ...\text{n times} = nk$; where k is a constant.

(iv) $\sum_{r=1}^{n} r = 1 + 2 + 3 + ...+ n = \frac{n(n+1)}{2}$

(v) $\sum_{r=1}^{n} r^2 = 1^2 + 2^2 + 3^2 + ...+ n^2 = \frac{n(n+1)(2n+1)}{6}$

(vi) $\sum_{r=1}^{n} r^3 = 1^3 + 2^3 + 3^3 + ...+ n^3 = \frac{n^2(n+1)^2}{4}$

Method of difference for finding n^{th} term :

Let u_1, u_2, u_3 ..... be a sequence, such that $u_2 - u_1, u_3 - u_2$, ... is either an A.P. or a G.P. then nth term u_n of this sequence is obtained as follows

$S = u_1 + u_2 + u_3 + ... + u_n$ .....(i)

$S = u_1 + u_2 + ... + u_{n-1} + u_n$ .....(ii)

(i) – (ii) $\Rightarrow$ $u_n = u_1 + (u_2 - u_1) + (u_3 - u_2) + ... + (u_n - u_{n-1})$

Where the series $(u_2 - u_1) + (u_3 - u_2) + ... + (u_n - u_{n-1})$ is

either in A.P. or in G.P. then we can find u_n.

So sum of series $S = \sum_{r=1}^{n} u_r$

Note : The above method can be generalised as follows :

Let $u_1, u_2, u_3, \ldots\ldots$ be a given sequence.

The first differences are $\Delta_1 u_1, \Delta_1 u_2, \Delta_1 u_3, \ldots\ldots$ where $\Delta_1 u_1 = u_2 - u_1$, $\Delta_1 u_2 = u_3 - u_2$ etc.

The second differences are $\Delta_2 u_1, \Delta_2 u_2, \Delta_2 u_3, \ldots\ldots$, where $\Delta_2 u_1 = \Delta_1 u_2 - \Delta_1 u_1$, $\Delta_2 u_2 = \Delta_1 u_3 - \Delta_1 u_2$ etc.

This process is continued untill the k^{th} differences $\Delta_k u_1, \Delta_k u_2, \ldots\ldots$ are obtained, where the k^{th} differences are all equal or they form a GP with common ratio different from 1.

Case - 1 : The k^{th} differences are all equal.

In this case the n^{th} term, u_n is given by

$u_n = a_0 n^k + a_1 n^{k-1} + \ldots\ldots + a_k$, where $a_0, a_1, \ldots\ldots, a_k$ are calculated by using first 'k + 1' terms of the sequence.

Case - 2 : The k^{th} differences are in GP with common ratio r ($r \neq 1$)

The n^{th} term is given by $u_n = \lambda r^{n-1} + a_0 n^{k-1} + a_1 n^{k-2} + \ldots\ldots + a_{k-1}$

Method of difference for finding s_n :

If possible express r^{th} term as difference of two terms as $t_r = \pm (f(r) - f(r \pm 1))$. This can be explained with the help of examples given below.

$t_1 = f(1) - f(0)$,

$t_2 = f(2) - f(1)$,

$\vdots \quad \vdots \quad \vdots$

$t_n = f(n) - f(n-1)$

$\Rightarrow \quad S_n = f(n) - f(0)$

* * * * *

MATRICES

Introduction :

Any rectangular arrangement of numbers (real or complex) (or of real valued or complex valued expressions) is called a **matrix**. If a matrix has m rows and n columns then the **order** of matrix is written as m × n and we call it as order m by n

The general m × n matrix is

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & ..... & a_{1j} & ..... & a_{1n} \\ a_{21} & a_{22} & a_{23} & ..... & a_{2j} & ..... & a_{2n} \\ ..... & ..... & ..... & ..... & ..... & ..... & ..... \\ a_{i1} & a_{i2} & a_{i3} & ..... & a_{ij} & ..... & a_{in} \\ ..... & ..... & ..... & ..... & ..... & ..... & ..... \\ a_{m1} & a_{m2} & a_{m3} & ..... & a_{mj} & ..... & a_{mn} \end{bmatrix}$$

where a_{ij} denote the element of i^{th} row & j^{th} column. The above matrix is usually denoted as $\mathbf{[a_{ij}]_{m \times n}}$.

Notes : (i) The elements a_{11}, a_{22}, a_{33},..... are called as **diagonal elements**. Their sum is called as **trace of A** denoted as tr(A)

(ii) Capital letters of English alphabets are used to denote matrices.

(iii) Order of a matrix : If a matrix has m rows and n columns, then we say that its order is "m by n", written as "m × n".

Types of Matrices :

Row matrix :

A matrix having only one row is called as row matrix (or row vector).General form of row matrix is $A = [a_{11}, a_{12}, a_{13}, ..., a_{1n}]$

This is a matrix of order "1 × n" (or a row matrix of order n)

Column matrix :

A matrix having only one column is called as column matrix (or column vector).

Column matrix is in the form $A = \begin{bmatrix} a_{11} \\ a_{21} \\ ... \\ a_{m1} \end{bmatrix}$

This is a matrix of order "m × 1" (or a column matrix of order m)

Zero matrix :

$A = [a_{ij}]_{m \times n}$ is called a zero matrix, if $a_{ij} = 0 \ \forall$ i & j.

e.g. : (i) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ (ii) $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

e.g. : (i) $\begin{bmatrix} a & 0 & 0 \\ b & c & 0 \\ x & y & z \end{bmatrix}$ (ii) $\begin{bmatrix} a & 0 & 0 & 0 \\ b & c & 0 & 0 \\ x & y & z & 0 \end{bmatrix}$

Comparable matrices : Two matrices A & B are said to be comparable, if they have the same order (i.e., number of rows of A & B are same and also the number of columns).

e.g. : (i) $A = \begin{bmatrix} 2 & 3 & 4 \\ 3 & -1 & 2 \end{bmatrix}$ & $B = \begin{bmatrix} 3 & 4 & 2 \\ 0 & 1 & 3 \end{bmatrix}$ are comparable

e.g. : (ii) $C = \begin{bmatrix} 2 & 3 & 4 \\ 3 & -1 & 2 \end{bmatrix}$ & $D = \begin{bmatrix} 3 & 0 \\ 4 & 1 \\ 2 & 3 \end{bmatrix}$ are not comparable

Equality of matrices :

Two matrices A and B are said to be equal if they are comparable and all the corresponding elements are equal.

Let $A = [a_{ij}]_{m \times n}$ & $B = [b_{ij}]_{p \times q}$

$A = B$ iff (i) $m = p,\ n = q$

(ii) $a_{ij} = b_{ij}\ \forall$ i & j.

Multiplication of matrix by scalar :

Let λ be a scalar (real or complex number) & $A = [a_{ij}]_{m \times n}$ be a matrix. Thus the product λA is defined as $\lambda A = [b_{ij}]_{m \times n}$ where $b_{ij} = \lambda a_{ij}\ \forall$ i & j.

e.g. : $A = \begin{bmatrix} 2 & -1 & 3 & 5 \\ 0 & 2 & 1 & -3 \\ 0 & 0 & -1 & -2 \end{bmatrix}$ & $-3A \equiv (-3)A = \begin{bmatrix} -6 & 3 & -9 & -15 \\ 0 & -6 & -3 & 9 \\ 0 & 0 & 3 & 6 \end{bmatrix}$

Note : If A is a scalar matrix, then $A = \lambda I$, where λ is a diagonal entry of A

Addition of matrices :

Let A and B be two matrices of same order (i.e. comparable matrices). Then A + B is defined to be.

$A + B = [a_{ij}]_{m \times n} + [b_{ij}]_{m \times n}$.

$= [c_{ij}]_{m \times n}$ where $c_{ij} = a_{ij} + b_{ij}\ \forall$ i & j.

e.g. : $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 2 \\ -2 & -3 \\ 5 & 7 \end{bmatrix}$, $A + B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \\ 6 & 7 \end{bmatrix}$

Substraction of matrices :

Let A & B be two matrices of same order. Then A – B is defined as A + (– B) where – B is (– 1) B.

Properties of addition & scalar multiplication :

Consider all matrices of order m × n, whose elements are from a set F (F denote Q, R or C). Let $M_{m \times n}(F)$ denote the set of all such matrices.

Then

(a) $A \in M_{m \times n}(F)$ & $B \in M_{m \times n}(F)$ $\Rightarrow$ $A + B \in M_{m \times n}(F)$

(b) $A + B = B + A$

(c) $(A + B) + C = A + (B + C)$

e.g. : $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ a & b & c & d \\ x & y & z & w \end{bmatrix}$, $A' = \begin{bmatrix} 1 & a & x \\ 2 & b & y \\ 3 & c & z \\ 4 & d & w \end{bmatrix}$

Results : (i) For any matrix $A = [a_{ij}]_{m \times n}$, $(A')' = A$

(ii) Let λ be a scalar & A be a matrix. Then $(\lambda A)' = \lambda A'$

(iii) $(A + B)' = A' + B'$ & $(A - B)' = A' - B'$ for two comparable matrices A and B.

(iv) $(A_1 \pm A_2 \pm ..... \pm A_n)' = A_1' \pm A_2' \pm ..... \pm A_n'$, where A_i are comparable.

(v) Let $A = [a_{ij}]_{m \times p}$ & $B = [b_{ij}]_{p \times n}$, then $(AB)' = B'A'$

(vi) $(A_1 A_2 .....A_n)' = A_n'. A_{n-1}' .....A_2' . A_1'$, provided the product is defined.

Symmetric & skew-symmetric matrix :

A square matrix A is said to be symmetric if $A' = A$

i.e. Let $A = [a_{ij}]_n$. A is symmetric iff $a_{ij} = a_{ji}$ $\forall$ i & j.

A square matrix A is said to be skew-symmetric if $A' = -A$

i.e. Let $A = [a_{ij}]_n$. A is skew-symmetric iff $a_{ij} = -a_{ji}$ $\forall$ i & j.

e.g. $A = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$ is a symmetric matrix.

$B = \begin{bmatrix} o & x & y \\ -x & o & z \\ -y & -z & 0 \end{bmatrix}$ is a skew-symmetric matrix.

Notes : (1) In a skew-symmetric matrix all the diagonal elements are zero.

$(\because \; a_{ii} = -a_{ii} \quad \Rightarrow \quad a_{ii} = 0)$

(2) For any square matrix A, $A + A'$ is symmetric & $A - A'$ is skew-symmetric.

(3) Every square matrix can be uniquely expressed as a sum of two square matrices of which one is symmetric and the other is skew-symmetric.

$A = B + C$, where $B = \frac{1}{2}(A + A')$ & $C = \frac{1}{2}(A - A')$.

Submatrix : Let A be a given matrix. The matrix obtained by deleting some rows or columns of A is called as submatrix of A.

eg. $A = \begin{bmatrix} a & b & c & d \\ x & y & z & w \\ p & q & r & s \end{bmatrix}$

Then $\begin{bmatrix} a & c \\ x & z \\ p & r \end{bmatrix}$, $\begin{bmatrix} a & b & d \\ p & q & s \end{bmatrix}$, $\begin{bmatrix} a & b & c \\ x & y & z \\ p & q & r \end{bmatrix}$ are all submatrices of A.

Minors & Cofactors :

Let Δ be a determinant. Then minor of element a_{ij}, denoted by M_{ij}, is defined as the determinant of the submatrix obtained by deleting i^{th} row & j^{th} column of Δ. Cofactor of element a_{ij}, denoted by C_{ij}, is defined as $C_{ij} = (-1)^{i+j} M_{ij}$.

e.g. 1 $\Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

Remarks :

1. The necessary and sufficient condition for existence of inverse of A is that A is non-singular.

2. A^{-1} is always non-singular.

3. If A = dia (a_{11}, a_{22}, ..., a_{nn}) where $a_{ii} \neq 0 \ \forall$ i, then A^{-1} = diag (a_{11}^{-1}, a_{22}^{-1}, ..., a_{nn}^{-1}).

4. $(A^{-1})' = (A')^{-1}$ for any non-singular matrix A. Also adj (A′) = (adj A)′.

5. $(A^{-1})^{-1} = A$ if A is non-singular.

6. Let k be a non-zero scalar & A be a non-singular matrix. Then $(kA)^{-1} = \frac{1}{k} A^{-1}$.

7. $|A^{-1}| = \frac{1}{|A|}$ for $|A| \neq 0$.

8. Let A be a non-singular matrix. Then AB = AC $\Rightarrow$ B = C & BA = CA $\Rightarrow$ B= C.

9. A is non-singular and symmetric $\Rightarrow$ A^{-1} is symmetric.

10. $(AB)^{-1} = B^{-1} A^{-1}$ if A and B are non- singular.

11. In general AB = **0** does not imply A = **0** or B = **0**. But if A is non-singular and AB = **0**, then B = **0**. Similarly B is non-singular and AB = **0** $\Rightarrow$ A = **0**. Therefore, AB = **0** $\Rightarrow$ either both are singular or one of them is **0**.

Elementary row transformation of matrix :

The following operations on a matrix are called as elementary row transformations.

(a) Interchanging two rows.

(b) Multiplications of all the elements of row by a nonzero scalar.

(c) Addition of constant multiple of a row to another row.

Note : Similar to above we have elementary column transformations also.

Remarks : Two matrices A & B are said to be equivalent if one is obtained from other using elementary transformations. We write A $\approx$ B.

Finding inverse using Elementry operations

(i) Using row transformations :

If A is a matrix such that A^{-1} exists, then to find A^{-1} using elementary row operations,

Step I : Write A = IA and

Step II : Apply a sequence of row operation on A = IA till we get, I = BA.

The matrix B will be inverse of A.

Note : In order to apply a sequence of elementary row operations on the matrix equation X = AB, we will apply these row operatdions simultaneously on X and on the first matrix A of the product AB on RHS.

(ii) Using column transformations :

If A is a matrix such that A^{-1} exists, then to find A^{-1} using elementary column operations,

Step I : Write A = AI and

Step II : Apply a sequence of column operations on A = AI till we get, I = AB.

The matrix B will be inverse of A.

Note : In order to apply a sequence of elementary column operations on the matrix equation X = AB, we will apply these row operatdions simultaneously on X and on the second matrix B of the product AB on RHS.

System of linear equations & matrices :

Consider the system

$$a_{11} x_1 + a_{12}x_2 + \ldots\ldots\ldots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22} x_2 + \ldots\ldots\ldots + a_{2n} x_n = b_2$$
$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$a_{m1}x_1 + a_{m2}x_2 + \ldots\ldots\ldots + a_{mn}x_n = b_n.$$

STRAIGHT LINE

Distance Formula :

The distance between the points $A(x_1,y_1)$ and $B(x_2,y_2)$ is = $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2}$.

Section Formula :

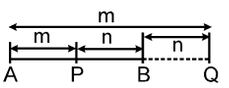
If P(x, y) divides the line joining $A(x_1, y_1)$ & $B(x_2, y_2)$ in the ratio m : n, then;

$$x = \frac{mx_2+nx_1}{m+n} \; ; \; y = \frac{my_2+ny_1}{m+n} .$$

Notes : (i) If $\frac{m}{n}$ is positive, the division is internal, but if $\frac{m}{n}$ is negative, the division is external.

(ii) If P divides AB internally in the ratio m : n & Q divides AB externally in the ratio m : n then P & Q are said to be harmonic conjugate of each other w.r.t. AB.

Mathematically, $\frac{2}{AB}=\frac{1}{AP}+\frac{1}{AQ}$ i.e. AP, AB & AQ are in H.P.



The ratio in which a given line divides the line segment joining two points :

Let the given line ax + by + c = 0 divide the line segment joining $A(x_1, y_1)$ & $B(x_2, y_2)$ in the ratio m : n, then $\frac{m}{n}=-\frac{ax_1+by_1+c}{ax_2+by_2+c}$. If A & B are on the same side of the given line then m/n is negative but if A & B are on opposite sides of the given line, then m/n is positive

Slope Formula :

If θ is the angle at which a straight line is inclined to the positive direction of x–axis, & $0° \le \theta < 180°$, $\theta \ne 90°$, then the slope of the line, denoted by m, is defined by $m = \tan\theta$. If θ is 90°, m does not exist, but the line is parallel to the y–axis. If θ = 0, then m = 0 & the line is parallel to the x-axis.

If $A(x_1, y_1)$ & $B(x_2, y_2)$, $x_1 \ne x_2$, are points on a straight line, then the slope m of the line is given by :

$$m = \left(\frac{y_1-y_2}{x_1-x_2}\right).$$

Condition of collinearity of three points :

Points $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are collinear if

(i) $m_{AB} = m_{BC} = m_{CA}$ i.e. $\left(\frac{y_1-y_2}{x_1-x_2}\right) = \left(\frac{y_2-y_3}{x_2-x_3}\right)$

(ii) $\Delta ABC = 0$ i.e. $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(iii) AC = AB + BC or AB ~ BC

(iv) A divides the line segment BC in some ratio.

Area of a Triangle :

If $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$ are the vertices of triangle ABC, then its area is equal to

$\Delta ABC = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$, provided the vertices are considered in the counter clockwise sense. The above formula will give a $(-)$ve area if the vertices (x_i, y_i), i = 1, 2, 3 are placed in the clockwise sense.

Note: Area of n-sided polygon formed by points (x_1, y_1) ; (x_2, y_2); .....;(x_n, y_n) is given by

$$\frac{1}{2}\left(\begin{vmatrix} x_1 & x_2 \\ y_1 & y_2 \end{vmatrix} + \begin{vmatrix} x_2 & x_3 \\ y_2 & y_3 \end{vmatrix} + \ldots\ldots\ldots\ldots \begin{vmatrix} x_{n-1} & x_n \\ y_{n-1} & y_n \end{vmatrix} + \begin{vmatrix} x_n & x_1 \\ y_n & y_1 \end{vmatrix}\right).$$

Here vertices are taken in order.

Equation of a Straight Line in various forms :

Point - Slope form :

$y - y_1 = m(x - x_1)$ is the equation of a straight line whose slope is m & which passes through the point (x_1, y_1).

Slope-intercept form :

$y = mx + c$ is the equation of a straight line whose slope is m & which makes an intercept c on the y–axis.

Two point form : $y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$ is the equation of a straight line which passes through the points (x_1, y_1) & (x_2, y_2).

Determinant form : Equation of line passing through (x_1, y_1) and (x_2, y_2) is $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

Intercept form : $\frac{x}{a} + \frac{y}{b} = 1$ is the equation of a straight line which makes intercepts a & b on OX & OY respectively.

Perpendicular/Normal form :

$x\cos\alpha + y\sin\alpha = p$ (where $p > 0$, $0 \le \alpha < 2\pi$) is the equation of the straight line where the length of the perpendicular from the origin O on the line is p and this perpendicular makes an angle α with positive x–axis.

General Form : $ax + by + c = 0$ is the equation of a straight line in the general form

In this case, slope of line $= -\frac{a}{b}$

x - intercept $= -\frac{c}{a}$, y - intercept $= -\frac{c}{b}$

Parametric form :

$P(r) = (x, y) = (x_1 + r\cos\theta, y_1 + r\sin\theta)$ or $\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$ is the equation of the line in parametric

form, where 'r' is the parameter whose absolute value is the distance of any point (x, y) on the line from the fixed point (x_1, y_1) on the line.

Remark : The above form is derived from point-slope form of line.

$y - y_1 = m\,(x - x_1)$ where $m = \tan\theta$

$\Rightarrow \quad y - y_1 = \dfrac{\sin\theta}{\cos\theta}\,(x - x_1)$

Angle between two straight lines in terms of their slopes:

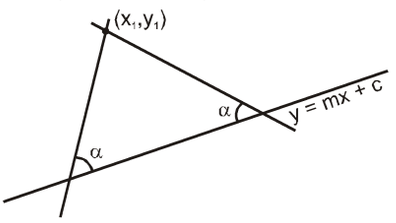
If m_1 & m_2 are the slopes of two intersecting straight lines ($m_1 m_2 \neq -1$) & θ is the acute angle between them, then $\tan\theta = \left|\dfrac{m_1 - m_2}{1 + m_1 m_2}\right|$.

Notes : (i) Let m_1, m_2, m_3 are the slopes of three lines $L_1 = 0; L_2 = 0; L_3 = 0$ where $m_1 > m_2 > m_3$, then the tangent of interior angles of the Δ ABC formed by these lines are given by,

$$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2};\ \tan B = \frac{m_2 - m_3}{1 + m_2 m_3} \ \&\ \tan C = \frac{m_3 - m_1}{1 + m_3 m_1}$$

(ii) The equation of lines passing through point (x_1, y_1) and making angle α with the line $y = mx + c$ are given by :

$(y - y_1) = \tan(\theta - \alpha)(x - x_1)$ & $(y - y_1) = \tan(\theta + \alpha)(x - x_1)$, where $\tan\theta = m$.



Parallel Lines :

(i) When two straight lines are parallel their slopes are equal. Thus any line parallel to $y = mx + c$ is of the type $y = mx + d$, where 'd' is a parameter.

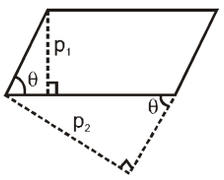
(ii) Two lines $ax + by + c = 0$ and $a'x + b'y + c' = 0$ are parallel if $\dfrac{a}{a'} = \dfrac{b}{b'} \neq \dfrac{c}{c'}$.

Thus any line parallel to $ax + by + c = 0$ is of the type $ax + by + k = 0$, where k is a parameter.

(iii) The distance between two parallel lines with equations $ax + by + c_1 = 0$ & $ax + by + c_2 = 0$ is $= \left|\dfrac{c_1 - c_2}{\sqrt{a^2 + b^2}}\right|$.

Note that coefficients of x & y in both the equations must be same.

(iv) The area of the parallelogram $= \dfrac{p_1 p_2}{\sin\theta}$, where p_1 & p_2 are distances between two pairs of opposite sides & θ is the angle between any two adjacent sides. Note that area of the parallelogram bounded by the lines $y = m_1x + c_1$, $y = m_1x + c_2$ and $y = m_2x + d_1$, $y = m_2x + d_2$ is given by $\left|\dfrac{(c_1 - c_2)(d_1 - d_2)}{m_1 - m_2}\right|$.



Perpendicular Lines:

(i) When two lines of slopes m_1 & m_2 are at right angles, the product of their slopes is -1, i.e. $m_1 m_2 = -1$. Thus any line perpendicular to y = mx + c is of the form

$y = -\frac{1}{m}x + d$, where 'd' is any parameter.

(ii) Two lines ax + by + c = 0 and a′x + b′y + c′ = 0 are perpendicular if aa′ + bb′ = 0. Thus any line perpendicular to ax + by + c = 0 is of the form bx − ay + k = 0, where 'k' is any parameter.

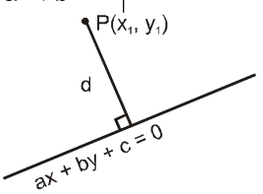
Position of the point (x_1, y_1) relative of the line ax + by + c = 0 :

In general two points (x_1, y_1) and (x_2, y_2) will lie on same side or opposite side of ax + by + c = 0 according as $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ are of same or opposite sign respectively.

Length of perpendicular from a point on a line :

The length of perpendicular from $P(x_1, y_1)$ on

ax + by + c = 0 is $\left|\frac{ax_1 + by_1 + c}{\sqrt{a^2+b^2}}\right|$.



Reflection of a point about a line :

(i) Foot of the perpendicular from a point (x_1, y_1) on the line ax + by + c = 0 is

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = -\left(\frac{ax_1 + by_1 + c}{a^2 + b^2}\right)$$

(ii) The image of a point (x_1, y_1) about the line ax + by + c = 0 is

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = -2\left(\frac{ax_1 + by_1 + c}{a^2 + b^2}\right).$$

Centroid, Incentre & Excentre :

If A (x_1, y_1), B(x_2, y_2), C(x_3, y_3) are the vertices of triangle ABC, whose sides BC, CA, AB are of lengths a, b, c respectively, then the co-ordinates of the special points of triangle ABC are as follows :

Centroid $G \equiv \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$

Incentre $I \equiv \left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c}\right)$, and

Excentre (to A) $I_1 \equiv \left(\frac{-ax_1+bx_2+cx_3}{-a+b+c}, \frac{-ay_1+by_2+cy_3}{-a+b+c}\right)$ and so on.

Bisectors of the angles between two lines:

Equations of the bisectors of angles between the lines $ax + by + c = 0$ & $a'x + b'y + c' = 0$ $(ab' \neq a'b)$ are :

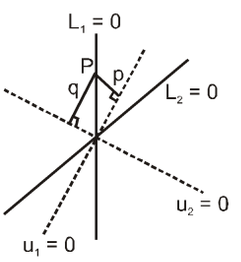
$$\frac{ax+by+c}{\sqrt{a^2+b^2}} = \pm \frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}}$$

Note : Equation of straight lines passing through $P(x_1, y_1)$ & equally inclined with the lines $a_1x + b_1y + c_1 = 0$ & $a_2x + b_2y + c_2 = 0$ are those which are parallel to the bisectors between these two lines & passing through the point P.

Methods to discriminate between the acute angle bisector & the obtuse angle bisector:

(i) If θ be the angle between one of the lines & one of the bisectors, find $\tan\theta$.
If $|\tan\theta| < 1$, then $2\theta < 90°$ so that this bisector is the acute angle bisector.
If $|\tan\theta| > 1$, then we get the bisector to be the obtuse angle bisector.

(ii) Let $L_1 = 0$ & $L_2 = 0$ are the given lines & $u_1 = 0$ and $u_2 = 0$ are the bisectors between $L_1 = 0$ & $L_2 = 0$. Take a point P on any one of the lines $L_1 = 0$ or $L_2 = 0$ and drop perpendicular on $u_1 = 0$ & $u_2 = 0$ as shown in figure. If,



$|p| < |q| \Rightarrow u_1$ is the acute angle bisector.
$|p| > |q| \Rightarrow u_1$ is the obtuse angle bisector.
$|p| = |q| \Rightarrow$ the lines L_1 & L_2 are perpendicular.

(iii) If $aa' + bb' < 0$, then the equation of the bisector of this acute angle is

$$\frac{ax+by+c}{\sqrt{a^2+b^2}} = + \frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}}$$

If, however, $aa' + bb' > 0$, the equation of the bisector of the obtuse angle is :

$$\frac{ax+by+c}{\sqrt{a^2+b^2}} = + \frac{a'x+b'y+c'}{\sqrt{a'^2+b'^2}}$$

Condition of Concurrency :

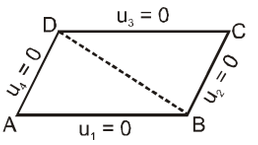
Three lines $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$ & $a_3x + b_3y + c_3 = 0$ are concurrent if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$$

Family of Straight Lines :

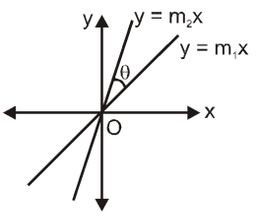
The equation of a family of straight lines passing through the point of intersection of the lines, $L_1 \equiv a_1x + b_1y + c_1 = 0$ & $L_2 \equiv a_2x + b_2y + c_2 = 0$ is given by $L_1 + k L_2 = 0$ i.e.
$(a_1x + b_1y + c_1) + k(a_2x + b_2y + c_2) = 0$, where k is an arbitrary real number.

Note :(i) If $u_1 = ax + by + c$, $u_2 = a'x + b'y + d$, $u_3 = ax + by + c'$, $u_4 = a'x + b'y + d'$ then $u_1 = 0; u_2 = 0; u_3 = 0; u_4 = 0$ form a parallelogram. The diagonal BD can be given by $u_2u_3 - u_1u_4 = 0$.



(ii) The diagonal AC is also given by $u_1 + \lambda u_4 = 0$ and $u_2 + \mu u_3 = 0$, if the two equations are identical for some real λ and μ.
[For getting the values of λ & μ compare the coefficients of x, y & the constant terms].

A Pair of straight lines through origin:



(i) A homogeneous equation of degree two,
"$ax^2 + 2hxy + by^2 = 0$" always represents a pair of straight lines passing through the origin if :

(a) $h^2 > ab \Rightarrow$ lines are real & distinct .

(b) $h^2 = ab \Rightarrow$ lines are coincident .

(c) $h^2 < ab \Rightarrow$ lines are imaginary with real point of intersection i.e. (0, 0)

This equation is obtained by multiplying the two equations of lines $(m_1x - y)(m_2x - y) = 0$
$\Rightarrow$ $m_1m_2x^2 - (m_1 + m_2)xy + y^2 = 0$

(ii) If $y = m_1x$ & $y = m_2x$ be the two equations represented by $ax^2 + 2hxy + by^2 = 0$, then;

$$m_1 + m_2 = -\frac{2h}{b} \quad \& \quad m_1 m_2 = \frac{a}{b}.$$

(iii) If θ is the acute angle between the pair of straight lines represented by,

$ax^2 + 2hxy + by^2 = 0$, then $\tan\theta = \left|\frac{2\sqrt{h^2 - ab}}{a + b}\right|$.

(iv) The condition that these lines are :

(a) at right angles to each other is $a + b = 0$. i.e. co–efficient of x^2 + co–efficient of $y^2 = 0$.

(b) coincident is $h^2 = ab$.

(c) equally inclined to the axis of x is $h = 0$.i.e. coeff. of $xy = 0$.

Note that a homogeneous equation of degree n represents n straight lines passing through origin.

(v) The equation to the pair of straight lines bisecting the angles between the straight lines $ax^2 + 2hxy + by^2 = 0$ is $\frac{x^2 - y^2}{a - b} = \frac{xy}{h}$.

General equation of second degree representing a pair of Straight lines :

(i) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if :

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0, \text{ i.e. if } \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0.$$

Such an equation is obtained again by multiplying the two equation of lines
$(a_1x + b_1y + c_1)(a_2x + b_2y + c_2) = 0$

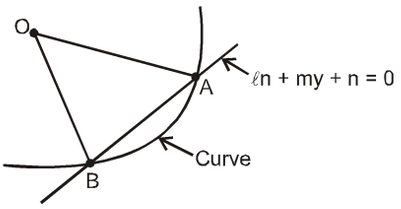
Homogenization : This method is used to write the joint equation of two lines connecting origin to the points of intersection of a given line and a given second degree curve.

The equation of a pair of straight lines joining origin to the points of intersection of the line $L \equiv \ell x + my + n = 0$ and a second degree curve

$S \equiv ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

is $ax^2 + 2hxy + by^2 + 2gx\left(\frac{\ell x+my}{-n}\right)+2fy\left(\frac{\ell x+my}{-n}\right)+c\left(\frac{\ell x+my}{-n}\right)^2 = 0.$

The equation is obtained by homogenizing the equation of curve with the help of equation of line.



Notes : (i) Here we have written 1 as $\frac{\ell x+my}{-n}$ and converted all terms of the curve to second degree expressions

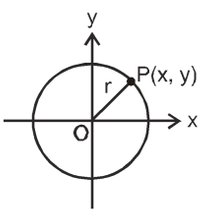
(ii) Equation of any curve passing through the points of intersection of two curves $C_1 = 0$ and $C_2 = 0$ is given by $\lambda C_1 + \mu C_2 = 0$, where λ & μ are parameters.

* * * * *

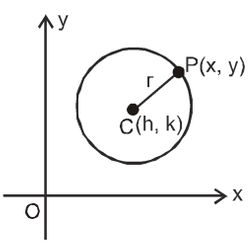
CIRCLE

Equation of a circle in various forms :

(a) The circle with centre as origin & radius 'r' has the equation; $x^2 + y^2 = r^2$.



(b) The circle with centre (h, k) & radius 'r' has the equation; $(x - h)^2 + (y - k)^2 = r^2$.



(c) The general equation of a circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

with centre as $(-g, -f)$ & radius = $\sqrt{g^2+f^2-c}$.

This can be obtained from the equation $(x - h)^2 + (y - k)^2 = r^2$

$\Rightarrow$ $x^2 + y^2 - 2hx - 2ky + h^2 + k^2 - r^2 = 0$

Take $-h = g, -k = f$ and $h^2 + k^2 - r^2 = c$

Condition to define circle :-

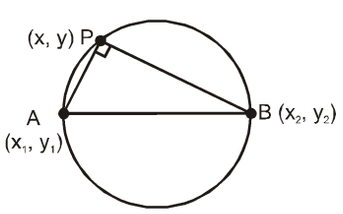
$g^2 + f^2 - c > 0 \Rightarrow$ real circle.

$g^2 + f^2 - c = 0 \Rightarrow$ point circle.

$g^2 + f^2 - c < 0 \Rightarrow$ imaginary circle, with real centre, that is $(-g, -f)$

Note : That every second degree equation in x & y, in which coefficient of x^2 is equal to coefficient of y^2 & the coefficient of xy is zero, always represents a circle.

(d) The equation of circle with (x_1, y_1) & (x_2, y_2) as extremeties of its diameter is:
$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$.



This is obtained by the fact that angle in a semicircle is a right angle.

$\therefore$ (Slope of PA) (Slope of PB) = – 1

$$\Rightarrow \quad \frac{y-y_1}{x-x_1}\cdot\frac{y-y_2}{x-x_2} = -1 \quad \Rightarrow \quad (x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$$

Note that this will be the circle of least radius passing through (x_1, y_1) & (x_2, y_2).

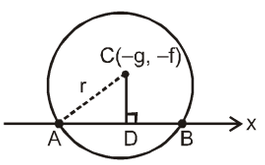
Intercepts made by a circle on the axes:

The intercepts made by the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ on the co–ordinate axes are $2\sqrt{g^2-c}$ (on x-axis) & $2\sqrt{f^2-c}$ (on y-axis) respectively.

If $g^2 > c \Rightarrow$ circle cuts the x axis at two distinct points.

$g^2 = c \Rightarrow$ circle touches the x–axis.

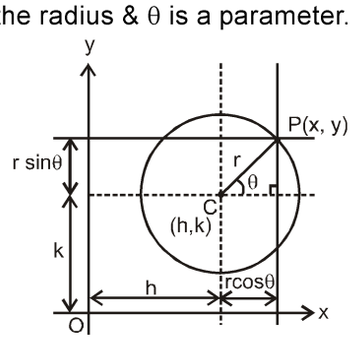
$g^2 < c \Rightarrow$ circle lies completely above or below the x–axis.



$$AB = 2AD = 2\sqrt{r^2-CD^2} = 2\sqrt{r^2-f^2} = 2\sqrt{g^2+f^2-c-f^2} = 2\sqrt{g^2-c}$$

Parametric equations of a circle:

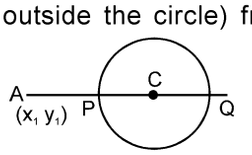
The parametric equations of $(x-h)^2 + (y-k)^2 = r^2$ are: $x = h + r\cos\theta$; $y = k + r\sin\theta$; $-\pi < \theta \le \pi$ where (h, k) is the centre, r is the radius & θ is a parameter.



Position of a point with respect to a circle:

The point (x_1, y_1) is inside, on or outside the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$.
according as $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c <, = \text{or} > 0$.

Note : The greatest & the least distance of a point A (lies outside the circle) from a circle with centre C & radius r is AC + r & AC – r respectively.



Line and a circle:

Let L = 0 be a line & S = 0 be a circle. If r is the radius of the circle & p is the length of the perpendicular from the centre on the line, then:

(i) $p > r \Leftrightarrow$ the line does not meet the circle i. e. passes out side the circle.

(ii) $p = r \Leftrightarrow$ the line touches the circle. (It is tangent to the circle)

(iii) $p < r \Leftrightarrow$ the line is a secant of the circle.

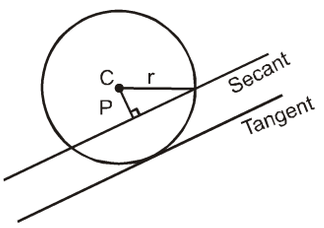
(iv) $p = 0 \Rightarrow$ the line is a diameter of the circle.

Also, if $y = mx + c$ is line and $x^2 + y^2 = a^2$ is circle then

(i) $c^2 < a^2 (1 + m^2) \Leftrightarrow$ the line is a secant of the circle.

(ii) $c^2 = a^2 (1 + m^2) \Leftrightarrow$ the line touches the circle. (It is tangent to the circle)

(iii) $c^2 > a^2 (1 + m^2) \Leftrightarrow$ the line does not meet the circle i. e. passes out side the circle.



These conditions can also be obtained by solving $y = mx + c$ with $x^2 + y^2 = a^2$ and making the discriminant of the quadratic greater than zero for secant, equal to zero for tangent and less the zero for the last case.

Slope form of tangent :

$y = mx + c$ is always a tangent to the circle $x^2 + y^2 = a^2$ if $c^2 = a^2 (1 + m^2)$. Hence, equation of tangent is $y = mx \pm a\sqrt{1+m^2}$ and the point of contact is $\left(-\frac{a^2 m}{c}, \frac{a^2}{c}\right)$.

Point form of tangent :

(i) The equation of the tangent to the circle $x^2 + y^2 = a^2$ at its point (x_1, y_1) is, $x x_1 + y y_1 = a^2$.

(ii) The equation of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at its point (x_1, y_1) is: $xx_1 + yy_1 + g(x+x_1) + f(y+y_1) + c = 0$.

Note : In general the equation of tangent to any second degree curve at point (x_1, y_1) on it can be obtained by replacing x^2 by $x x_1$, y^2 by yy_1, x by $\frac{x+x_1}{2}$, y by $\frac{y+y_1}{2}$, xy by $\frac{x_1 y + x y_1}{2}$ and c remains as c.

Parametric form of tangent :

The equation of a tangent to circle $x^2 + y^2 = a^2$ at $(a\cos\alpha, a\sin\alpha)$ is $x\cos\alpha + y\sin\alpha = a$.

NOTE : The point of intersection of the tangents at the points $P(\alpha)$ & $Q(\beta)$ is $\left(\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{a\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$

Normal :

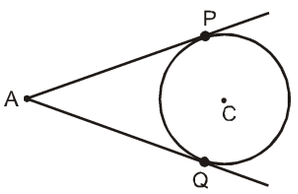
If a line is normal / orthogonal to a circle, then it must pass through the centre of the circle. Using this fact normal to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at (x_1, y_1) is; $y - y_1 = \frac{y_1 + f}{x_1 + g}(x - x_1)$.

Pair of tangents from a point :

The equation of a pair of tangents drawn from the point $A(x_1, y_1)$ to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is : $SS_1 = T^2$.

Where $S \equiv x^2 + y^2 + 2gx + 2fy + c$; $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

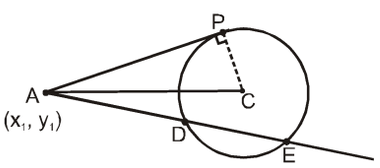
$T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c.$



Length of a tangent and power of a point :

The length of a tangent from an external point (x_1, y_1) to the circle

$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ is given by $L = \sqrt{x_1^2 + y_1^2 + 2gx_1 + 2f_1y + c} = \sqrt{S_1}$.



AP = length of tangent

$AP^2 = AD \cdot AE$

Square of length of the tangent from the point A is also called the power of point w.r.t. a circle.

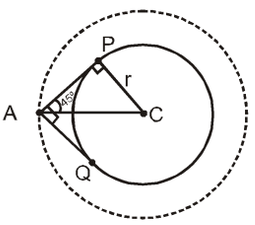
Power of a point w.r.t. a circle remains constant.

Power of a point P is positive, negative or zero according as the point 'A' is outside, inside or on the circle respectively.

Director circle :

The locus of the point of intersection of two perpendicular tangents is called the director circle of the given circle. The director circle of a circle is the concentric circle having radius equal to $\sqrt{2}$ times the original circle.

Proof :



$AC = r \operatorname{cosec} 45^\circ = r\sqrt{2}$

Chord of contact :

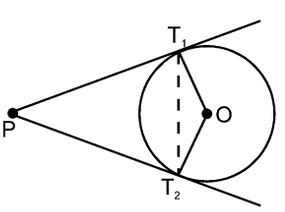
If two tangents PT_1 & PT_2 are drawn from the point $P(x_1, y_1)$ to the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, then the equation of the chord of contact T_1T_2 is:

$$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0.$$

Note : Here R = radius; L = length of tangent.

(a) Chord of contact exists only if the point 'P' is not inside.

(b) Length of chord of contact $T_1T_2 = \dfrac{2LR}{\sqrt{R^2 + L^2}}$.



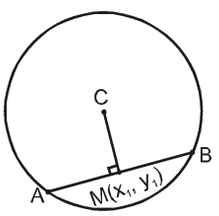
(c) Area of the triangle formed by the pair of the tangents & its chord of contact = $\frac{RL^3}{R^2+L^2}$

(d) Tangent of the angle between the pair of tangents from (x_1, y_1) = $\left(\frac{2RL}{L^2-R^2}\right)$

(e) Equation of the circle circumscribing the triangle PT_1T_2 is:
$(x - x_1)(x + g) + (y - y_1)(y + f) = 0$.

Equation of the chord with a given middle point:

The equation of the chord of the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ in terms of its mid point $M(x_1, y_1)$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$ which is designated by $T = S_1$.



Notes : (i) The shortest chord of a circle passing through a point 'M' inside the circle is one chord whose middle point is M.

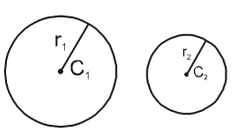
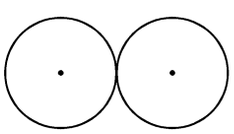
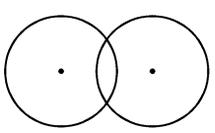
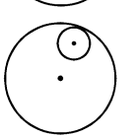
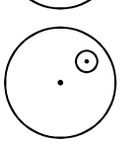
(ii) The chord passing through a point 'M' inside the circle and which is at a maximum distance from the centre is a chord with middle point M.

Equation of the chord joining two points of circle :

The equation of chord PQ to the circle $x^2 + y^2 = a^2$ joining two points $P(\alpha)$ and $Q(\beta)$ on it is given by the equation of a straight line joining two point α & β on the circle $x^2 + y^2 = a^2$ is

$x \cos\frac{\alpha+\beta}{2} + y \sin\frac{\alpha+\beta}{2} = a \cos\frac{\alpha-\beta}{2}$.

Common tangents to two circles:

Case		Number of Tangents	Condition
(i)		4 common tangents (2 direct and 2 transverse)	$r_1 + r_2 < c_1 c_2$.
(ii)		3 common tangents.	$r_1 + r_2 = c_1 c_2$.
(iii)		2 common tangents.	$\lvert r_1 - r_2 \rvert < c_1 c_2 < r_1 + r_2$
(iv)		1 common tangent.	$\lvert r_1 - r_2 \rvert = c_1 c_2$.
(v)		No common tangent.	$c_1 c_2 < \lvert r_1 - r_2 \rvert$.

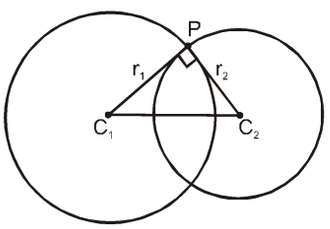
(Here C_1C_2 is distance between centres of two circles.)

Notes :(i) The direct common tangents meet at a point which divides the line joining centre of circles externally in the ratio of their radii.
Transverse common tangents meet at a point which divides the line joining centre of circles internally in the ratio of their radii.

(ii) Length of an external (or direct) common tangent & internal (or transverse) common tangent to the two circles are given by: $L_{ext} = \sqrt{d^2-(r_1-r_2)^2}$ & $L_{int} = \sqrt{d^2-(r_1+r_2)^2}$,
where d = distance between the centres of the two circles and r_1, r_2 are the radii of the two circles. Note that length of internal common tangent is always less than the length of the external or direct common tangent.

Orthogonality of two circles:

Two circles $S_1 = 0$ & $S_2 = 0$ are said to be orthogonal or said to intersect orthogonally if the tangents at their point of intersection include a right angle. The condition for two circles to be orthogonal is: $2 g_1 g_2 + 2 f_1 f_2 = c_1 + c_2$.
Proof :



$(C_1C_2)^2 = (C_1P)^2 + (C_2P)^2 \Rightarrow \quad (g_1 - g_2)^2 + (f_1 - f_2)^2 = g_1^2 + f_1^2 - c_1 + g_2^2 + f_2^2 - c_2$

$\Rightarrow \quad 2g_1g_2 + 2f_1f_2 = c_1 + c_2$

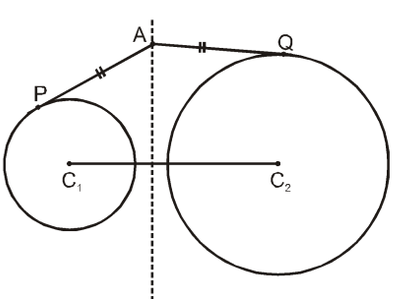
Notes : (a) The centre of a variable circle orthogonal to two fixed circles lies on the radical axis of two circles.

(b) If two circles are orthogonal, then the polar of a point 'P' on first circle w.r.t. the second circle passes through the point Q which is the other end of the diameter through P. Hence locus of a point which moves such that its polars w.r.t. the circles $S_1 = 0$, $S_2 = 0$ & $S_3 = 0$ are concurrent in a circle which is orthogonal to all the three circles.

(c) The centre of a circle which is orthogonal to three given circles is the radical centre provided the radical centre lies outside all the three circles.

Radical axis and radical centre:

The radical axis of two circles is the locus of points whose powers w.r.t. the two circles are equal. The equation of radical axis of the two circles $S_1 = 0$ & $S_2 = 0$ is given by
$S_1 - S_2 = 0$ i.e. $2(g_1 - g_2)x + 2(f_1 - f_2)y + (c_1 - c_2) = 0$.



The common point of intersection of the radical axes of three circles taken two at a time is called the radical centre of three circles. Note that the length of tangents from radical centre to the three circles are equal.

Notes : (a) If two circles intersect, then the radical axis is the common chord of the two circles.

(b) If two circles touch each other, then the radical axis is the common tangent of the two circles at the common point of contact.

(c) Radical axis is always perpendicular to the line joining the centres of the two circles.

(d) Radical axis will pass through the mid point of the line joining the centres of the two circles only if the two circles have equal radii.

(e) Radical axis bisects a common tangent between the two circles.

(f) A system of circles, every two which have the same radical axis, is called a coaxial system.

(g) Pairs of circles which do not have radical axis are concentric.

Family of Circles:

This article is aimed at obtaining the equation of a group of circles having a specific characteristic.

For example, the equation $x^2 + y^2 + 4x + 2y + \lambda = 0$ where λ is arbitrary, represents a family of circles with fixed centre (–2, –1) but variable radius. We have the following results for some other families of circles.

(a) The equation of the family of circles passing through the points of intersection of two circles $S_1 = 0$ & $S_2 = 0$ is : $S_1 + K S_2 = 0$ ($K \neq -1$ provided the co–efficient of x^2 & y^2 in S_1 & S_2 are same)

(b) The equation of the family of circles passing through the point of intersection of a circle S = 0 & a line L = 0 is given by S + KL = 0.

(c) The equation of a family of circles passing through two given points (x_1, y_1) & (x_2, y_2) can be written in the form:

$$(x - x_1)(x - x_2) + (y - y_1)(y - y_2) + K\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0, \text{ where K is a parameter.}$$

(d) The equation of a family of circles touching a fixed line $y - y_1 = m(x - x_1)$ at the fixed point (x_1, y_1) is $(x - x_1)^2 + (y - y_1)^2 + K(y - y_1 - m(x - x_1)) = 0$, where K is a parameter.

(e) Family of circles circumscribing a triangle whose sides are given by $L_1 = 0$, $L_2 = 0$ and $L_3 = 0$ is given by; $L_1L_2 + \lambda L_2L_3 + \mu L_3L_1 = 0$ provided co–efficient of xy = 0 and co–efficient of x^2 = co–efficient of y^2.

(f) Equation of circle circumscribing a quadrilateral whose side in order are represented by the lines $L_1 = 0$, $L_2 = 0$, $L_3 = 0$ & $L_4 = 0$ are $u L_1L_3 + \lambda L_2L_4 = 0$ where values of u & λ can be found out by using condition that co–efficient of x^2 = co–efficient of y^2 and co–efficient of xy = 0.

* * * * *

LIMITS

Definition : Limit of a function f(x) is said to exist, as $x \to a$ when,

$\lim_{h\to 0^+} f(a-h) = \lim_{h\to 0^+} f(a+h) = \text{Finite}$

(Left hand limit) (Right hand limit)

Note that we are not interested in knowing about what happens at x = a. Also note that if L.H.L. & R.H.L. are both tending towards '∞' or '$-\infty$', then it is said to be infinite limit.

Remember, '$x \to a$' means that x is approaching to 'a' but not equal to 'a'.

Fundamental theorems on limits:

Let $\lim_{x\to a} f(x) = \ell$ and $\lim_{x\to a} g(x) = m$. If ℓ & m are finite, then:

(A) $\lim_{x\to a} \{ f(x) \pm g(x) \} = \ell \pm m$

(B) $\lim_{x\to a} \{ f(x).\ g(x) \} = \ell \,.\, m$

(C) $\lim_{x\to a} \frac{f(x)}{g(x)} = \frac{\ell}{m}$, provided $m \neq 0$

(D) $\lim_{x\to a} k f(x) = k \lim_{x\to a} f(x) = k\ell$; where k is a constant.

(E) $\lim_{x\to a} f(g(x)) = f\left(\lim_{x\to a} g(x)\right) = f(m)$; provided f is continuous at $g(x) = m$.

Indeterminate forms :

If on putting x = a in f(x), any one of $\frac{0}{0}, \frac{\infty}{\infty}$, $\mathbf{0 \times \infty}$, $\mathbf{\infty - \infty}$, $\mathbf{\infty^0}$, $\mathbf{0^0}$, $\mathbf{1^\infty}$ form is obtained, then the limit has an indeterminate form. All the above forms are interchangeable, i.e. we can change one form to other by suitable substitutions etc.

In such cases $\lim_{x\to a} f(x)$ may exist.

Consider $f(x) = \frac{x^2-4}{x-2}$. Here $\lim_{x\to 2} x^2 - 4 = 0$ and $\lim_{x\to 2} x - 2 = 0$

$\therefore$ $\lim_{x\to 2} f(x)$ has an indeterminate form of the type $\frac{0}{0}$.

$\lim_{x\to\infty} \frac{\ell n\, x}{x}$ has an indeterminate form of type $\frac{\infty}{\infty}$.

$\lim_{x\to 0} (1+x)^{1/x}$ is an indeterminate form of the type 1^∞

NOTE :

(i) $\infty + \infty = \infty$

(ii) $\infty \times \infty = \infty$

(iii) $\frac{a}{\infty} = 0$, if a is finite.

(iv) $\frac{a}{0}$ is not defined for any $a \in R$.

(v) $\lim\limits_{x\to 0} \frac{x}{x}$ is an indeterminate form whereas $\lim\limits_{x\to 0} \frac{[x^2]}{x^2}$ is not an indeterminate form (where [.] represents greatest integer function)

Students may remember these forms alongwith the prefix 'tending to'

i.e. $\frac{\text{tending to zero}}{\text{tending to zero}}$ is an indeterminate form whereas $\frac{\text{exactly zero}}{\text{tending to zero}}$ is not an indeterminate form, its value is zero.

similarly $(\text{tending to one})^{\text{tending to } \infty}$ is indeterminate form whereas $(\text{exactly one})^{\text{tending to } \infty}$ is not an indeterminate form, its value is one.

To evaluate a limit, we must always put the value where 'x' is approaching to in the function. If we get a determinate form, then that value becomes the limit otherwise if an indeterminate form comes, we have to remove the indeterminancy, once the indeterminancy is removed the limit can be evaluated by putting the value of x, where it is approaching.

Methods of removing indeterminancy

Basic methods of removing indeterminancy are

(A) Factorisation (B) Rationalisation

(C) Using standard limits (D) Substitution (E) Expansion of functions.

Factorisation method :-

We can cancel out the factors which are leading to indeterminancy and find the limit of the remaining expression.

Rationalisation method :-

We can rationalise the irrational expression in numerator or denominator or in both to remove the indeterminancy.

Standard limits :

(a) (i) $\lim\limits_{x\to 0} \frac{\sin x}{x} = \lim\limits_{x\to 0} \frac{\tan x}{x} = 1$ [Where x is measured in radians]

(ii) $\lim\limits_{x\to 0} \frac{\tan^{-1} x}{x} = \lim\limits_{x\to 0} \frac{\sin^{-1} x}{x} = 1$

(iii) $\lim\limits_{x\to 0} (1+x)^{\frac{1}{x}} = e$; $\lim\limits_{x\to 0} (1+ax)^{\frac{1}{x}} = e^a$

(iv) $\lim\limits_{x\to \infty} \left(1+\frac{1}{x}\right)^x = e$; $\lim\limits_{x\to \infty} \left(1+\frac{a}{x}\right)^x = e^a$

(v) $\lim\limits_{x\to 0} \frac{e^x - 1}{x} = 1$; $\lim\limits_{x\to 0} \frac{a^x - 1}{x} = \log_e a = \ell na$, $a > 0$

(vi) $\lim\limits_{x\to 0} \frac{\ell n(1+x)}{x} = 1$

(vii) $\lim\limits_{x\to a} \frac{x^n - a^n}{x - a} = na^{n-1}$

(b) If $f(x) \to 0$, when $x \to a$, then

(i) $\lim\limits_{x\to a} \frac{\sin f(x)}{f(x)} = 1$ (ii) $\lim\limits_{x\to a} \cos f(x) = 1$

(iii) $\lim\limits_{x\to a} \frac{\tan f(x)}{f(x)} = 1$ (iv) $\lim\limits_{x\to a} \frac{e^{f(x)} - 1}{f(x)} = 1$

(v) $\lim\limits_{x\to a} \frac{b^{f(x)} - 1}{f(x)} = \ell n\, b,\ (b > 0)$ (vi) $\lim\limits_{x\to a} \frac{\ell n(1+f(x))}{f(x)} = 1$

(vii) $\lim_{x\to a}(1+f(x))^{\frac{1}{f(x)}} = e$

(c) $\lim_{x\to a} f(x) = A > 0$ and $\lim_{x\to a} \phi(x) = B$(a finite quantity), then $\lim_{x\to a}[f(x)]^{\phi(x)} = A^B$.

Use of substitution in solving limit problems

Sometimes in solving limit problem we convert $\lim_{x\to a} f(x)$ into $\lim_{h\to 0} f(a+h)$ or $\lim_{h\to 0} f(a-h)$ according as need of the problem. (here h is approaching to zero.)

Limits using expansion

(a) $a^x = 1+\frac{x\ell n a}{1!}+\frac{x^2\ell n^2 a}{2!}+\frac{x^3\ell n^3 a}{3!}+\ldots\ldots\ldots, a>0$

(b) $e^x = 1+\frac{x}{1!}+\frac{x^2}{2!}+\frac{x^3}{3!}+\ldots\ldots$

(c) $\ell n\,(1+x) = x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\ldots\ldots\ldots$, for $-1<x\le 1$

(d) $\sin x = x-\frac{x^3}{3!}+\frac{x^5}{5!}-\frac{x^7}{7!}+\ldots\ldots$

(e) $\cos x = 1-\frac{x^2}{2!}+\frac{x^4}{4!}-\frac{x^6}{6!}+\ldots\ldots$

(f) $\tan x = x+\frac{x^3}{3}+\frac{2x^5}{15}+\ldots\ldots$

(g) $\tan^{-1}x = x-\frac{x^3}{3}+\frac{x^5}{5}-\frac{x^7}{7}+\ldots\ldots$

(h) $\sin^{-1}x = x+\frac{1^2}{3!}x^3+\frac{1^2.3^2}{5!}x^5+\frac{1^2.3^2.5^2}{7!}x^7+\ldots\ldots$

(i) $\sec^{-1}x = 1+\frac{x^2}{2!}+\frac{5x^4}{4!}+\frac{61x^6}{6!}+\ldots\ldots$

(j) for $|x|<1$, $n\in R$; $(1+x)^n = 1+nx+\frac{n(n-1)}{1.2}x^2+\frac{n(n-1)(n-2)}{1.2.3}x^3+\ldots\ldots\ldots\infty$

(k) $(1+x)^{\frac{1}{x}} = e\left(1-\frac{x}{2}+\frac{11}{24}x^2-\ldots\ldots\ldots\ldots\right)$

Limit when $x \to \infty$

In these types of problems we usually cancel out the greatest power of x common in numerator and denominator both. Also sometime when $x \to \infty$, we use to substitute $y = \frac{1}{x}$ and in this case $y \to 0^+$.

Some important notes :

(i) $\lim_{x\to\infty}\frac{\ell nx}{x} = 0$ (ii) $\lim_{x\to\infty}\frac{x}{e^x} = 0$ (iii) $\lim_{x\to\infty}\frac{x^n}{e^x} = 0$

(iv) $\lim_{x\to\infty}\frac{(\ell nx)^n}{x} = 0$ (v) $\lim_{x\to 0^+} x(\ell nx)^n = 0$

As $x \to \infty$, $\ell n\, x$ increases much slower than any (positive) power of x where as e^x increases much faster than any (positive) power of x.

(vi) $\lim_{n\to\infty} (1-h)^n = 0$ and $\lim_{n\to\infty} (1+h)^n \to \infty$, where $h \to 0^+$.

Limits of form 1^∞, 0^0, ∞^0

(A) All these forms can be converted into $\frac{0}{0}$ form in the following ways

(a) If $x \to 1$, $y \to \infty$, then $z = (x)^y$ is of 1^∞ form

$\Rightarrow \quad \ell n\, z = y\, \ell n\, x$

$\Rightarrow \quad \ell n\, z = \dfrac{\ell n x}{\frac{1}{y}} \qquad \left(\frac{0}{0} \text{ form}\right)$

As $y \to \infty \Rightarrow \frac{1}{y} \to 0$ and $x \to 1 \Rightarrow \ell n x \to 0$

(b) If $x \to 0$, $y \to 0$, then $z = x^y$ is of (0^0) form

$\Rightarrow \quad \ell n\, z = y\, \ell n\, x$

$\Rightarrow \quad \ell n\, z = \dfrac{y}{\frac{1}{\ell n x}} \qquad \left(\frac{0}{0} \text{ form}\right)$

(c) If $x \to \infty$, $y \to 0$, then $z = x^y$ is of $(\infty)^0$ form

$\Rightarrow \quad \ell n\, z = y\, \ell n\, x$

$\Rightarrow \quad \ell n\, z = \dfrac{y}{\frac{1}{\ell n x}} \qquad \left(\frac{0}{0} \text{ form}\right)$

(B) $(1)^\infty$ type of problems can be solved by the following method

(a) $\lim_{x\to 0} (1+x)^{\frac{1}{x}} = e$

(b) $\lim_{x\to a} [f(x)]^{g(x)}$; where $f(x) \to 1$; $g(x) \to \infty$ as $x \to a$

$$= \lim_{x\to a} [1+f(x)-1]^{\frac{1}{f(x)-1}\{f(x)-1\}.g(x)} = \lim_{x\to a} \left([1+(f(x)-1)]^{\frac{1}{f(x)-1}}\right)^{(f(x)-1)g(x)}$$

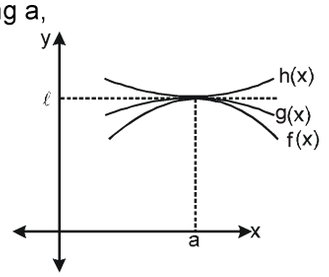
$$= e^{\lim_{x\to a}[f(x)-1]\,g(x)}$$

Sandwitch theorem or squeeze play theorem:

Suppose that $f(x) \le g(x) \le h(x)$ for all x in some open interval containing a, except possibly at x = a itself. Suppose also that

$\lim_{x\to a} f(x) = \ell = \lim_{x\to a} h(x)$,

Then $\lim_{x\to a} g(x) = \ell$.



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CONTINUITY & DERIVABILITY

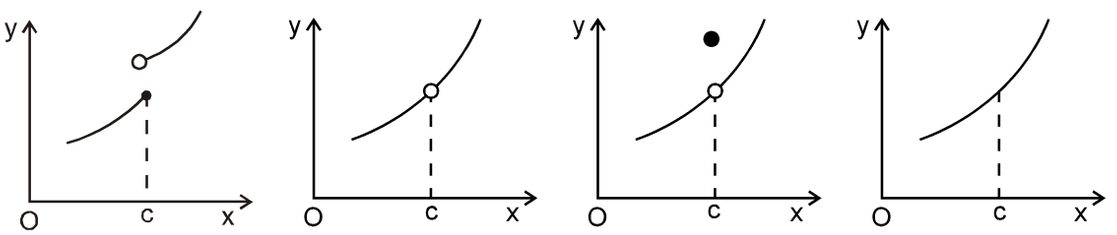
Continuity & Derivability :

A function f(x) is said to be continuous at x = c, if $\lim_{x\to c} f(x) = f(c)$

i.e. f is continuous at x = c

if $\lim_{h\to 0^+} f(c-h) = \lim_{h\to 0^+} f(c+h) = f(c)$.

If a function f(x) is continuous at x = c, the graph of f(x) at the corresponding point (c, f(c)) will not be broken. But if f(x) is discontinuous at x = c, the graph will be broken when x = c



(i) **(ii)** **(iii)** **(iv)**

((i), (ii) and (iii) are discontinuous at x = c)
((iv) is continuous at x = c)

A function f can be discontinuous due to any of the following three reasons:

(i) $\lim_{x\to c} f(x)$ does not exist i.e. $\lim_{x\to c^-} f(x) \neq \lim_{x\to c^+} f(x)$ **[figure (i)]**

(ii) f(x) is not defined at x = c **[figure (ii)]**

(iii) $\lim_{x\to c} f(x) \neq f(c)$ **[figure (iii)]**

Geometrically, the graph of the function will exhibit a break at x= c.

Theorems on continuity :

(i) If f & g are two functions which are continuous at x = c, then the functions defined by:
$F_1(x) = f(x) \pm g(x)$; $F_2(x) = K f(x)$, K is any real number ; $F_3(x) = f(x).g(x)$ are also continuous at x = c. Further, if g (c) is not zero, then $F_4(x) = \frac{f(x)}{g(x)}$ is also continuous at x = c.

(ii) If f(x) is continuous & g(x) is discontinuous at x = a, then the product function $\phi(x) = f(x).g(x)$ may or may not be continuous but sum or difference function $\phi(x) = f(x) \pm g(x)$ will necessarily be discontinuous at x = a.

(iii) If f (x) and g(x) both are discontinuous at x = a, then the product function $\phi(x) = f(x).\ g(x)$ is not necessarily be discontinuous at x = a.
and atmost one out of f(x) + g(x) and f(x) – g(x) is continuous at x = a.

Continuity of composite functions :

If f is continuous at x = c and g is continuous at x = f(c), then the composite g[f(x)] is continuous at x = c. eg. $f(x) = \frac{x\sin x}{x^2+2}$ & $g(x) = |x|$ are continuous at x = 0, hence the composite function $(gof)(x) = \left|\frac{x\sin x}{x^2+2}\right|$ will also be continuous at x = 0.

Continuity in an Interval :

(a) A function f is said to be continuous in (a, b) if f is continuous at each & every point $\in$(a, b).

(b) A function f is said to be continuous in a closed interval [a, b] if:

(i) f is continuous in the open interval (a, b),

(ii) f is right continuous at 'a' i.e. $\underset{x\to a^+}{\text{Limit}}$ f(x) = f(a) = a finite quantity and

(iii) f is left continuous at 'b' i.e. $\underset{x\to b^-}{\text{Limit}}$ f(x) = f(b) = a finite quantity.

(c) All Polynomial functions, Trigonometrical functions, Exponential and Logarithmic functions are continuous at every point of their respective domains.

On the basis of above facts continuity of a function should be checked at the following points

(i) Continuity of a function should be checked at the points where definition of a function changes.

(ii) Continuity of {f(x)} and [f(x)] should be checked at all points where f(x) becomes integer.

(iii) Continuity of sgn (f(x)) should be checked at the points where f(x) = 0 (if f(x) = 0 in any open interval containing a, then x = a is not a point of discontinuity)

(iv) In case of composite function f(g(x)) continuity should be checked at all possible points of discontinuity of g(x) and at the points where g(x) = c, where x = c is a possible point of discontinuity of f(x).

Intermediate value theorem :

A function f which is continuous in $[a, b]$ possesses the following properties:

(i) If f(a) & f(b) possess opposite signs, then there exists at least one solution of the equation f(x) = 0 in the open interval (a, b).

(ii) If K is any real number between f(a) & f(b), then there exists at least one solution of the equation f(x) = K in the open interval (a, b).

Differentiability of a function at a point :

(i) The right hand derivative of f(x) at x = a denoted by $f'(a^+)$ is defined by:

R.H.D. = $f'(a^+) = \underset{h\to 0^+}{\text{Limit}} \frac{f(a+h)-f(a)}{h}$, provided the limit exists.

(ii) The left hand derivative of f(x) at x = a denoted by $f'(a^-)$ is defined by:

L.H.D. = $f'(a^-) = \underset{h\to 0^+}{\text{Limit}} \frac{f(a-h)-f(a)}{-h}$, provided the limit exists.

A function f(x) is said to be differentiable at x = a if $f'(a^+) = f'(a^-) =$ finite

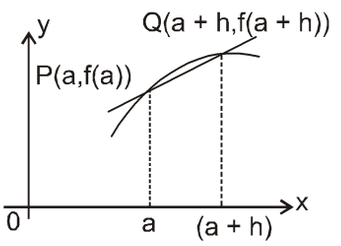
By definition $f'(a) = \underset{h\to 0^+}{\text{Limit}} \frac{f(a+h)-f(a)}{h}$

Concept of tangent and its association with derivability :

Tangent :- The tangent is defined as the limiting case of a chord or a secant.

slope of the line joining (a,f(a)) and (a + h, f(a + h)) = $\frac{f(a+h)-f(a)}{h}$

Slope of tangent at P = $f'(a) = \underset{h\to 0}{\text{Lim}} \frac{f(a+h)-f(a)}{h}$



The tangent to the graph of a continuous function f at the point P(a, f(a)) is

(i) the line through P with slope f′(a) if f′(a) exists ;

(ii) the line x = a if L.H.D. and R.H.D. both are either ∞ or $-\infty$.

If neither (i) nor (ii) holds then the graph of f does not have a tangent at the point P.

In case (i) the equation of tangent is y – f(a) = f′(a) (x – a).

In case (ii) it is x = a

Note : (i) tangent is also defined as the line joining two infinitesimally close points on a curve.

(ii) A function is said to be derivable at x = a if there exist a tangent of finite slope at that point. $f'(a^+) = f'(a^-)$ = finite value

(iii) $y = x^3$ has x-axis as tangent at origin.

(iv) y = |x| does not have tangent at x = 0 as L.H.D. ≠ R.H.D.

Relation between differentiability & continuity:

(i) If f ′(a) exists, then f(x) is continuous at x = a.

(ii) If f(x) is differentiable at every point of its domain of definition, then it is continuous in that domain.

Note : The converse of the above result is not true i.e. "If 'f' is continuous at x = a, then 'f' is differentiable at x = a is not true.

e.g. the functions f(x) = | x – 2 | is continuous at x = 2 but not differentiable at x = 2.

If f(x) is a function such that R.H.D = $f'(a^+) = \ell$ and L.H.D. = $f'(a^-) = m$.

Case - I

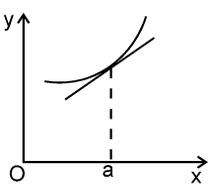
If ℓ = m = some finite value, then the function f(x) is differentiable as well as continuous.

Case - II

if $\ell \neq m$ = but both have some finite value, then the function f(x) is non differentiable but it is continuous.

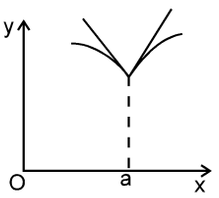
Case - III

If at least one of the ℓ or m is infinite, then the function is non differentiable but we can not say about continuity of f(x).



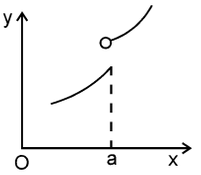
(i)

continuous and differentiable



(ii)

continuous but not differentiable



(iii)

neither continuous nor differentiable

Differentiability of sum, product & composition of functions :

(i) If f(x) & g(x) are differentiable at x = a, then the functions $f(x) \pm g(x)$, f(x). g(x) will also be differentiable at x = a & if $g(a) \neq 0$, then the function f(x)/g(x) will also be differentiable at x = a.

(ii) If f(x) is not differentiable at x = a & g(x) is differentiable at x = a, then the product function F(x) = f(x) . g(x) can still be differentiable at x = a
e.g. $f(x) = |x|$ and $g(x) = x^2$.

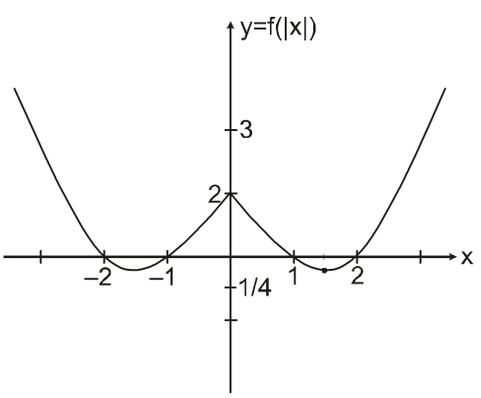
(iii) If f(x) & g(x) both are not differentiable at x = a, then the product function F(x) = f(x) . g(x) can still be differentiable at x = a e.g. $f(x) = |x|$ & $g(x) = |x|$.

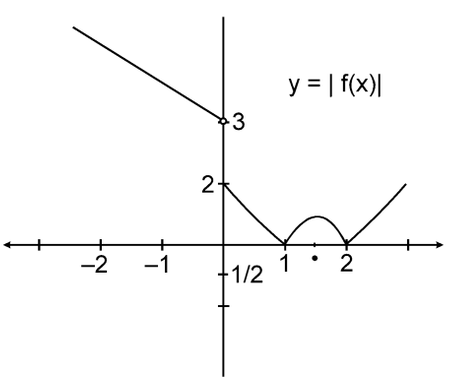
(iv) If f(x) & g(x) both are non-differentiable at x = a, then the sum function F(x) = f(x) + g(x) may be a differentiable function. e.g. $f(x) = |x|$ & $g(x) = -|x|$.

(v) If f is differentiable at x = a, then $\lim\limits_{h\to 0} \dfrac{f(a+g(h))-f(a+p(h))}{g(h)-p(h)} = f'(a)$, where

$\lim\limits_{h\to 0} p(h) = \lim\limits_{h\to 0} g(h) = 0$

Graph of f(|x|) and |f(x)|





If f(|x|) and |f(x)| are continous, then g(x) is continuous. At x = 0 f(|x|) is continuous, and |f(x)| is discontinuous therefore g(x) is discontineous at x = 0.

$\therefore$ g(x) is non differentiable at x = 0, 1, 2, (find the reason yourself).

Differentiability over an Interval :

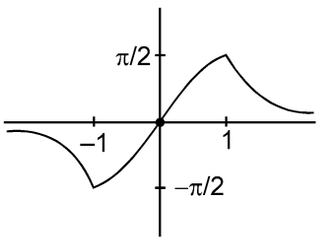
f (x) is said to be differentiable over an open interval if it is differentiable at each point of the interval and f(x) is said to be differentiable over a closed interval [a, b] if:

(i) for the points a and b, $f'(a^+)$ and $f'(b^-)$ exist finitely

(ii) for any point c such that $a < c < b$, $f'(c^+)$ & $f'(c^-)$ exist finitely and are equal.

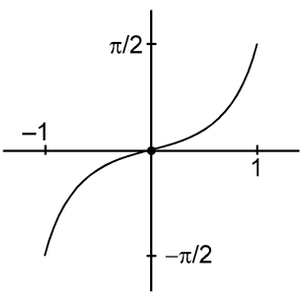
All polynomial, exponential, logarithmic and trigonometric (inverse trigonometric not included) functions are differentiable in their domain.

Graph of $y = \sin^{-1}\frac{2x}{1+x^2}$



Non differentiable at x = 1 & x = – 1

Graph of $y = \sin^{-1} x$.



Non differentiable at x = 1 & x = – 1

Note : Derivability should be checked at following points

(i) At all points where continuity is required to be checked.

(ii) At the critical points of modulus and inverse trigonometric function.

Problems of finding functions satisfying given conditions :

Important formula :

For finding limit $\lim_{h\to 0}\frac{f(a+g(h))-f(a+p(h))}{g(h)-p(h)} = f'(a)$,

if $\lim_{h\to 0} p(h) = \lim_{h\to 0} g(h) = 0$ and f(x) is differentiable at x = a

General tips :

If x, y are independent variables, then:

(i) $f(xy) = f(x) + f(y) \Rightarrow f(x) = k \ln x$ or $f(x) = 0$.

(ii) $f(xy) = f(x).\ f(y) \Rightarrow f(x) = x^n$, $n \in R$

(iii) $f(x + y) = f(x).\ f(y) \Rightarrow f(x) = a^{kx}$.

(iv) $f(x + y) = f(x) + f(y) \Rightarrow f(x) = kx$, where k is a constant.

* * * * *

METHOD OF DIFFERENTIATION

First principle of differentiation

1. The derivative of a given function f at a point x = a in its domain represent the slope of the tangent at that point, and it is defined as:

$\underset{h\to 0}{\text{Limit}} \frac{f(a+h)-f(a)}{h}$, provided the limit exists & is denoted by f′(a).

i.e. $f'(a) = \underset{x\to a}{\text{Limit}} \frac{f(x)-f(a)}{x-a}$, provided the limit exists.

2. If x and x + h belong to the domain of a function f defined by y = f(x), then

$\underset{h\to 0}{\text{Limit}} \frac{f(x+h)-f(x)}{h}$ if it exists, is called the Derivative of f at x & is denoted by f′(x) or $\frac{dy}{dx}$.

i.e., $f'(x) = \underset{h\to 0}{\text{Limit}} \frac{f(x+h)-f(x)}{h}$

This method of differentiation is also called ab-initio method or first principle method.

Derivative of some elementary functions :

	f(x)	f′(x)	
1.	x^n	nx^{n-1} ,	$(x \in R, n \in R)$
2.	a^x	$a^x \ell n\, a$,	$a > 0$
3.	$\ell n\, \lvert x \rvert$	$\frac{1}{x}$	
4.	$\log_a x$	$\frac{1}{x\, \ell n\, a}$	
5.	sin x	cos x	
6.	cos x	– sin x	
7.	sec x	sec x tan x	
8.	cosec x	– cosec x cot x	
9.	tan x	$\sec^2 x$	
10.	cot x	$-\text{cosec}^2 x$	

Basic theorems :

Sum of two differentiable functions is always differentiable.
Sum of two non-differentiable functions may be differentiable.

There are certain basic theorems in differentiation:

1. $\frac{d}{dx}(f \pm g) = f'(x) \pm g'(x)$

2. $\frac{d}{dx}(k\,f(x)) = k\,\frac{d}{dx}\,f(x)$

3. $\frac{d}{dx}\,(f(x)\,.\,g(x)) = f(x)\,g'(x) + g(x)\,f'(x)$

4. $\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)\,f'(x) - f(x)g'(x)}{g^2(x)}$

5. $\frac{d}{dx}\,(f(g(x))) = f'(g(x))\,g'(x)$

This rule is also called the chain rule of differentiation and can be written as

$$\frac{dy}{dx} = \frac{dy}{dz}\cdot\frac{dz}{dx}$$

Note that an important inference obtained from the chain rule is that

$$\frac{dy}{dy} = 1 = \frac{dy}{dx}\cdot\frac{dx}{dy}$$

$$\Rightarrow \quad \frac{dy}{dx} = \frac{1}{dx/dy}$$

another way of expressing the same concept is by considering y = f(x) and x = g(y) as inverse functions of each other.

$$\frac{dy}{dx} = f'(x) \quad \text{and} \quad \frac{dx}{dy} = g'(y)$$

$$\Rightarrow \quad g'(y) = \frac{1}{f'(x)}$$

Derivative of inverse trigonometric functions :

1. $y = \sin^{-1} x \quad \Rightarrow \quad -\frac{\pi}{2} \le y \le \frac{\pi}{2}$ and $x = \sin y$

$$\frac{dx}{dy} = \cos y \quad \Rightarrow \quad \frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-\sin^2 y}}\,,\ \sin y \neq \pm 1$$

$$\frac{dy}{dx} = \frac{1}{\sqrt{1-x^2}}\,,\ -1 < x < 1.$$

In general $|\cos y| = \sqrt{1-\sin^2 y}$. But here since $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$ $\quad \therefore \quad |\cos y| = \cos y$

2. $y = \tan^{-1}x \quad \Rightarrow \quad x = \tan y$ and $-\frac{\pi}{2} < y < \frac{\pi}{2}$

$$\frac{dx}{dy} = \sec^2 y = 1 + \tan^2 y$$

$$\Rightarrow \quad \frac{dx}{dy} = 1 + x^2 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{1}{1+x^2} \qquad (x \in R)$$

3. $y = \sec^{-1}x \quad \Rightarrow \quad y \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$ and $x = \sec y$

$$\frac{dx}{dy} = \sec y \tan y \quad \Rightarrow \quad \frac{dy}{dx} = \frac{1}{\sec y . \tan y} \ , \quad \tan y \neq 0$$

$$= \frac{1}{|\sec y . \tan y|} , \ y \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$$

$$= \frac{1}{|\sec y||\tan y|} = \frac{1}{|x|\sqrt{x^2-1}} \Rightarrow \quad \frac{dy}{dx} = \frac{1}{|x|\sqrt{x^2-1}} \qquad x \in (-\infty, -1) \cup (1, \infty)$$

Results for the derivative of inverse trigonometric functions can be summarized as :

f(x)	f′(x)	
$\sin^{-1}x$	$\frac{1}{\sqrt{1-x^2}}$	; $\quad \lvert x\rvert < 1$
$\cos^{-1}x$	$\frac{-1}{\sqrt{1-x^2}}$	; $\quad \lvert x\rvert < 1$
$\tan^{-1}x$	$\frac{1}{1+x^2}$	; $\quad x \in R$
$\cot^{-1}x$	$\frac{-1}{1+x^2}$	; $\quad x \in R$
$\sec^{-1}x$	$\frac{1}{\lvert x\rvert\sqrt{x^2-1}}$	; $\quad \lvert x\rvert > 1$
$\text{cosec}^{-1}x$	$\frac{-1}{\lvert x\rvert\sqrt{x^2-1}}$	; $\quad \lvert x\rvert > 1$

Logarithmic differentiation :

The process of taking logarithm of the function first and then differentiate is called **Logarithmic differentiation**. It is often useful in situations when

(i) a function is the product or quotient of a number of functions OR

(ii) a function is of the form $[f(x)]^{g(x)}$ where f & g are both derivable,

Implicit differentiation :

If f(x, y) = 0, is an implicit function i.e. y can't be expressed explicitly as a function of x then in order to find dy/dx, we differentiate each term w.r.t. x regarding y as a function of x and then collect terms in dy/dx.

Differentiation using substitution :

In certain situations as mentioned below, substitution simplifies differentiation. For each of the following expression, appropriate substitution is as follows

(i) $\sqrt{x^2+a^2}$: $x = a \tan\theta$, where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ or $x = a \cot\theta$, where $0 < \theta < \pi$

(ii) $\sqrt{a^2 - x^2}$: $x = a \sin\theta$, where $-\frac{\pi}{2} \le \theta \le \frac{\pi}{2}$ or $x = a \cos\theta$, where $0 \le \theta \le \pi$

(iii) $\sqrt{x^2 - a^2}$: $x = a \sec\theta$, where $\theta \in [0, \pi] - \left\{\frac{\pi}{2}\right\}$

or $x = a \operatorname{cosec}\theta$, where $\theta \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

(iv) $\sqrt{\frac{x+a}{a-x}}$: $x = a \cos\theta$, where $0 < \theta \le \pi$

Parametric differentiation :

If $y = f(\theta)$ & $x = g(\theta)$ where θ is a parameter, then $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$.

Derivative of one function with respect to another :

Let $y = f(x)$; $z = g(x)$ then $\frac{dy}{dz} = \frac{dy/dx}{dz/dx} = \frac{f'(x)}{g'(x)}$.

Higher order derivatives :

Let a function $y = f(x)$ be defined on an open interval (a, b). It's derivative, if it exists in (a, b) is a certain function $f'(x)$ [or (dy/dx) or y'] & is called the first derivative of y w. r. t. x.

If it happens that the first derivative has a derivative in (a, b) then this derivative is called the second derivative of y w. r. t. x & is denoted by $f''(x)$ or (d^2y/dx^2) or y''. While the first derivative denotes slope of the graph, the second derivative denotes it's concavity.

Similarly, the 3rd order derivative of y w. r. t. x, if it exists, is defined by $\frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right)$, it is also denoted by $f'''(x)$ or y'''.

It must be carefully noted that in case of parametric functions

although $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$ but $\frac{d^2y}{dx^2} \ne \frac{d^2y/dt^2}{dx^2/dt^2}$ rather $\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy/dt}{dx/dt}\right)$

which on applying chain rule can be resolved as

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left\{\frac{dy/dt}{dx/dt}\right\} \cdot \frac{dt}{dx}$$

$$\Rightarrow \quad \frac{d^2y}{dx^2} = \frac{\left(\frac{dx}{dt}\cdot\frac{d^2y}{dt^2} - \frac{dy}{dt}\cdot\frac{d^2x}{dt^2}\right)}{\left(\frac{dx}{dt}\right)^2} \cdot \frac{dt}{dx}$$

$$\frac{d^2y}{dx^2} = \frac{\left[\frac{dx}{dt}\cdot\frac{d^2y}{dt^2} - \frac{dy}{dt}\cdot\frac{d^2x}{dt^2}\right]}{\left(\frac{dx}{dt}\right)^3}$$

Derivative of a determinant :

If $F(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ l(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix}$, where f, g, h, l, m, n, u, v, w are differentiable functions of x, then

$$F'(x) = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ l(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ l'(x) & m'(x) & n'(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ l(x) & m(x) & n(x) \\ u'(x) & v'(x) & w'(x) \end{vmatrix}$$

L' hospital's rule :

If f(x) & g(x) are functions of x such that:

(i) $\lim\limits_{x\to a} f(x) = \infty = \lim\limits_{x\to a} g(x)$

OR

(ii) $\lim\limits_{x\to a} f(x) = 0 = \lim\limits_{x\to a} g(x)$, both f(x) and g(x) are continuous at x = a, both f(x) and g(x) are differentiable at x = a and both $f'(x)$ and $g'(x)$ are continuous at x = a,

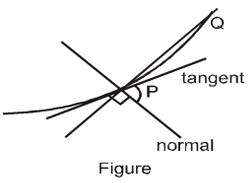
then $\lim\limits_{x\to a} \frac{f(x)}{g(x)} = \lim\limits_{x\to a} \frac{f'(x)}{g'(x)}$

* * * * *

(TANGENT & NORMAL)

Tangent and Normal

Let y = f(x) be function with graph as shown in figure. Consider secant PQ. If Q tends to P along the curve passing through the points Q_1, Q_2, ...
I.e. $Q \to P$, secant PQ will become tangent at P. A line through P perpendicular to tangent is called normal at P.



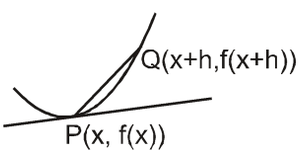
Figure

Geometrical Meaning of $\frac{dy}{dx}$

As $Q \to P$, $h \to 0$ and slope of chord PQ tends to slope of tangent at P (see figure).

Slope of chord PQ = $\frac{f(x+h)-f(x)}{h}$

$\lim\limits_{Q \to P}$ slope of chord PQ = $\lim\limits_{h \to 0} \frac{f(x+h)-f(x)}{h}$



$\Rightarrow$ slope of tangent at P = $f'(x) = \frac{dy}{dx}$

Equation of tangent and normal

$\left.\frac{dy}{dx}\right]_{(x_1, y_1)} = f'(x_1)$ denotes the slope of tangent at point (x_1, y_1) on the curve y = f(x). Hence the equation of tangent at (x_1, y_1) is given by

$(y - y_1) = f'(x_1)\,(x - x_1)$; when, $f'(x_1)$ is real.

Also, since normal is a line perpendicular to tangent at (x_1, y_1) so its equation is given by

$(y - y_1) = -\frac{1}{f'(x_1)}\,(x - x_1)$, when $f'(x_1)$ is nonzero real.

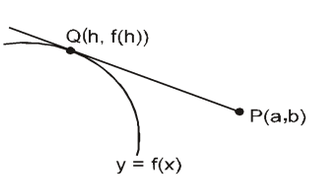
If $f'(x_1) = 0$, then tangent is the line $y = y_1$ and normal is the line $x = x_1$.

If $\lim\limits_{h \to 0} \frac{f(x_1+h)-f(x_1)}{h} = \infty$ or $-\infty$, then $x = x_1$ is tangent **(VERTICAL TANGENT)** and $y = y_1$ is normal.

Tangent and Normal from an external point

Given a point P(a, b) which does not lie on the curve y = f(x), then the equation of possible tangents to the curve y = f(x), passing through (a, b) can be found by solving for the point of contact Q.

$f'(h) = \frac{f(h)-b}{h-a}$

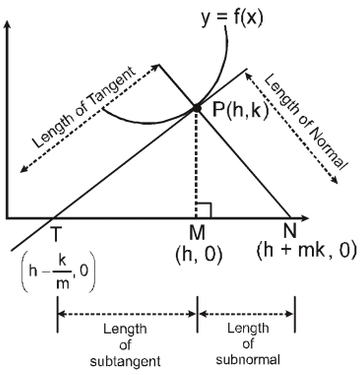


Figure

And equation of tangent is $y - b = \frac{f(h)-b}{h-a}\,(x - a)$

Lengths of tangent, normal, subtangent and subnormal :

Let P (h, k) be any point on curve y = f(x). Let tangent drawn at point P meets x-axis at T & normal at point P meets x-axis at N. Then the length PT is called the length of tangent and PN is called length of normal. (as shown in figure)



Projection of segment PT on x-axis, TM, is called the subtangent and similarly projection of line segment PN on x axis, MN is called subnormal.

Let $m = \left.\frac{dy}{dx}\right]_{(h,k)}$ = slope of tangent.

Hence equation of tangent is $m(x - h) = (y - k)$.

Putting y = 0, we get x - intercept of tangent is $x = h - \frac{k}{m}$

Similarly, the x-intercept of normal is x = h + km

Now, length PT, PN, TM, MN can be easily evaluated using distance formula

(i) $PT = |k|\sqrt{1+\frac{1}{m^2}}$ = Length of Tangent (ii) $PN = |k|\sqrt{1+m^2}$ = Length of Normal

(iii) $TM = \left|\frac{k}{m}\right|$ = Length of subtangent (iv) MN = |km| = Length of subnormal.

* * * * *

APPLICATION OF DERIVATIVES

Derivative as rate of change

In various fields of applied mathematics one has the quest to know the rate at which one variable is changing, with respect to other. The rate of change naturally refers to time. But we can have rate of change with respect to other variables also.

An economist may want to study how the investment changes with respect to variations in interest rates.

Definition : The average rate of change of a function f(x) with respect to x over an interval [a, a + h] is defined as $\frac{f(a+h)-f(a)}{h}$.

Definition : The instantaneous rate of change of f(x) with respect to x is defined as $f'(x) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h}$, provided the limit exists.

Note : To use the word 'instantaneous', x may not be representing time. We usually use the word 'rate of change' to mean 'instantaneous rate of change'.

Error and Approximation :

Let y = f(x) be a function. If these is an error δx in x then corresponding error in y is $\delta y = f(x + \delta x) - f(x)$.

We have $\lim_{\delta x\to 0} \frac{f(x+\delta x)-f(x)}{\delta x} = \frac{dy}{dx} = f'(x)$

We define the differential of y, at point x, corresponding to the increment δx as $f'(x)\,\delta x$ and denote it by dy.

i.e. $dy = f'(x)\,\delta x$.

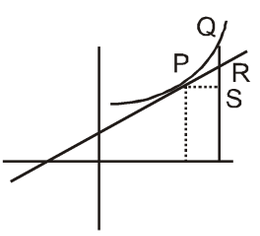
Let P(x, f(x)), Q((x + δx), f(x + δx)) (as shown in figure)

δy = QS,

δx = PS,

dy = RS

In many practical situations, it is easier to evaluate dy but not δy.



Monotonicity of a function :

Let f be a real valued function having domain D($D \subset R$) and S be a subset of D. f is said to be monotonically increasing (non decreasing) (increasing) in S if for every $x_1, x_2 \in S$, $x_1 < x_2 \Rightarrow f(x_1) \le f(x_2)$. f is said to be monotonically decreasing (non increasing) (decreasing) in S if for every $x_1, x_2 \in S$, $x_1 < x_2 \Rightarrow f(x_1) \ge f(x_2)$

f is said to be strictly increasing in S if for $x_1, x_2 \in S$, $x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$. Similarly, f is said to be strictly decreasing in S if for $x_1, x_2 \in S$, $x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$.

Notes : (i) f is strictly increasing $\Rightarrow$ f is monotonically increasing (non decreasing). But converse need not be true.

(ii) f is strictly decreasing $\Rightarrow$ f is monotonically decreasing (non increasing). Again, converse need not be true.

(iii) If f(x) = constant in S, then f is increasing as well as decreasing in S

(iv) A function f is said to be an increasing function if it is increasing in the domain. Similarly, if f is decreasing in the domain, we say that f is monotonically decreasing

(v) f is said to be a monotonic function if either it is monotonically increasing or monotonically decreasing

(vi) If f is increasing in a subset of S and decreasing in another subset of S, then f is non monotonic in S.

Application of differentiation for detecting monotonicity :

Let I be an interval (open or closed or semi open and semi closed)

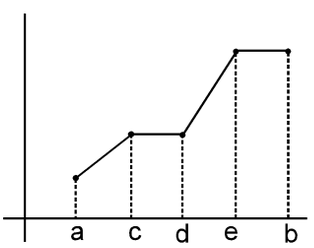
(i) If $f'(x) > 0 \ \forall \ x \in I$, then f is strictly increasing in I

(ii) If $f'(x) < 0 \ \forall \ x \in I$, then f is strictly decreasing in I

Note : Let I be an interval (or ray) which is a subset of domain of f. If $f'(x) > 0, \ \forall \ x \in I$, except for countably many points where $f'(x) = 0$, then f(x) is strictly increasing in I.

{$f'(x) = 0$ at countably many points $\Rightarrow f'(x) = 0$ does not occur on an interval which is a subset of I }

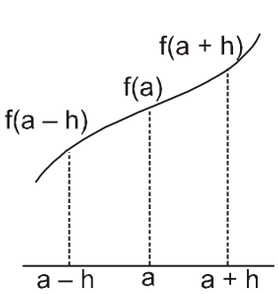
Let us consider another function whose graph is shown below for $x \in (a, b)$.

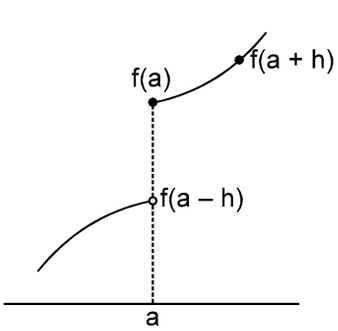


Here also $f'(x) \geq 0$ for all $x \in (a, b)$. But, note that in this case, $f'(x) = 0$ holds for all $x \in (c, d)$ and (e,b). Thus the given function is increasing (monotonically increasing) in (a, b), but not strictly increasing.

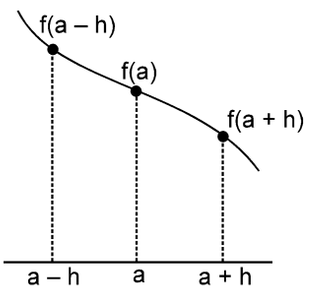
Monotonicity of function about a point :

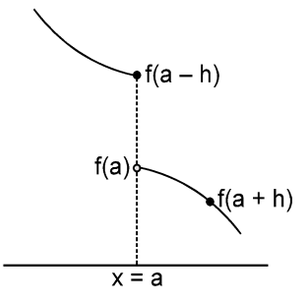
1. A function f(x) is called as a strictly increasing function about a point (or at a point) $a \in D_f$ if it is strictly increasing in an open interval containing a (as shown in figure).



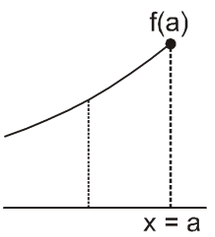


2. A function f(x) is called a strictly decreasing function about a point x = a, if it is strictly decreasing in an open interval containing a (as shown in figure).

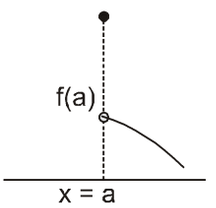




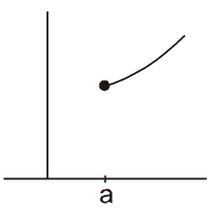
Note : If x = a is a boundary point then use the appropriate one sided inequality to test monotonicity of f(x).



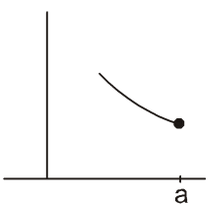
f is increasing at x = a



f is decreasing at x = a



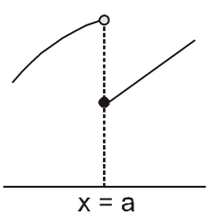
f is increasing at x = a



f is decreasing at x = a

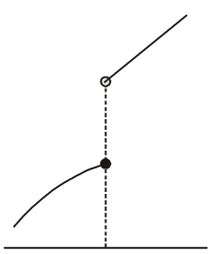
e.g. : Which of the following functions (as shown in figure) is increasing, decreasing or neither increasing nor decreasing at x = a.

(i)



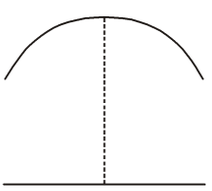
neither increasing
nor decreasing at x = a

(ii)



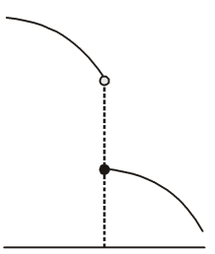
Increasing at x = a

(iii)



neither increasing
nor decreasing at x = a

(iv)



decreasing at x = a

Test for increasing and decreasing functions about a point

Let f(x) be differentiable.

(1) If $f'(a) > 0$ then f(x) is increasing at x = a.

(2) If $f'(a) < 0$ then f(x) is decreasing at x = a.

(3) If $f'(a) = 0$ then examine the sign of $f'(x)$ on the left neighbourhood and the right neighbourhood of a.

(i) If $f'(x)$ is positive on both the neighbourhoods, then f is increasing at x = a.

(ii) If $f'(x)$ is negative on both the neighbourhoods, then f is decreasing at x = a.

(iii) If $f'(x)$ have opposite signs on these neighbourhoods, then f is non-monotonic at x = a.

Note : Above method is applicable only for functions those are continuous at x = a.

Global Maximum :

A function f(x) is said to have global maximum on a set E if there exists at least one $c \in E$ such that $f(x) \leq f(c)$ for all $x \in E$.

We say global maximum occurs at x = c and global maximum (or global maximum value) is f(c).

Local Maxima :

A function f(x) is said to have a local maximum at x = c if f(c) is the greatest value of the function in a small neighbourhood (c – h, c + h), h > 0 of c.

i.e. for all $x \in (c - h, c + h)$, $x \neq c$, we have $f(x) \leq f(c)$.

i.e. $f(c - \delta) < f(c) > f(c + \delta)$, $0 < \delta < h$

Note : If x = c is a boundary point then consider (c – h, c) or (c, c + h) (h > 0) appropriately.

Global Minimum :

A function f(x) is said to have a global minimum on a set E if there exists at least one $c \in E$ such that $f(x) \geq f(c)$ for all $x \in E$.

Local Minima :

A function f(x) is said to have a local minimum at x = c if f(c) is the least value of the function in a small neighbourhood (c – h, c + h), h > 0 of c.

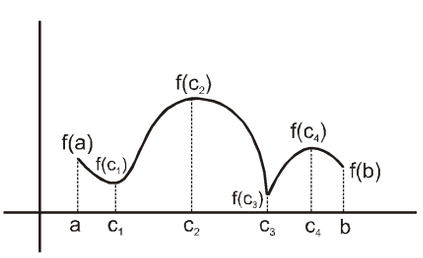
i.e. for all $x \in (c - h, c + h)$, $x \neq c$, we have $f(x) \geq f(c)$.

i.e. $f(c - \delta) > f(c) < f(c + \delta)$, $0 < \delta < h$

Extrema :

A maxima or a minima is called an extrema.

Explanation : Consider graph of y = f(x), $x \in [a, b]$



$x = a$, $x = c_2$, $x = c_4$ are points of local maxima, with maximum values $f(a)$, $f(c_2)$, $f(c_4)$ respectively.

$x = c_1$, $x = c_3$, $x = b$ are points of local minima, with minimum values $f(c_1)$, $f(c_3)$, $f(b)$ respectively
$x = c_2$ is a point of global maximum
$x = c_3$ is a point of global minimum

Maxima, Minima for differentiable functions :

Mere definition of maxima, minima becomes tedious in solving problems. We use derivative as a tool to overcome this difficulty.

A necessary condition for an extrema :

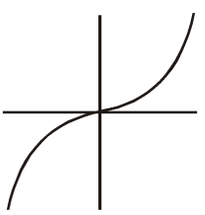
Let f(x) be differentiable at x = c.

Theorem : A necessary condition for f(c) to be an extremum of f(x) is that $f'(c) = 0$.

i.e. f(c) is extremum $\Rightarrow$ $f'(c) = 0$

Note : $f'(c) = 0$ is only a necessary condition but not sufficient

i.e. $f'(c) = 0$ $\not\Rightarrow$ f(c) is extremum.



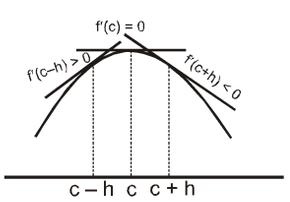
Consider $f(x) = x^3$
$f'(0) = 0$
but f(0) is not an extremum (see figure).

Sufficient condition for an extrema :

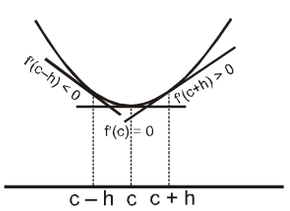
Let f(x) be a differentiable function.

Theorem : A sufficient condition for f(c) to be an extremum of f(x) is that $f'(x)$ changes sign as x passes through c.

i.e. f(c) is an extrema (see figure) $\Leftrightarrow$ $f'(x)$ changes sign as x passes through c.



x = c is a point of maxima. $f'(x)$ changes sign from positive to negative.



x = c is a point of local minima (see figure), $f'(x)$ changes sign from negative to positive.

Stationary points :

The points on graph of function f(x) where $f'(x) = 0$ are called stationary points.
Rate of change of f(x) is zero at a stationary point.

First Derivative Test :

Let f(x) be continuous and differentiable function.

Step - I. Find f′(x)

Step - II. Solve f′(x) = 0, let x = c be a solution. (i.e. Find stationary points)

Step - III. Observe change of sign

(i) If f′(x) changes sign from negative to positive as x crosses c from left to right then x = c is a point of local minima

(ii) If f′(x) changes sign from positive to negative as x crosses c from left to right then x = c is a point of local maxima.

(iii) If f′(x) does not changes sign as x crosses c then x = c is neither a point of maxima nor minima.

Maxima, Minima for continuous functions :

Let f(x) be a continuous function.

Critical points :

The points where f′(x) = 0 or f(x) is not differentiable are called critical points.

Stationary points $\subseteq$ Critical points.

Important Note :

For f(x) defined on a subset of R, points of extrema (if exists) occur at critical points

Maxima, Minima by higher order derivatives :

Second derivative test :

Let f(x) have derivatives up to second order

Step - I. Find f′(x)

Step - II. Solve f′(x) = 0. Let x = c be a solution

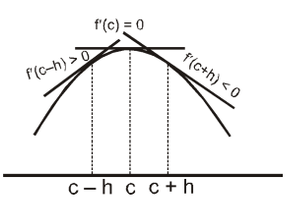
Step - III. Find f′′(c)

Step - IV.

(i) If f′′(c) = 0 then further investigation is required

(ii) If f′′(c) > 0 then x = c is a point of minima.

(iii) If f′′(c) < 0 then x = c is a point of maxima.



For maxima f′(x) changes from positive to negative (as shown in figure).

⇒ f′(x) is decreasing hence f′′(c) < 0

n^{th} Derivative test :

Let f(x) have derivatives up to nth order

If $f'(c) = f''(c) = ...= f^{n-1}(c) = 0$ and

$f^n(c) \neq 0$ then we have following possibilities

(i) n is even, $f^{(n)}(c) < 0 \Rightarrow x = c$ is point of maxima

(ii) n is even, $f^{(n)}(c) > 0 \Rightarrow x = c$ is point of minima.

(iii) n is odd, $f^{(n)}(c) < 0 \Rightarrow f(x)$ is decreasing about $x = c$

(iv) n is odd, $f^{(n)} > 0 \Rightarrow f(x)$ is increasing about $x = c$.

Application of Maxima, Minima :

For a given problem, an objective function can be constructed in terms of one parameter and then extremum value can be evaluated by equating the differential to zero. As discussed in n^{th} derivative test maxima/minima can be identified.

Useful Formulae of Mensuration to Remember :

1. Volume of a cuboid = ℓbh.

2. Surface area of cuboid = $2(\ell b + bh + h\ell)$.

3. Volume of cube = a^3

4. Surface area of cube = $6a^2$

5. Volume of a cone = $\frac{1}{3}\pi r^2 h$.

6. Curved surface area of cone = $\pi r\ell$ (ℓ = slant height)

7. Curved surface area of a cylinder = $2\pi rh$.

8. Total surface area of a cylinder = $2\pi rh + 2\pi r^2$.

9. Volume of a sphere = $\frac{4}{3}\pi r^3$.

10. Surface area of a sphere = $4\pi r^2$.

11. Area of a circular sector = $\frac{1}{2} r^2 \theta$, when θ is in radians.

12. Volume of a prism = (area of the base) × (height).

13. Lateral surface area of a prism = (perimeter of the base) × (height).

14. Total surface area of a prism = (lateral surface area) + 2 (area of the base)
(Note that lateral surfaces of a prism are all rectangle).

15. Volume of a pyramid = $\frac{1}{3}$ (area of the base) × (height).

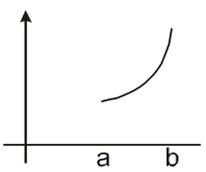
16. Curved surface area of a pyramid = $\frac{1}{2}$ (perimeter of the base) × (slant height).

(Note that slant surfaces of a pyramid are triangles).

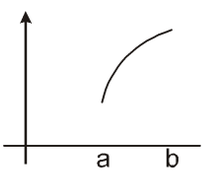
Concavity, Convexity

A function f(x) is concave in (a, b) if tangent drawn at every point $(x_0, (f(x_0))$, for $x_0 \in (a, b)$ lie below the curve. f(x) is convex in (a,b) if tangent drawn at each point $(x_0, f(x_0))$, $x_0 \in (a, b)$ lie above the curve.

Results : **1.** If $f''(x) > 0 \ \forall \ x \in (a, b)$, then the curve y = f(x) is concave in (a, b)



2. If $f''(x) < 0 \ \forall \ x \in (a, b)$ then the curve y = f(x) is convex in (a, b)



Use of monotonicity for proving inequalities

Comparison of two functions f(x) and g(x) can be done by analysing the monotonic behaviour of h(x) = f(x) – g(x)

NOTE : In proving inequalities, we must always check when does the equality takes place because the point of equality is very important in this method. Normally point of equality occur at end point of the interval or will be easily predicted by hit and trial.

Rolle's Theorem :

If a function f defined on [a, b] is

(i) continuous on [a, b]

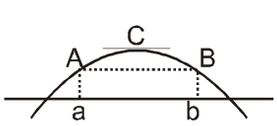
(ii) derivable on (a, b) and

(iii) f(a) = f(b),

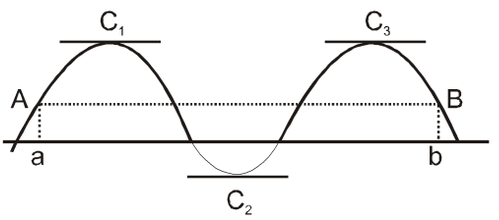
then there exists at least one real number c between a and b (a < c < b) such that f'(c) = 0

Geometrical Explanation of Rolle's Theorem :

Let the curve y = f(x), which is continuous on [a, b] and derivable on (a, b), be drawn (as shown in figure).



A(a, f(a)), B(b, f(b)), f(a) = f(b), C(c, f(c)), f'(c) = 0.



$C_1\ (c_1, f(c_1)),\ f'(c_1) = 0$

$C_2\ (c_2, f(c_2)),\ f'(c_2) = 0$

$C_3\ (c_3, f(c_3)),\ f'(c_3) = 0$

The theorem simply states that between two points with equal ordinates on the graph of f(x), there exists at least one point where the tangent is parallel to x-axis.

Algebraic Interpretation of Rolle's Theorem :

Between two zeros a and b of f(x) (i.e. between two roots a and b of f(x) = 0) there exists at least one zero of f′(x)

Lagrange's Mean Value Theorem (LMVT) :

If a function f defined on [a, b] is

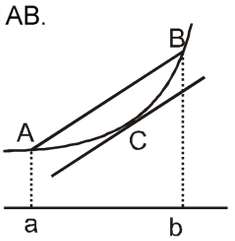
(i) continuous on [a, b] and

(ii) derivable on (a, b)

then there exists at least one real numbers between a and b (a < c < b) such that $\frac{f(b)-f(a)}{b-a} = f'(c)$

Geometrical Interpretation of LMVT :

The theorem simply states that between two points A and B of the graph of f(x) there exists at least one point where tangent is parallel to chord AB.



C(c, f(c)), f′(c) = slope of AB.

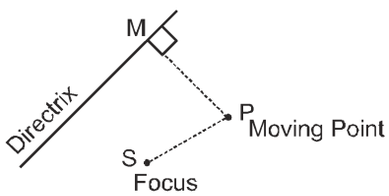
* * * * *

PARABOLA

Conic Sections:

A conic section, or conic is the locus of a point which moves in a plane so that its distance from a fixed point is in a constant ratio to its perpendicular distance from a fixed straight line.

- The fixed point is called the **Focus.**
- The fixed straight line is called the **Directrix.**
- The constant ratio is called the **Eccentricity** denoted by e.



$\frac{PS}{PM} = e$

- The line passing through the focus & perpendicular to the directrix is called the **Axis**.
- A point of intersection of a conic with its axis is called a **Vertex**.

If S is (p, q) & directrix is $\ell x + my + n = 0$

then $PS = \sqrt{(x-\alpha)^2 + (y-\beta)^2}$ & $PM = \frac{|\ell x + my + n|}{\sqrt{\ell^2 + m^2}}$

$\frac{PS}{PM} = e \quad \Rightarrow \quad (\ell^2 + m^2)[(x-p)^2 + (y-q)^2] = e^2(\ell x + my + n)^2$

Which is of the form $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

Distinguishing various conics :

The nature of the conic section depends upon the position of the focus S w.r.t. the directrix & also upon the value of the eccentricity e. Two different cases arise.

Case (I) When The Focus Lies On The Directrix.

In this case $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ & the general equation of a conic represents a pair of straight lines if:

$e > 1 \equiv h^2 > ab$ the lines will be real & distinct intersecting at S.

$e = 1 \equiv h^2 \geq ab$ the lines will coincident.

$e < 1 \equiv h^2 < ab$ the lines will be imaginary.

Case (II) When The Focus Does Not Lie On Directrix.

a parabola	an ellipse	a hyperbola	rectangular hyperbola
$e = 1; \Delta \neq 0,$	$0 < e < 1; \Delta \neq 0;$	$e > 1; \Delta \neq 0;$	$e > 1; \Delta \neq 0$
$h^2 = ab$	$h^2 < ab$	$h^2 > ab$	$h^2 > ab; a + b = 0$

PARABOLA

Definition and terminology

A parabola is the locus of a point, whose distance from a fixed point (focus) is equal to perpendicular distance from a fixed straight line (directrix).

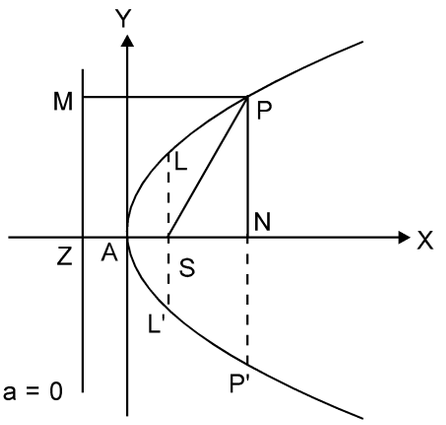
Four standard forms of the parabola are

$y^2 = 4ax$; $y^2 = -4ax$; $x^2 = 4ay$; $x^2 = -4ay$

For parabola $y^2 = 4ax$:

(i) Vertex is (0, 0) **(ii)** focus is (a, 0)

(iii) Axis is y = 0 **(iv)** Directrix is x + a = 0



Focal Distance: The distance of a point on the parabola from the focus. $y^2 = 4ax$

Focal Chord : A chord of the parabola, which passes through the focus.

Double Ordinate: A chord of the parabola perpendicular to the axis of the symmetry.

Latus Rectum: A double ordinate passing through the focus or a focal chord perpendicular to the axis of parabola is called the Latus Rectum (L.R.).

For $y^2 = 4ax$. $\Rightarrow$ Length of the latus rectum = 4a.

$\Rightarrow$ ends of the latus rectum are L(a, 2a) & L' (a, – 2a).

NOTE :

(i) Perpendicular distance from focus on directrix = half the latus rectum.

(ii) Vertex is middle point of the focus & the point of intersection of directrix & axis.

(iii) Two parabolas are said to be equal if they have the same latus rectum.

Parametric representation:

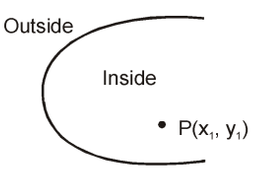
The simplest & the best form of representing the co–ordinates of a point on the parabola is $(at^2, 2at)$ i.e. the equations $x = at^2$ & $y = 2at$ together represents the parabola $y^2 = 4ax$, t being the parameter.

Parametric form for :

$y^2 = -4ax$	$(-at^2, 2at)$
$x^2 = 4ay$	$(2at , at^2)$
$x^2 = -4ay$	$(2at , -at^2)$

Position of a point relative to a parabola:

The point (x_1 , y_1) lies outside, on or inside the parabola $y^2 = 4ax$ according as the expression $y_1^2 - 4ax_1$ is positive, zero or negative.



$S_1 : y_1^2 - 4ax_1$

$S_1 < 0 \rightarrow$ Inside

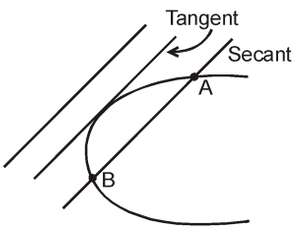
$S_1 > 0 \rightarrow$ Outside

Line & a parabola:

The line y = mx + c meets the parabola $y^2 = 4ax$ in two points real, coincident or imaginary according as $a \gtreqless cm \Rightarrow$ condition of tangency is, c = a/m.

Length of the chord intercepted by the parabola on the line y = m x + c is :

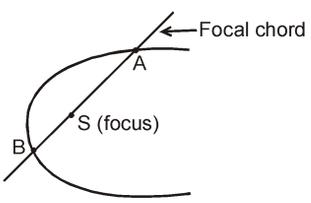
$$\left(\frac{4}{m^2}\right)\sqrt{a(1+m^2)(a-mc)}\ .$$



NOTE :

1. The equation of a chord joining t_1 & t_2 is $2x - (t_1 + t_2)\, y + 2\, at_1\, t_2 = 0$.

2. If t_1 & t_2 are the ends of a focal chord of the parabola $y^2 = 4ax$ then $t_1t_2 = -1$. Hence the co–ordinates at the extremities of a focal chord can be taken as $(at^2, 2at)$ & $\left(\frac{a}{t^2}, -\frac{2a}{t}\right)$



3. Length of the focal chord making an angle α with the x– axis is $4a\,\text{cosec}^2\,\alpha$.

Tangents to the parabola $y^2 = 4ax$:

Equation of tangent at a point on the parabola can be obtained by replacement method or using derivatives.
In replacement method, following changes are made to the second degree equation to obtain T.
$x^2 \to x\,x_1$, $y^2 \to y\,y_1$, $2xy \to xy_1 + x_1y$, $2x \to x + x_1$, $2y \to y + y_1$
So, it follows that the targents are :

(i) $y\,y_1 = 2\,a\,(x + x_1)$ at the point (x_1, y_1) ;

(ii) $y = mx + \frac{a}{m}$ $(m \neq 0)$ at $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$

(iii) $t\,y = x + a\,t^2$ at $(at^2, 2at)$.

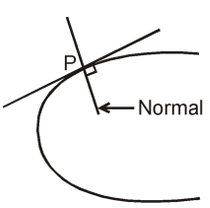
(iv) Point of intersection of the tangents at the point t_1 & t_2 is $\{ at_1\,t_2, a(t_1 + t_2) \}$.

Normals to the parabola $y^2 = 4ax$:

Normal is obtained using the slope of tangent.

Slope of tangent at $(x_1, y_1) = \frac{2a}{y_1}$

$\Rightarrow$ Slope of normal $= -\frac{y_1}{2a}$



(i) $y - y_1 = -\frac{y_1}{2a}\,(x - x_1)$ at (x_1, y_1) ;

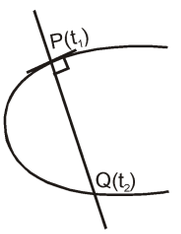
(ii) $y = mx - 2am - am^3$ at $(am^2, -2am)$

(iii) $y + tx = 2at + at^3$ at $(at^2, 2at)$.

NOTE :

(i) Point of intersection of normals at t_1 & t_2 is $(a(t_1^2 + t_2^2 + t_1t_2 + 2), -at_1t_2(t_1 + t_2))$.

(ii) If the normals to the parabola $y^2 = 4ax$ at the point t_1, meets the parabola again at the point t_2, then $t_2 = -\left(t_1 + \frac{2}{t_1}\right)$.

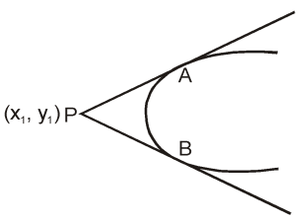


(iii) If the normals to the parabola $y^2 = 4ax$ at the points t_1 & t_2 intersect again on the parabola at the point 't_3' then $t_1t_2 = 2$; $t_3 = -(t_1 + t_2)$ and the line joining t_1 & t_2 passes through a fixed point $(-2a, 0)$.

Pair of tangents:

The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the parabola $y^2 = 4ax$ is given by: $SS_1 = T^2$ where :

$S \equiv y^2 - 4ax$; $S_1 = y_1^2 - 4ax_1$; $T \equiv yy_1 - 2a(x + x_1)$.

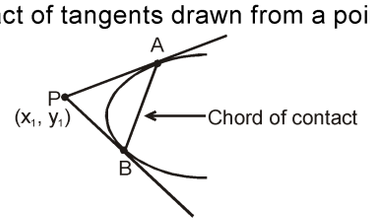


Director circle:

Locus of the point of intersection of the perpendicular tangents to a curve is called the Director Circle. For parabola $y^2 = 4ax$ it's equation is $x + a = 0$ which is parabola's own directrix.

Chord of contact:

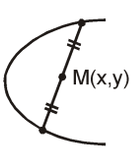
Equation to the chord of contact of tangents drawn from a point $P(x_1, y_1)$ is $yy_1 = 2a(x + x_1)$.



NOTE : The area of the triangle formed by the tangents from the point (x_1, y_1) & the chord of contact is $\frac{1}{2a}(y_1^2 - 4ax_1)^{3/2}$

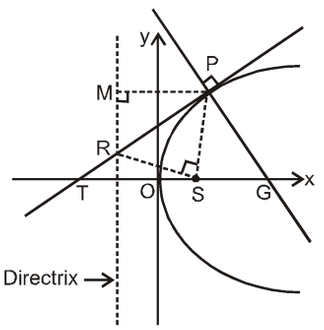
Chord with a given middle point:

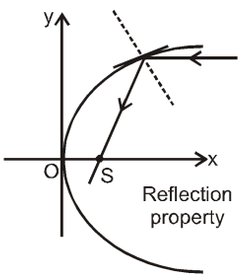
Equation of the chord of the parabola $y^2 = 4ax$ whose middle point is (x_1, y_1) is $y - y_1 = \frac{2a}{y_1}(x - x_1) \equiv T = S_1$



Important Highlights:

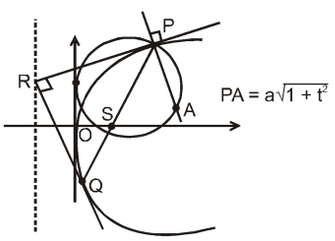
(i) If the tangent & normal at any point 'P' of the parabola intersect the axis at T & G then ST = SG = SP where 'S' is the focus. In other words the tangent and the normal at a point P on the parabola are the bisectors of the angle between the focal radius SP & the perpendicular from P on the directrix. From this we conclude that all rays emanating from S will become parallel to the axis of theparabola after reflection.



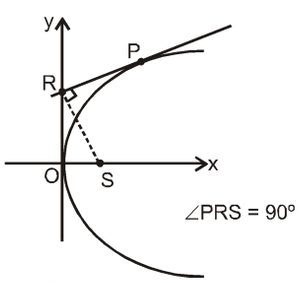


(ii) The portion of a tangent to a parabola cut off between the directrix & the curve subtends a right angle at the focus.
See figure above.

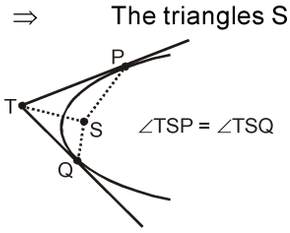
(iii) The tangents at the extremities of a focal chord intersect at right angles on the directrix, and hence a circle on any focal chord as diameter touches the directrix. Also a circle on any focal radii of a point P $(at^2, 2at)$ as diameter touches the tangent at the vertex and intercepts a chord of length $a\sqrt{1+t^2}$ on a normal at the point P.



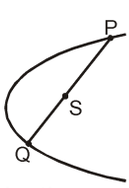
(iv) Any tangent to a parabola & the perpendicular on it from the focus meet on the tangent at the vertex.



(v) If the tangents at P and Q meet in T, then:
⇒ TP and TQ subtend equal angles at the focus S.
⇒ $ST^2 = SP.\ SQ$ & ⇒ The triangles SPT and STQ are similar.



(vi) Semi latus rectum of the parabola $y^2 = 4ax$, is the harmonic mean between segments of any focal chord of the parabola.



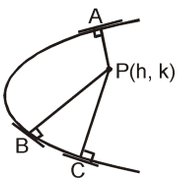
$$\frac{2(PS)(SQ)}{PS+SQ} = 2a$$

(vii) The area of the triangle formed by three points on a parabola is twice the area of the triangle formed by the tangents at these points.

(viii) If normal are drawn from a point P(h, k) to the parabola $y^2 = 4ax$ then
$k = mh - 2am - am^3$ i.e. $am^3 + m(2a - h) + k = 0$.

$m_1 + m_2 + m_3 = 0$; $m_1m_2 + m_2m_3 + m_3m_1 = \frac{2a-h}{a}$; $m_1 m_2 m_3 = -\frac{k}{a}$.

Where m_1, m_2, & m_3 are the slopes of the three concurrent normals. Note that



A, B, C → Conormal points

⇒ algebraic sum of the slopes of the three concurrent normals is zero.

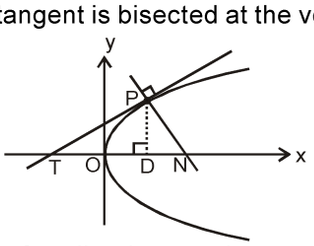
⇒ algebraic sum of the ordinates of the three conormal points on the parabola is zero

⇒ Centroid of the Δ formed by three co–normal points lies on the x–axis.

⇒ Condition for three real and distinct normals to be drawn froma point P (h, k) is

$h > 2a$ & $k^2 < \frac{4}{27a}(h - 2a)^3$.

(ix) Length of subtangent at any point P(x, y) on the parabola $y^2 = 4ax$ equals twice the abscissa of the point P. Note that the subtangent is bisected at the vertex.



TD = 2(OD) , DN = 2a

(x) Length of subnormal is constant for all points on the parabola & is equal to the semi latus rectum.
See figure above.

* * * * *

ELLIPSE

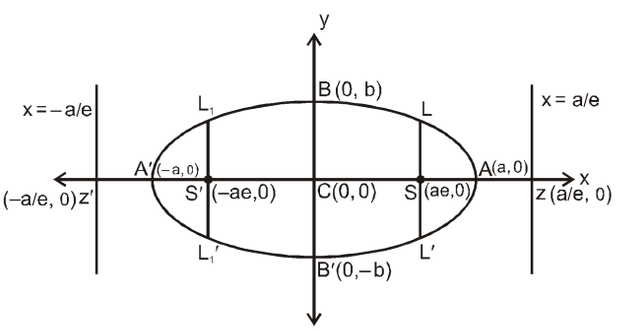
In this chapter we are going to discuss in detail the nature of path in which on planets move around the sun. They follow on elliptical path with the sun at one of its foci. Let us look at the definition of ellipse.

Definitions : -

It is locus of a point which moves in such a way that the ratio of its distance from a fixed point and a fixed line (not passes through fixed point and all points and line lies in same plane) is constant (e), which is less than one.
The fixed point is called - **focus**
The fixed line is called -**directrix.**
The constant ratio is called - **eccentricity**, it is denoted by **'e'**.

Standard Equation

Standard equation of an ellipse referred to its principal axes along the co–ordinate axes is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$,
where $a > b$ & $b^2 = a^2(1-e^2)$.



Eccentricity: $e=\sqrt{1-\frac{b^2}{a^2}}$, $(0 < e < 1)$

Focii : $S \equiv (ae, 0)$ & $S' \equiv (-ae, 0)$.

Equations of Directrices : $x=\frac{a}{e}$ & $x=-\frac{a}{e}$.

Major Axis : The line segment A′A in which the focii S′ & S lie is of length 2a & is called the major axis ($a > b$) of the ellipse. Point of intersection of major axis with directrix is called the foot of the directrix (Z).

Minor Axis : The y–axis intersects the ellipse in the points $B' \equiv (0, -b)$ & $B \equiv (0, b)$. The line segment B′B is of length 2b ($b < a$) is called the minor axis of the ellipse.

Principal Axis : The major & minor axes together are called principal axis of the ellipse.

Vertices : Point of intersection of ellipse with major axis. $A' \equiv (-a, 0)$ & $A \equiv (a, 0)$.

Focal Chord : A chord which passes through a focus is called a focal chord.

Double Ordinate : A chord perpendicular to the major axis is called a double ordinate.

Latus Rectum : The focal chord perpendicular to the major axis is called the latus rectum.

Length of latus rectum **(LL′)** $=\frac{2b^2}{a}=\frac{(\text{minor axis})^2}{\text{major axis}}=2a\left(1-e^2\right)$

$= 2e$ (distance from focus to the corresponding directrix)

Centre : The point which bisects every chord of the conic drawn through it, is called the centre of the conic. $C \equiv (0, 0)$ the origin is the centre of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$.

Note : (i) If the equation of the ellipse is given as $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ and nothing is mentioned, then the rule is to assume that $a > b$.

(ii) If b > a is given, then the y–axis will become major axis and x-axis will become the minor axis and all other points and lines will change accordingly.

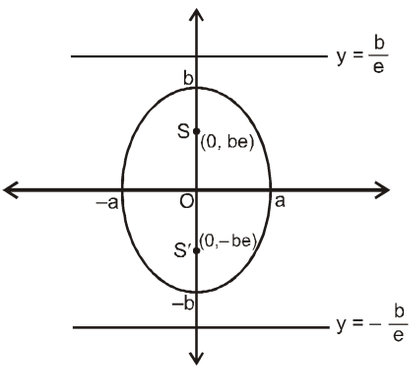
Equation : $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Foci $(0, \pm be)$ Directrices : $y = \pm \frac{b}{e}$

$a^2 = b^2 (1 - e^2)$, $a < b$. $\Rightarrow$ $e = \sqrt{1 - \frac{a^2}{b^2}}$

Vertices $(0, \pm b)$; L.R. $y = \pm be$

ℓ (L·R.) $= \frac{2a^2}{b}$, centre : (0, 0)

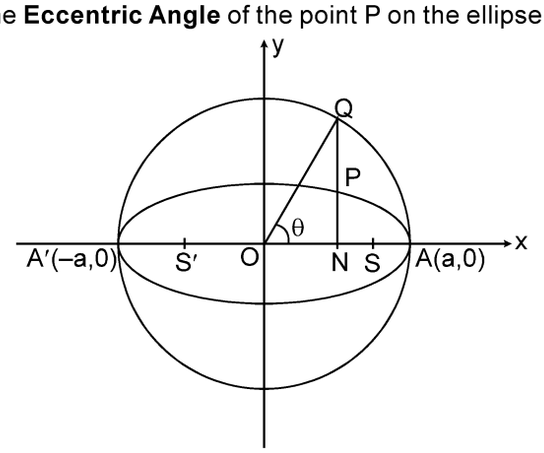


Auxiliary Circle / Eccentric Angle :

A circle described on major axis of ellipse as diameter is called the **auxiliary circle**.

Let Q be a point on the auxiliary circle $x^2 + y^2 = a^2$ such that line through Q perpendicular to the x – axis on the way intersects the ellipse at P, then P & Q are called as the **Corresponding Points** on the ellipse & the auxiliary circle respectively. 'θ' is called the **Eccentric Angle** of the point P on the ellipse $(-\pi < \theta \le \pi)$. $Q \equiv (a\cos\theta , a\sin\theta)$

$P \equiv (a\cos\theta , b\sin\theta)$



Note that :

$$\frac{\ell(PN)}{\ell(QN)} = \frac{b}{a} = \frac{\text{Semi minor axis}}{\text{Semi major axis}}$$

NOTE : If from each point of a circle perpendiculars are drawn upon a fixed diameter then the locus of the points dividing these perpendiculars in a given ratio is an ellipse of which the given circle is the auxiliary circle.

Parametric Representation :

The equations $x = a\cos\theta$ & $y = b\sin\theta$ together represent the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

Where θ is a parameter. Note that if $P(\theta) \equiv (a\cos\theta, b\sin\theta)$ is on the ellipse then; $Q(\theta) \equiv (a\cos\theta, a\sin\theta)$ is on the auxiliary circle.

The equation to the chord of the ellipse joining two points with eccentric angles α & β is given by $\frac{x}{a}\cos\frac{\alpha+\beta}{2} + \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha-\beta}{2}$.

Position of a Point w.r.t. an Ellipse :

The point $P(x_1, y_1)$ lies outside, inside or on the ellipse according as ; $\frac{x_1^2}{a^2} + \frac{y_1^2}{b^2} - 1 > < \text{or} = 0$.

Line and an Ellipse :

The line $y = mx + c$ meets the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in two points real, coincident or imaginary according as c^2 is < = or > $a^2m^2 + b^2$.

Hence $y = mx + c$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ if $c^2 = a^2m^2 + b^2$.

Tangents :

(a) Slope form: $y = mx \pm \sqrt{a^2m^2+b^2}$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ for all values of m.

(b) Point form : $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at (x_1, y_1).

(c) Parametric form: $\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$ is tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $(a\cos\theta, b\sin\theta)$.

Note : (i) There are two tangents to the ellipse having the same m, i.e. there are two tangents parallel to any given direction. These tangents touches the ellipse at extremities of a diameter.

(ii) Point of intersection of the tangents at the point α & β is, $\left(a\frac{\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, b\frac{\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$

(iii) The eccentric angles of the points of contact of two parallel tangents differ by π.

Normals :

(i) Equation of the normal at (x_1, y_1) to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $\frac{a^2x}{x_1}-\frac{b^2y}{y_1} = a^2 - b^2$.

(ii) Equation of the normal at the point $(a\cos\theta, b\sin\theta)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is;

$ax.\sec\theta - by.\operatorname{cosec}\theta = (a^2 - b^2)$.

(iii) Equation of a normal in terms of its slope 'm' is $y = mx - \frac{(a^2-b^2)m}{\sqrt{a^2+b^2m^2}}$.

Pair of Tangents :

The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$ is given by: $SS_1 = T^2$ where :

$S \equiv \frac{x^2}{a^2}+\frac{y^2}{b^2} - 1$; $S_1 = \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2} - 1$; $T \equiv \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$.

Director Circle :

Locus of the point of intersection of the tangents which meet at right angles is called the **Director Circle.** The equation to this locus is $x^2 + y^2 = a^2 + b^2$ i.e. a circle whose centre is the centre of the ellipse & whose radius is the length of the line joining the ends of the major & minor axes.

Chord of Contact :

Equation to the chord of contact of tangents drawn from a point $P(x_1, y_1)$ to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $T = 0$, where $T = \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$

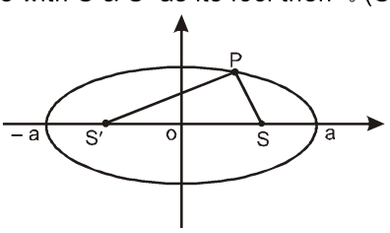
Chord with a given middle point :

Equation of the chord of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ whose middle point is (x_1, y_1) is $T = S_1$, where $S_1 = \frac{x_1^2}{a^2}+\frac{y_1^2}{b^2} - 1$; $T \equiv \frac{xx_1}{a^2} + \frac{yy_1}{b^2} - 1$.

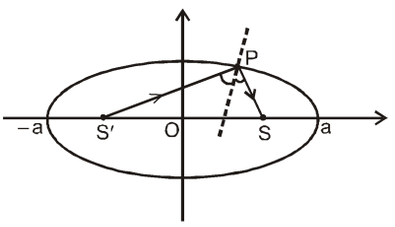
Important Highlights :

Refering to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

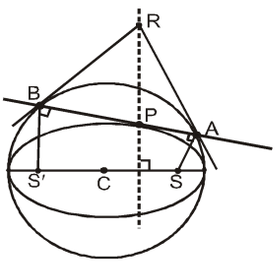
(1) If P be any point on the ellipse with S & S′ as its foci then ℓ (SP) + ℓ (S′P) = 2a.



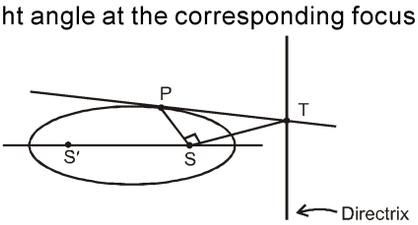
(2) The tangent & normal at a point P on the ellipse bisect the external & internal angles between the focal distances of P. This refers to the well known reflection property of the ellipse which states that rays from one focus are reflected through other focus & vice–versa. Hence we can deduce that the straight lines joining each focus to the foot of the perpendicular from the other focus upon the tangent at any point P meet on the normal PG and bisects it where G is the point where normal at P meets the major axis.



(3) The product of the length's of the perpendicular segments from the foci on any tangent to the ellipse is b² and the feet of these perpendiculars lie on its auxiliary circle and the tangents at these feet to the auxiliary circle meet on the ordinate of P and that the locus of their point of intersection is a similiar ellipse as that of the original one.

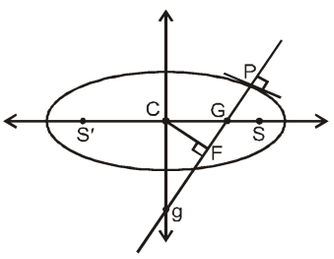


(4) The portion of the tangent to an ellipse between the point of contact & the directrix subtends a right angle at the corresponding focus.



(5) If the normal at any point P on the ellipse with centre C meet the major & minor axes in G & g respectively, & if CF be perpendicular upon this normal, then

(i) PF. PG = b²
(ii) PF. Pg = a²
(iii) PG. Pg = SP. S′P
(iv) CG. CT = CS²
(v) locus of the mid point of Gg is another ellipse having the same eccentricity as that of the original ellipse.



[where S and S′ are the focii of the ellipse and T is the point where tangent at P meet the major axis]

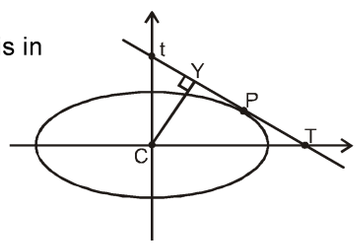
(6) The circle on any focal distance as diameter touches the auxiliary circle. Perpendiculars from the centre upon all chords which join the ends of any perpendicular diameters of the ellipse are of constant length.

(7) If the tangent at the point P of a standard ellipse meets the axis in T and t and CY is the perpendicular on it from the centre then,

(i) T t. PY = $a^2 - b^2$

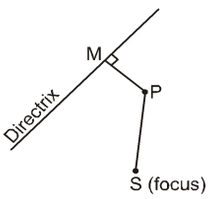
and

(ii) least value of T t is a + b.



HYPERBOLA

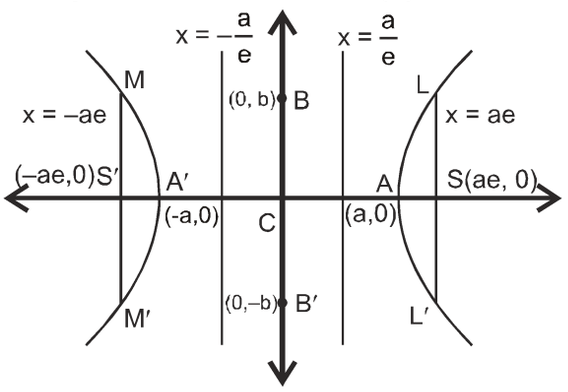
Hyperbolic curves are of special importance in the field of science and technology especially astronomy and space studies. In this chapter we are going to study the characteristics of such curves.



Definition : A hyperbola is defined as the locus of a point moving in a plane in such a way that the ratio of its distance from a fixed point to that from a fixed line (the point does not lie on the line) is a fixed constant greater than 1.

$\frac{PS}{PM} = e > 1$, e – eccentricity

Standard equation & definition(s)



Standard equation of hyperbola is $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$,

where $b^2 = a^2(e^2 - 1)$.

- **Eccentricity (e) :**

  $e^2 = 1 + \frac{b^2}{a^2}$

- **Foci :**

  S ≡ (ae, 0) & S′ ≡ (– ae, 0).

- **Equations of directrices :**

  $x = \frac{a}{e}$ & $x = -\frac{a}{e}$.

- **Transverse axis :**

  The line segment A′A of length 2a in which the foci S′ & S both lie is called the transverse axis of the hyperbola.

- **Conjugate axis :**

  The line segment B′B of length 2b between the two points B′ ≡ (0, – b) & B ≡ (0, b) is called as the conjugate axis of the hyperbola.

- **Principal axes :**

  The transverse & conjugate axis together are called principal axes of the hyperbola.

- **Vertices :**

  A ≡ (a, 0) & A′ ≡ (– a, 0)

- **Focal chord :**
  A chord which passes through a focus is called a focal chord.
- **Double ordinate :**
  A chord perpendicular to the transverse axis is called a double ordinate.
- **Latus rectum :**
  Focal chord perpendicular to the transverse axis is called latus rectum. Its length (ℓ) is given by

$$\ell = \frac{2b^2}{a} = \frac{(C.A.)^2}{T.A.} = 2a\,(e^2 - 1).$$

Note : (i) Length of latus rectum = 2 e × (distance of focus from corresponding directrix)

(ii) End points of latus rectum are $L \equiv \left(ae, \frac{b^2}{a}\right)$, $L' \equiv \left(ae, -\frac{b^2}{a}\right)$, $M \equiv \left(-ae, \frac{b^2}{a}\right)$, $M' \equiv \left(-ae, -\frac{b^2}{a}\right)$

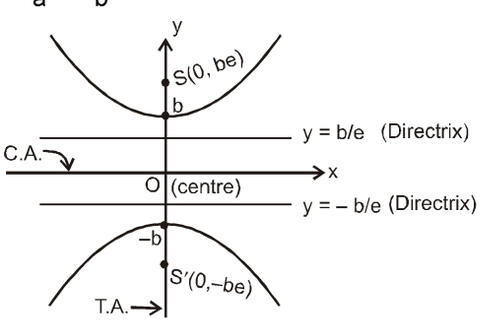
- **Centre :**
  The point which bisects every chord of the conic, drawn through it, is called the centre of the conic. $C \equiv (0,0)$ the origin is the centre of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$.

General note :
Since the fundamental equation to hyperbola only differs from that to ellipse in having $-b^2$ instead of b^2 it will be found that many propositions for hyperbola are derived from those for ellipse by simply changing the sign of b^2.

Conjugate hyperbola :

Two hyperbolas such that transverse & conjugate axes of one hyperbola are respectively the conjugate & the transverse axes of the other are called **conjugate hyperbolas** of each other.

eg. $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ & $-\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are conjugate hyperbolas of each other.



Equation : $\frac{y^2}{b^2} - \frac{x^2}{a^2} = 1$

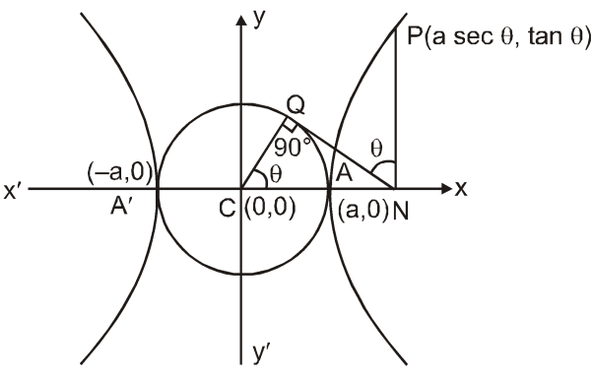
$a^2 = b^2\,(e^2 - 1) \quad \Rightarrow \quad e = \sqrt{1 + \frac{a^2}{b^2}}$

Vertices $(0, \pm b)$; ℓ (L.R.) $= \frac{2a^2}{b}$

Note : **(a)** If e_1 & e_2 are the eccentrcities of the hyperbola & its conjugate then $e_1^{-2} + e_2^{-2} = 1$.
(b) The foci of a hyperbola and its conjugate are concyclic and form the vertices of a square.
(c) Two hyperbolas are said to be similar if they have the same eccentricity.
(d) Two similar hyperbolas are said to be equal if they have same latus rectum.
(e) If a hyperbola is equilateral then the conjugate hyperbola is also equilateral.

Auxiliary Circle :

A circle drawn with centre C and transverse axis as a diameter is called the **auxiliary circle** of the hyperbola. Equation of the auxiliary circle is $x^2 + y^2 = a^2$.
Note from the following figure that P & Q are called the **"corresponding points"** of the hyperbola & the auxiliary circle.



Parametric representation :

The equations $x = a \sec\theta$ & $y = b \tan\theta$ together represent the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where θ is a parameter.
Note that if $P(\theta) \equiv (a \sec\theta, b \tan\theta)$ is on the hyperbola then,
$Q(\theta) \equiv (a \cos\theta, a \sin\theta)$ is on the auxiliary circle.
The equation to the chord of the hyperbola joining the two points $P(\alpha)$ & $Q(\beta)$ is given by

$$\frac{x}{a}\cos\frac{\alpha-\beta}{2} - \frac{y}{b}\sin\frac{\alpha+\beta}{2} = \cos\frac{\alpha+\beta}{2}.$$

Position of a point 'P' w.r.t. a hyperbola :

The quantity $S_1 \equiv \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$ is positive, zero or negative according as the point (x_1, y_1) lies inside, on or outside the curve.

Line and a hyperbola :

The straight line $y = mx + c$ is a secant, a tangent or passes outside the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ according as : $c^2 >$ or $=$ or $< a^2m^2 - b^2$, respectively.

Tangents :

(i) Slope form : $y = mx \pm \sqrt{a^2m^2 - b^2}$ can be taken as the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, having slope 'm'.

(ii) Point form : Equation of tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point (x_1, y_1) is $\frac{xx_1}{a^2} - \frac{yy_1}{b^2} = 1$.

(iii) Parametric form : Equation of the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point. $(a \sec\theta, b \tan\theta)$ is $\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1$.

Note : **(i)** Point of intersection of the tangents at $P(\theta_1)$ & $Q(\theta_2)$ is $\left(a\frac{\cos\frac{\theta_1-\theta_2}{2}}{\cos\frac{\theta_1+\theta_2}{2}},\ b\tan\left(\frac{\theta_1+\theta_2}{2}\right)\right)$

(ii) If $|\theta_1 + \theta_2| = \pi$, then tangents at these points (θ_1 & θ_2) are parallel.

(iii) There are two parallel tangents having the same slope m. These tangents touches the hyperbola at the extremities of a diameter.

Normals :

(a) The equation of the normal to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ at the point P (x_1, y_1) on it is

$\frac{a^2x}{x_1} + \frac{b^2y}{y_1} = a^2 + b^2 = a^2e^2$.

(b) The equation of the normal at the point P (a sec θ, b tan θ) on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is

$\frac{ax}{\sec\theta} + \frac{by}{\tan\theta} = a^2 + b^2 = a^2e^2$.

(c) Equation of normals in terms of its slope 'm' are $y = mx \pm \frac{(a^2+b^2)m}{\sqrt{a^2-b^2m^2}}$.

Pair of tangents:

The equation to the pair of tangents which can be drawn from any point (x_1, y_1) to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is given by: $SS_1 = T^2$ where :

$S \equiv \frac{x^2}{a^2} - \frac{y^2}{b^2} - 1$; $S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$; $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$.

Director circle :

The locus of the point of intersection of the tangents which are at right angle is known as the director circle of the hyperbola. The equation to the director circle is : $x^2 + y^2 = a^2 - b^2$.
If $b^2 < a^2$, then the director circle is real.
If $b^2 = a^2$ (i.e. rectangular hyperbola), then the radius of the director circle is zero and it reduces to a point circle at the origin. In this case centre is the only point from which two perpendicular tangents can be drawn on the curve.
If $b^2 > a^2$, then the radius of the director circle is imaginary, so that there is no such circle and so no pair of tangents at right angle can be drawn to the curve.

Chord of contact :

Equation to the chord of contact of tangents drawn from a point $P(x_1, y_1)$ to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is , T = 0, where $T = \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$

Chord with a given middle point :

Equation of the chord of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ whose middle point is (x_1, y_1) is $\mathbf{T = S_1}$,

where $S_1 = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} - 1$; $T \equiv \frac{xx_1}{a^2} - \frac{yy_1}{b^2} - 1$.

Diameter :

The locus of the middle points of a system of parallel chords with slope 'm' of hyperbola is called its diameter. It is a straight line passing through the centre of the hyperbola and has the equation

$y = -\frac{b^2}{a^2 m}x$.

NOTE : All diameters of the hyperbola pass through its centre.

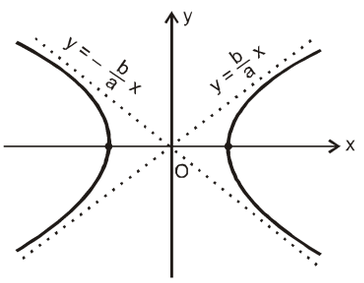
Asymptotes :

Definition : If the length of the perpendicular let fall from a point on a hyperbola to a straight line tends to zero as the point on the hyperbola moves to infinity along the hyperbola, then the straight line is called **asymptote** of the hyperbola.

- **Equations of asymptote** :

$$\frac{x}{a}+\frac{y}{b}=0 \quad \text{and} \quad \frac{x}{a}-\frac{y}{b}=0 .$$

- **Pair of asymptotes** : $\frac{x^2}{a^2}-\frac{y^2}{b^2}=0$.



Note : **(i)** A hyperbola and its conjugate have the same asymptote.

(ii) The equation of the pair of asymptotes differs from the equation of hyperbola (or conjugate hyperbola) by the constant term only.

(iii) The asymptotes pass through the centre of the hyperbola & are equally inclined to the transverse axis of the hyperbola. Hence the bisectors of the angles between the asymptotes are the principle axes of the hyperbola.

(iv) The asymptotes of a hyperbola are the diagonals of the rectangle formed by the lines drawn through the extremities of each axis parallel to the other axis.

(v) A simple method to find the co-ordinates of the centre of the hyperbola expressed as a general equation of degree 2 should be remembered as:

Let $f(x, y) = 0$ represents a hyperbola.

Find $\frac{\partial f}{\partial x}$ & $\frac{\partial f}{\partial y}$. Then the point of intersection of $\frac{\partial f}{\partial x} = 0$ & $\frac{\partial f}{\partial y} = 0$ gives the centre of the hyperbola.

Remarks :

(i) No tangent to the hyperbola can be drawn from its centre.

(ii) Only one tangent to the hyperbola can be drawn from a point lying on its asymptotes other than centre

(iii) Two tangents can be drawn to the hyperbola from any of its external points which does not lie at its asymptotes

Rectangular hyperbola (equilateral hyperbola) :

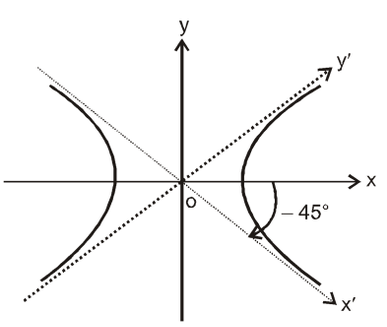
The particular kind of hyperbola in which the lengths of the transverse & conjugate axis are equal is called an Equilateral Hyperbola. Note that the eccentricity of the rectangular hyperbola is $\sqrt{2}$.

Since $a = b$

equation becomes $x^2 - y^2 = a^2$

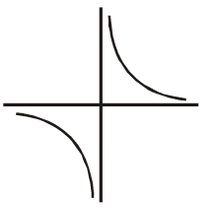
whose asymptotes are $y = \pm x$.

$e = \sqrt{1+\frac{b^2}{a^2}} = \sqrt{1+1} = \sqrt{2}$



Rotation of this system through an angle of 45° in clockwise direction gives another form to the equation of rectangular hyperbola.

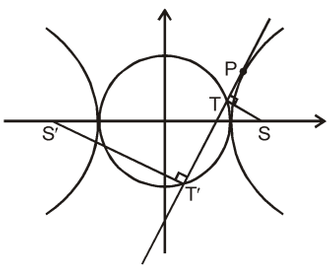
which is $xy = c^2$ where $c^2 = \frac{a^2}{2}$.



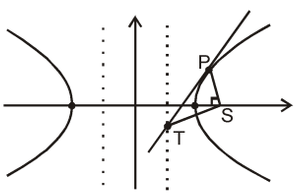
Rectangular hyperbola ($xy = c^2$) :
It is referred to its asymptotes as axes of co–ordinates.
Vertices : (c, c) & $(-c, -c)$;
Foci : $\left(\sqrt{2}\,c, \sqrt{2}\,c\right)$ & $\left(-\sqrt{2}\,c, -\sqrt{2}\,c\right)$,
Directrices : $x + y = \pm\sqrt{2}\,c$
Latus Rectum (l) : $\ell = 2\sqrt{2}\,c$ = T.A. = C.A.
Parametric equation $x = ct$, $y = c/t$, $t \in R - \{0\}$
Equation of a chord joining the points $P(t_1)$ & $Q(t_2)$ is $x + t_1 t_2 y = c(t_1 + t_2)$.
Equation of the tangent at $P(x_1, y_1)$ is $\frac{x}{x_1} + \frac{y}{y_1} = 2$ & at $P(t)$ is $\frac{x}{t} + ty = 2c$.
Equation of the normal at $P(t)$ is $xt^3 - yt = c(t^4 - 1)$.
Chord with a given middle point as (h, k) is $kx + hy = 2hk$.

Important results :

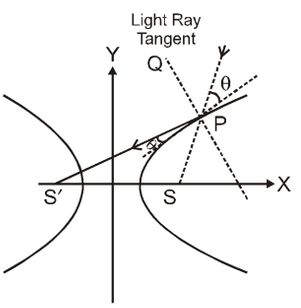
- Difference of focal distances is a constant, i.e. $|PS - PS'| = 2a$
- Locus of the feet of the perpendicular drawn from focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ upon any tangent is its auxiliary circle i.e. $x^2 + y^2 = a^2$ & the product of these perpendiculars is b^2.



- The portion of the tangent between the point of contact & the directrix subtends a right angle at the corresponding focus.

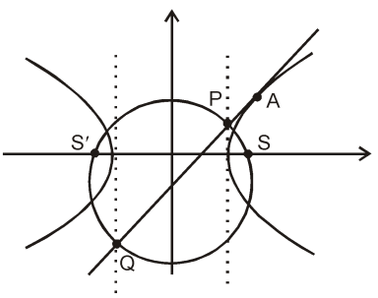


- The tangent & normal at any point of a hyperbola bisect the angle between the focal radii. This explains the reflection property of the hyperbola as **"An incoming light ray"** aimed towards one focus is reflected from the outer surface of the hyperbola towards the other focus. It follows that if an ellipse and a hyperbola have the same foci, they cut at right angles at any of their common point.

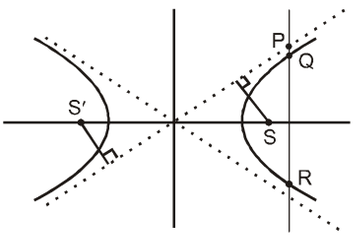


Note that the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ & the hyperbola $\frac{x^2}{a^2-k^2}-\frac{y^2}{k^2-b^2}=1$ $(a > k > b > 0)$ are confocal and therefore orthogonal.

- The foci of the hyperbola and the points P and Q in which any tangent meets the tangents at the vertices are concyclic with PQ as diameter of the circle.



- If from any point on the asymptote a straight line be drawn perpendicular to the transverse axis, the product of the segments of this line, intercepted between the point & the curve is always equal to the square of the semi conjugate axis.



$(PQ)\,(PR) = b^2$

- Perpendicular from the foci on either asymptote meet it in the same points as the corresponding directrix & the common points of intersection lie on the auxiliary circle.
- The tangent at any point P on a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ with centre C, meets the asymptotes in Q and R and cuts off a Δ CQR of constant area equal to ab from the asymptotes & the portion of the tangent intercepted between the asymptote is bisected at the point of contact. This implies that locus of the centre of the circle circumscribing the Δ CQR in case of a rectangular hyperbola is the hyperbola itself & for a standard hyperbola the locus would be the curve, $4(a^2x^2-b^2y^2)=(a^2+b^2)^2$.
- If the angle between the asymptote of a hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is 2θ then the eccentricity of the hyperbola is $\sec\theta$.
- A rectangular hyperbola circumscribing a triangle also passes through the orthocentre of this triangle.

If $\left(ct_i, \frac{c}{t_i}\right)$ i = 1, 2, 3 be the angular points P, Q, R then orthocentre is $\left(\frac{-c}{t_1 t_2 t_3}, -ct_1 t_2 t_3\right)$.

- If a circle and the rectangular hyperbola $xy = c^2$ meet in the four points t_1, t_2, t_3 & t_4, then

(a) $t_1 t_2 t_3 t_4 = 1$

(b) the centre of the mean position of the four points bisects the distance between the centres of the two curves.

(c) the centre of the circle through the points t_1, t_2 & t_3 is :

$$\left\{\frac{c}{2}\left(t_1+t_2+t_3+\frac{1}{t_1t_2t_3}\right), \frac{c}{2}\left(\frac{1}{t_1}+\frac{1}{t_2}+\frac{1}{t_3}+t_1t_2t_3\right)\right\}$$

* * * * *

INDEFINITE INTEGRATION

If f & g are functions of x such that $g'(x) = f(x)$, then indefinite integration of f(x) with respect to x is defined and denoted as $\int f(x)\,dx = g(x) + C$, where C is called the **constant of integration**.

Standard Formula:

(i) $\int (ax+b)^n\,dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C,\ n \neq -1$

(ii) $\int \frac{dx}{ax+b} = \frac{1}{a}\,\ell n\,|ax+b| + C$

(iii) $\int e^{ax+b}\,dx = \frac{1}{a}\,e^{ax+b} + C$

(iv) $\int a^{px+q}\,dx = \frac{1}{p}\,\frac{a^{px+q}}{\ell n a} + C;\ a > 0$

(v) $\int \sin(ax+b)\,dx = -\frac{1}{a}\cos(ax+b) + C$

(vi) $\int \cos(ax+b)\,dx = \frac{1}{a}\sin(ax+b) + C$

(vii) $\int \tan(ax+b)\,dx = \frac{1}{a}\,\ell n\,|\sec(ax+b)| + C$

(viii) $\int \cot(ax+b)\,dx = \frac{1}{a}\,\ell n\,|\sin(ax+b)| + C$

(ix) $\int \sec^2(ax+b)\,dx = \frac{1}{a}\tan(ax+b) + C$

(x) $\int \operatorname{cosec}^2(ax+b)\,dx = -\frac{1}{a}\cot(ax+b) + C$

(xi) $\int \sec(ax+b).\tan(ax+b)\,dx = \frac{1}{a}\sec(ax+b) + C$

(xii) $\int \operatorname{cosec}(ax+b).\cot(ax+b)\,dx = -\frac{1}{a}\operatorname{cosec}(ax+b) + C$

(xiii) $\int \sec x\,dx = \ell n\,|\sec x + \tan x| + C$ **OR** $\ell n\left|\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)\right| + C$

(xiv) $\int \text{cosec } x\,dx = \ell n\,|\text{cosec}x - \text{cot}x| + C$ **OR** $\ell n\left|\tan\frac{x}{2}\right| + C$

OR $-\ell n\,|\text{cosec}x + \text{cot}x| + C$

(xv) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$

(xvi) $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$

(xvii) $\int \frac{dx}{x\sqrt{x^2-a^2}} = \frac{1}{a}\sec^{-1}\frac{x}{a} + C$

(xviii) $\int \frac{dx}{\sqrt{x^2+a^2}} = \ell n\left|x+\sqrt{x^2+a^2}\right| + C$ **OR** $\sinh^{-1}\frac{x}{a} + C$

(xix) $\int \frac{dx}{\sqrt{x^2-a^2}} = \ell n\left|x+\sqrt{x^2-a^2}\right| + C$ **OR** $\cosh^{-1}\frac{x}{a} + C$

(xx) $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\ln\left|\frac{a+x}{a-x}\right| + C$

(xxi) $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C$

(xxii) $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$

(xxiii) $\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\ell n\left|\frac{x+\sqrt{x^2+a^2}}{a}\right| + C$

(xxiv) $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\ell n\left|\frac{x+\sqrt{x^2-a^2}}{a}\right| + C$

(xxv) $\int e^{ax}.\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + C$

(xxvi) $\int e^{ax}.\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + C$

Theorems on integration

(i) $\int C\,f(x).dx = C\int f(x).dx$ (ii) $\int (f(x)\pm g(x))dx = \int f(x)dx \pm \int g(x)\,dx$

(iii) $\int f(x)dx = g(x)+C_1 \Rightarrow \int f(ax+b)dx = \frac{g(ax+b)}{a} + C_2$

Integration by Substitution

If we substitution $\phi(x) = t$ in an integral then

(i) everywhere x will be replaced in terms of new variable t.

(ii) dx also gets converted in terms of dt.

Integration by Parts : Product of two functions f(x) and g(x) can be integrate using formula :

$$\int (f(x)\, g(x))\, dx = f(x) \int (g(x))\, dx - \int \left(\frac{d}{dx}(f(x)) \int (g(x))\, dx \right) dx$$

(i) when you find integral $\int g(x)\, dx$ then it will **not** contain arbitarary constant.

(ii) $\int g(x)\, dx$ should be taken as same at both places.

(iii) The choice of f(x) and g(x) can be decided by ILATE guideline.
the function will come later is taken an integral function (g(x)).

I → Inverse function
L → Logarithmic function
A → Algebraic function
T → Trigonometric function
E → Exponential function

Integration of Rational Algebraic Functions by using Partial Fractions:

PARTIAL FRACTIONS :

If f(x) and g(x) are two polynomials, then $\frac{f(x)}{g(x)}$ defines a rational algebraic function of x.

If degree of f(x) < degree of g(x), then $\frac{f(x)}{g(x)}$ is called a proper rational function.

If degree of f(x) ≥ degree of g(x) then $\frac{f(x)}{g(x)}$ is called an improper rational function.

If $\frac{f(x)}{g(x)}$ is an improper rational function, we divide f(x) by g(x) so that the rational function $\frac{f(x)}{g(x)}$ is expressed in the form $\phi(x) + \frac{\Psi(x)}{g(x)}$, where $\phi(x)$ and $\Psi(x)$ are polynomials such that the degree of $\Psi(x)$ is less than that of g(x). Thus, $\frac{f(x)}{g(x)}$ is expressible as the sum of a polynomial and a proper rational function.

Any proper rational function $\frac{f(x)}{g(x)}$ can be expressed as the sum of rational functions, each having a simple factor of g(x). Each such fraction is called a partial fraction and the process of obtained them is called the resolutions or decomposition of $\frac{f(x)}{g(x)}$ into partial fractions.

The resolution of $\frac{f(x)}{g(x)}$ into partial fractions depends mainly upon the nature of the factors of g(x) as discussed below :

CASE I When denominator is expressible as the product of non-repeating linear factors.
Let $g(x) = (x - a_1)(x - a_2) \ldots (x - a_n)$. Then, we assume that

$$\frac{f(x)}{g(x)} \equiv \frac{A_1}{x-a_1} + \frac{A_2}{x-a_2} + ... + \frac{A_n}{x-a_n}$$

where $A_1, A_2, ... A_n$ are constants and can be determined by equating the numerator on R.H.S. to the numerator on L.H.S. and then substituting $x = a_1, a_2, ..., a_n$.

CASE II When the denominator g(x) is expressible as the product of the linear factors such that some of them are repeating.

Example $\frac{1}{g(x)} = \frac{1}{(x-a)^k(x-a_1)(x-a_2).....(x-a_r)}$ this can be expressed as

$$\frac{A_1}{x-a} + \frac{A_2}{(x-a)^2} + \frac{A_3}{(x-a)^3} + ...+ \frac{A_k}{(x-a)^k} + \frac{B_1}{(x-a_1)} + \frac{B_2}{(x-a_2)} + ... + \frac{B_r}{(x-a_r)}$$

Now to determine constants we equate numerators on both sides. Some of the constants are determined by substitution as in case I and remaining are obtained by equating the coefficient of same power of x.

The following example illustrate the procedure.

CASE III When some of the factors of denominator g(x) are quadratic but non-repeating. Corresponding to each quadratic factor $ax^2 + bx + c$, we assume partial fraction of the type $\frac{Ax+B}{ax^2+bx+c}$, where A and B are constants to be determined by comparing coefficients of similar powers of x in the numerator of both sides. In practice it is advisable to assume partial fractions of the type $\frac{A(2ax+b)}{ax^2+bx+c} + \frac{B}{ax^2+bx+c}$

The following example illustrates the procedure

CASE IV When some of the factors of the denominator g(x) are quadratic and repeating fractions of the form $\left\{\frac{A_0(2ax+b)}{ax^2+bx+c} + \frac{A_1}{ax^2+bx+c}\right\} + \left\{\frac{A_1(2ax+b)}{(ax^2+bx+c)^2} + \frac{A_2}{(ax^2+bx+c)^2}\right\}$

$$+ .....+ \left\{\frac{A_{2k-1}(2ax+b)}{(ax^2+bx+c)^k} + \frac{A_{2k}}{(ax^2+bx+c)^k}\right\}$$

The following example illustrates the procedure.

Integration of type $\int \frac{dx}{ax^2+bx+c}, \int \frac{dx}{\sqrt{ax^2+bx+c}}, \int \sqrt{ax^2+bx+c}\ dx$

Express $ax^2 + bx + c$ in the form of perfect square & then apply the standard results.

Integration of type

$$\int \frac{px+q}{ax^2+bx+c}\ dx, \int \frac{px+q}{\sqrt{ax^2+bx+c}}\ dx, \int (px+q)\sqrt{ax^2+bx+c}\ dx$$

Express px + q = A (differential co–efficient of denominator) + B.

Integration of trigonometric functions

(i) $\int \frac{dx}{a + b\sin^2 x}$ OR $\int \frac{dx}{a + b\cos^2 x}$ OR $\int \frac{dx}{a\sin^2 x + b\sin x\cos x + c\cos^2 x}$

Multiply Nr & Dr by $\sec^2 x$ & put $\tan x = t$.

(ii) $\int \frac{dx}{a+b\sin x}$ OR $\int \frac{dx}{a+b\cos x}$ OR $\int \frac{dx}{a+b\sin x+c\cos x}$

Convert sines & cosines into their respective tangents of half the angles and then, put $\tan\frac{x}{2}=t$

(iii) $\int \frac{a.\cos x+b.\sin x+c}{\ell.\cos x+m.\sin x+n}\,dx.$

Express Nr $\equiv$ A(Dr) + B $\frac{d}{dx}$ (Dr) + C & proceed.

Integration of type $\int \sin^m x.\cos^n x\,dx$

Case - I
If m and n are even natural number then converts higher power into higher angles.
Case - II
If at least one of m or n is odd natural number then if m is odd put cosx = t and vice-versa.
Case - III
When m + n is a negative even integer then put tan x = t.

Integration of type

$\int \frac{x^2 \pm 1}{x^4+Kx^2+1}\,dx$ **where K is any constant.**

Divide Nr & Dr by x^2 & put $x \mp \frac{1}{x} = t$.

Integration of type

$\int \frac{dx}{(ax+b)\sqrt{px+q}}$ OR $\int \frac{dx}{(ax^2+bx+c)\sqrt{px+q}}$.

Put $px + q = t^2$.

Integration of type

$\int \frac{dx}{(ax+b)\sqrt{px^2+qx+r}}$, **put** $ax + b = \frac{1}{t}$;

$\int \frac{dx}{(ax^2+b)\sqrt{px^2+q}}$, **put** $x = \frac{1}{t}$

Integration of type

$\int \sqrt{\frac{x-\alpha}{\beta-x}}\,dx$ **or** $\int \sqrt{(x-\alpha)(\beta-x)}\,dx$; **put** $x = \alpha\cos^2\theta + \beta\sin^2\theta$

$\int \sqrt{\frac{x-\alpha}{x-\beta}}\,dx$ **or** $\int \sqrt{(x-\alpha)(x-\beta)}\,dx$; **put** $x = \alpha\sec^2\theta - \beta\tan^2\theta$

$\int \frac{dx}{\sqrt{(x-\alpha)(x-\beta)}}$; put $x - \alpha = t^2$ or $x - \beta = t^2$.

Reduction formula of $\int \tan^n x\,dx$, $\int \cot^n x\,dx$, $\int \sec^n x\,dx$, $\int \operatorname{cosec}^n x\,dx$

1. $I_n = \int \tan^n x\,dx = \int \tan^2 x \tan^{n-2} x\,dx = \int (\sec^2 x - 1)\tan^{n-2} x\,dx$

$\Rightarrow \quad I_n = \int \sec^2 x \tan^{n-2} x\,dx - I_{n-2} \quad \Rightarrow \quad I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$, $n \geq 2$

2. $I_n = \int \cot^n x\,dx = \int \cot^2 x \,.\, \cot^{n-2} x\,dx = \int (\operatorname{cosec}^2 x - 1)\cot^{n-2} x\,dx$

$\Rightarrow \quad I_n = \int \operatorname{cosec}^2 x \cot^{n-2} x\,dx - I_{n-2} \quad \Rightarrow \quad I_n = -\frac{\cot^{n-1} x}{n-1} - I_{n-2}$, $n \geq 2$

3. $I_n = \int \sec^n x\,dx = \int \sec^2 x \sec^{n-2} x\,dx$

$\Rightarrow \quad I_n = \tan x \sec^{n-2} x - \int (\tan x)(n-2) \sec^{n-3} x.\ \sec x \tan x\,dx.$

$\Rightarrow \quad I_n = \tan x \sec^{n-2} x - (n-2) \int (\sec^2 x - 1)\sec^{n-2} x\,dx$

$\Rightarrow \quad (n-1) I_n = \tan x \sec^{n-2} x + (n-2) I_{n-2}$

$$I_n = \frac{\tan x \sec^{n-2} x}{n-1} + \frac{n-2}{n-1} I_{n-2}$$

4. $I_n = \int \operatorname{cosec}^n x\,dx = \int \operatorname{cosec}^2 x \operatorname{cosec}^{n-2} x\,dx$

$\Rightarrow \quad I_n = -\cot x \operatorname{cosec}^{n-2} x + \int (\cot x)(n-2)(-\operatorname{cosec}^{n-3} x \operatorname{cosec} x \cot x)\,dx$

$\Rightarrow \quad -\cot x \operatorname{cosec}^{n-2} x - (n-2) \int \cot^2 x \operatorname{cosec}^{n-2} x\,dx$

$\Rightarrow \quad I_n = -\cot x \operatorname{cosec}^{n-2} x - (n-2) \int (\operatorname{cosec}^2 x - 1) \operatorname{cosec}^{n-2} x\,dx$

$\Rightarrow \quad (n-1) I_n = -\cot x \operatorname{cosec}^{n-2} x + (n-2) I_{n-2}$

$$I_n = \frac{\cot x \operatorname{cosec}^{n-2} x}{-(n-1)} + \frac{n-2}{n-1} I_{n-2}$$

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DEFINITE INTEGRATION

Newton-Leibnitz formula.

Let $\frac{d}{dx}(F(x)) = f(x)\ \forall\ x \in (a, b)$. Then $\int_a^b f(x)\,dx = \lim_{x\to b^-} F(x) - \lim_{x\to a^+} F(x)$.

Note : 1. If $a > b$, then $\int_a^b f(x)\,dx = \lim_{x\to b^+} F(x) - \lim_{x\to a^-} F(x)$.

2. If F(x) is continuous at a and b, then $\int_a^b f(x)\,dx = F(b) - F(a)$

Property (1) $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$ i.e. definite integral is independent of variable of integration.

Property (2) $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

Property (3) $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$, where c may lie inside or outside the interval [a, b].

Property (4) $$\int_{-a}^{a} f(x)\,dx = \int_0^a (f(x)+f(-x))\,dx = \begin{cases} 2\int_0^a f(x)dx, & \text{if } f(-x) = f(x) \text{ i.e. } f(x) \text{ is even} \\ 0, & \text{if } f(-x) = -f(x) \text{ i.e. } f(x) \text{ is odd} \end{cases}$$

Property (5) $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$. Further $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Property (6) $$\int_0^{2a} f(x)\,dx = \int_0^a (f(x)+f(2a-x))\,dx = \begin{cases} 2\int_0^a f(x)dx, & \text{if } f(2a-x) = f(x) \\ 0, & \text{if } f(2a-x) = -f(x) \end{cases}$$

Property (7) If f(x) is a periodic function with period T, then

(i) $\int_0^{nT} f(x)\,dx = n\int_0^T f(x)\,dx,\ n \in z$

(ii) $\int_a^{a+nT} f(x)\,dx = n\int_0^T f(x)\,dx,\ n \in z,\ a \in R$

(iii) $\int_{mT}^{nT} f(x)\,dx = (n-m)\int_0^T f(x)\,dx,\ m, n \in z$

(iv) $\int_{nT}^{a+nT} f(x)\,dx = \int_0^a f(x)\,dx,\ n \in z,\ a \in R$

(v) $\int_{a+nT}^{b+nT} f(x)\ dx = \int_{a}^{b} f(x)\,dx,\ n \in z,\ a, b \in R$

Property (8) If $\psi(x) \le f(x) \le \phi(x)$ for $a \le x \le b$, then $\int_a^b \psi(x)\ dx \le \int_a^b f(x)\ dx \le \int_a^b \phi(x)\ dx$

Property (9) If $m \le f(x) \le M$ for $a \le x \le b$, then $m(b-a) \le \int_a^b f(x)\,dx \le M(b-a)$

Further if f(x) is monotonically decreasing in (a, b), then $f(b)(b-a) < \int_a^b f(x)\,dx < f(a)(b-a)$ and if f(x)

is monotonically increasing in (a, b), then $f(a)(b-a) < \int_a^b f(x)\ dx < f(b)(b-a)$

Property (10) $\left|\int_a^b f(x)\,dx\right| \le \int_a^b |f(x)|\,dx$

Property (11) If $f(x) \ge 0$ on [a, b], then $\int_a^b f(x)\,dx \ge 0$

Leibnitz Theorem : If $F(x) = \int_{g(x)}^{h(x)} f(t)\,dt$, then $\frac{dF(x)}{dx} = h'(x)\,f(h(x)) - g'(x)\,f(g(x))$

Proof : Let $P(t) = \int f(t)\,dt \Rightarrow F(x) = \int_{g(x)}^{h(x)} f(t)\,dt = P(h(x)) - P(g(x))$

$\Rightarrow \frac{dF(x)}{dx} = P'(h(x))\,h'(x) - P'(g(x))\,g'(x) = f(h(x))\,h'(x) - f(g(x))\,g'(x)$

Steps to express the limit of sum as definte integral :

Step 1. Replace $\frac{r}{n}$ by x, $\frac{1}{n}$ by dx and $\underset{n\to\infty}{Lt}\ \Sigma$ by $\int$

Step 2. Evaluate $\underset{n\to\infty}{Lt}\left(\frac{r}{n}\right)$ by putting least and greatest values of r as lower and upper limits respectively.

For example $\underset{n\to\infty}{Lt}\sum_{r=1}^{pn}\frac{1}{n}f\left(\frac{r}{n}\right) = \int_0^p f(x)\,dx$ $\left(\because \underset{n\to\infty}{Lt}\left(\frac{r}{n}\right)\Big|_{r=1} = 0,\ \underset{n\to\infty}{Lt}\left(\frac{r}{n}\right)\Big|_{r=np} = p\right)$

Reduction formulae in definite Integrals:

1. If $I_n = \int_0^{\frac{\pi}{2}} \sin^n x\ dx$, then show that $I_n = \left(\frac{n-1}{n}\right) I_{n-2}$

Note : **1.** $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \int_0^{\frac{\pi}{2}} \cos^n x\,dx$

2. $I_n = \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\left(\frac{n-5}{n-4}\right)$ ..... I_0 or I_1 according as n is even or odd. $I_0 = \frac{\pi}{2}$, $I_1 = 1$

Hence $I_n = \begin{cases} \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\left(\frac{n-5}{n-4}\right)\cdots\cdots\left(\frac{1}{2}\right).\frac{\pi}{2}, & \text{if n is even} \\ \left(\frac{n-1}{n}\right)\left(\frac{n-3}{n-2}\right)\left(\frac{n-5}{n-4}\right)\cdots\cdots\left(\frac{2}{3}\right).1, & \text{if n is odd} \end{cases}$

2. If $I_n = \int_0^{\frac{\pi}{4}} \tan^n x\,dx$, then show that $I_n + I_{n-2} = \frac{1}{n-1}$

3. If $I_{m,n} = \int_0^{\frac{\pi}{2}} \sin^m x \cdot \cos^n x\,dx$, then show that $I_{m,n} = \frac{m-1}{m+n} I_{m-2,\,n}$

Note : **1.** $I_{m,n} = \left(\frac{m-1}{m+n}\right)\left(\frac{m-3}{m+n-2}\right)\left(\frac{m-5}{m+n-4}\right)$ ..... $I_{0,n}$ or $I_{1,n}$ according as m is even or odd.

$I_{0,n} = \int_0^{\frac{\pi}{2}} \cos^n x\,dx$ and $I_{1,n} = \int_0^{\frac{\pi}{2}} \sin x \cdot \cos^n x\,dx = \frac{1}{n+1}$

2. **<u>Walli's Formula</u>**

$$I_{m,n} = \begin{cases} \frac{(m-1)(m-3)(m-5)\cdots\cdots(n-1)(n-3)(n-5)\cdots\cdots}{(m+n)(m+n-2)(m+n-4)\cdots\cdots}\frac{\pi}{2} & \text{when both m, n are even} \\ \frac{(m-1)(m-3)(m-5)\cdots\cdots(n-1)(n-3)(n-5)\cdots\cdots}{(m+n)(m+n-2)(m+n-4)\cdots\cdots} & \text{otherwise} \end{cases}$$

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DETERMINANT

Determinant of a square matrix :

To every square matrix $A = [a_{ij}]$ of order n, we can associate a number (real or complex) called determinant of the square matrix.

Let $A = [a]_{1\times1}$ be a 1×1 matrix. Determinant A is defined as $|A| = a$.

e.g. $A = [-3]_{1\times1}$ $\quad |A| = -3$

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then $|A|$ is defined as $ad - bc$.

e.g. $A = \begin{bmatrix} 5 & 3 \\ -1 & 4 \end{bmatrix}$, $|A| = 23$

Minors & Cofactors :

Let Δ be a determinant. Then minor of element a_{ij}, denoted by M_{ij}, is defined as the determinant of the submatrix obtained by deleting i^{th} row & j^{th} column of Δ. Cofactor of element a_{ij}, denoted by C_{ij}, is defined as $C_{ij} = (-1)^{i+j} M_{ij}$.

e.g. 1 $\quad \Delta = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$

$M_{11} = d = C_{11}$
$M_{12} = c,\ C_{12} = -c$
$M_{21} = b,\ C_{21} = -b$
$M_{22} = a = C_{22}$

e.g. 2 $\quad \Delta = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$

$M_{11} = \begin{vmatrix} q & r \\ y & z \end{vmatrix} = qz - yr = C_{11}.$

$M_{23} = \begin{vmatrix} a & b \\ x & y \end{vmatrix} = ay - bx,\ C_{23} = -(ay - bx) = bx - ay$ $\quad$ etc.

Determinant of any order : Let $A = [a_{ij}]_n$ be a square matrix (n > 1). Determinant of A is defined as the sum of products of elements of any one row (or any one column) with corresponding cofactors.

e.g.1 $\quad A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$|A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$ (using first row).

the determinant can be expressed as the sum of two determinants, i.e.

$$\begin{vmatrix} a_1+x & b_1+y & c_1+z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

(10) The value of a determinant is not altered by adding to the elements of any row (or column) a constant multiple of the corresponding elements of any other row (or column),

i.e. $D_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D_2 = \begin{vmatrix} a_1+ma_2 & b_1+mb_2 & c_1+mc_2 \\ a_2 & b_2 & c_2 \\ a_3+na_1 & b_3+nb_1 & c_3+nc_1 \end{vmatrix}$. Then $D_2 = D_1$

(11) Let $A = [a_{ij}]_n$. The sum of the products of elements of any row with corresponding cofactors of any other row is zero. (Similarly the sum of the products of elements of any column with corresponding cofactors of any other column is zero).

Application of determinants : Following examples of short hand writing large expressions are:

(i) Area of a triangle whose vertices are (x_r, y_r); r = 1, 2, 3 is:

$$D = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

If D = 0 then the three points are collinear.

(ii) Equation of a straight line passing through (x_1, y_1) & (x_2, y_2) is $\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$

(iii) The lines: $a_1x + b_1y + c_1 = 0$..... (1)
$a_2x + b_2y + c_2 = 0$..... (2)
$a_3x + b_3y + c_3 = 0$..... (3)

are concurrent if, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$

Condition for the consistency of three simultaneous linear equations in 2 variables.

(iv) $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines if:

$$abc + 2fgh - af^2 - bg^2 - ch^2 = 0 = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

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DIFFERENTIAL EQUATION

Introduction :

An equation involving independent and dependent variables and the derivatives of the dependent variables is called a **differential equation**.

1. **Ordinary Differential Equation :** If the dependent variables depend on one independent variable x, then the differential equation is said to be ordinary.

for example $\frac{dy}{dx} + \frac{dz}{dx} = y + z,$

$\frac{dy}{dx} + xy = \sin x\,, \qquad \frac{d^3y}{dx^3} + 2\frac{dy}{dx} + y = e^x\,,$

$k\frac{d^2y}{dx^2} = \left\{1+\left(\frac{dy}{dx}\right)^2\right\}^{3/2}\,, \quad y = x\frac{dy}{dx} + k\sqrt{\left\{1+\left(\frac{dy}{dx}\right)^2\right\}}$

Order and Degree of a Differential Equation:

2.1 **Order :** Order is the highest differential appearing in a differential equation.

2.2 **Degree :**

It is determined by the highest degree of the highest order derivative present in it after the differential equation is cleared of radicals and fractions so far as the derivatives are concerned.

Note : In the differential equation, all the derivatives should be expressed in the polynomial form

$$f_1(x, y)\left[\frac{d^m y}{dx^m}\right]^{n_1} + f_2(x, y)\left[\frac{d^{m-1}y}{dx^{m-1}}\right]^{n_2} + \ldots\ldots\ldots f_k(x, y)\left[\frac{dy}{dx}\right]^{n_k} = 0$$

The above differential equation has the order m and degree n_1.

Formation of Differential Equation:

Differential equation corresponding to a family of curve will have :

(a) Order exactly same as number of essential arbitrary constants in the equation of curve.
(b) No arbitrary constant present in it.

The differential equation corresponding to a family of curve can be obtained by using the following steps:

(i) Identify the number of essential arbitrary constants in equation of curve.

NOTE : If arbitrary constants appear in addition, subtraction, multiplication or division, then we can club them to reduce into one new arbitrary constant.

(ii) Differentiate the equation of curve till the required order.
(iii) Eliminate the arbitrary constant from the equation of curve and additional equations obtained in step (ii) above.

Solution of a Differential Equation:

Finding the dependent variable from the differential equation is called solving or integrating it. The solution or the integral of a differential equation is, therefore, a relation between dependent and independent variables (free from derivatives) such that it satisfies the given differential equation

NOTE : The solution of the differential equation is also called its primitive, because the differential equation can be regarded as a relation derived from it.

There can be three types of solution of a differential equation:

(i) General solution (or complete integral or complete primitive) : A relation in x and y satisfying a given differential equation and involving exactly same number of arbitrary constants as order of differential equation.

(ii) Particular Solution : A solution obtained by assigning values to one or more than one arbitrary constant of general solution.

(iii) Singular Solution : It is not obtainable from general solution. Geometrically, **General solution** acts as an envelope to **singular solution**.

4.1. Differential Equation of First Order and First Degree :

A differential equation of first order and first degree is of the type $\frac{dy}{dx} + f(x, y) = 0$, which can also be written as : $Mdx + Ndy = 0$, where M and N are functions of x and y.

Solution methods of First Order and First Degree Differential Equations :

5.1 Variables separable : If the differential equation can be put in the form, $f(x)\,dx = \phi(y)\,dy$ we say that variables are separable and solution can be obtained by integrating each side separately.

A general solution of this will be $\int f(x)\,dx = \int \phi(y)\,dy + c$, where c is an arbitrary constant

5.1.1 Polar coordinates transformations :

Sometimes transformation to the polar co-ordinates facilitates separation of variables. In this connection it is convenient to remember the following differentials:

(a) If $x = r\cos\theta$; $y = r\sin\theta$ then,

(i) $x\,dx + y\,dy = r\,dr$ (ii) $dx^2 + dy^2 = dr^2 + r^2d\theta^2$ (iii) $x\,dy - y\,dx = r^2d\theta$

(b) If $x = r\sec\theta$ & $y = r\tan\theta$ then

(i) $x\,dx - y\,dy = r\,dr$ (ii) $x\,dy - y\,dx = r^2\sec\theta\,d\theta$.

5.1.2 Equations Reducible to the Variables Separable form : If a differential equation can be reduced into a variables separable form by a proper substitution, then it is said to be

"Reducible to the variables separable type". Its general form is $\frac{dy}{dx} = f(ax + by + c)$ $a, b \neq 0$.

To solve this, put $ax + by + c = t$.

5.2 Homogeneous Differential Equations :

A differential equation of the form $\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)}$ where f and g are homogeneous function of x and y, and of the same degree, is called homogeneous differential equaiton and can be solved easily by putting $y = vx$.

5.2.1 Equations Reducible to the Homogeneous form

Equations of the form $\frac{dy}{dx} = \frac{ax + by + c}{Ax + By + C}$ .....(1)

can be made homogeneous (in new variables X and Y) by substituting x = X + h and y = Y + k,

where h and k are constants to obtain, $\frac{dY}{dX} = \frac{aX + bY + (ah + bk + c)}{AX + BY + (Ah + Bk + C)}$ . .....(2)

These constants are chosen such that ah + bk + c = 0, and Ah + Bk + C = 0. Thus we obtain

the following differential equation $\frac{dY}{dX} = \frac{aX + bY}{AX + BY}$.

The differential equation can now be solved by substituting Y = vX.

5.3 Exact Differential Equation :

The differential equation $M + N\frac{dy}{dx} = 0$ .....(1)

Where M and N are functions of x and y is said to be exact if it can be derived by direct differentiation (without any subsequent multiplication, elimination etc.) of an equation of the form f(x, y) = c

e.g. $y^2\,dy + x\,dx + \frac{dx}{x} = 0$ is an exact differential equation.

NOTE : (i) The necessary condition for (1) to be exact is $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

(ii) For finding the solution of exact differential equation, following results on exact differentials should be remembered :

(a) $xdy + y\,dx = d(xy)$ (b) $\frac{xdy - ydx}{x^2} = d\left(\frac{y}{x}\right)$ (c) $2(x\,dx + y\,dy) = d(x^2 + y^2)$

(d) $\frac{xdy - ydx}{xy} = d\left(\ln\frac{y}{x}\right)$ (e) $\frac{xdy - ydx}{x^2 + y^2} = d\left(\tan^{-1}\frac{y}{x}\right)$ (f) $\frac{xdy + ydx}{xy} = d(\ln xy)$

(g) $\frac{xdy + ydx}{x^2\,y^2} = d\left(-\frac{1}{xy}\right)$

Linear Differential Equation :

A linear differential equation has the following characteristics :

(i) Dependent variable and its derivative in first degree only and are not multiplied together

(ii) All the derivatives should be in a polynomial form

(iii) The order may be more than one

The m^{th} order linear differential equation is of the form.

$$P_0(x)\frac{d^m y}{dx^m} + P_1(x)\frac{d^{m-1}y}{dx^{m-1}} + \ldots\ldots\ldots\ldots\ldots + P_{m-1}(x)\frac{dy}{dx} + P_m(x)\,y = \phi(x),$$

where $P_0(x)$, $P_1(x)$ .....$P_m(x)$ are called the coefficients of the differential equation.

NOTE : $\frac{dy}{dx} + y^2 \sin x = \ln x$ is not a Linear differential equation.

6.1 Linear differential equations of first order :

The differential equation $\frac{dy}{dx} + Py = Q$, is linear in y. (Where P and Q are functions of x only).

Integrating Factor (I.F.) : It is an expression which when multiplied to a differential equation converts it into an exact form.

I.F for linear differential equation = $e^{\int Pdx}$ (constant of integration will not be considered)

$\therefore$ after multiplying above equation by I.F it becomes;

$$\frac{dy}{dx} \cdot e^{\int Pdx} + Py \cdot e^{\int Pdx} = Q \cdot e^{\int Pdx}$$

$$\Rightarrow \quad \frac{d}{dx}(y.e^{\int Pdx}) = Q.e^{\int Pdx}$$

$$\Rightarrow \quad y.e^{\int Pdx} = \int Q.e^{\int Pdx} + C .$$

NOTE : Some times differential equation becomes linear, if x is taken as the dependent variable, and y as independent variable. The differential equation has then the following form : $\frac{dx}{dy} + P_1 x = Q_1$.

where P_1 and Q_1 are functions of y. The I.F. now is $e^{\int P_1 dy}$

6.2 Equations reducible to linear form

6.2.1 By change of variable.

Often differential equation can be reduced to linear form by appropriate substitution of the non-linear term

6.2.2 Bernoulli's equation :

Equations of the form $\frac{dy}{dx} + Py = Q.y^n$, $n \neq 0$ and $n \neq 1$

where P and Q are functions of x, is called Bernoulli's equation and can be made linear in v by dividing by y^n and putting $y^{-n+1} = v$. Now its solution can be obtained as in (v).

e.g. : $2 \sin x \frac{dy}{dx} - y \cos x = xy^3 e^x$.

Orthogonal Trajectory :

An orthogonal trajectory of a given system of curves is defined to be a curve which cuts every member of a given family of curve at right angle.

Steps to find orthogonal trajectory :

(i) Let f (x, y, c) = 0 be the equation of the given family of curves, where 'c' is an arbitrary constant.

(ii) Differentiate the given equation w.r.t. x and then eliminate c.

(iii) Replace $\frac{dy}{dx}$ by $-\frac{dx}{dy}$ in the equation obtained in (ii).

(iv) Solve the differential equation obtained in (iii).
Hence solution obtained in (iv) is the required orthogonal trajectory.

Geometrical application of differential equation :

Form a differential equation from a given geometrical problem. Often following formulae are useful to remember

(i) Length of tangent $(L_T) = \left|\frac{y\sqrt{1+m^2}}{m}\right|$.

(ii) Length of normal $(L_N) = \left|y\sqrt{1+m^2}\right|$.

(iii) Length of sub-tangent $(L_{ST}) = \left|\frac{y}{m}\right|$.

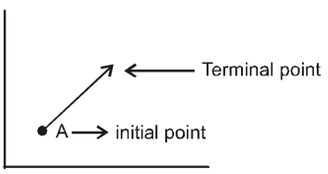
(iv) Length of subnormal $(L_{SN}) = |my|$.

where y is the ordinate of the point, m is the slope of the tangent $\left(\frac{dy}{dx}\right)$.

* * * * *

VECTORS

Vectors and their representation :



Vector quantities are specified by definite magnitude and definite direction. A vector is generally represented by a directed line segment, say $\overrightarrow{AB}$. A is called the **initial point** and B is called the **terminal point**. The magnitude of vector $\overrightarrow{AB}$ is expressed by $|\overrightarrow{AB}|$.

Zero vector:
A vector of zero magnitude i.e. which has the same initial and terminal point, is called a **zero vector.** It is denoted by **O**. **The direction of zero vector is indeterminate.**

Unit vector:
A vector of unit magnitude in the direction of a vector $\vec{a}$ is called unit vector along $\vec{a}$ and is denoted by $\hat{a}$, symbolically $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$.

Equal vectors:

Two vectors are said to be equal if they have the same magnitude, direction and represent the same physical quantity.

Collinear vectors:

Two vectors are said to be collinear if their directed line segments are parallel irrespective of their directions. Collinear vectors are also called **parallel vectors**. If they have the same direction they are named as **like vectors** otherwise **unlike vectors**.

Symbolically, two non-zero vectors $\vec{a}$ and $\vec{b}$ are collinear if and only if, $\vec{a} = \lambda\vec{b}$, where $\lambda \in R$

Vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are collinear if $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$

Coplanar vectors:

A given number of vectors are called coplanar if their line segments are all parallel to the same plane. Note that **"two vectors are always coplanar"**.

Multiplication of a vector by a scalar :

$m(\vec{a}) = (\vec{a})m = m\vec{a}$, $m(n\vec{a}) = n(m\vec{a}) = (mn)\vec{a}$

$(m+n)\vec{a} = m\vec{a} + n\vec{a}$, $m(\vec{a}+\vec{b}) = m\vec{a} + m\vec{b}$

Addition of vectors :

(i) If two vectors $\vec{a}$ and $\vec{b}$ are represented by $\overrightarrow{OA}$ and $\overrightarrow{OB}$, then their sum $\vec{a}+\vec{b}$ is a vector represented by $\overrightarrow{OC}$, where OC is the diagonal of the parallelogram OACB.

(ii) $\vec{a}+\vec{b} = \vec{b}+\vec{a}$ **(commutative)** (iii) $(\vec{a}+\vec{b})+\vec{c} = \vec{a}+(\vec{b}+\vec{c})$ **(associative)**

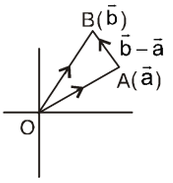
(iv) $\vec{a}+\vec{0} = \vec{a} = \vec{0}+\vec{a}$ (v) $\vec{a}+(-\vec{a}) = \vec{0} = (-\vec{a})+\vec{a}$

(vi) $|\vec{a}+\vec{b}| \le |\vec{a}|+|\vec{b}|$ (vii) $|\vec{a}-\vec{b}| \ge ||\vec{a}|-|\vec{b}||$

Position vector of a point:

Let O be a fixed origin, then the position vector of a point P is the vector $\overrightarrow{OP}$. If $\vec{a}$ and $\vec{b}$ are position vectors of two points A and B, then

$\overrightarrow{AB} = \vec{b}-\vec{a}$ = position vector **(p.v.)** of B – position vector **(p.v.)** of A.

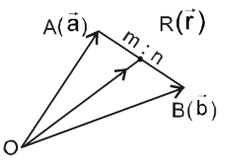


DISTANCE FORMULA

Distance between the two points A$(\vec{a})$ and B$(\vec{b})$ is AB = $|\vec{a}-\vec{b}|$

SECTION FORMULA

If $\vec{a}$ and $\vec{b}$ are the position vectors of two points A and B, then the p.v. of a point which divides AB in the ratio m: n is given by $\vec{r} = \frac{n\vec{a}+m\vec{b}}{m+n}$.



Note : Position vector of mid point of AB = $\frac{\vec{a}+\vec{b}}{2}$.

Vector Method

We know that if position vector of two points A and B are given as $\overrightarrow{OA}$ and $\overrightarrow{OB}$ then

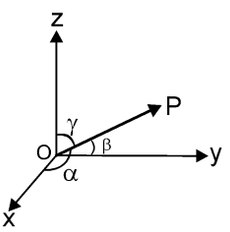
AB = $|\overrightarrow{OB} - \overrightarrow{OA}|$ $\Rightarrow$ AB = $|(x_2\hat{i} + y_2\hat{j} + z_2\hat{k}) - (x_1\hat{i} + y_1\hat{j} + z_1\hat{k})|$

$\Rightarrow$ AB = $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$

Direction cosines and direction ratios

(i) **Direction cosines :** Let α, β, γ be the angles which a directed line makes with the positive directions of the axes of x, y and z respectively, then cos α, cosβ, cos γ are called the direction cosines of the line. The direction cosines are usually denoted by ℓ, m, n.

Thus ℓ = cos α, m = cos β, n = cos γ.



(ii) If ℓ, m, n be the direction cosines of a line, then $\ell^2 + m^2 + n^2 = 1$

(iii) **Direction ratios :** Let a, b, c be proportional to the direction cosines ℓ, m, n then a, b, c are called the direction ratios.

If a, b, c, are the direction ratios of any line L, then $a\hat{i}+b\hat{j}+c\hat{k}$ will be a vector parallel to the line L.

If ℓ, m, n are direction cosines of line L, then $\ell\hat{i} + m\hat{j} + n\hat{k}$ is a unit vector parallel to the line L.

(iv) If ℓ, m, n be the direction cosines and a, b, c be the direction ratios of a vector, then

$$\left(\ell = \frac{a}{\sqrt{a^2+b^2+c^2}}, m = \frac{b}{\sqrt{a^2+b^2+c^2}}, n = \frac{c}{\sqrt{a^2+b^2+c^2}}\right)$$

or $\ell = \frac{-a}{\sqrt{a^2+b^2+c^2}}$, $m = \frac{-b}{\sqrt{a^2+b^2+c^2}}$, $n = \frac{-c}{\sqrt{a^2+b^2+c^2}}$

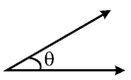
(v) If OP = r, when O is the origin and the direction cosines of OP are ℓ, m, n then the coordinates of P are (ℓr, mr, nr).
If direction cosines of the line AB are ℓ, m,n, |AB| = r, and the coordinates of A is (x_1, y_1, z_1) then the coordinates of B is given as $(x_1 + r\ell, y_1 + rm, z_1 + rn)$

(vi) If the coordinates P and Q are (x_1, y_1, z_1) and (x_2, y_2, z_2), then the direction ratios of line PQ are, $a = x_2 - x_1$, $b = y_2 - y_1$ & $c = z_2 - z_1$ and the direction cosines of line PQ are $\ell = \frac{x_2 - x_1}{|PQ|}$, $m = \frac{y_2 - y_1}{|PQ|}$ and $n = \frac{z_2 - z_1}{|PQ|}$.

(vii) **Direction cosines of axes :** Since the positive x–axis makes angles 0º, 90º, 90º with axes of x, y and z respectively. Therefore
Direction cosines of x–axis are (1, 0, 0)
Direction cosines of y–axis are (0, 1, 0)
Direction cosines of z–axis are (0, 0, 1)

Angle between two vectors :

It is the smaller angle formed when the initial points or the terminal points of the two vectors are brought together. Note that $0^\circ \le \theta \le 180^\circ$.



Vector equation of a line :

Parametric vector equation of a line passing through two point $A(\vec{a})$ and $B(\vec{b})$ is given by $\vec{r} = \vec{a} + t(\vec{b} - \vec{a})$, where 't' is a parameter. If the line passes through the point $A(\vec{a})$ and is parallel to the vector $\vec{b}$, then its equation is $\vec{r} = \vec{a} + t\vec{b}$. Note : $\vec{r}$ is the p.v. of the point on the line.

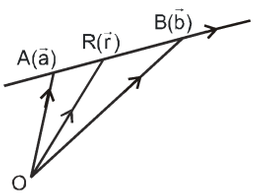


Fig. 1

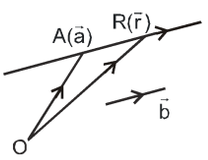


Fig. 2

A vector in the direction of the bisector of the angle between the two vectors $\vec{a}$ and $\vec{b}$ is $\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}$.
Hence bisector of the angle between the two vectors $\vec{a}$ and $\vec{b}$ is $\lambda\left(\hat{a} + \hat{b}\right)$, where $\lambda \in R^+$. Bisector of the exterior angle between $\vec{a}$ and $\vec{b}$ is $\lambda\left(\hat{a} - \hat{b}\right)$, $\lambda \in R^+$.

Note that the equations of the bisectors of the angles between the lines $\vec{r} = \vec{a} + \lambda\vec{b}$ and $\vec{r} = \vec{a} + \mu\vec{c}$ are :

$\vec{r} = \vec{a} + t\left(\hat{b}+\hat{c}\right)$ and $\vec{r} = \vec{a} + p\left(\hat{c}-\hat{b}\right)$.

Scalar product (Dot Product) of two vectors :

$$\vec{a}\,.\,\vec{b} = |\vec{a}|\,|\vec{b}|\cos\theta,\ (0 \le \theta \le \pi)$$

Note: (a) If θ is acute, then $\vec{a}\,.\,\vec{b} > 0$ and if θ is obtuse, then $\vec{a}.\vec{b} < 0$.

(b) $\vec{a}\,.\,\vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$ $\quad (\vec{a} \neq 0\,,\,\vec{b} \neq 0)$

(c) Maximum value of $\vec{a}\,.\,\vec{b}$ is $|\vec{a}|\,|\vec{b}|$

(d) Minimum value of $\vec{a}\,.\,\vec{b}$ is $-|\vec{a}|\,|\vec{b}|$

Geometrical interpretation of scalar product :
Geometrically interpreted, the scalar product of two vectors is the product of modulus of either vector and the projection of the other in its direction.

(i) Projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a}.\vec{b}}{|\vec{b}|}$

(ii) $\vec{a}\,.\,\vec{b} = \vec{b}\,.\,\vec{a}$ **(commutative)**

(iii) $\vec{a}\,.\,(\vec{b}+\vec{c}) = \vec{a}\,.\,\vec{b} + \vec{a}\,.\,\vec{c}$ **(distributive)**

(iv) $(m\vec{a})\,.\,\vec{b} = \vec{a}\,.\,(m\vec{b}) = m(\vec{a}\,.\,\vec{b})$, where m is a scalar.

(v) $\hat{i}\,.\,\hat{i} = \hat{j}\,.\,\hat{j} = \hat{k}\,.\,\hat{k} = 1;\ \ \hat{i}\,.\,\hat{j} = \hat{j}\,.\,\hat{k} = \hat{k}\,.\,\hat{i} = 0$

(vi) $\vec{a}\,.\,\vec{a} = |\vec{a}|^2 = \vec{a}^2$

(vii) If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, then $\vec{a}\,.\,\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$

$$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}\ , \qquad |\vec{b}| = \sqrt{b_1^2 + b_2^2 + b_3^2}$$

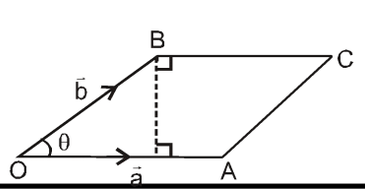
(viii) $|\vec{a} \pm \vec{b}| = \sqrt{|\vec{a}|^2 + |\vec{b}|^2 \pm 2|\vec{a}||\vec{b}|\cos\theta}$, where θ is the angle between the vectors

(ix) Any vector $\vec{a}$ can be written as $\vec{a} = (\vec{a}\,.\,\hat{i})\,\hat{i} + (\vec{a}\,.\,\hat{j})\,\hat{j} + (\vec{a}\,.\,\hat{k})\,\hat{k}$.

Vector product (Cross Product) of two vectors:

(i) If $\vec{a}, \vec{b}$ are two vectors and θ is the angle between them, then $\vec{a} \times \vec{b} = |\vec{a}|\,|\vec{b}|\sin\theta\,\hat{n}$, where $\hat{n}$ is the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ such that $\vec{a}, \vec{b}$ and $\hat{n}$ forms a right handed screw system.

(ii) Geometrically $|\vec{a} \times \vec{b}|$ = area of the parallelogram whose two adjacent sides are represented by $\vec{a}$ and $\vec{b}$.



(iii) $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$ **(not commutative)**

(iv) $(m\vec{a}) \times \vec{b} = \vec{a} \times (m\vec{b}) = m(\vec{a} \times \vec{b})$, where m is a scalar.

(v) $\vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})$ **(distributive)**

(vi) $\vec{a} \times \vec{b} = \vec{0} \Leftrightarrow \vec{a}$ and $\vec{b}$ are parallel (collinear) $(\vec{a} \neq 0 \ , \ \vec{b} \neq 0)$ i.e. $\vec{a} = K\vec{b}$, where K is a scalar.

(vii) $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$; $\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$

(viii) If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, then $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$

(ix) Unit vector perpendicular to the plane of $\vec{a}$ and $\vec{b}$ is $\hat{n} = \pm \dfrac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$

(x) A vector of magnitude 'r' and perpendicular to the plane of $\vec{a}$ and $\vec{b}$ is $\pm \dfrac{r(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$

(xi) If θ is the angle between $\vec{a}$ and $\vec{b}$, then $\sin\theta = \dfrac{|\vec{a} \times \vec{b}|}{|\vec{a}||\vec{b}|}$

(xii) If $\vec{a}, \vec{b}$ and $\vec{c}$ are the position vectors of 3 points A, B and C respectively, then the vector area of $\Delta ABC = \dfrac{1}{2}\left[\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right]$. The points A, B and C are collinear if $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = \vec{0}$

(xiii) Area of any quadrilateral whose diagonal vectors are $\vec{d}_1$ and $\vec{d}_2$ is given by $\dfrac{1}{2}\left|\vec{d}_1 \times \vec{d}_2\right|$

(xiv) **Lagrange's Identity :** For any two vectors $\vec{a}$ and $\vec{b}$; $(\vec{a} \times \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{b} \end{vmatrix}$

Projection of a line segment on a line

(i) **Vector form :** projection of a vector $\vec{a}$ on another vector $\vec{b}$ is $\vec{a} \cdot \hat{b} = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

In the above case we can consider $\overrightarrow{PQ}$ as $(x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$ in place of $\vec{a}$ and $\ell\hat{i} + m\hat{j} + n\hat{k}$ in place of $\vec{b}$.

(ii) $\ell|\vec{r}|$, $m|\vec{r}|$ & $n|\vec{r}|$ are the projection of $\vec{r}$ in OX, OY & OZ axes.

(iii) $\vec{r} = |\vec{r}|(\ell\hat{i} + m\hat{j} + n\hat{k})$

A LINE

Equation of a line

(i) Vector equation: Vector equation of a straight line passing through a fixed point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r} = \vec{a} + \lambda\vec{b}$ where λ is a scalar.

(ii) Vector equation of a straight line passing through two points with position vectors $\vec{a}$ & $\vec{b}$ is $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$.

(iii) Reduction of cartesion form of equation of a line to vector form & vice versa

$$\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} \quad \Leftrightarrow \quad \vec{r} = (x_1\hat{i} + y_1\hat{j} + z_1\hat{k}) + \lambda(a\hat{i} + b\hat{j} + c\hat{k}).$$

Foot, length and equation of perpendicular from a point to a line :

(i) **Vector Form :** Equation of a line passing through a point having position vector $\vec{\alpha}$ and perpendicular to the lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \lambda\vec{b}_2$ is parallel to $\vec{b}_1 \times \vec{b}_2$. So the vector equation of such a line is $\vec{r} = \vec{\alpha} + \lambda(\vec{b}_1 \times \vec{b}_2)$. Position vector $\vec{\beta}$ of the image of a point $\vec{\alpha}$ in a straight line $\vec{r} = \vec{a} + \lambda\vec{b}$ is given by $\vec{\beta} = 2\vec{a} - \left[\frac{2(\vec{a}-\vec{\alpha}).\vec{b}}{|\vec{b}|^2}\right]\vec{b} - \vec{\alpha}$. Position vector of the foot of the perpendicular on line is $\vec{f} = \vec{a} - \left[\frac{(\vec{a}-\vec{\alpha}).\vec{b}}{|\vec{b}|^2}\right]\vec{b}$. The equation of the perpendicular is $\vec{r} = \vec{\alpha} + \mu\left[(\vec{a}-\vec{\alpha}) - \left(\frac{(\vec{a}-\vec{\alpha}).\vec{b}}{|\vec{b}|^2}\right)\vec{b}\right]$.

Angle between two line segments :

If two lines have direction ratios a_1, b_1, c_1 and a_2, b_2, c_2 respectively, then we can consider two vectors parallel to the lines as $a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and $a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$ and angle between them can be given as.

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}.$$

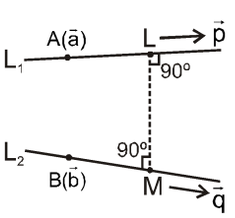
(i) The lines will be perpendicular if $a_1a_2 + b_1b_2 + c_1c_2 = 0$

(ii) The lines will be parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(iii) Two parallel lines have same direction cosines i.e. $\ell_1 = \ell_2$, $m_1 = m_2$, $n_1 = n_2$

Skew Lines and Shortest distance between two lines :

If two lines in space intersect at a point, then obviously the shortest distance between them is zero. Lines which do not intersect and are also not parallel are called **skew line**. For Skew lines the direction of the shortest distance would be perpendicular to both the lines.



Let $\overrightarrow{LM}$ be the shortest distance vector between the lines L_1 and L_2. Then $\overrightarrow{LM}$ is perpendicular to both $\vec{p}$ and $\vec{q}$ i.e. $\overrightarrow{LM}$ is parallel to $\vec{p} \times \vec{q}$. Therefore the magnitude of the shortest distance vector (i.e. $|\overrightarrow{LM}|$) would be equal to that of the projection of $\overrightarrow{AB}$ along the direction of the line of shortest distance.

$\therefore \quad |\overrightarrow{LM}| = \left|\text{Projection of } \overrightarrow{AB} \text{ on } \overrightarrow{LM}\right| = \left|\text{Projection of } \overrightarrow{AB} \text{ on } \vec{p} \times \vec{q}\right|$

$$= \left|\frac{\overrightarrow{AB} \cdot (\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}\right| = \left|\frac{(\vec{b} - \vec{a}) \cdot (\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|}\right|$$

(i) The two lines directed along $\vec{p}$ and $\vec{q}$ will intersect only if shortest distance = 0

i.e. $(\vec{b} - \vec{a}) \cdot (\vec{p} \times \vec{q}) = 0$ i.e. $(\vec{b} - \vec{a})$ lies in the plane containing $\vec{p}$ and $\vec{q}$. $\Rightarrow (\vec{b} - \vec{a}) \cdot \vec{p} \times \vec{q} = 0$.

(ii) If two parallel lines are given by $\vec{r}_1 = \vec{a}_1 + K\vec{b}$ and $\vec{r}_2 = \vec{a}_2 + K\vec{b}$, then distance (d) between them is given by $d = \left|\frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|}\right|$

(iii) Vector Form: For lines $\vec{r} = \vec{a}_1 + \lambda_1 \vec{b}_1$ & $\vec{r} = \vec{a}_2 + \lambda_2 \vec{b}_2$ to be skew

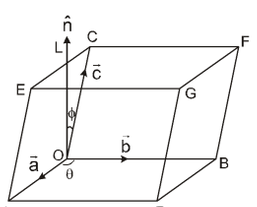
$(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1) \neq 0$ or $[\vec{b}_1 \ \vec{b}_2 \ (\vec{a}_2 - \vec{a}_1)] \neq 0$.

(iv) Shortest distance between the two parallel lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}$ & $\vec{r} = \vec{a}_2 + \mu \vec{b}$ is $d = \left|\frac{(\vec{a}_2 - \vec{a}_1) \times \vec{b}}{|\vec{b}|}\right|$.

Scalar triple product (Box Product) (S.T.P.) :

(i) The scalar triple product of three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is defined as: $\vec{a} \times \vec{b} \cdot \vec{c} = |\vec{a}| \, |\vec{b}| \, |\vec{c}| \sin\theta \cdot \cos\phi$, where θ is the angle between $\vec{a}, \vec{b}$ **(i.e. $\vec{a} \wedge \vec{b} = \theta$)** and ϕ is the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ **(i.e. $(\vec{a} \times \vec{b}) \wedge \vec{c} = \phi$)**. It is (i.e. $\vec{a} \times \vec{b} \cdot \vec{c}$) also written as $[\vec{a} \ \vec{b} \ \vec{c}]$ and spelled as box product.

(ii) Scalar triple product geometrically represents the volume of the parallelopiped whose three coterminous edges are represented by $\vec{a}, \vec{b}$ and $\vec{c}$ i.e. $V = |[\vec{a} \ \vec{b} \ \vec{c}]|$



(iii) In a scalar triple product the position of dot and cross can be interchanged i.e.

$\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c} \quad \Rightarrow \quad [\vec{a}\,\vec{b}\,\vec{c}] = [\vec{b}\,\vec{c}\,\vec{a}] = [\vec{c}\,\vec{a}\,\vec{b}]$

(iv) $\vec{a} \cdot (\vec{b} \times \vec{c}) = -\vec{a} \cdot (\vec{c} \times \vec{b})$ i.e. $[\vec{a}\,\vec{b}\,\vec{c}] = -[\vec{a}\,\vec{c}\,\vec{b}]$

(v) If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$, then $[\vec{a}\,\vec{b}\,\vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.

In general, if $\vec{a} = a_1\vec{\ell} + a_2\vec{m} + a_3\vec{n}$; $\vec{b} = b_1\vec{\ell} + b_2\vec{m} + b_3\vec{n}$ and $\vec{c} = c_1\vec{\ell} + c_2\vec{m} + c_3\vec{n}$

then $\left[\vec{a}\ \vec{b}\ \vec{c}\right]=\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}\left[\vec{\ell}\,\vec{m}\,\vec{n}\right]$, where $\vec{\ell}$, $\vec{m}$ and $\vec{n}$ are non-coplanar vectors.

(vi) If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar $\Leftrightarrow [\vec{a}\ \vec{b}\ \vec{c}]=0$.

(vii) Scalar product of three vectors, two of which are equal or parallel is 0 $\Rightarrow [\vec{a}\ \vec{b}\ \vec{c}]=0$,

(viii) If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-coplanar, then $[\vec{a}\ \vec{b}\ \vec{c}] > 0$ for right handed system and $[\vec{a}\ \vec{b}\ \vec{c}] < 0$ for left handed system.

(ix) $[\hat{i}\ \hat{j}\ \hat{k}] = 1$

(x) $[K\vec{a}\ \vec{b}\ \vec{c}]= K[\vec{a}\ \vec{b}\ \vec{c}]$

(xi) $[(\vec{a}+\vec{b})\ \vec{c}\ \vec{d}\,] =[\vec{a}\ \vec{c}\ \vec{d}] + [\vec{b}\ \vec{c}\ \vec{d}]$

(xii) $\left[\vec{a}-\vec{b}\quad \vec{b}-\vec{c}\quad \vec{c}-\vec{a}\right] = 0$ and $\left[\vec{a}+\vec{b}\quad \vec{b}+\vec{c}\quad \vec{c}+\vec{a}\right] = 2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$.

(xiii) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix}$

Tetrahedron and its properties :

(a) The volume of the tetrahedron OABC with O as origin and the position vectors of A, B and C being $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively is given by $V = \frac{1}{6}\left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right|$

(b) If the position vectors of the vertices of tetrahedron are $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$, then the position vector of its centroid is given by $\frac{1}{4}(\vec{a}+\vec{b}+\vec{c}+\vec{d})$.

note that this is also the point of concurrency of the lines joining the vertices to the centroids of the opposite faces and is also called the centre of the tetrahedron. In case the tetrahedron is regular it is equidistant from the vertices and the four faces of the tetrahedron.

Vector triple product :

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be any three vectors, then the expression $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector & is called a vector triple product.

Geometrical interpretation of $\vec{a} \times (\vec{b} \times \vec{c})$

Consider the expression $\vec{a} \times (\vec{b} \times \vec{c})$ which itself is a vector, since it is a cross product of two vectors $\vec{a}$ and $(\vec{b} \times \vec{c})$. Now $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector perpendicular to the plane containing $\vec{a}$ and $(\vec{b} \times \vec{c})$ but $\vec{b} \times \vec{c}$ is a vector perpendicular to the plane containing $\vec{b}$ and $\vec{c}$, therefore $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector which lies in the plane of $\vec{b}$ and $\vec{c}$ and perpendicular to $\vec{a}$. Hence we can express $\vec{a} \times (\vec{b} \times \vec{c})$ in terms of $\vec{b}$ and $\vec{c}$ i.e.

$\vec{a} \times (\vec{b} \times \vec{c}) = x\vec{b} + y\vec{c}$, where x , y are scalars.

- $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$
- $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$
- In general $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$

Linear combinations :

Given a finite set of vectors $\vec{a},\vec{b},\vec{c},.....$, then the vector $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c} + ...$ is called a linear combination of $\vec{a},\vec{b},\vec{c},.....$ for any x, y, z..... $\in$ R. We have the following results:

(a) If $\vec{a},\vec{b}$ are non zero, non–collinear vectors, then $x\vec{a} + y\vec{b} = x'\vec{a} + y'\vec{b} \Rightarrow x = x',\ y = y'$

(b) **Fundamental Theorem in plane :** Let $\vec{a},\vec{b}$ be non zero, non collinear vectors, then any vector $\vec{r}$ coplanar

with $\vec{a},\vec{b}$ can be expressed uniquely as a linear combination of $\vec{a}$ and $\vec{b}$

i.e. there exist some unique x, y $\in$ R such that $x\vec{a} + y\vec{b} = \vec{r}$.

(c) If $\vec{a},\vec{b},\vec{c}$ are non–zero, non–coplanar vectors, then

$x\vec{a} + y\vec{b} + z\vec{c} = x'\vec{a} + y'\vec{b} + z'\vec{c} \Rightarrow x = x'\ ,\ y = y'\ ,\ z = z'$

(d) **Fundamental theorem in space:** Let $\vec{a},\vec{b},\vec{c}$ be non–zero, non–coplanar vectors in space.

Then any vector $\vec{r}$ can be uniquely expressed as a linear combination of $\vec{a},\vec{b},\vec{c}$ i.e. there exist some unique x,y, z $\in$ R such that $x\vec{a} + y\vec{b} + z\vec{c} = \vec{r}$.

(e) If $\vec{x}_1,\vec{x}_2,.....,\ \vec{x}_n$ are n non zero vectors and $k_1,k_2,.....,k_n$ are n scalars and if the linear combination $k_1\vec{x}_1 + k_2\vec{x}_2 + ... + k_n\vec{x}_n = \vec{0} \Rightarrow k_1 = 0, k_2 = 0,.....,k_n = 0$, then we say that vectors $\vec{x}_1, \vec{x}_2, ..., \vec{x}_n$ are **linearly independent vectors**.

(f) If $\vec{x}_1,\vec{x}_2,.....,\ \vec{x}_n$ are not **linearly independent** then they are said to be **linearly dependent vectors**.

i.e. if $k_1\vec{x}_1 + k_2\vec{x}_2 + k_3\vec{x}_3 ..... + k_r\vec{x}_r + ... + k_n\vec{x}_n = \vec{0}$ and if there exists at least one $k_r \neq 0$, then $\vec{x}_1, \vec{x}_2, ..., \vec{x}_n$ are said to be **linearly dependent vectors.**

Note 2:

☞ If $\vec{a} = 3\hat{i} + 2\hat{j} + 5\hat{k}$ then $\vec{a}$ is expressed as a **Linear Combination** of vectors $\hat{i}$, $\hat{j}$, $\hat{k}$. Also $\vec{a}$, $\hat{i}$, $\hat{j}$, $\hat{k}$ form a linearly dependent set of vectors. In general, in 3 dimensional space every set of four vectors is a linearly dependent system.

☞ $\hat{i},\hat{j},\hat{k}$ are **Linearly Independent** set of vectors. For $K_1\hat{i} + K_2\hat{j} + K_3\hat{k} = \vec{0} \Rightarrow K_1 = K_2 = K_3 = 0$

☞ Two vectors $\vec{a}$ and $\vec{b}$ are linearly dependent $\Rightarrow \vec{a}$ is parallel to $\vec{b}$ i.e. $\vec{a} \times \vec{b} = \vec{0} \Rightarrow$ linear dependence of $\vec{a}$ and $\vec{b}$. Conversely if $\vec{a} \times \vec{b} \neq \vec{0}$ then $\vec{a}$ and $\vec{b}$ are linearly independent.

☞ If three vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ are linearly dependent, then they are coplanar i.e. $[\vec{a}\ \ \vec{b}\ \ \vec{c}] = 0$. Conversely if

$[\vec{a}\ \vec{b}\ \vec{c}] \neq 0$ then the vectors are linearly independent.

Test of collinearity :

Three points A,B,C with position vectors $\vec{a}, \vec{b}, \vec{c}$ respectively are collinear, if & only if there exist scalars x, y, z not all zero simultaneously such that $x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$, where x + y + z = 0.

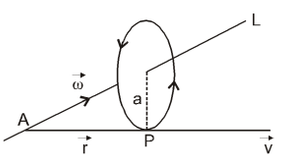
Test of coplanarity :

Four points A, B, C, D with position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ respectively are coplanar if and only if there exist scalars x, y, z, w not all zero simultaneously such that $x\vec{a} + y\vec{b} + z\vec{c} + w\vec{d} = \vec{0}$, where x + y + z + w = 0.

Application of vectors :

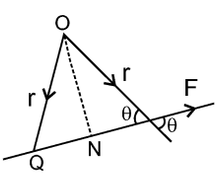
(a) Work done against a constant force $\vec{F}$ over a displacement $\vec{s}$ is defined as $W = \vec{F}.\vec{s}$

(b) The tangential velocity $\vec{V}$ of a body moving in a circle is given by $\vec{V} = \vec{\omega} \times \vec{r}$, where $\vec{r}$ is the position vector of the point P.



(c) The moment of $\vec{F}$ about 'O' is defined as $\vec{M} = \vec{r} \times \vec{F}$, where $\vec{r}$ is the position vector of P w.r.t. 'O'. The direction of $\vec{M}$ is along the normal to the plane OPN such that $\vec{r}, \vec{F}$ & $\vec{M}$ form a right handed system.

(d) Moment of the couple = $(\vec{r}_1 - \vec{r}_2) \times \vec{F}$, where $\vec{r}_1$ and $\vec{r}_2$ are position vectors of the point of the application of the forces $\vec{F}$ and $-\vec{F}$.



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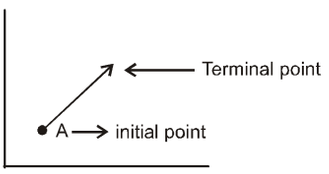
VECTORS & THREE DIMENSIONAL GEOMETRY

Vectors and their representation :



Vector quantities are specified by definite magnitude and definite direction. A vector is generally represented by a directed line segment, say $\overrightarrow{AB}$. A is called the **initial point** and B is called the **terminal point**. The magnitude of vector $\overrightarrow{AB}$ is expressed by $|\overrightarrow{AB}|$.

Zero vector:
A vector of zero magnitude i.e. which has the same initial and terminal point, is called a **zero vector.** It is denoted by **O**. **The direction of zero vector is indeterminate.**

Unit vector:
A vector of unit magnitude in the direction of a vector $\vec{a}$ is called unit vector along $\vec{a}$ and is denoted by $\hat{a}$, symbolically $\hat{a} = \frac{\vec{a}}{|\vec{a}|}$.

Equal vectors:

Two vectors are said to be equal if they have the same magnitude, direction and represent the same physical quantity.

Collinear vectors:

Two vectors are said to be collinear if their directed line segments are parallel irrespective of their directions. Collinear vectors are also called **parallel vectors**. If they have the same direction they are named as **like vectors** otherwise **unlike vectors**.

Symbolically, two non-zero vectors $\vec{a}$ and $\vec{b}$ are collinear if and only if, $\vec{a} = \lambda\vec{b}$, where $\lambda \in R$

$$\vec{a} = \lambda\vec{b} \Leftrightarrow \left(a_1\hat{i} + a_2\hat{j} + a_3\hat{k}\right) = \lambda\left(b_1\hat{i} + b_2\hat{j} + b_3\hat{k}\right) \Leftrightarrow \quad a_1 = \lambda b_1,\ a_2 = \lambda b_2,\ a_3 = \lambda b_3$$

$$\Leftrightarrow \frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} \ (= \lambda)$$

Vectors $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are collinear if $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$

Coplanar vectors:

A given number of vectors are called coplanar if their line segments are all parallel to the same plane. Note that **"two vectors are always coplanar"**.

Multiplication of a vector by a scalar :

If $\vec{a}$ is a vector and m is a scalar, then $m\vec{a}$ is a vector parallel to $\vec{a}$ whose magnitude is $|m|$ times that of $\vec{a}$. This multiplication is called **scalar multiplication**. If $\vec{a}$ and $\vec{b}$ are vectors and m, n are scalars, then :

$m(\vec{a}) = (\vec{a})m = m\vec{a}$, $m(n\vec{a}) = n(m\vec{a}) = (mn)\vec{a}$

$(m+n)\vec{a} = m\vec{a} + n\vec{a}$, $m(\vec{a}+\vec{b}) = m\vec{a} + m\vec{b}$

Addition of vectors :

(i) If two vectors $\vec{a}$ and $\vec{b}$ are represented by $\overrightarrow{OA}$ and $\overrightarrow{OB}$, then their sum $\vec{a}+\vec{b}$ is a vector represented by $\overrightarrow{OC}$, where OC is the diagonal of the parallelogram OACB.

(ii) $\vec{a}+\vec{b} = \vec{b}+\vec{a}$ **(commutative)** (iii) $(\vec{a}+\vec{b})+\vec{c} = \vec{a}+(\vec{b}+\vec{c})$ **(associative)**

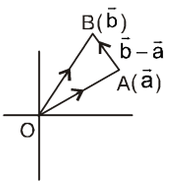
(iv) $\vec{a}+\vec{0} = \vec{a} = \vec{0}+\vec{a}$ (v) $\vec{a}+(-\vec{a}) = \vec{0} = (-\vec{a})+\vec{a}$

(vi) $|\vec{a}+\vec{b}| \le |\vec{a}|+|\vec{b}|$ (vii) $|\vec{a}-\vec{b}| \ge ||\vec{a}|-|\vec{b}||$

Position vector of a point:

Let O be a fixed origin, then the position vector of a point P is the vector $\overrightarrow{OP}$. If $\vec{a}$ and $\vec{b}$ are position vectors of two points A and B, then

$\overrightarrow{AB}$ = $\vec{b}-\vec{a}$ = position vector **(p.v.)** of B – position vector **(p.v.)** of A.

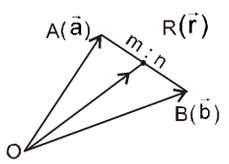


DISTANCE FORMULA

Distance between the two points A$(\vec{a})$ and B$(\vec{b})$ is AB = $|\vec{a}-\vec{b}|$

SECTION FORMULA

If $\vec{a}$ and $\vec{b}$ are the position vectors of two points A and B, then the p.v. of a point which divides AB in the ratio m: n is given by $\vec{r} = \frac{n\vec{a}+m\vec{b}}{m+n}$.

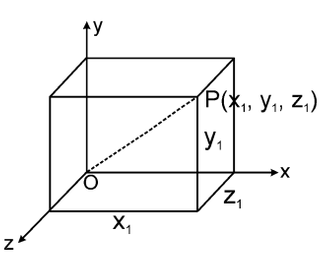


Note : Position vector of mid point of AB = $\frac{\vec{a}+\vec{b}}{2}$.

Coordinate of a point in space

There are infinite number of points in space. We want to identify each and every point of space with the help of three mutually perpendicular coordinate axes OX, OY and OZ.Three mutually perpendicular lines OX, OY, OZ are considered as the three axes. The plane formed with the help of x & y axes is called x-y plane, similarly y & z axes form y-z plane and z & x axes form z - x plane.

Consider any point P in the space, Drop a perpendicular from that point to x -y plane, then the algebraic length of this perpendicular is considered as z-coordinate and from foot of the perpendicular drop perpendiculars to x and y axes. These algebraic lengths of perpendiculars are considered as y and x coordinates respectively.



Vector representation of a point in space

If coordinate of a point P in space is (x, y, z), then the position vector of the point P with respect to the same origin is $x\hat{i} + y\hat{j} + z\hat{k}$.

Distance formula

Distance between any two points (x_1, y_1, z_1) and (x_2, y_2, z_2) is given as $\sqrt{(x_1-x_2)^2+(y_1-y_2)^2+(z_1-z_2)^2}$

Vector method

We know that if position vector of two points A and B are given as $\overrightarrow{OA}$ and $\overrightarrow{OB}$ then

$$AB = |\overrightarrow{OB} - \overrightarrow{OA}| \quad \Rightarrow \quad AB = |(x_2\hat{i} + y_2\hat{j} + z_2\hat{k}) - (x_1\hat{i} + y_1\hat{j} + z_1\hat{k})|$$

$$\Rightarrow \quad AB = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

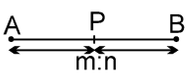
Distance of a point P from coordinate axes

Let PA, PB and PC are distances of the point P(x, y, z) from the coordinate axes OX, OY and OZ respectively then

$$PA = \sqrt{y^2+z^2}\ ,\ PB = \sqrt{z^2+x^2}\ ,\ PC = \sqrt{x^2+y^2}$$

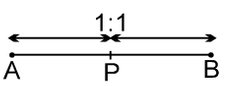
Section formula

If point P divides the distance between the points A (x_1, y_1, z_1) and B (x_2, y_2, z_2) in the ratio of m : n, internally then coordinates of P are given as $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}, \frac{mz_2+nz_1}{m+n}\right)$



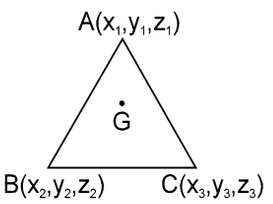
Note :- Mid point

$$\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}, \frac{z_1+z_2}{2}\right)$$



Centroid of a triangle

$$G \equiv \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}, \frac{z_1+z_2+z_3}{3}\right)$$



Incentre of triangle ABC

$$\left(\frac{ax_1+bx_2+cx_3}{a+b+c}, \frac{ay_1+by_2+cy_3}{a+b+c}, \frac{az_1+bz_2+cz_3}{a+b+c}\right)$$

Where AB = c, BC = a, CA = b

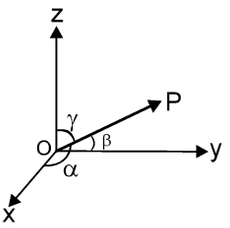
Centroid of a tetrahedron

A (x_1, y_1, z_1) B (x_2, y_2, z_2) C (x_3, y_3, z_3) and D (x_4, y_4, z_4) are the vertices of a tetrahedron, then coordinate of its centroid (G) is given as $\left(\frac{\sum_{i=1}^{4} x_i}{4}, \frac{\sum_{i=1}^{4} y_i}{4}, \frac{\sum_{i=1}^{4} z_i}{4}\right)$

Direction cosines and direction ratios

(i) **Direction cosines :** Let α, β, γ be the angles which a directed line makes with the positive directions of the axes of x, y and z respectively, then $\cos\alpha, \cos\beta, \cos\gamma$ are called the direction cosines of the line. The direction cosines are usually denoted by ℓ, m, n.

Thus $\ell = \cos\alpha$, $m = \cos\beta$, $n = \cos\gamma$.



(ii) If ℓ, m, n be the direction cosines of a line, then $\ell^2 + m^2 + n^2 = 1$

(iii) **Direction ratios :** Let a, b, c be proportional to the direction cosines ℓ, m, n then a, b, c are called the direction ratios.

If a, b, c, are the direction ratios of any line L, then $a\hat{i} + b\hat{j} + c\hat{k}$ will be a vector parallel to the line L.

If ℓ, m, n are direction cosines of line L, then $\ell\,\hat{i} + m\,\hat{j} + n\,\hat{k}$ is a unit vector parallel to the line L.

(iv) If ℓ, m, n be the direction cosines and a, b, c be the direction ratios of a vector, then

$$\left(\ell = \frac{a}{\sqrt{a^2+b^2+c^2}}, m = \frac{b}{\sqrt{a^2+b^2+c^2}}, n = \frac{c}{\sqrt{a^2+b^2+c^2}}\right)$$

or $\quad \ell = \dfrac{-a}{\sqrt{a^2+b^2+c^2}}$, $m = \dfrac{-b}{\sqrt{a^2+b^2+c^2}}$, $n = \dfrac{-c}{\sqrt{a^2+b^2+c^2}}$

(v) If OP = r, when O is the origin and the direction cosines of OP are ℓ, m, n then the coordinates of P are $(\ell r, mr, nr)$.

If direction cosines of the line AB are ℓ, m,n, |AB| = r, and the coordinates of A is (x_1, y_1, z_1) then the coordinates of B is given as $(x_1 + r\ell, y_1 + rm, z_1 + rn)$

(vi) If the coordinates P and Q are (x_1, y_1, z_1) and (x_2, y_2, z_2), then the direction ratios of line PQ are, $a = x_2 - x_1$, $b = y_2 - y_1$ & $c = z_2 - z_1$ and the direction cosines of line PQ are $\ell = \dfrac{x_2 - x_1}{|PQ|}$,

$m = \dfrac{y_2 - y_1}{|PQ|}$ and $n = \dfrac{z_2 - z_1}{|PQ|}$.

(vii) **Direction cosines of axes :** Since the positive x–axis makes angles 0º, 90º, 90º with axes of x, y and z respectively. Therefore

Direction cosines of x–axis are (1, 0, 0)

Direction cosines of y–axis are (0, 1, 0)

Direction cosines of z–axis are (0, 0, 1)

Case I. when $\dfrac{\ell}{m} = 1$: In this case $m = \ell$

From (1), $2\ell + n = 0 \quad \Rightarrow \quad n = -2\ell$

$\therefore$ $\ell : m : n = 1 : 1 : -2$

$\therefore$ Direction ratios of the line are 1, 1, – 2

$\therefore$ Direction cosines are

$$\pm \frac{1}{\sqrt{1^2+1^2+(-2)^2}}, \pm \frac{1}{\sqrt{1^2+1^2+(-2)^2}}, \pm \frac{-2}{\sqrt{1^2+1^2+(-2)^2}}$$

$$\Rightarrow \quad \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}} \text{ or } -\frac{1}{\sqrt{6}}, -\frac{1}{\sqrt{6}}, \frac{2}{\sqrt{6}}$$

Case II. When $\frac{\ell}{m} = -2$: In this case $\ell = -2m$

From (1), $-2m + m + n = 0 \quad \Rightarrow \quad n = m$

$\therefore \quad \ell : m : n = -2m : m : m$

$= -2 : 1 : 1$

$\therefore$ Direction ratios of the line are $-2, 1, 1$.

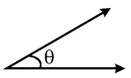
$\therefore$ Direction cosines are

$$\pm\frac{-2}{\sqrt{(-2)^2+1^2+1^2}}, \pm\frac{1}{\sqrt{(-2)^2+1^2+1^2}}, \pm\frac{1}{\sqrt{(-2)^2+1^2+1^2}}$$

$$\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}} \quad \text{or} \quad \frac{2}{\sqrt{6}}, \frac{-1}{\sqrt{6}}, \frac{-1}{\sqrt{6}}$$

Angle between two vectors :

It is the smaller angle formed when the initial points or the terminal points of the two vectors are brought together. Note that $0^\circ \le \theta \le 180^\circ$.



Vector equation of a line :

Parametric vector equation of a line passing through two point $A(\vec{a})$ and $B(\vec{b})$ is given by $\vec{r} = \vec{a} + t(\vec{b} - \vec{a})$, where 't' is a parameter. If the line passes through the point $A(\vec{a})$ and is parallel to the vector $\vec{b}$, then its equation is $\vec{r} = \vec{a} + t\,\vec{b}$. Note : $\vec{r}$ is the p.v. of the point on the line.

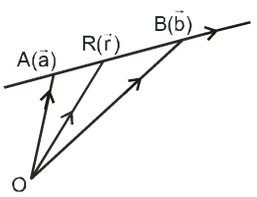


Fig. 1

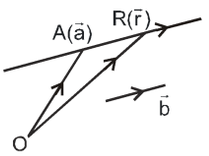


Fig. 2

A vector in the direction of the bisector of the angle between the two vectors $\vec{a}$ and $\vec{b}$ is $\frac{\vec{a}}{|\vec{a}|} + \frac{\vec{b}}{|\vec{b}|}$.

Hence bisector of the angle between the two vectors $\vec{a}$ and $\vec{b}$ is $\lambda\left(\hat{a} + \hat{b}\right)$, where $\lambda \in R^+$. Bisector of the exterior angle between $\vec{a}$ and $\vec{b}$ is $\lambda\left(\hat{a} - \hat{b}\right)$, $\lambda \in R^+$.

Note that the equations of the bisectors of the angles between the lines $\vec{r} = \vec{a} + \lambda\vec{b}$ and $\vec{r} = \vec{a} + \mu\vec{c}$ are :

$\vec{r} = \vec{a} + t\left(\hat{b} + \hat{c}\right)$ and $\vec{r} = \vec{a} + p\left(\hat{c} - \hat{b}\right)$.

Scalar product (Dot Product) of two vectors :

$$\vec{a} \cdot \vec{b} = |\vec{a}|\,|\vec{b}| \cos\theta, \ (0 \le \theta \le \pi)$$

Note: (a) If θ is acute, then $\vec{a}.\vec{b} > 0$ and if θ is obtuse, then $\vec{a}.\vec{b} < 0$.

(b) $\vec{a}.\vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$ $\quad (\vec{a} \neq 0, \vec{b} \neq 0)$

(c) Maximum value of $\vec{a}.\vec{b}$ is $|\vec{a}||\vec{b}|$

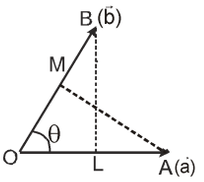
(d) Minimum value of $\vec{a}.\vec{b}$ is $-|\vec{a}||\vec{b}|$

Geometrical interpretation of scalar product :

Let $\vec{a}$ and $\vec{b}$ be vectors represented by $\overrightarrow{OA}$ and $\overrightarrow{OB}$ respectively. Let θ be the angle between $\overrightarrow{OA}$ and $\overrightarrow{OB}$. Draw BL $\perp$ OA and AM $\perp$ OB.

From ΔOBL and ΔOAM, we have OL = OB cos θ and OM = OA cos θ.

Here OL and OM are known as projections of $\vec{b}$ on $\vec{a}$ and $\vec{a}$ on $\vec{b}$ respectively.



Now, $\vec{a}.\vec{b} = |\vec{a}||\vec{b}|\cos\theta = |\vec{a}|(|\vec{b}|\cos\theta)$

$= |\vec{a}|(OB\cos\theta) = |\vec{a}|(OL)$

= (Magnitude of $\vec{a}$) (Projection of $\vec{b}$ on $\vec{a}$) .....(i)

Again $\vec{a}.\vec{b} = |\vec{a}||\vec{b}|\cos\theta = |\vec{b}|(|\vec{a}|\cos\theta) = |\vec{b}|(OA\cos\theta) = |\vec{b}|(OM)$

= (magnitude of $\vec{b}$) (Projection of $\vec{a}$ on $\vec{b}$) .....(ii)

Thus geometrically interpreted, the scalar product of two vectors is the product of modulus of either vector and the projection of the other in its direction.

(i) Projection of $\vec{a}$ on $\vec{b} = \dfrac{\vec{a}.\vec{b}}{|\vec{b}|}$

(ii) $\vec{a}.\vec{b} = \vec{b}.\vec{a}$ **(commutative)**

(iii) $\vec{a}.(\vec{b}+\vec{c}) = \vec{a}.\vec{b} + \vec{a}.\vec{c}$ **(distributive)**

(iv) $(m\vec{a}).\vec{b} = \vec{a}.(m\vec{b}) = m(\vec{a}.\vec{b})$, where m is a scalar.

(v) $\hat{i}.\hat{i} = \hat{j}.\hat{j} = \hat{k}.\hat{k} = 1$; $\hat{i}.\hat{j} = \hat{j}.\hat{k} = \hat{k}.\hat{i} = 0$

(vi) $\vec{a}.\vec{a} = |\vec{a}|^2 = \vec{a}^2$

(vii) If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, then $\vec{a}.\vec{b} = a_1b_1 + a_2b_2 + a_3b_3$

$|\vec{a}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$, $\quad |\vec{b}| = \sqrt{b_1^2 + b_2^2 + b_3^2}$

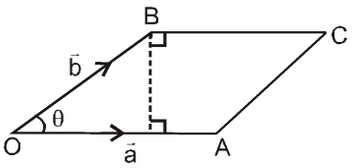
(viii) $|\vec{a} \pm \vec{b}| = \sqrt{|\vec{a}|^2 + |\vec{b}|^2 \pm 2|\vec{a}||\vec{b}|\cos\theta}$, where θ is the angle between the vectors

(ix) Any vector $\vec{a}$ can be written as $\vec{a} = (\vec{a}.\hat{i})\hat{i} + (\vec{a}.\hat{j})\hat{j} + (\vec{a}.\hat{k})\hat{k}$.

Vector product (Cross Product) of two vectors:

(i) If $\vec{a}, \vec{b}$ are two vectors and θ is the angle between them, then $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{n}$, where $\hat{n}$ is the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ such that $\vec{a}, \vec{b}$ and $\hat{n}$ forms a right handed screw system.

(ii) Geometrically $|\vec{a} \times \vec{b}|$ = area of the parallelogram whose two adjacent sides are represented by $\vec{a}$ and $\vec{b}$.



(iii) $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$ **(not commutative)**

(iv) $(m\vec{a}) \times \vec{b} = \vec{a} \times (m\vec{b}) = m(\vec{a} \times \vec{b})$, where m is a scalar.

(v) $\vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b}) + (\vec{a} \times \vec{c})$ **(distributive)**

(vi) $\vec{a} \times \vec{b} = \vec{0} \Leftrightarrow \vec{a}$ and $\vec{b}$ are parallel (collinear) $(\vec{a} \neq 0 \ , \ \vec{b} \neq 0)$ i.e. $\vec{a} = K\vec{b}$, where K is a scalar.

(vii) $\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = \vec{0}$; $\hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$

(viii) If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$, then $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$

(ix) Unit vector perpendicular to the plane of $\vec{a}$ and $\vec{b}$ is $\hat{n} = \pm \dfrac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$

(x) A vector of magnitude 'r' and perpendicular to the plane of $\vec{a}$ and $\vec{b}$ is $\pm \dfrac{r(\vec{a} \times \vec{b})}{|\vec{a} \times \vec{b}|}$

(xi) If θ is the angle between $\vec{a}$ and $\vec{b}$, then $\sin\theta = \dfrac{|\vec{a} \times \vec{b}|}{|\vec{a}|\,|\vec{b}|}$

(xii) If $\vec{a}, \vec{b}$ and $\vec{c}$ are the position vectors of 3 points A, B and C respectively, then the vector area of $\Delta ABC = \dfrac{1}{2}\left[\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a}\right]$. The points A, B and C are collinear if $\vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = \vec{0}$

(xiii) Area of any quadrilateral whose diagonal vectors are $\vec{d}_1$ and $\vec{d}_2$ is given by $\dfrac{1}{2}\left|\vec{d}_1 \times \vec{d}_2\right|$

(xiv) **Lagrange's Identity :** For any two vectors $\vec{a}$ and $\vec{b}$; $(\vec{a} \times \vec{b})^2 = |\vec{a}|^2 |\vec{b}|^2 - (\vec{a} \cdot \vec{b})^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} \\ \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{b} \end{vmatrix}$

Projection of a line segment on a line

(i) If the coordinates of P and Q are (x_1, y_1, z_1) and (x_2, y_2, z_2), then the projection of the line segments PQ on a line having direction cosines ℓ, m, n is
$\left|\ell(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)\right|$

(ii) **Vector form :** projection of a vector $\vec{a}$ on another vector $\vec{b}$ is $\vec{a} \cdot \hat{b} = \dfrac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$

In the above case we can consider $\overrightarrow{PQ}$ as $(x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$ in place of $\vec{a}$ and $\ell\hat{i} + m\hat{j} + n\hat{k}$ in place of $\vec{b}$.

(iii) $\ell|\vec{r}|$, $m|\vec{r}|$ & $n|\vec{r}|$ are the projection of $\vec{r}$ in OX, OY & OZ axes.

(iv) $\vec{r} = |\vec{r}|(\ell\hat{i} + m\hat{j} + n\hat{k})$

A LINE

Equation of a line

(i) A straight line in space is characterised by the intersection of two planes which are not parallel and therefore, the equation of a straight line is a solution of the system constituted by the equations of the two planes, $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$.
This form is also known as non–symmetrical form.

(ii) The equation of a line passing through the point (x_1, y_1, z_1) and having direction ratios a, b, c is $\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = r$. This form is called symmetric form. A general point on the line is given by $(x_1 + ar, y_1 + br, z_1 + cr)$.

(iii) Vector equation: Vector equation of a straight line passing through a fixed point with position vector $\vec{a}$ and parallel to a given vector $\vec{b}$ is $\vec{r} = \vec{a} + \lambda\vec{b}$ where λ is a scalar.

(iv) The equation of the line passing through the points (x_1, y_1, z_1) and (x_2, y_2, z_2) is

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

(v) Vector equation of a straight line passing through two points with position vectors $\vec{a}$ & $\vec{b}$ is $\vec{r} = \vec{a} + \lambda(\vec{b} - \vec{a})$.

(vi) Reduction of cartesion form of equation of a line to vector form & vice versa

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} \quad \Leftrightarrow \quad \vec{r} = (x_1\hat{i} + y_1\hat{j} + z_1\hat{k}) + \lambda(a\hat{i} + b\hat{j} + c\hat{k}).$$

Note: Straight lines parallel to co-ordinate axes:

Straight lines		Equation	Straight lines		Equation
(i)	Through origin	y = mx and z = nx	**(v)**	Parallel to x–axis	y = p, z = q
(ii)	x–axis	y = 0 and z = 0	**(vi)**	Parallel to y–axis	x = h, z = q
(iii)	y–axis	x = 0 and z = 0	**(vii)**	Parallel to z–axis	x = h, y = p
(iv)	z–axis	x = 0 and y = 0			

Reduction of non-symmetrical form to symmetrical form :

Let equation of the line in non–symmetrical form be $a_1x + b_1y + c_1z + d_1 = 0$, $a_2x + b_2y + c_2z + d_2 = 0$. To find the equation of the line in symmetrical form, we must know (i) its direction ratios (ii) coordinate of any point on it.

(i) **Direction ratios:** Let ℓ, m, n be the direction ratios of the line. Since the line lies in both the planes, it must be perpendicular to normals of both planes. So $a_1\ell + b_1m + c_1n = 0$, $a_2\ell + b_2m + c_2n = 0$. From these equations, proportional values of ℓ, m, n can be found by cross–multiplication as $\frac{\ell}{b_1c_2 - b_2c_1} = \frac{m}{c_1a_2 - c_2a_1} = \frac{n}{a_1b_2 - a_2b_1}$

Alternative method

The vector $\begin{vmatrix} i & j & k \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = i(b_1c_2 - b_2c_1) + j(c_1a_2 - c_2a_1) + k(a_1b_2 - a_2b_1)$ will be parallel to the line of

intersection of the two given planes. hence $\ell : m: n = (b_1c_2 - b_2c_1): (c_1a_2 - c_2a_1): (a_1b_2 - a_2b_1)$

(ii) **Point on the line** – Note that as ℓ, m, n cannot be zero simultaneously, so at least one must be non–zero. Let $a_1b_2 - a_2b_1 \neq 0$, then the line cannot be parallel to xy plane, so it intersect it. Let it intersect xy–plane in $(x_1, y_1, 0)$. Then $a_1x_1 + b_1y_1 + d_1 = 0$ and $a_2x_1 + b_2y_1 + d_2 = 0$. Solving these,we get a point on the line. Then its equation becomes.

$$\frac{x - x_1}{b_1c_2 - b_2c_1} = \frac{y - y_1}{c_1a_2 - c_2a_1} = \frac{z - 0}{a_1b_2 - a_2b_1} \text{ or } \frac{x - \frac{b_1d_2 - b_2d_1}{a_1b_2 - a_2b_1}}{b_1c_2 - b_2c_1} = \frac{y - \frac{d_1a_2 - d_2a_1}{a_1b_2 - a_2b_1}}{c_1a_2 - c_2a_1} = \frac{z - 0}{a_1b_2 - a_2b_1}$$

Note: If $\ell \neq 0$, take a point on yz–plane as $(0, y_1, z_1)$ and if $m \neq 0$, take a point on xz–plane as $(x_1, 0, z_1)$.

Alternative method

If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ Put z = 0 in both the equations and solve the equations $a_1x + b_1y + d_1 = 0, a_2x + b_2y + d_2 = 0$

otherwise Put y = 0 and solve the equations $a_1x + c_1z + d_1 = 0$ and $a_2x + c_2z + d_2 = 0$

Foot, length and equation of perpendicular from a point to a line :

(i) **Cartesian form :** Let equation of the line be $\frac{x-a}{\ell} = \frac{y-b}{m} = \frac{z-c}{n} = r$ (say) .....(i)

and A (α, β, γ) be the point.

Any point on line (i) is P $(\ell r + a, mr + b, nr + c)$ ..... (ii)

If it is the foot of the perpendicular from A on the line, then AP is perpendicular to the line. So $\ell(\ell r + a - \alpha) + m(mr + b - \beta) + n(nr + c - \gamma) = 0$ i.e. $r = (\alpha - a)\ell + (\beta - b)m + (\gamma - c)n$ since $\ell^2 + m^2 + n^2 = 1$. Putting this value of r in (ii), we get the foot of perpendicular from point A on the given line. Since foot of perpendicular P is known, then the length of perpendicular is given by $AP = \sqrt{(\ell r + a - \alpha)^2 + (mr + b - \beta)^2 + (nr + c - \gamma)^2}$ the equation of perpendicular is given by $\frac{x - \alpha}{\ell r + a - \alpha} = \frac{y - \beta}{mr + b - \beta} = \frac{z - \gamma}{nr + c - \gamma}$

(ii) **Vector Form :** Equation of a line passing through a point having position vector $\vec{\alpha}$ and perpendicular to the lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \lambda\vec{b}_2$ is parallel to $\vec{b}_1 \times \vec{b}_2$. So the vector equation of such a line is $\vec{r} = \vec{\alpha} + \lambda(\vec{b}_1 \times \vec{b}_2)$. Position vector $\vec{\beta}$ of the image of a point $\vec{\alpha}$ in a straight line $\vec{r} = \vec{a} + \lambda\vec{b}$ is given by $\vec{\beta} = 2\vec{a} - \left[\frac{2(\vec{a} - \vec{\alpha}).\vec{b}}{|\vec{b}|^2}\right]\vec{b} - \vec{\alpha}$. Position vector of the foot of the perpendicular on line is $\vec{f} = \vec{a} - \left[\frac{(\vec{a} - \vec{\alpha}).\vec{b}}{|\vec{b}|^2}\right]\vec{b}$. The equation of the perpendicular is $\vec{r} = \vec{\alpha} + \mu\left[(\vec{a} - \vec{\alpha}) - \left(\frac{(\vec{a} - \vec{\alpha}).\vec{b}}{|\vec{b}|^2}\right)\vec{b}\right]$.

To find image of a point w. r. t a line

Let $L \equiv \frac{x - x_2}{a} = \frac{y - y_2}{b} = \frac{z - z_2}{c}$ is a given line

Let (x', y', z') is the image of the point P (x_1, y_1, z_1) with respect to the line L. Then

(i) $a(x_1 - x') + b(y_1 - y') + c(z_1 - z') = 0$

(ii) $$\frac{\frac{x_1 + x'}{2} - x_2}{a} = \frac{\frac{y_1 + y'}{2} - y_2}{b} = \frac{\frac{z_1 + z'}{2} - z_2}{c} = \lambda$$

from (ii) get the value of x', y', z' in terms of λ as $x' = 2a\lambda + 2x_2 - x_1$,
$y' = 2b\lambda + 2y_2 - y_1$, $z' = 2c\lambda + 2z_2 - z_1$
now put the values of x', y', z' in (i) get λ and resubtitute the value of λ to to get $(x'\ y'\ z')$.

Angle between two line segments :

If two lines have direction ratios a_1, b_1, c_1 and a_2, b_2, c_2 respectively, then we can consider two vectors parallel to the lines as $a_1\hat{i} + b_1\hat{j} + c_1\hat{k}$ and $a_2\hat{i} + b_2\hat{j} + c_2\hat{k}$ and angle between them can be given as.

$$\cos\theta = \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}.$$

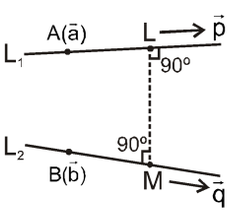
(i) The lines will be perpendicular if $a_1a_2 + b_1b_2 + c_1c_2 = 0$

(ii) The lines will be parallel if $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(iii) Two parallel lines have same direction cosines i.e. $\ell_1 = \ell_2$, $m_1 = m_2$, $n_1 = n_2$

Skew Lines and Shortest distance between two lines :

If two lines in space intersect at a point, then obviously the shortest distance between them is zero. Lines which do not intersect and are also not parallel are called **skew line**. For Skew lines the direction of the shortest distance would be perpendicular to both the lines.



Let $\overrightarrow{LM}$ be the shortest distance vector between the lines L_1 and L_2. Then $\overrightarrow{LM}$ is perpendicular to both $\vec{p}$ and $\vec{q}$ i.e. $\overrightarrow{LM}$ is parallel to $\vec{p} \times \vec{q}$. Therefore the magnitude of the shortest distance vector (i.e. $|\overrightarrow{LM}|$) would be equal to that of the projection of $\overrightarrow{AB}$ along the direction of the line of shortest distance.

$$\therefore \quad |\overrightarrow{LM}| = \left| \text{Projection of } \overrightarrow{AB} \text{ on } \overrightarrow{LM} \right| = \left| \text{Projection of } \overrightarrow{AB} \text{ on } \vec{p} \times \vec{q} \right|$$

$$= \left| \frac{\overrightarrow{AB} \cdot (\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|} \right| = \left| \frac{(\vec{b} - \vec{a}) \cdot (\vec{p} \times \vec{q})}{|\vec{p} \times \vec{q}|} \right|$$

(i) The two lines directed along $\vec{p}$ and $\vec{q}$ will intersect only if shortest distance = 0

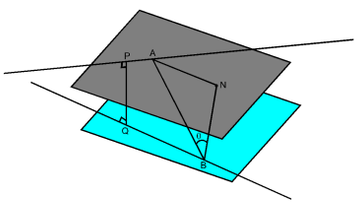
i.e. $(\vec{b} - \vec{a}) \cdot (\vec{p} \times \vec{q}) = 0$ i.e. $(\vec{b} - \vec{a})$ lies in the plane containing $\vec{p}$ and $\vec{q}$. $\Rightarrow (\vec{b} - \vec{a}) \cdot \vec{p} \times \vec{q} = 0$.

(ii) If two parallel lines are given by $\vec{r}_1 = \vec{a}_1 + K\vec{b}$ and $\vec{r}_2 = \vec{a}_2 + K\vec{b}$, then distance (d) between them is given by $d = \left| \frac{\vec{b} \times (\vec{a}_2 - \vec{a}_1)}{|\vec{b}|} \right|$

(iii) The straight lines which are not parallel and non–coplanar i.e. non–intersecting are called skew lines. If $\Delta = \begin{vmatrix} \alpha'-\alpha & \beta'-\beta & \gamma'-\gamma \\ \ell & m & n \\ \ell' & m' & n' \end{vmatrix} \neq 0$,

then lines $\frac{x-\alpha}{\ell} = \frac{y-\beta}{m} = \frac{z-\gamma}{n}$ & $\frac{x-\alpha'}{\ell'} = \frac{y-\beta'}{m'} = \frac{z-\gamma'}{n'}$ are skew lines.

(iv) Shortest distance: Suppose the equation of the lines are



$$\frac{x-\alpha}{\ell} = \frac{y-\beta}{m} = \frac{z-\gamma}{n} \text{ and } \frac{x-\alpha'}{\ell'} = \frac{y-\beta'}{m'} = \frac{z-\gamma'}{n'}$$

$$\text{S.D.} = \left| \frac{(\alpha-\alpha')(mn'-m'n)+(\beta-\beta')(n\ell'-n'\ell)+(\gamma-\gamma')(\ell m'-\ell' m)}{\sqrt{\sum (mn'-m'n)^2}} \right|$$

$$= \begin{vmatrix} \alpha'-\alpha & \beta'-\beta & \gamma'-\gamma \\ \ell & m & n \\ \ell' & m' & n' \end{vmatrix} \div \sqrt{\sum (mn' - m'n)^2}$$

(v) Vector Form: For lines $\vec{r} = \vec{a}_1 + \lambda_1 \vec{b}_1$ & $\vec{r} = \vec{a}_2 + \lambda_2 \vec{b}_2$ to be skew

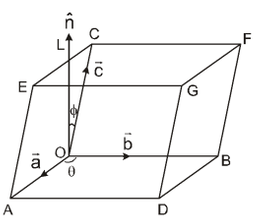
$(\vec{b}_1 \times \vec{b}_2).(\vec{a}_2 - \vec{a}_1) \neq 0$ or $[\vec{b}_1\ \vec{b}_2\ (\vec{a}_2 - \vec{a}_1)] \neq 0$.

(vi) Shortest distance between the two parallel lines $\vec{r} = \vec{a}_1 + \lambda \vec{b}$ & $\vec{r} = \vec{a}_2 + \mu \vec{b}$ is $d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \times \vec{b}}{|\vec{b}|} \right|$.

Scalar triple product (Box Product) (S.T.P.) :

(i) The scalar triple product of three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ is defined as: $\vec{a} \times \vec{b} . \vec{c} = |\vec{a}|\,|\vec{b}|\,|\vec{c}| \sin\theta . \cos\phi$, where θ is the angle between $\vec{a}, \vec{b}$ **(i.e.** $\vec{a} \wedge \vec{b} = \theta$**)** and ϕ is the angle between $\vec{a} \times \vec{b}$ and $\vec{c}$ **(i.e.** $(\vec{a} \times \vec{b}) \wedge \vec{c} = \phi$**)**. It is (i.e. $\vec{a} \times \vec{b} . \vec{c}$) also written as $[\vec{a}\ \vec{b}\ \vec{c}]$ and spelled as box product.

(ii) Scalar triple product geometrically represents the volume of the parallelopiped whose three coterminous edges are represented by $\vec{a}, \vec{b}$ and $\vec{c}$ i.e. $V = |[\vec{a}\ \vec{b}\ \vec{c}]|$



(iii) In a scalar triple product the position of dot and cross can be interchanged i.e.

$\vec{a}.(\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}).\vec{c} \Rightarrow [\vec{a}\vec{b}\vec{c}] = [\vec{b}\vec{c}\vec{a}] = [\vec{c}\vec{a}\vec{b}]$

(iv) $\vec{a}.(\vec{b} \times \vec{c}) = -\vec{a}.(\vec{c} \times \vec{b})$ i.e. $[\vec{a}\vec{b}\vec{c}] = -[\vec{a}\vec{c}\vec{b}]$

(v) If $\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$; $\vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$, then $[\vec{a}\ \vec{b}\ \vec{c}] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.

In general, if $\vec{a} = a_1\vec{\ell} + a_2\vec{m} + a_3\vec{n}$; $\vec{b} = b_1\vec{\ell} + b_2\vec{m} + b_3\vec{n}$ and $\vec{c} = c_1\vec{\ell} + c_2\vec{m} + c_3\vec{n}$

then $\left[\vec{a}\ \vec{b}\ \vec{c}\right] = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \left[\vec{\ell}\ \vec{m}\ \vec{n}\right]$, where $\vec{\ell}$, $\vec{m}$ and $\vec{n}$ are non-coplanar vectors.

(vi) If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are coplanar $\Leftrightarrow [\vec{a}\ \vec{b}\ \vec{c}] = 0$.

(vii) Scalar product of three vectors, two of which are equal or parallel is 0 $\Rightarrow [\vec{a}\ \vec{b}\ \vec{c}] = 0$,

(viii) If $\vec{a}$, $\vec{b}$, $\vec{c}$ are non-coplanar, then $[\vec{a}\ \vec{b}\ \vec{c}] > 0$ for right handed system and $[\vec{a}\ \vec{b}\ \vec{c}] < 0$ for left handed system.

(ix) $[\hat{i}\ \hat{j}\ \hat{k}] = 1$

(x) $[K\vec{a}\ \vec{b}\ \vec{c}] = K[\vec{a}\ \vec{b}\ \vec{c}]$

(xi) $[(\vec{a}+\vec{b})\ \vec{c}\ \vec{d}] = [\vec{a}\ \vec{c}\ \vec{d}] + [\vec{b}\ \vec{c}\ \vec{d}]$

(xii) $\left[\vec{a}-\vec{b}\quad \vec{b}-\vec{c}\quad \vec{c}-\vec{a}\right] = 0$ and $\left[\vec{a}+\vec{b}\quad \vec{b}+\vec{c}\quad \vec{c}+\vec{a}\right] = 2\left[\vec{a}\ \vec{b}\ \vec{c}\right]$.

(xiii) $\left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{vmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{vmatrix}$

Tetrahedron and its properties :

(a) The volume of the tetrahedron OABC with O as origin and the position vectors of A, B and C being $\vec{a}$, $\vec{b}$ and $\vec{c}$ respectively is given by $V = \frac{1}{6}\left|\left[\vec{a}\ \vec{b}\ \vec{c}\right]\right|$

(b) If the position vectors of the vertices of tetrahedron are $\vec{a}$, $\vec{b}$, $\vec{c}$ and $\vec{d}$, then the position vector of its centroid is given by $\frac{1}{4}(\vec{a} + \vec{b} + \vec{c} + \vec{d})$.

note that this is also the point of concurrency of the lines joining the vertices to the centroids of the opposite faces and is also called the centre of the tetrahedron. In case the tetrahedron is regular it is equidistant from the vertices and the four faces of the tetrahedron.

Vector triple product :

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be any three vectors, then the expression $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector & is called a vector triple product.

Geometrical interpretation of $\vec{a} \times (\vec{b} \times \vec{c})$

Consider the expression $\vec{a} \times (\vec{b} \times \vec{c})$ which itself is a vector, since it is a cross product of two vectors $\vec{a}$ and $(\vec{b} \times \vec{c})$. Now $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector perpendicular to the plane containing $\vec{a}$ and $(\vec{b} \times \vec{c})$ but $\vec{b} \times \vec{c}$

is a vector perpendicular to the plane containing $\vec{b}$ and $\vec{c}$, therefore $\vec{a} \times (\vec{b} \times \vec{c})$ is a vector which lies in the plane of $\vec{b}$ and $\vec{c}$ and perpendicular to $\vec{a}$. Hence we can express $\vec{a} \times (\vec{b} \times \vec{c})$ in terms of $\vec{b}$ and $\vec{c}$ i.e. $\vec{a} \times (\vec{b} \times \vec{c}) = x\vec{b} + y\vec{c}$, where x, y are scalars.

- $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$
- $(\vec{a} \times \vec{b}) \times \vec{c} = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{b} \cdot \vec{c})\vec{a}$
- In general $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$

Linear combinations :

Given a finite set of vectors $\vec{a}, \vec{b}, \vec{c}, \ldots\ldots$, then the vector $\vec{r} = x\vec{a} + y\vec{b} + z\vec{c} + \ldots\ldots$ is called a linear combination of $\vec{a}, \vec{b}, \vec{c}, \ldots\ldots$ for any x, y, z..... $\in$ R. We have the following results:

(a) If $\vec{a}, \vec{b}$ are non zero, non–collinear vectors, then $x\vec{a} + y\vec{b} = x'\vec{a} + y'\vec{b} \Rightarrow x = x',\ y = y'$

(b) Fundamental Theorem in plane : Let $\vec{a}, \vec{b}$ be non zero, non collinear vectors, then any vector $\vec{r}$ coplanar with $\vec{a}, \vec{b}$ can be expressed uniquely as a linear combination of $\vec{a}$ and $\vec{b}$

i.e. there exist some unique x, y $\in$ R such that $x\vec{a} + y\vec{b} = \vec{r}$.

(c) If $\vec{a}, \vec{b}, \vec{c}$ are non–zero, non–coplanar vectors, then

$x\vec{a} + y\vec{b} + z\vec{c} = x'\vec{a} + y'\vec{b} + z'\vec{c} \Rightarrow x = x',\ y = y',\ z = z'$

(d) **Fundamental theorem in space:** Let $\vec{a}, \vec{b}, \vec{c}$ be non–zero, non–coplanar vectors in space.

Then any vector $\vec{r}$ can be uniquely expressed as a linear combination of $\vec{a}, \vec{b}, \vec{c}$ i.e. there exist some unique x,y, z $\in$ R such that $x\vec{a} + y\vec{b} + z\vec{c} = \vec{r}$.

(e) If $\vec{x}_1, \vec{x}_2, \ldots\ldots, \vec{x}_n$ are n non zero vectors and $k_1, k_2, \ldots\ldots, k_n$ are n scalars and if the linear combination $k_1\vec{x}_1 + k_2\vec{x}_2 + \ldots\ldots + k_n\vec{x}_n = \vec{0} \Rightarrow k_1 = 0, k_2 = 0, \ldots\ldots, k_n = 0$, then we say that vectors $\vec{x}_1, \vec{x}_2, \ldots\ldots, \vec{x}_n$ are **linearly independent vectors**.

(f) If $\vec{x}_1, \vec{x}_2, \ldots\ldots, \vec{x}_n$ are not **linearly independent** then they are said to be **linearly dependent vectors**.

i.e. if $k_1\vec{x}_1 + k_2\vec{x}_2 + k_3\vec{x}_3 \ldots\ldots + k_r\vec{x}_r + \ldots\ldots + k_n\vec{x}_n = \vec{0}$ and if there exists at least one $k_r \neq 0$, then $\vec{x}_1, \vec{x}_2, \ldots\ldots, \vec{x}_n$ are said to be **linearly dependent vectors.**

Note 1: If $k_r \neq 0$; $k_1\vec{x}_1 + k_2\vec{x}_2 + k_3\vec{x}_3 \ldots\ldots + k_r\vec{x}_r + \ldots\ldots + k_n\vec{x}_n = \vec{0}$

$$\Rightarrow \quad -k_r\vec{x}_r = k_1\vec{x}_1 + k_2\vec{x}_2 + \ldots\ldots + k_{r-1}.\vec{x}_{r-1} + k_{r+1}.\vec{x}_{r+1} + \ldots\ldots + k_n\vec{x}_n$$

$$\Rightarrow \quad -k_r\frac{1}{k_r}\vec{x}_r = k_1\frac{1}{k_r}\vec{x}_1 + k_2\frac{1}{k_r}\vec{x}_2 + \ldots\ldots + k_{r-1}.\frac{1}{k_r}\vec{x}_{r-1} + \ldots\ldots + k_n\frac{1}{k_r}\vec{x}_n$$

$$\Rightarrow \quad \vec{x}_r = c_1\vec{x}_1 + c_2\vec{x}_2 + \ldots\ldots + c_{r-1}\vec{x}_{r-1} + c_{r+1}\vec{x}_{r+1} + \ldots\ldots + c_n\vec{x}_n$$

i.e. $\vec{x}_r$ is expressed as a linear combination of vectors $\vec{x}_1, \vec{x}_2, \ldots\ldots, \vec{x}_{r-1}, \vec{x}_{r+1}, \ldots\ldots, \vec{x}_n$

Hence $\vec{x}_r$ with $\vec{x}_1, \vec{x}_2, \ldots\ldots, \vec{x}_{r-1}, \vec{x}_{r+1}, \ldots\ldots, \vec{x}_n$ forms a linearly dependent set of vectors.

Note 2:

☞ If $\vec{a} = 3\hat{i} + 2\hat{j} + 5\hat{k}$ then $\vec{a}$ is expressed as a **Linear Combination** of vectors $\hat{i}, \hat{j}, \hat{k}$. Also $\vec{a}, \hat{i}, \hat{j}, \hat{k}$ form a linearly dependent set of vectors. In general, in 3 dimensional space every set of four vectors is a linearly dependent system.

☞ $\hat{i}, \hat{j}, \hat{k}$ are **Linearly Independent** set of vectors. For $K_1\hat{i} + K_2\hat{j} + K_3\hat{k} = \vec{0} \Rightarrow K_1 = K_2 = K_3 = 0$

☞ Two vectors $\vec{a}$ and $\vec{b}$ are linearly dependent $\Rightarrow \vec{a}$ is parallel to $\vec{b}$ i.e. $\vec{a} \times \vec{b} = \vec{0} \Rightarrow$ linear dependence of $\vec{a}$ and $\vec{b}$. Conversely if $\vec{a} \times \vec{b} \neq \vec{0}$ then $\vec{a}$ and $\vec{b}$ are linearly independent.

☞ If three vectors $\vec{a}, \vec{b}, \vec{c}$ are linearly dependent, then they are coplanar i.e. $[\vec{a}\ \vec{b}\ \vec{c}] = 0$. Conversely if $[\vec{a}\ \vec{b}\ \vec{c}] \neq 0$ then the vectors are linearly independent.

Test of collinearity :

Three points A,B,C with position vectors $\vec{a}, \vec{b}, \vec{c}$ respectively are collinear, if & only if there exist scalars x, y, z not all zero simultaneously such that $x\vec{a} + y\vec{b} + z\vec{c} = \vec{0}$, where x + y + z = 0.

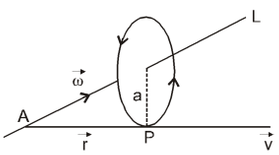
Test of coplanarity :

Four points A, B, C, D with position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ respectively are coplanar if and only if there exist scalars x, y, z, w not all zero simultaneously such that $x\vec{a} + y\vec{b} + z\vec{c} + w\vec{d} = \vec{0}$, where x + y + z + w = 0.

Application of vectors :

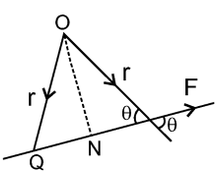
(a) Work done against a constant force $\vec{F}$ over a displacement $\vec{s}$ is defined as $W = \vec{F}.\vec{s}$

(b) The tangential velocity $\vec{V}$ of a body moving in a circle is given by $\vec{V} = \vec{\omega} \times \vec{r}$, where $\vec{r}$ is the position vector of the point P.



(c) The moment of $\vec{F}$ about 'O' is defined as $\vec{M} = \vec{r} \times \vec{F}$, where $\vec{r}$ is the position vector of P w.r.t. 'O'. The direction of $\vec{M}$ is along the normal to the plane OPN such that $\vec{r}, \vec{F}$ & $\vec{M}$ form a right handed system.

(d) Moment of the couple = $(\vec{r}_1 - \vec{r}_2) \times \vec{F}$, where $\vec{r}_1$ and $\vec{r}_2$ are position vectors of the point of the application of the forces $\vec{F}$ and $-\vec{F}$.



STATISTICS

MEASURES OF CENTRAL TENDENCY :

An average value or central value of a distribution is the value of variable which is representative of the entire distribution, this repsresentive value are called the measures of central tendency are of following type.

(A) Mathematical average
- (i) Arithmetic mean or mean
- (ii) Geometrical mean
- (iii) Harmonic mean

(B) Positional average
- (i) Median
- (ii) Mode

(I) MEAN :

Mathematical average is mean.

(A) Arithmetic Mean / or Mean

If $x_1, x_2, x_3, ...x_n$ are n values of variate x_i then their A.M. $\bar{x}$ is defined as

$$\bar{x} = \frac{x_1 + x_2 + x_3 + ... + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

If $x_1, x_2, x_3, ... x_n$ are values of veriate with frequencies $f_1, f_2, f_3, ...f_n$ then their A.M. is given by

$$\bar{x} = \frac{f_1x_1 \times f_2x_2 + f_3x_3 + ...f_nf_n}{f_1 + f_2 + f_3 + ... + f_n} = \frac{\sum_{i=1}^{n} f_ix_i}{N}, \text{ where } N = \sum_{i=1}^{n} f_i$$

Properties of Arithmetic Mean :

(i) Sum of deviation of variate from their A.M. is always zero that is $\Sigma(x_i - \bar{x}) = 0$.

(ii) Sum of square of deviation of variate from their A.M. is minimum that is $\Sigma(x_i - \bar{x})^2$ is minimum

(iii) If $\bar{x}$ is mean of variate x_i then

A.M. of $(x_i + \lambda) = \bar{x} + \lambda$

A.M. of $\lambda_i . x_i = \lambda . \bar{x}$

A.M. of $(ax_i + b) = a\bar{x} + b$

Merits of Arithmetic Mean :

(i) It is rigidly defined.

(ii) It is based on all the observation taken.

(iii) It is calculated with reasonable ease.

(iv) It is least affected by fluctuations in sampling.

(v) It is based on each observation and so it is a better representative of the data.

(vi) It is relatively reliable

(vii) Mathematical analysis of mean is possible.

Demerits of Arithmetic Mean :

(i) It is severely affected by the extreme values.

(ii) It cannot be represented in the actual data since the mean does not coincide with any of the observed value.

(iii) It cannot be computed unless all the items are known.

(B) Geometric Mean :

If $x_1, x_2, x_3, ...x_3$ are n positive values of variate then their geometric mean G is given by

$$G = (x_1\, x_2\, x_3 \ldots\ldots\, x_n)^{1/n}$$

$$\Rightarrow \quad G = \text{antilog}\left[\frac{1}{n}\sum_{i=1}^{n}\log x_i\right]$$

(II) MEDIAN :

The median of a series is values of middle term of series when the values are written is ascending order or descending order. Therefore median, divide on arranged series in two equal parts

(A) For ungrouped distribution :

If n be number of variates in a series then

$$\text{Median} = \begin{cases} \left(\frac{n+1}{2}\right)^{th} \text{term, (when n is odd)} \\ \text{Mean of } \left(\frac{n}{2}\right)^{th} \text{ and } \left(\frac{n}{2}+2\right)^{th} \text{term (when n is even)} \end{cases}$$

(B) For ungrouped frequency distribution :

First we calculate cumulative frequency (sum of all frequencies). Let it be N then

$$\text{Median} \begin{cases} \left(\frac{N+1}{2}\right)^{th} \text{term (when n odd)} \\ \text{Mean of } \left(\frac{N}{2}\right) \& \left(\frac{N+2}{2}\right) \text{(when n is even)} \end{cases}$$

Merits And DemeriTs of Median :

The following are some merits and demerits of median :

Merits :

(i) It is easy to compute and understand.

(ii) It is well defined an ideal average should be

(iii) It can also be computed in case of frequency distribution with open ended classes.

(iv) It is not affected by extreme values.

(v) It can be determined graphically.

(vi) It is proper average for qualitative data where items are not measured but are scored.

Demerits :

(i) For computing median data needs to be arranged in ascending or descending order.

(ii) It is not based on all the observations of the data.

(iii) It cannot be given further algebraic treatment.

(iv) It is affected by fluctuations of sampling.

(v) It is not accurate when the data is not large.

(vi) In some cases median is determined approximately as the mid-point of two observations whereas for mean this does not happen.

(c) Harmonic Mean :

If $x_1, x_2, x_3, ...x_3$ are n non-zero values of variate then their harmonic mean H is defined as

$$H = \frac{n}{\frac{1}{x_i}+\frac{1}{x_2}+....+\frac{1}{x_n}} = \frac{n}{\sum_{i=1}^{n}\frac{1}{x_i}}$$

(III) MODE :

If a frequency distribution the mode is the value of that variate which have the maximum frequency. Mode for

(A) For ungrouped distribution :

The value of variate which has maximum frequency.

(b) For ungrouped frequency distribution :

The value of that variate which have maximum frequency.

Relationship between mean, median and mode.

(i) In symmetric distribution, mean = mode = median

(ii) In skew (moderately asymmetrical) distribution,

median divides mean and mode internally in 1 : 2 ratio.

$$\Rightarrow \quad \text{median} = \frac{2(\text{Mean})+(\text{Mode})}{3}$$

Merits, Demerits of Mode :

The following are some merits and demerits of mode :

Merits :

(i) It is readily comprehensible and easy to compute. In some case it can be computed merely by inspection.

(ii) It is not affected by extreme values. It can be obtained even if the extreme values are not known.

(iii) Mode can be determined in distributions with open classes.

(iv) Mode can be located on graph also.

Demerits :

(i) It is ill-defined. It is not always possible to find a clearly defined mode. In some cases, we may come across distributions with two modes. Such distributions are called bimodal. If a distribution has more than two modes, it is said to be multimodal.

(ii) It is not based upon all the observation.

(iii) Mode can be calculated by various formulae as such the value may differ from one to other. Therefore, it is not rigidly defined.

(iv) It is affected to a greater extent by fluctuations of sampling.

Measure of Dispersion :

It is measure of deviation of its value about their central values. It gives an idea of scatterdness of different values from the central values.

Types :

(I) RANGE :

Difference between greatest values & least values of variates of a distribution are called the range of distribution.

$$\text{Also coefficient of range} = \frac{\text{difference of extreme values}}{\text{sum of extreme values}} = \frac{L-S}{L+S}$$

where L = largest value and S = smallest value

(II) MEAN DEVIATION :

Mean deviation of a distribution is, the mean of absolute value of deviation of variate from their statistical average (median, mean or mode).

If A is any statistical average then mean deviation about A is defined as

$$\text{Mean deviation} = \frac{\sum_{i=1}^{n} |x_i - A|}{n}$$

$$\text{Mean deviation} = \frac{\sum_{i=1}^{n} f_i |x_i - A|}{N} \quad \text{(for frequency distribution)}$$

(III) VARIANCE :

It is the mean of squares of deviation of variate from their mean. It is denoted by σ^2 or var(x). The positive square root of the variance are called the standard deviation. It is denoted by σ or S.D.

so standard deviation = $+\sqrt{\text{variance}}$

formula

$$\sigma_x^2 = \frac{\Sigma(x_i - \bar{x})^2}{n}$$

$$\sigma_x^2 = \frac{\sum_{i=1}^{n} x_i^2}{n} - \left(\frac{\sum_{i=1}^{n} x_i}{n}\right)^2 = \frac{\sum_{i=1}^{n} x_i^2}{n} - (\bar{x})^2$$

$$\sigma_d^2 = \frac{\Sigma d_i^2}{n} - \left(\frac{\Sigma di}{n}\right)^2, \text{ where } d_i = x_i - a \text{ , where a = assumed mean}$$

(ii) coefficient of S.D. = $\left(\frac{\sigma}{\bar{x}}\right)$

coefficient of variation = $\left(\frac{\sigma}{\bar{x}}\right)$ × 100 (in percentage)

Properties of variance :

(i) $var(x_i + \lambda) = var(x_i)$

(ii) $var(\lambda.x_i) = \lambda^2(var\ x_i)$

(iii) $var(a\ x_i + b) = a^2(var\ x_i)$

where λ, a, b are constant.

MATHEMATICAL REASONING

STATEMENTS (Proposition) :

In reasoning we communicate our ideas or thoughts with the help of sentences in a particular language. "A sentence is called a mathematically acceptable statement or proposition if it is either true or false but not both". A statement is assumed to be either true or false. A true statement is known as a **valid statement** and a false statement is known as an **invalid statement**.

Note : (i) Imperative (expresses a request or command), exclaimatory sentences (expresses some strong feeling), Interrogative sentences (asks some question), Optative sentences (blessing & wishes) are not considered as a statement in mathematical language.

(ii) Sentences involving variable time such as "today", "tomorrow" or "yesterday" are not statements.

(iii) Scientifically established facts are considered true.

(iv) Sentences involving word "here", "there" are not statements.

Self Practice Problems :

(1) Which of the following are statements :
(i) open the door (ii) square of an odd number is even

Ans. (1) (i) is not a statement, (ii) is a statement

TRUTH TABLE :

Truth table is that which gives truth values (The truth or falsity of a statement is called its truth value) of statements. It has a number of rows and columns.

Note that for **n** statements, there are $\mathbf{2^n}$ rows.

(i) Truth table for single statement p:

Number of rows = $2^1 = 2$

p
T
F

(ii) Truth table for two statements p and q :

Number of rows = $2^2 = 4$

p	q
T	T
T	F
F	T
F	F

(iii) Truth table for three statements p, q and r.
Number of rows = $2^3 = 8$

p	q	r
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

NEGATION OF A STATEMENT :

The denial of a statement p is called its negation and is written as ~ p and read as 'not p'. Negation of any statement p is formed by writing "It is not the case that ....."
or "It is false that ....."
or inserting the word "not" in p.

p	~p
T	F
F	T

Truth table

Self Practice Problems :

(2) Write negation of statement '2 + 2 = 7'

Ans. (2) $2 + 2 \neq 7$

COMPOUND STATEMENTS :

If a statement is combination of two or more statements, then it is said to be a compound statement. Each statement which form a compound statement is known as its sub-statement or component statement.

BASIC CONNECTIVES :

In the compound statement, two or more statements are connected by words like 'and', 'or', 'if . . . . . then', 'only if', 'if and only if', 'there exists', 'for all' etc. These are called connectives. When we use these compound statements, it is necessary to understand the role of these words.

Basic logical connective	*Symbol*	*Compound statement*
AND	$\wedge$	Conjunction
OR	$\vee$	Disjunction
IF..... THEN	$\rightarrow$	Conditional statement
IF AND ONLY IF	$\leftrightarrow$	Biconditional statement

THE WORD "AND" (CONJUCTION) :

Any two statements can be connected by the word "and" to form a compound statement. The compound statement with word "and" is true if all its component statements are true. The compound statement with word "and" is false if any or all of its component statements are false. The compound statement "p and q" is denoted by "$p \wedge q$".

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Truth table

THE WORD "OR" (DISJUNCTION) :

Any two statements can be connected by the word "OR"

to form a compound statement. The compound statement with word "or" is true if any or all of its component statements are true. The compound statement with word "or" is false if all its component statement are false. The compound statement "p or q" is denoted by "p v q."

TYPES OF "OR" :

(i) Exclusive OR : If in statement $p \vee q$ i.e. p or q, happening of any one of p, q excludes the happening of the other then it is exclusive OR. Here both p and q cannot occur together. For example in statement "I will go to delhi either by bus or by train", the use of 'or' is exclusive.

p	q	p v q
T	T	F
T	F	T
F	T	T
F	F	F

Truth table

(ii) Inclusive OR : If in statement p or q, both p and q can also occur together then it is inclusive OR. The statement 'In senior secondary exam, you can take optional subject as physical education or computers' is an example of use of inclusive OR.

p	q	p v q
T	T	T
T	F	T
F	T	T
F	F	F

Truth table

Note : In general truth table of inclusive OR is used for compound statement $p \vee q$.

Self practice problems :

(3) Find the truth values of $\sim p \vee q$

(4) Find the truth values of the compound statement $(p \vee \sim r) \wedge (q \vee \sim r)$

Ans. (3)

p	q	~ p	~ p ∨ q
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

(4)

p	q	r	~ r	p∨ ~ r	q∨ ~ r	(p∨ ~ r) ∧ (q∨ ~ r)
T	T	T	F	T	T	T
T	T	F	T	T	T	T
T	F	T	F	T	F	F
F	T	T	F	F	T	F
T	F	F	T	T	T	T
F	T	F	T	T	T	T
F	F	T	F	F	F	F
F	F	F	T	T	T	T

QUANTIFIERS :

There exists (atleast one), For all (for every)

Quantifiers are phrases like "There exists" ($\exists$) and " For all" ($\forall$). In general, a mathematical statement that says "for every" can be interpreted as saying that all the members of the given set S satisfies that property. While "There exists" can be interpreted as saying that "atleast one" member of the given set S satisfies that property.

Self practice problems :

(5) Negate the following sentences using quantifiers.
(i) There exists $x \in Z$, such that $x^2 + x = 0$
(ii) For all $x \in R$, $\sqrt{x}$ is a real number.

Ans. (5) (i) For all $x \in Z$, $x^2 + x \neq 0$.
(ii) There exists $x \in R$ such that $\sqrt{x}$ is not a real number.

IMPLICATION :

There are three types of implications which are "if . . . . . then", "only if" and "if and only if".

CONDITIONAL CONNECTIVE 'IF . . . . THEN' :

If p and q are any two statements then the compound statement in the form "If p then q" is called a conditional statement. The statement "If p then q" is denoted by $p \rightarrow q$ or $p \Rightarrow q$ (to be read as p implies q). In the implication $p \rightarrow q$, p is called the antecedent (or the hypothesis) and q the consequent (or the conclusion)
If p then q reveals the following facts :
(i) p is a sufficient condition for q
(ii) q is a necessary condition for p
(iii) 'If p then q' has same meaning as that of 'p only if q'
(iv) $p \rightarrow q$ has same meaning as that of $\sim q \rightarrow \sim p$

p	q	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T
T	F	F	T
F	T	T	F
F	F	T	T

Truth table

Note : The conditional statement $p \rightarrow q$ is defined to be true except in case p is true and q is false.

Examples :
(i) If $x = 4$, then $x^2 = 16$
(ii) If ABCD is a parallelogram, then AB = CD
(iii) If Mumbai is in England, then 2 + 2 = 5
(iv) If Shikha works hard, then it will rain today.

We also need to be aware that in the English language, there are other ways for expressing the conditional statement $P \rightarrow Q$ other than "If P, then Q." Following are some common ways to express the conditional statement $P \rightarrow Q$ in the English language :

- If P, then Q.
- P implies Q.
- P only if Q.
- Q if P.
- Whenever P is true, Q is true.
- Q is true whenever P is true.
- Q is necessary for P. (This means that if P is true, then Q is necessarily true.)
- P is sufficient for Q. (This means that if you want Q to be true, it is sufficient to show that P is true.)

CONTRAPOSITIVE AND CONVERSE OF A CONDITIONAL STATEMENT :

If p and q are two statements then

Let $p \Rightarrow q$ Then
(i) (Contrapositive of $p \Rightarrow q$) is ($\sim q \Rightarrow \sim p$)
(ii) (Converse of $p \Rightarrow q$) is ($q \Rightarrow p$)

Note : A statement and its contrapositive convey the same meaning.

BICONDITIONAL CONNECTIVE "IF AND ONLY IF" :

If p and q are any two statements then the compound statement in the form of "p if and only if q" is called a biconditional statement and is written in symbolic form as $p \leftrightarrow q$ or $p \Leftrightarrow q$.

Statement $p \leftrightarrow q$ reveals the following facts :

(i) p if and only if q

(ii) q if and only if p

(iii) p is necessary and sufficient condition for q

(iv) q is necessary and sufficient condition for p

p	q	$p \leftrightarrow q$	$q \leftrightarrow p$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	T	T

Truth table

Note : The biconditional statement $p \leftrightarrow q$ is defined to be true only when p and q have same truth value.

For Example The following statements are biconditional statements

(i) A number is divisible by 3 if and only if the sum of the digits forming the number is divisible by 3.

(ii) One is less than seven if and only if two is less than eight.

(iii) A triangle is equilateral if and only if it is equiangular.

Self Practice Problems :

(6) If statements p and q are respectively : 4 + 5 = 9 and 2 + 3 = 5, then write the conditional statement $p \rightarrow q$.

(7) Find the truth values of the compound statement $(p \wedge q) \rightarrow \sim p$

Ans. (6) If 4 + 5 = 9, then 2 + 3 = 5

(7)

p	q	$(p \wedge q) \rightarrow \sim p$
T	T	F
T	F	T
F	T	T
F	F	T

TAUTOLOGY AND FALLACY / CONTRADICTION :

(a) **Tautology :** This is a statement which is true for all truth values of its components. It is denoted by t. Consider truth table of $p \vee \sim p$

p	$\sim p$	$p \vee \sim p$
T	F	T
F	T	T

we observe that last column is always true. Hence $p \vee \sim p$ is a tautology.

(b) **Fallacy :** This is statement which is false for all truth values of its components. It is denoted by f or c. Consider truth table of $p \wedge \sim p$

p	$\sim p$	$p \wedge \sim p$
T	F	F
F	T	F

We observe that last column is always false. Hence $p \wedge \sim p$ is a fallacy (contradiction).

LOGICALLY EQUIVALENT STATEMENTS :

If truth values of statements p and q are same then they are logically equivalent and written as $p \equiv q$.

ALGEBRA OF STATEMENTS :

If p, q, r are any three statements and t is a tautology, c is a contradiction then

(1) Commutative Law :

(i) $p \vee q = q \vee p$ (ii) $p \wedge q = q \wedge p$

p	q	$(p \wedge q)$	$(q \wedge p)$	$(p \vee q)$	$(q \vee p)$
T	T	T	T	T	T
T	F	F	F	T	T
F	T	F	F	T	T
F	F	F	F	F	F

(2) Associative Law :

(i) $p \vee (q \vee r) = (p \vee q) \vee r$ (ii) $p \wedge (q \wedge r) = (p \wedge q) \wedge r$

p	q	r	$(p \wedge q)$	$(q \wedge r)$	$(p \wedge q) \wedge r$	$p \wedge (q \wedge r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	F	T	F	F
F	T	F	F	F	F	F
F	F	T	F	F	F	F
F	F	F	F	F	F	F

(3) Distributive Law :

(i) $p \wedge (q \vee r) = (p \wedge q) \vee (p \wedge r)$ (ii) $p \vee (q \wedge r) = (p \vee q) \wedge (p \vee r)$

(iii) $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge (p \wedge r)$ (iv) $p \vee (q \vee r) \equiv (p \vee q) \vee (p \vee r)$

p	q	r	$(q \vee r)$	$(p \wedge q)$	$(p \wedge r)$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	F	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

(4) Identity Law :

(i) $p \vee t = t$ (ii) $p \wedge t = p$ (iii) $p \vee c = p$

(iv) $p \wedge c = c$

p	t	c	$(p \wedge t)$	$(p \vee t)$	$(p \wedge c)$	$(p \vee c)$
T	T	F	T	T	F	T
F	T	F	F	T	F	F

(5) Complement Law :

(i) $p \vee (\sim p) = t$ (ii) $p \wedge (\sim p) = c$ (iii) $\sim t = c$

(iv) $\sim c = t$ (v) $\sim(\sim p) = p$

p	$\sim p$	$(p \wedge \sim p)$	$(p \vee \sim p)$
T	F	F	T
F	T	F	T

(6) Idempotent Law :

(i) $p \vee p = p$ (ii) $p \wedge p = p$

p	$(p \wedge p)$	$(p \vee p)$
T	T	T
F	F	F

(7) De Morgan's law :

(i) $\sim (p \vee q) = (\sim p) \wedge (\sim q)$

(ii) $\sim (p \wedge q) = (\sim p) \vee (\sim q)$

p	q	$\sim p$	$\sim q$	$(p \wedge q)$	$\sim (p \wedge q)$	$(\sim p \vee \sim q)$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

(8) Involution laws (or Double negation laws) :

$\sim(\sim p) \equiv p$

p	$\sim q$	$\sim (\sim p)$
T	F	T
F	T	F

(9) Contrapositive Laws : $p \rightarrow q \equiv \sim q \rightarrow \sim p$

p	q	$\sim p$	$\sim q$	$p \rightarrow q$	$\sim q \rightarrow \sim p$
T	T	F	F	T	T
T	F	F	T	F	F
F	T	T	F	T	T
F	F	T	T	T	T

(10) Truth table of biconditional statement :

p	q	$p \leftrightarrow q$	$q \leftrightarrow p$	$p \rightarrow q$	$q \rightarrow p$	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
T	T	T	T	T	T	T
T	F	F	F	F	T	F
F	T	F	F	T	F	F
F	F	T	T	T	T	T

NEGATION OF COMPOUND STATEMENTS :

If p and q are two statements then

(i) Negation of conjunction : $\sim(p \wedge q) \equiv \sim p \vee \sim q$

p	q	$\sim p$	$\sim q$	$(p \wedge q)$	$\sim (p \wedge q)$	$(\sim p \vee \sim q)$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

(ii) Negation of disjunction : $\sim(p \vee q) \equiv \sim p \wedge \sim q$

p	q	~ p	~ q	(p ∨ q)	~ (p ∨ q)	(~ p∧ ~ q)
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

(iii) Negation of conditional : $\sim(p \to q) \equiv p \wedge \sim q$

p	q	~ q	(p → q)	~ (p → q)	(p∧ ~ q)
T	T	F	T	F	F
T	F	T	F	T	T
F	T	F	T	F	F
F	F	T	T	F	F

(iv) Negation of biconditional : $\sim(p \leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$ or $p \leftrightarrow \sim q$

p	q	~ p	~ q	(p → q)	~ (p → q)	(p∧ ~ q)	(p ↔ q)	~ (p ↔ q)	p ↔~ q	q∧ ~ p	(p∧ ~ q) ∨ (q∧ ~ p)
T	T	F	F	T	F	F	T	F	F	F	F
T	F	F	T	F	T	T	F	T	T	F	T
F	T	T	F	T	F	F	F	T	T	T	T
F	F	T	T	T	F	F	T	F	F	F	F

we know that $p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$

$\therefore$ $\sim(p \leftrightarrow q) \equiv \sim[(p \to q) \wedge (q \to p)]$

$\equiv \sim(p \to q) \vee \sim(q \to p) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$

SUMMARY

(i) $\sim (p \wedge q) \equiv (\sim p) \vee (\sim q)$

(ii) $\sim (p \vee q) \equiv (\sim p) \wedge (\sim q)$

(iii) $\sim (p \Rightarrow q) \equiv \sim (\sim p \vee q) = p \wedge (\sim q)$

(iv) $\sim (p \Leftrightarrow q) \equiv (p \wedge \sim q) \vee (q \wedge \sim p)$ or $p \Leftrightarrow \sim q$ or $\sim p \Leftrightarrow q$

Note : $p \to q \equiv \sim p \vee q$

$p \to q \equiv (\sim p \vee q) \wedge (p \vee \sim q)$

Self Practice Problems :

(8) Show that $[(\sim p \wedge q) \wedge (q \wedge r)] \wedge \sim q$ is a fallacy.

(9) By using laws of algebra of statements show that $p \wedge (p \vee q) \equiv p \vee (p \wedge q)$.

(10) By using laws of algebra of statements show that $p \vee (p \wedge q) \equiv p$.

(11) Find the negation of statement of $(p \to q) \to (q \to p)$.

Ans. (11) $(\sim p \vee q) \wedge (q \wedge \sim p)$

* * * * *

COMPLEX NUMBER

The Complex Number System

z = a + ib. 'a' is called as real part of z which is denoted by (Re z) and 'b' is called as imaginary part of z, which is denoted by (Im z).

Any complex number is :

(i) Purely real, if b = 0 ;

(ii) Imaginary, if $b \neq 0$.

(iii) Purely imaginary, if a = 0

Note : (a) The set R of real numbers is a proper subset of the Complex Numbers. Hence the complete number system is $N \subset W \subset I \subset Q \subset R \subset C$.

(b) Zero is purely real as well as purely imaginary but not imaginary.

(c) $i = \sqrt{-1}$ is called the imaginary unit.

Also $i^2 = -1$; $i^3 = -i$; $i^4 = 1$ etc.

(d) $\sqrt{a}\ \sqrt{b} = \sqrt{ab}$ only if atleast one of a or b is non - negative.

(e) If z = a + ib, then a – ib is called complex conjugate of z and written as $\bar{z}$ = a – ib

(f) Real numbers satisfy order relations where as imaginary numbers do not satisfy order relations i.e. i > 0, 3 + i < 2 are meaningless.

Algebraic Operations:

Fundamental operations with complex numbers

In performing operations with complex numbers we can proceed as in the algebra of real numbers, replacing i^2 by – 1 when it occurs.

1. Addition (a + bi) + (c + di) = a + bi + c + di = (a + c) + (b + d) i
2. Subtraction (a + bi) – (c + di) = a + bi – c – di = (a – c) + (b – d) i
3. Multiplication (a + bi) (c + di) = ac + adi + bci + bdi^2 = (ac – bd) + (ad+ bc)i
4. Division $$\frac{a+bi}{c+di} = \frac{a+bi}{c+di}\cdot\frac{c-di}{c-di} = \frac{ac-adi+bci-bdi^2}{c^2-d^2i^2}$$

$$= \frac{ac+bd+(bc-ad)i}{c^2+d^2} = \frac{ac+bd}{c^2+d^2} + \frac{bc-ad}{c^2+d^2}i$$

Inequalities in imaginary numbers are not defined. There is no validity if we say that imaginary number is positive or negative.

e.g. z > 0, 4 + 2i < 2 + 4 i are meaningless.

In real numbers if $a^2 + b^2 = 0$ then a = 0 = b however in complex numbers,

$z_1^2 + z_2^2 = 0$ does not imply $z_1 = z_2 = 0$.

Equality In Complex Number :

Two complex numbers $z_1 = a_1 + ib_1$ & $z_2 = a_2 + ib_2$ are equal if and only if their real and imaginary parts

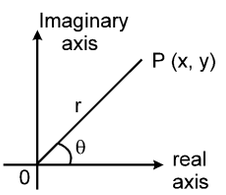
are equal respectively

i.e. $z_1 = z_2 \Leftrightarrow Re(z_1) = Re(z_2)$ and $I_m(z_1) = I_m(z_2)$.

Representation of a Complex Number :

(a) Cartesian Form (Geometric Representation) :

Every complex number z = x + i y can be represented by a point on the Cartesian plane known as complex plane (Argand diagram) by the ordered pair (x, y).



Length OP is called modulus of the complex number which is denoted by $|z|$ & θ is called the argument or amplitude.

$|z| = \sqrt{x^2 + y^2}$ and $\tan\theta = \left(\frac{y}{x}\right)$ (angle made by OP with positive x–axis)

Note : (i) Argument of a complex number is a many valued function. If θ is the argument of a complex number then $2n\pi + \theta$; $n \in I$ will also be the argument of that complex number. Any two arguments of a complex number differ by $2n\pi$.

(ii) The unique value of θ such that $-\pi < \theta \leq \pi$ is called the principal value of the argument. Unless otherwise stated, amp z implies principal value of the argument.

(iii) By specifying the modulus & argument a complex number is defined completely. For the complex number 0 + 0 i the argument is not defined and this is the only complex number which is only given by its modulus.

(b) Trignometric/Polar Representation :

$z = r(\cos\theta + i\sin\theta)$ where $|z| = r$; $\arg z = \theta$; $\bar{z} = r(\cos\theta - i\sin\theta)$

Note : $\cos\theta + i\sin\theta$ is also written as CiS θ

(c) Euler's Formula :

$z = re^{i\theta}$, $|z| = r$, $\arg z = \theta$

$\bar{z} = re^{-i\theta}$

Note: If θ is real then $\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$; $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$

(d) **Vectorial Representation :**

Every complex number can be considered as the position vector of a point. If the point P represents the complex number z then, $\vec{OP} = z$ & $|\vec{OP}| = |z|$.

Agrument of a Complex Number :

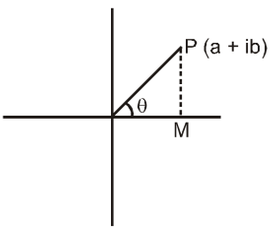
Argument of a non-zero complex number P(z) is denoted and defined by arg(z) = angle which OP makes with the positive direction of real axis.

If OP = |z| = r and arg(z) = θ, then obviously $z = r(\cos\theta + i\sin\theta)$, called the polar form of z. 'Argument of z' would mean principal argument of z(i.e. argument lying in $(-\pi, \pi]$ unless the context requires otherwise.

Thus argument of a complex number $z = a + ib = r(\cos\theta + i\sin\theta)$ is the value of θ satisfying $r\cos\theta = a$ and $r\sin\theta = b$.

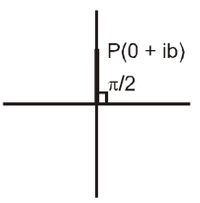
Let $\theta = \tan^{-1}\left|\frac{b}{a}\right|$

(i) $a > 0, b > 0$



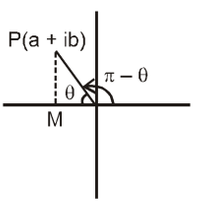
p.v. arg $z = \theta$

(ii) $a = 0, b > 0$



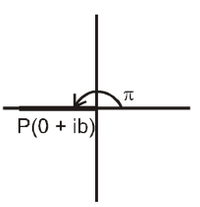
p.v. arg $z = \frac{\pi}{2}$

(iii) $a < 0, b > 0$



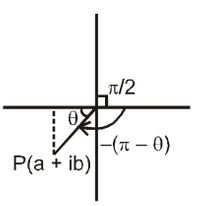
p.v. arg $z = \pi - \theta$

(iv) $a < 0, b = 0$



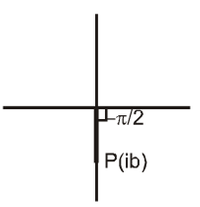
p.v. arg $z = \pi$

(v) $a < 0, b < 0$

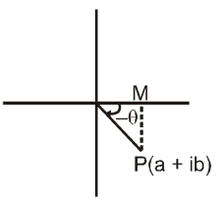


p.v. arg $z = -(\pi - \theta)$

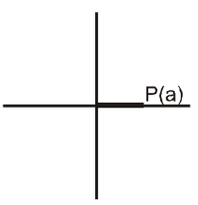
(vi) $a = 0, b < 0$



p.v. arg $z = -\frac{\pi}{2}$

(vii) $a > 0, b < 0$ 

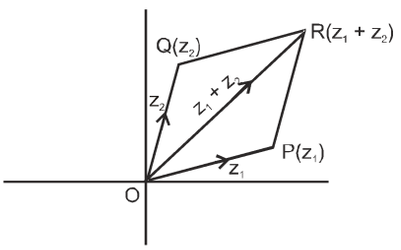
p.v. arg $z = -\theta$

(viii) $a > 0, b = 0$ 

p.v. arg $z = 0$

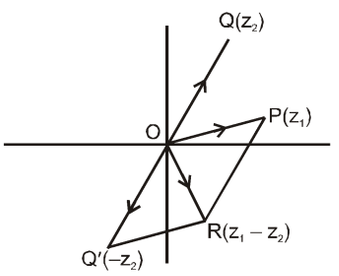
Geometrical Representation of Fundamental Operations :

(i) Geometrical representation of addition.



If two points P and Q represent complex numbers z_1 and z_2 respectively in the Argand plane, then the sum $z_1 + z_2$ is represented by the extremity R of the diagonal OR of parallelogram OPRQ having OP and OQ as two adjacent sides.

(ii) Geometric representation of substraction.



(iii) Modulus and argument of multiplication of two complex numbers.

Theorem : For any two complex numbers z_1, z_2 we have $|z_1 z_2| = |z_1| |z_2|$ and $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$.

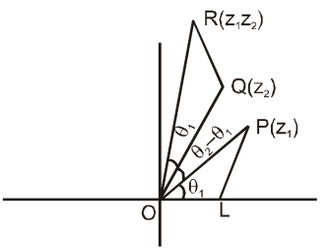
Note : (i) P.V. $\arg(z_1 z_2) \neq$ P.V. $\arg(z_1)$ + P.V. $\arg(z_2)$

(ii) $|z_1 z_2 \ldots z_n| = |z_1| |z_2| \ldots |z_n|$

(iii) $\arg(z_1 z_2 \ldots z_n) = \arg z_1 + \arg z_2 + \ldots + \arg z_n$

(iv) Geometrical representation of multiplication of complex numbers.

Let P, Q be represented by $z_1 = r_1\ e^{i\theta_1}$, $z_2 = r_2\ e^{i\theta_2}$ repectively. To find point R representing complex number z_1z_2, we take a point L on real axis such that OL = 1 and draw triangle OQR similar to triangle OLP. Therefore



R is represented by $z_1z_2 = r_1r_2\ e^{i(\theta_1+\theta_2)}$

(v) Modulus and argument of division of two complex numbers.

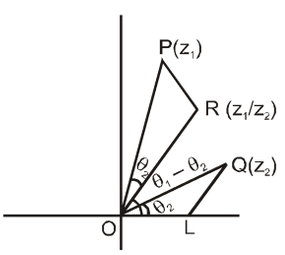
Theorem : If z_1 and z_2 $(\neq 0)$ are two complex numbers, then $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$ and

$$\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$$

Note : P.V. $\arg\left(\frac{z_1}{z_2}\right) \neq$ P.V. $\arg(z_1)$ – P.V. $\arg(z_2)$

(vi) Geometrical representation of the division of complex numbers.

Let P, Q be represented by $z_1 = r_1e^{i\theta_1}$, $z_2 = r_2e^{i\theta_2}$ respectively. To find point R representing complex number $\frac{z_1}{z_2}$, we take a point L on real axis such that OL = 1 and draw a triangle OPR similar to OQL. Therefore $\frac{OP}{OQ} = \frac{OR}{OL} \Rightarrow OR = \frac{r_1}{r_2}$



and $L\hat{O}R = L\hat{O}P - R\hat{O}P = \theta_1 - \theta_2$

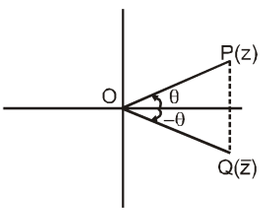
Hence, R is represented by $\frac{z_1}{z_2} = \frac{r_1}{r_2}\ e^{i(\theta_1-\theta_2)}$.

Conjugate of a complex Number :

Conjugate of a complex number z = a + ib is denoted and defined by $\bar{z}$ = a – ib.

In a complex number if we replace i by – i, we get conjugate of the complex number. $\bar{z}$ is the mirror image of z about real axis on Argand's Plane.

Geometrical representation of conjugate of complex number.



$|z| = |\bar{z}|$

$\arg(\bar{z}) = -\arg(z)$

General value of $\arg(\bar{z}) = 2n\pi - \text{P.V. arg}(z)$

Properties

(i) If $z = x + iy$, then $x = \frac{z+\bar{z}}{2}$, $y = \frac{z-\bar{z}}{2i}$

(ii) $z = \bar{z} \Leftrightarrow$ z is purely real

(iii) $z + \bar{z} = 0 \Leftrightarrow$ z is purely imaginary

(iv) Relation between modulus and conjugate. $|z|^2 = z\bar{z}$

(v) $\bar{\bar{z}} = z$

(vi) $\overline{(z_1 \pm z_2)} = \bar{z}_1 \pm \bar{z}_2$

(vii) $\overline{(z_1 z_2)} = \bar{z}_1 \bar{z}_2$

(viii) $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{(\bar{z}_1)}{(\bar{z}_2)}$ $(z_2 \neq 0)$

Theorem : Imaginary roots of polynomial equations with real coefficients occur in conjugate pairs

Note : If $w = f(z)$, then $\bar{w} = f(\bar{z})$

Theorem : $|z_1 \pm z_2|^2 = |z_1|^2 + |z_2|^2 \pm (z_1\bar{z}_2 + \bar{z}_1 z_2)$

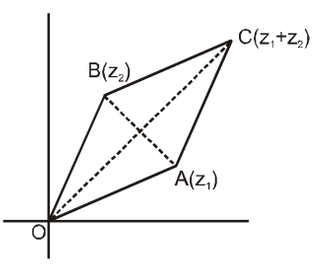
$= |z_1|^2 + |z_2|^2 \pm 2\,\text{Re}(z_1\bar{z}_2)$

$= |z_1|^2 + |z_2|^2 \pm 2|z_1||z_2|\cos(\theta_1 - \theta_2)$

Distance, Triangular Inequality

If $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$, then distance between points z_1, z_2 in argand plane is

$|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$



In triangle OAC

$OC \leq OA + AC$

$OA \leq AC + OC$

$AC \leq OA + OC$

using these in equalities we have $||z_1| - |z_2|| \leq |z_1 + z_2| \leq |z_1| + |z_2|$

Similarly from triangle OAB

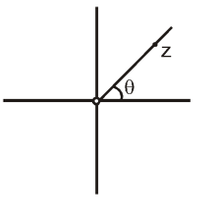
we have $||z_1| - |z_2|| \leq |z_1 - z_2| \leq |z_1| + |z_2|$

Note : (a) $||z_1| - |z_2|| = |z_1 + z_2|$, $|z_1 - z_2| = |z_1| + |z_2|$ iff origin, z_1 and z_2 are collinear and origin lies between z_1 and z_2.

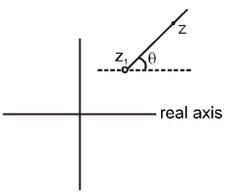
(b) $|z_1 + z_2| = |z_1| + |z_2|$, $||z_1| - |z_2|| = |z_1 - z_2|$ iff origin, z_1 and z_2 are collinear and z_1 and z_2 lies on the same side of origin.

Important results :

(i) arg z = θ represnets points (non-zero) on ray eminating from origin making an angle θ with positive direction of real axis



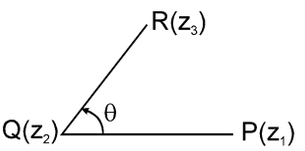
(ii) arg $(z - z_1) = \theta$ represents points $(\neq z_1)$ on ray eminating from z_1 making an angle θ with positive direction of real axis



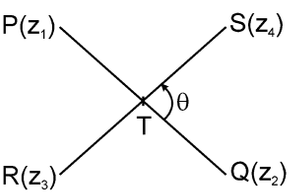
Rotation theorem :

(i) If $P(z_1)$ and $Q(z_z)$ are two complex numbers such that $|z_1| = |z_2|$, then $z_2 = z_1 e^{i\theta}$ where $\theta = \angle POQ$

(ii) If $P(z_1)$, $Q(z_2)$ and $R(z_3)$ are three complex numbers and $\angle PQR = \theta$, then $\left(\frac{z_3 - z_2}{z_1 - z_2}\right) = \left|\frac{z_3 - z_2}{z_1 - z_2}\right| e^{i\theta}$



(iii) If $P(z_1)$, $Q(z_2)$, $R(z_3)$ and $S(z_4)$ are four complex numbers and $\angle STQ = \theta$, then $\frac{z_3 - z_4}{z_1 - z_2} = \left|\frac{z_3 - z_4}{z_1 - z_2}\right| e^{i\theta}$



Demoivre's Theorem:

Case I

Statement :

If n is any integer then

(i) $(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$

(ii) $(\cos\theta_1 + i\sin\theta_1)(\cos\theta_2 + i\sin\theta_2)(\cos\theta_3 + i\sin\theta_2)(\cos\theta_3 + i\sin\theta_3) \ldots\ldots(\cos\theta_n + i\sin\theta_n)$
$= \cos(\theta_1 + \theta_2 + \theta_3 + \ldots\ldots\ldots \theta_n) + i\sin(\theta_1 + \theta_2 + \theta_3 + \ldots\ldots + \theta_n)$

Case II

Statement : If $p, q \in Z$ and $q \neq 0$ then

$$(\cos\theta + i\sin\theta)^{p/q} = \cos\left(\frac{2k\pi + p\theta}{q}\right) + i\sin\left(\frac{2k\pi + p\theta}{q}\right)$$

where k = 0, 1, 2, 3, ..., q – 1

Note : Continued product of the roots of a complex quantity should be determined using theory of equations.

Cube Root of Unity :

(i) The cube roots of unity are 1, $\frac{-1 + i\sqrt{3}}{2}$, $\frac{-1 - i\sqrt{3}}{2}$.

(ii) If ω is one of the imaginary cube roots of unity then $1 + \omega + \omega^2 = 0$. In general $1 + \omega^r + \omega^{2r} = 0$; where $r \in I$ but is not the multiple of 3.

(iii) In polar form the cube roots of unity are :$\cos 0 + i \sin 0$; $\cos\frac{2\pi}{3} + i \sin\frac{2\pi}{3}$, $\cos\frac{4\pi}{3} + i \sin\frac{4\pi}{3}$

(iv) The three cube roots of unity when plotted on the argand plane constitute the verties of an equilateral triangle.

(v) The following factorisation should be remembered :

(a, b, c $\in$ R & ω is the cube root of unity)

$a^3 - b^3 = (a - b)(a - \omega b)(a - \omega^2 b)$; $x^2 + x + 1 = (x - \omega)(x - \omega^2)$;

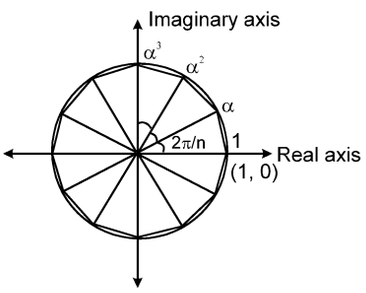
$a^3 + b^3 = (a + b)(a + \omega b)(a + \omega^2 b)$; $a^2 + ab + b^2 = (a - b\omega)(a - b\omega^2)$

$a^3 + b^3 + c^3 - 3abc = (a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c)$

n^{th} Roots of Unity :

If 1, α_1, α_2, α_3..... α_{n-1} are the n, n^{th} root of unity then :

(i) They are in G.P. with common ratio $e^{i(2\pi/n)}$



(ii) $1^p + \alpha_1^p + \alpha_2^p + ... + \alpha_{n-1}^p = 0$ if p is not an integral multiple of n

= n if p is an integral multiple of n

(iii) $(1 - \alpha_1)(1 - \alpha_2)..... (1 - \alpha_{n-1}) = n$ &

$(1 + \alpha_1)(1 + \alpha_2)..... (1 + \alpha_{n-1}) = 0$ if n is even and 1 if n is odd.

(iv) $1.\ \alpha_1.\ \alpha_2.\ \alpha_3..... \alpha_{n-1} = 1$ or -1 according as n is odd or even.

The Sum Of The Following Series Should Be Remembered :

(i) $\cos\theta + \cos 2\theta + \cos 3\theta + ... + \cos n\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\left(\frac{n+1}{2}\right)\theta.$

(ii) $\sin\theta + \sin 2\theta + \sin 3\theta + ... + \sin n\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \sin\left(\frac{n+1}{2}\right)\theta.$

Note : If $\theta = (2\pi/n)$ then the sum of the above series vanishes.

Logarithm Of A Complex Quantity :

$\text{Log}_e(\alpha + i\beta) = \frac{1}{2}\text{Log}_e(\alpha^2 + \beta^2) + i\left(2n\pi + \tan^{-1}\frac{\beta}{\alpha}\right)$ where $n \in I$.

Geometrical Properties :

Section formula

If z_1 and z_2 are affixes of the two points P and Q respectively and point C divides the line segment joining P and Q internally in the ratio m : n then affix z of C is given by

$$z = \frac{mz_2 + nz_1}{m+n} \qquad \text{where } m, n > 0$$

If C divides PQ in the ratio m : n externally then $z = \frac{mz_2 - nz_1}{m-n}$

Note : If a, b, c are three real numbers such that $az_1 + bz_2 + cz_3 = 0$; where $a + b + c = 0$ and a,b,c are not all simultaneously zero, then the complex numbers z_1, z_2 & z_3 are collinear.

(1) If the vertices A, B, C of a Δ are represented by complex numbers z_1, z_2, z_3 respectively and a, b, c are the length of sides then,

(i) Centroid of the Δ ABC = $\frac{z_1 + z_2 + z_3}{3}$:

(ii) Orthocentre of the Δ ABC =

$$\frac{(a\sec A)z_1 + (b\sec B)z_2 + (c\sec C)z_3}{a\sec A + b\sec B + c\sec C} \text{ or } \frac{z_1\tan A + z_2\tan B + z_3\tan C}{\tan A + \tan B + \tan C}$$

(iii) Incentre of the Δ ABC = $(az_1 + bz_2 + cz_3) \div (a + b + c)$.

(iv) Circumcentre of the Δ ABC = :
$(Z_1 \sin 2A + Z_2 \sin 2B + Z_3 \sin 2C) \div (\sin 2A + \sin 2B + \sin 2C)$.

(2) amp(z) = θ is a ray emanating from the origin inclined at an angle θ to the positive x– axis.

(3) $|z - a| = |z - b|$ is the perpendicular bisector of the line joining a to b.

(4) The equation of a line joining z_1 & z_2 is given by, $z = z_1 + t(z_1 - z_2)$ where t is a real parameter.

(5) $z = z_1(1 + it)$ where t is a real parameter is a line through the point z_1 & perpendicular to the line joining z_1 to the origin.

(6) The equation of a line passing through z_1 & z_2 can be expressed in the determinant form as

$$\begin{vmatrix} z & \bar{z} & 1 \\ z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \end{vmatrix} = 0.$$ This is also the condition for three complex numbers z, z_1, z_2 to be collinear.

The above equation on manipulating, takes the form $\bar{\alpha}z + \alpha\bar{z} + r = 0$ where r is real and α is a non zero complex constant.

Note : If we replace z by $ze^{i\theta}$ and $\bar{z}$ by $\bar{z}e^{-i\theta}$ then we get equation of a straight line which makes an angle θ with the given straight line.

(7) The equation of circle having centre z_0 & radius ρ is :

$|z - z_0| = \rho$ or $z\bar{z} - z_0\bar{z} - \bar{z}_0 z + \bar{z}_0 z_0 - \rho^2 = 0$ which is of the form

$z\bar{z} + \bar{\alpha}z + \alpha\bar{z} + k = 0$, k is real. Centre is $-\alpha$ & radius = $\sqrt{|\alpha|^2 - k}$.

Circle will be real if $|\alpha|^2 - k \geq 0$..

(8) The equation of the circle described on the line segment joining z_1 & z_2 as diameter is

$\arg\frac{z-z_2}{z-z_1} = \pm\frac{\pi}{2}$ or $(z-z_1)(\bar{z}-\bar{z}_2)+(z-z_2)(\bar{z}-\bar{z}_1)=0$.

(9) Condition for four given points z_1, z_2, z_3 & z_4 to be concyclic is the number

$\frac{z_3-z_1}{z_3-z_2}.\frac{z_4-z_2}{z_4-z_1}$ should be real.

(10) $\text{Arg}\left(\frac{z-z_1}{z-z_2}\right) = \theta$ represent (i) a line segment if $\theta = \pi$

(ii) Pair of ray if $\theta = 0$ (iii) a part of circle, if $0 < \theta < \pi$.

(11) Area of triangle formed by the points z_1, z_2 & z_3 is $\left|\frac{1}{4i}\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix}\right|$

(12) Perpendicular distance of a point z_0 from the line $\bar{\alpha}z+\alpha\bar{z}+r=0$ is $\frac{|\bar{\alpha}z_0+\alpha\bar{z}_0+r|}{2|\alpha|}$

(13) **Dot and cross product**

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers [vectors]. The dot product [also called the scalar product] of z_1 and z_2 is defined by

$z_1 \cdot z_2 = |z_1|\,|z_2| \cos\theta = x_1x_2 + y_1y_2 = \text{Re}\{\bar{z}_1z_2\} = \frac{1}{2}\{\bar{z}_1z_2+z_1\bar{z}_2\}$

where θ is the angle between z_1 and z_2 which lies between 0 and π

If vectors z_1, z_2 are perpendicular then $z_1 \cdot z_2 = 0 \Rightarrow \frac{z_1}{\bar{z}_1} + \frac{z_2}{\bar{z}_2} = 0$.

i.e. Sum of complex slopes = 0

The cross product of z_1 and z_2 is defined by

$z_1 \times z_2 = |z_1|\,|z_2| \sin\theta = x_1y_2 - y_1x_2 = \text{Im}\{\bar{z}_1z_2\} = \frac{1}{2i}\{\bar{z}_1z_2 - z_1\bar{z}_2\}$

If vectors z_1, z_2 are parallel then $z_1 \times z_2 = 0 \Rightarrow \frac{z_1}{\bar{z}_1} = \frac{z_2}{\bar{z}_2}$.

(14) If $|z-z_1|+|z-z_2| = K > |z_1-z_2|$ then locus of z is an ellipse whose focii are z_1 & z_2

(15) If $|z-z_0| = \left|\frac{\bar{\alpha}z+\alpha\bar{z}+r}{2|\alpha|}\right|$ then locus of z is parabola whose focus is z_0 and directrix is the line $\bar{\alpha}z + \alpha\bar{z} + r = 0$ (Provided $\bar{\alpha}z_0+\alpha\bar{z}_0+r \neq 0$)

(16) If $\left|\frac{z-z_1}{z-z_2}\right| = k \neq 1, 0$, then locus of z is circle.

(17) If $\big||z-z_1|-|z-z_2|\big| = K < |z_1-z_2|$ then locus of z is a hyperbola, whose focii are z_1 & z_2.

* * * * *

BINOMIAL THEOREM

Binomial expression :

Any algebraic expression which contains two dissimilar terms is called binomial expression.

For example : $x + y$, $x^2y + \frac{1}{xy^2}$, $3 - x$, $\sqrt{x^2+1} + \frac{1}{(x^3+1)^{1/3}}$ etc.

Terminology used in binomial theorem :

Factorial notation : $\lfloor n$ or $n!$ is pronounced as factorial n and is defined as

$$n! = \begin{cases} n(n-1)(n-2).....3.2.1 & ; \text{ if } n \in N \\ 1 & ; \text{ if } n = 0 \end{cases}$$

Note : $n! = n\,.\,(n-1)!$; $n \in N$

Mathematical meaning of ${}^nC_r$: The term ${}^nC_r$ denotes number of combinations of r things choosen from n distinct things mathematically, ${}^nC_r = \frac{n!}{(n-r)!\,r!}$, $n \in N$, $r \in W$, $0 \le r \le n$

Note : Other symbols of of ${}^nC_r$ are $\binom{n}{r}$ and C(n, r).

Properties related to ${}^nC_r$:

(i) ${}^nC_r = {}^nC_{n-r}$

Note : If ${}^nC_x = {}^nC_y \Rightarrow$ Either $x = y$ or $x + y = n$

(ii) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

(iii) $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$

(iv) ${}^nC_r = \frac{n}{r}\,{}^{n-1}C_{r-1} = \frac{n(n-1)}{r(r-1)}\,{}^{n-2}C_{r-2} = \ldots\ldots\ldots\ldots = \frac{n(n-1)(n-2).....(n-(r-1))}{r\,(r-1)(r-2).....2\,.1}$

(v) If n and r are relatively prime, then ${}^nC_r$ is divisible by n. But converse is not necessarily true.

Statement of binomial theorem :

$(a + b)^n = {}^nC_0\,a^nb^0 + {}^nC_1\,a^{n-1}b^1 + {}^nC_2\,a^{n-2}b^2 + \ldots + {}^nC_r\,a^{n-r}b^r + \ldots\ldots + {}^nC_n\,a^0b^n$

where $n \in N$

or $(a + b)^n = \sum_{r=0}^{n} {}^nC_r\,a^{n-r}b^r$

Note : If we put a = 1 and b = x in the above binomial expansion, then

or $(1 + x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \ldots + {}^nC_r x^r + \ldots + {}^nC_n x^n$

or $(1 + x)^n = \sum_{r=0}^{n} {}^nC_r \, x^r$

Observations :

(i) The number of terms in the binomial expansion $(a + b)^n$ is $n + 1$.

(ii) The sum of the indices of a and b in each term is n.

(iii) The binomial coefficients $({}^nC_0, {}^nC_1 \ldots\ldots {}^nC_n)$ of the terms equidistant from the beginning and the end are equal, i.e. ${}^nC_0 = {}^nC_n$, ${}^nC_1 = {}^nC_{n-1}$ etc. $\{\because {}^nC_r = {}^nC_{n-r}\}$

(iv) The binomial coefficient can be remembered with the help of the following pascal's Triangle (also known as Meru Prastra provided by Pingla)

Index of the binomial	The binomial coefficient
0	1
1	1 1
2	1 2 1
3	1 3 3 1
4	1 4 6 4 1
5	1 5 10 10 5 1

Regarding Pascal's Triangle, we note the following :

(a) Each row of the triangle begins with 1 and ends with 1.

(b) Any entry in a row is the sum of two entries in the preceding row, one on the immediate left and the other on the immediate right.

General term :

$(x + y)^n = {}^nC_0 x^n y^0 + {}^nC_1 x^{n-1} y^1 + \ldots\ldots\ldots + {}^nC_r x^{n-r} y^r + \ldots\ldots\ldots + {}^nC_n x^0 y^n$

$(r + 1)^{th}$ term is called general term and denoted by T_{r+1}.

$T_{r+1} = {}^nC_r x^{n-r} y^r$

Note : The r^{th} term from the end is equal to the $(n - r + 2)^{th}$ term from the begining, i.e. ${}^nC_{n-r+1} \, x^{r-1} \, y^{n-r+1}$

Middle term(s) :

(a) If n is even, there is only one middle term, which is $\left(\frac{n+2}{2}\right)^{th}$ term.

(b) If n is odd, there are two middle terms, which are $\left(\frac{n+1}{2}\right)^{th}$ and $\left(\frac{n+1}{2}+1\right)^{th}$ terms.

Numerically greatest term in the expansion of $(a + b)^n$, $n \in N$

Binomial expansion of $(a + b)^n$ is as follows : –

$(a + b)^n = {}^nC_0 a^n b^0 + {}^nC_1 a^{n-1} b^1 + {}^nC_2 a^{n-2} b^2 + \ldots + {}^nC_r a^{n-r} b^r + \ldots\ldots + {}^nC_n a^0 b^n$

If we put certain values of a and b in RHS, then each term of binomial expansion will have certain value. The term having numerically greatest value is said to be numerically greatest term.

Let T_r and T_{r+1} be the r^{th} and $(r+1)^{th}$ terms respectively

$T_r = {}^nC_{r-1}\ a^{n-(r-1)}\ b^{r-1}$

$T_{r+1} = {}^nC_r\ a^{n-r}\ b^r$

Now, $\left|\frac{T_{r+1}}{T_r}\right| = \left|\frac{{}^nC_r}{{}^nC_{r-1}}\frac{a^{n-r}\ b^r}{a^{n-r+1}b^{r-1}}\right| = \frac{n-r+1}{r}\ .\ \left|\frac{b}{a}\right|$

Consider $\left|\frac{T_{r+1}}{T_r}\right| \geq 1$

$\left(\frac{n-r+1}{r}\right)\left|\frac{b}{a}\right| \geq 1 \qquad \frac{n+1}{r} - 1 \geq \left|\frac{a}{b}\right|$

$r \leq \frac{n+1}{1+\left|\frac{a}{b}\right|}$

Case - I When $\frac{n+1}{1+\left|\frac{a}{b}\right|}$ is an integer (say m), then

(i) $T_{r+1} > T_r$ when $r < m$ $(r = 1, 2, 3 \ldots., m-1)$
i.e. $T_2 > T_1, T_3 > T_2, \ldots\ldots, T_m > T_{m-1}$

(ii) $T_{r+1} = T_r$ when $r = m$
i.e. $T_{m+1} = T_m$

(iii) $T_{r+1} < T_r$ when $r > m$ $(r = m+1, m+2, \ldots\ldots\ldots n)$
i.e. $T_{m+2} < T_{m+1}\ ,\ T_{m+3} < T_{m+2}\ ,\ \ldots\ldots\ldots T_{n+1} < T_n$

Conclusion :

When $\frac{n+1}{1+\left|\frac{a}{b}\right|}$ is an integer, say m, then T_m and T_{m+1} will be numerically greatest terms (both terms are equal in magnitude)

Case - II

When $\frac{n+1}{1+\left|\frac{a}{b}\right|}$ is not an integer (Let its integral part be m), then

(i) $T_{r+1} > T_r$ when $r < \frac{n+1}{1+\left|\frac{a}{b}\right|}$ $(r = 1, 2, 3, \ldots\ldots\ldots, m-1, m)$

i.e. $T_2 > T_1\ ,\ T_3 > T_2, \ldots\ldots\ldots\ldots\ldots, T_{m+1} > T_m$

(ii) $T_{r+1} < T_r$ when $r > \frac{n+1}{1+\left|\frac{a}{b}\right|}$ $(r = m+1, m+2, \ldots\ldots\ldots\ldots n)$

i.e. $T_{m+2} < T_{m+1}\ ,\ T_{m+3} < T_{m+2}\ ,\ \ldots\ldots\ldots\ldots\ldots, T_{n+1} < T_n$

Conclusion :

When $\frac{n+1}{1+\left|\frac{a}{b}\right|}$ is not an integer and its integral part is m, then T_{m+1} will be the numerically greatest term.

Note : (i) In any binomial expansion, the middle term(s) has greatest binomial coefficient.
In the expansion of $(a + b)^n$

If	n	No. of greatest binomial coefficient	Greatest binomial coefficient
	Even	1	${}^nC_{n/2}$
	Odd	2	${}^nC_{(n-1)/2}$ and ${}^nC_{(n+1)/2}$

(Values of both these coefficients are equal)

(ii) In order to obtain the term having numerically greatest coefficient, put a = b = 1, and proceed as discussed above.

Explanation:

For $r \le 4$, $T_{r+1} \ge T_r$

$\Rightarrow$ $T_2 > T_1$

$T_3 > T_2$

$T_4 > T_3$

$T_5 = T_4$

For $r > 5$, $T_{r+1} < T_r$

$T_6 < T_5$

$T_7 < T_6$

and so on

Hence, T_4 and T_5 are numerically greatest terms and both are equal.

Result : If $(\sqrt{A} + B)^n = I + f$, where I and n are positive integers, n being odd and $0 < f < 1$, then $(I + f)$ $f = k^n$ where $A - B^2 = k > 0$ and $\sqrt{A} - B < 1$. If n is an even integer, then $(I + f)(1 - f) = k^n$

Some standard expansions :

(i) Consider the expansion

$$(x + y)^n = \sum_{r=0}^{n} {}^nC_r \; x^{n-r} y^r = {}^nC_0 \, x^n y^0 + {}^nC_1 \, x^{n-1} y^1 + \ldots\ldots\ldots + {}^nC_r \, x^{n-r} y^r + \ldots\ldots\ldots + {}^nC_n \, x^0 y^n \quad \ldots.(i)$$

(ii) Now replace $y \to -y$ we get

$$(x - y)^n = \sum_{r=0}^{n} {}^nC_r \, (-1)^r x^{n-r} y^r = {}^nC_0 \, x^n y^0 - {}^nC_1 \, x^{n-1} y^1 + \ldots + {}^nC_r \, (-1)^r x^{n-r} y^r + \ldots + {}^nC_n \, (-1)^n x^0 y^n \quad \ldots.(ii)$$

(iii) Adding (i) & (ii), we get

$(x + y)^n + (x - y)^n = 2[{}^nC_0 \, x^n y^0 + {}^nC_2 \, x^{n-2} y^2 + \ldots\ldots\ldots]$

(iv) Subtracting (ii) from (i), we get

$(x + y)^n - (x - y)^n = 2[{}^nC_1 \, x^{n-1} y^1 + {}^nC_3 \, x^{n-3} y^3 + \ldots\ldots\ldots]$

Properties of binomial coefficients :

$(1 + x)^n = C_0 + C_1 x + C_2 x^2 + \ldots\ldots\ldots + C_r x^r + \ldots\ldots\ldots + C_n x^n$ .....(1)

where C_r denotes ${}^nC_r$

(1) The sum of the binomial coefficients in the expansion of $(1 + x)^n$ is 2^n

Putting x = 1 in (1)

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \ldots\ldots + {}^nC_n = 2^n \qquad \ldots\ldots(2)$$

or $$\sum_{r=0}^{n} {}^nC_r = 2^n$$

(2) Again putting x = –1 in (1), we get

$${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \ldots\ldots\ldots\ldots + (-1)^n\,{}^nC_n = 0 \qquad \ldots\ldots(3)$$

or $$\sum_{r=0}^{n} (-1)^r\,{}^nC_r = 0$$

(3) The sum of the binomial coefficients at odd position is equal to the sum of the binomial coefficients at even position and each is equal to 2^{n-1}.
from (2) and (3)

$${}^nC_0 + {}^nC_2 + {}^nC_4 + \ldots\ldots\ldots\ldots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \ldots\ldots\ldots\ldots = 2^{n-1}$$

(4) Sum of two consecutive binomial coefficients

$${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$$

$$\text{L.H.S.} = {}^nC_r + {}^nC_{r-1} = \frac{n!}{(n-r)!\,r!} + \frac{n!}{(n-r+1)!\,(r-1)!}$$

$$= \frac{n!}{(n-r)!\,(r-1)!}\left[\frac{1}{r} + \frac{1}{n-r+1}\right]$$

$$= \frac{n!}{(n-r)!\,(r-1)!}\,\frac{(n+1)}{r(n-r+1)}$$

$$= \frac{(n+1)!}{(n-r+1)!\,r!} = {}^{n+1}C_r = \text{R.H.S.}$$

(5) Ratio of two consecutive binomial coefficients

$$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$$

(6) $${}^nC_r = \frac{n}{r}\,{}^{n-1}C_{r-1} = \frac{n(n-1)}{r(r-1)}\,{}^{n-2}C_{r-2} = \ldots\ldots\ldots\ldots = \frac{n(n-1)(n-2)\ldots\ldots(n-(r-1))}{r\,(r-1)(r-2)\ldots\ldots.2\,.1}$$

Multinomial theorem :

As we know the Binomial Theorem –

$$(x+y)^n = \sum_{r=0}^{n} {}^nC_r\; x^{n-r}\,y^r = \sum_{r=0}^{n} \frac{n!}{(n-r)!\,r!}\; x^{n-r}\,y^r$$

putting $n - r = r_1$, $r = r_2$

therefore, $$(x+y)^n = \sum_{r_1+r_2=n} \frac{n!}{r_1!\;r_2!}\; x^{r_1}\,.\,y^{r_2}$$

Total number of terms in the expansion of $(x+y)^n$ is equal to number of non-negative integral solution

of $r_1 + r_2 = n$ i.e. ${}^{n+2-1}C_{2-1} = {}^{n+1}C_1 = n + 1$

In the same fashion we can write the multinomial theorem

$$(x_1 + x_2 + x_3 + \ldots\ldots\ldots x_k)^n = \sum_{r_1+r_2+\ldots+r_k=n} \frac{n!}{r_1!\, r_2!\ldots r_k!}\; x_1^{r_1} \cdot x_2^{r_2}\ldots x_k^{r_k}$$

Here total number of terms in the expansion of $(x_1 + x_2 + \ldots\ldots\ldots + x_k)^n$ is equal to number of non-negative integral solution of $r_1 + r_2 + \ldots\ldots\ldots + r_k = n$ i.e. ${}^{n+k-1}C_{k-1}$

Binomial theorem for negative and fractional indices :

If $n \in R$, then

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}\, x^2 + \frac{n(n-1)(n-2)}{3!}\, x^3 + \ldots\ldots\ldots\ldots$$

$$\ldots\ldots\ldots\ldots + \frac{n(n-1)(n-2)\ldots\ldots(n-r+1)}{r!}\, x^r + \ldots\ldots\ldots\ldots\ldots \infty.$$

Remarks

(i) The above expansion is valid for any rational number other than a whole number if $|x| < 1$.

(ii) When the index is a negative integer or a fraction then number of terms in the expansion of $(1 + x)^n$ is infinite, and the symbol ${}^nC_r$ cannot be used to denote the coefficient of the general term.

(iii) The first term must be unity in the expansion, when index 'n' is a negative integer or fraction

$$(x + y)^n = \begin{cases} x^n\left(1+\frac{y}{x}\right)^n = x^n\left\{1 + n\cdot\frac{y}{x} + \frac{n(n-1)}{2!}\left(\frac{y}{x}\right)^2 + \ldots\ldots\right\} \text{ if } \left|\frac{y}{x}\right| < 1 \\ y^n\left(1+\frac{x}{y}\right)^n = y^n\left\{1 + n\cdot\frac{x}{y} + \frac{n(n-1)}{2!}\left(\frac{x}{y}\right)^2 + \ldots\ldots\right\} \text{if } \left|\frac{x}{y}\right| < 1 \end{cases}$$

(iv) The general term in the expansion of $(1 + x)^n$ is $T_{r+1} = \frac{n(n-1)(n-2)\ldots\ldots(n-r+1)}{r!}\, x^r$

(v) When 'n' is any rational number other than whole number then approximate value of $(1 + x)^n$ is $1 + nx$ (x^2 and higher powers of x can be neglected)

(vi) Expansions to be remembered ($|x| < 1$)

(a) $(1 + x)^{-1} = 1 - x + x^2 - x^3 + \ldots\ldots\ldots + (-1)^r x^r + \ldots\ldots\ldots\infty$

(b) $(1 - x)^{-1} = 1 + x + x^2 + x^3 + \ldots\ldots\ldots + x^r + \ldots\ldots\ldots\infty$

(c) $(1 + x)^{-2} = 1 - 2x + 3x^2 - 4x^3 + \ldots\ldots\ldots + (-1)^r (r + 1) x^r + \ldots\ldots\ldots\infty$

(d) $(1 - x)^{-2} = 1 + 2x + 3x^2 + 4x^3 + \ldots\ldots\ldots\ldots + (r + 1)x^r + \ldots\ldots\ldots \infty$

* * * * *

PERMUTATION AND COMBINATION

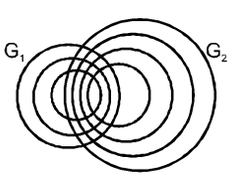
The most fundamental application of mathematics is counting. There are many natural methods used for counting

This chapter is dealing with various known techniques those are much faster than the usual counting methods.

We mainly focus, our methods, on counting the number of arrangements (Permutations) and the number of selections (combinations), even although we may use these techniques for counting in some other situations also .

Let us start with a simple problem

A group G_1 of 3 circles C_1, C_2, C_3 having different centers are situated in such a way that C_2 lie entirely inside C_1 ; C_3 lie entirely inside C_2. Another group G_2 of 4 circles C_1', C_2', C_3', C_4' are also situated in a similar fashion. The two groups of circles are in such a way that each member of G_1 intersect with every member of G_2, as shown in the following figure



(i) How many centres the circles altogether has ?

(ii) How many common chords are obtained ?

The answer to the first part is "3 + 4 = 7" and answer to the second part is "3 × 4 = 12" . The method in which we calculated first part of the problem is called as "addition rule" and the method we used to calculate its second part is called as the "multiplication rule". These rules altogether are the most important tools in counting, popularly known as "the fundamental counting principle".

Fundamental counting principle :

Suppose that an operation O_1 can be done in m different ways and another operation O_2 can be done in n different ways.

(i) **Addition rule :** The number of ways in which we can do exactly one of the operations O_1, O_2 is m + n

(ii) **Multiplication rule :** The number of ways in which we can do both the operations O_1, O_2 is mn.

Note : The addition rule is true only when O_1 & O_2 are mutually exclusive and multiplication rule is true only when O_1 & O_2 are independent (The reader will understand the concepts of mutual exclusiveness and independence, in the due course)

Arrangements :

If ${}^nP_r$ denotes the number of permutations (arrangements) of n different things, taking r at a time, then

$${}^nP_r = n\,(n-1)\,(n-2)\ldots\ldots\,(n-r+1) = \frac{n!}{(n-r)!}$$

NOTE : (i) Factorials of negative integers are not defined.

(ii) $0! = 1! = 1$

(iii) ${}^nP_n = n! = n.\,(n-1)!$

(iv) $(2n)! = 2^n.\,n!\,[1.\,3.\,5.\,7\ldots(2n-1)]$

Circular Permutation :

The number of circular permutations of n different things taken all at a time is $(n-1)!$.

If clockwise & anti–clockwise circular permutations are considered to be same, then it is $\frac{(n-1)!}{2}$.

Note: Number of circular permutations of n things when p are alike and the rest are different, taken all at a time, distinguishing clockwise and anticlockwise arrangement is $\frac{(n-1)!}{p!}$.

Combination :

If ${}^nC_r$ denotes the number of combinations (selections) of n different things taken r at a time, then

$${}^nC_r = \frac{n!}{r!\,(n-r)!} = \frac{{}^nP_r}{r!} \text{ where } r \le n \text{ ; } n \in N \text{ and } r \in W.$$

NOTE : (i) ${}^nC_r = {}^nC_{n-r}$

(ii) ${}^nC_r + {}^nC_{r-1} = {}^{n+1}C_r$

(iii) ${}^nC_r = 0$ if $r \notin \{0, 1, 2, 3\ldots\ldots, n\}$

Arrangement of n things, those are not all different :

The number of permutations of 'n' things, taken all at a time, when 'p' of them are same & of one type, q of them are same & of second type, 'r' of them are same & of a third type & the remaining $n - (p + q + r)$ things are all different, is $\frac{n!}{p!\,q!\,r!}$.

Formation of Groups :

Number of ways in which (m + n + p) different things can be divided into three different groups containing m, n & p things respectively is $\frac{(m+n+p)!}{m!\,n!\,p!}$,

If m = n = p and the groups have identical qualitative characteristic then the number of groups = $\frac{(3n)!}{n!\,n!\,n!\,3!}$.

Note : If 3n different things are to be distributed equally among three people then the number of ways = $\frac{(3n)!}{(n!)^3}$.

Selection of one or more objects

(a) Number of ways in which atleast one object may be selected out of 'n' distinct objects, is

$${}^nC_1 + {}^nC_2 + {}^nC_3 + \ldots\ldots\ldots\ldots + {}^nC_n = 2^n - 1$$

(b) Number of ways in which atleast one object may be selected out of 'p' alike objects of one type, 'q' alike objects of second type and 'r' alike objects of third type, is

$$(p + 1)(q + 1)(r + 1) - 1$$

(c) Number of ways in which atleast one object may be selected from 'n' objects where 'p' alike of one type, 'q' alike of second type and 'r' alike of third type and rest $n - (p + q + r)$ are different, is $(p + 1)(q + 1)(r + 1)\,2^{n-(p+q+r)} - 1$

Negative binomial expansion :

$(1 - x)^{-n} = 1 + {}^{n}C_1 x + {}^{n+1}C_2\, x^2 + {}^{n+2}C_3 x^3 + \ldots\ldots$ to ∞ , if $-1 < x < 1$.

Coefficient of x^r in this expansion = ${}^{n+r-1}C_r$ $(n \in N)$

Result : Number of ways in which it is possible to make a selection from m + n + p = N things, where p are alike of one kind, m alike of second kind & n alike of third kind, taken r at a time is given by coefficient of x^r in the expansion of

$(1 + x + x^2 + \ldots\ldots + x^p)\ (1 + x + x^2 + \ldots\ldots + x^m)\ (1 + x + x^2 + \ldots\ldots + x^n)$.

For example the number of ways in which a selection of four letters can be made from the letters of the word **PROPORTION** is given by coefficient of x^4 in

$(1 + x + x^2 + x^3)\ (1 + x + x^2)\ (1 + x + x^2)\ (1 + x)\ (1 + x)\ (1 + x)$.

Results : Let $N = p^a \cdot q^b \cdot r^c \ldots\ldots$ where p, q, r..... are distinct primes & a, b, c..... are natural numbers then:

(a) The total numbers of divisors of N including 1 & N is = $(a + 1)\ (b + 1)\ (c + 1)\ldots\ldots\ldots$

(b) The sum of these divisors is =

$(p^0 + p^1 + p^2 + \ldots. + p^a)\ (q^0 + q^1 + q^2 + \ldots. + q^b)\ (r^0 + r^1 + r^2 + \ldots. + r^c)\ldots\ldots\ldots$

(c) Number of ways in which N can be resolved as a product of two factors is

$$= \begin{cases} \frac{1}{2}(a+1)(b+1)(c+1)\ldots. & \text{if N is not a perfect square} \\ \frac{1}{2}\left[(a+1)(b+1)(c+1)\ldots.+1\right] & \text{if N is a perfect square} \end{cases}$$

(d) Number of ways in which a composite number N can be resolved into two factors which are relatively prime (or coprime) to each other is equal to 2^{n-1} where n is the number of different prime factors in N.

Result : Let there be 'n' types of objects, with each type containing atleast r objects. Then the number of ways of arranging r objects in a row is n^r.

Method of fictious partition :

Number of ways in which n identical things may be distributed among p persons if each person may receive none, one or more things is ${}^{n+p-1}C_n$.

Derrangements :

Number of ways in which 'n' letters can be put in 'n' corresponding envelopes such that no letter goes to correct envelope is $n!\left(1-\frac{1}{1!}+\frac{1}{2!}-\frac{1}{3!}+\frac{1}{4!}-\ldots\ldots\ldots+(-1)^n\frac{1}{n!}\right)$

* * * * *

PROBABILITY

(I) Important terminology :

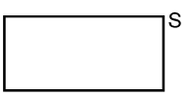
(i) Random experiment :

It is a process which results in an outcome which is one of the various possible outcomes that are known to us before hand e.g. throwing of a die is a random experiment as it leads to fall of one of the outcome from {1, 2, 3, 4, 5, 6}. Similarly taking a card from a pack of 52 cards is also a random experiment.

(ii) Sample space :

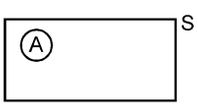
It is the set of all possible outcomes of a random experiment e.g. {H, T} is the sample space associated with tossing of a coin.

In set notation it can be interpreted as the universal set.



(iii) Event :

It is subset of sample space. e.g. getting a head in tossing a coin or getting a prime number in throwing a die. In general if a sample space consists 'n' elements, then a maximum of 2^n events can be associated with it.



(iv) Complement of event :

The complement of an event 'A' with respect to a sample space S is the set of all elements of 'S' which are not in A. It is usually denoted by A′, $\overline{A}$ or A^C.

(v) Simple event :

If an event covers only one point of sample space, then it is called a simple event e.g. getting a head followed by a tail in throwing of a coin 2 times is a simple event.

(vi) Compound event :

When two or more than two events occur simultaneously, the event is said to be a compound event. Symbolically A ∩ B or AB represent the occurrence of both A & B simultaneously.

Note : "A ∪ B" or A + B represent the occurrence of either A or B.

(vii) Equally likely events :

If events have same chance of occurrence, then they are said to be equally likely.

e.g

(i) In a single toss of a fair coin, the events {H} and {T} are equally likely.

(ii) In a single throw of an unbiased die the events {1}, {2}, {3} and {4}, are equally likely.

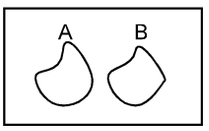
(iii) In tossing a biased coin the events {H} and {T} are not equally likely.

(viii) Mutually exclusive / disjoint / incompatible events :

Two events are said to be mutually exclusive if occurrence of one of them rejects the possibility of occurrence of the other i.e. both cannot occur simultaneously.

In the vein diagram the events A and B are mutually exclusive. Mathematically, we write $A \cap B = \phi$

Events $A_1, A_2, A_3, \ldots\ldots A_n$ are said to be mutually exclusive events iff $A_i \cap A_j = \phi \ \forall\ i, j \in \{1, 2, \ldots, n\}$ where $i \neq j$



Note : If $A_i \cap A_j = \phi \ \forall\ i, j \in \{1, 2, \ldots, n\}$ where $i \neq j$,
then $A_1 \cap A_2 \cap A_3 \ldots\ldots \cap A_n = \phi$ but converse need not to be true.

(ix) Exhaustive system of events :

If each outcome of an experiment is associated with at least one of the events $E_1, E_2, E_3, \ldots\ldots E_n$, then collectively the events are said to be exhaustive. Mathematically we write
$E_1 \cup E_2 \cup E_3 \ldots\ldots E_n = S$. (Sample space)

(II) Classical (a priori) definition of probability :

If an experiment results in a total of (m + n) outcomes which are equally likely and if 'm' outcomes are favorable to an event 'A' while 'n' are unfavorable, then the probability of occurrence of the event 'A', denoted by P(A), is defined by $\frac{m}{m+n} = \frac{\text{number of favourable outcomes}}{\text{total number of outcomes}}$

i.e. $P(A) = \frac{m}{m+n}$.

We say that odds in favour of 'A' are m : n, while odds against 'A' are n : m.

Note that $P(\overline{A})$ or $P(A')$ or $P(A^C)$, i.e. probability of non-occurrence of A = $\frac{n}{m+n} = 1 - P(A)$

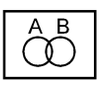
In the above we shall denote the number of out comes favourable to the event A by n(A) and the total number of out comes in the sample space S by n(S).

$\therefore \quad P(A) = \frac{n(A)}{n(S)}$.

(III) Addition theorem of probability :

If 'A' and 'B' are any two events associated with an experiment, then

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$



De Morgan's laws : If A & B are two subsets of a universal set U, then

(a) $(A \cup B)^c = A^c \cap B^c$

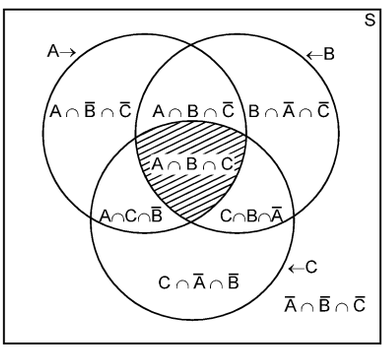
(b) $(A \cap B)^c = A^c \cup B^c$

Distributive laws :

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

For any three events A, B and C we have the figure



(i) $P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$

(ii) $P(\text{at least two of A, B, C occur}) = P(B \cap C) + P(C \cap A) + P(A \cap B) - 2P(A \cap B \cap C)$

(iii) $P(\text{exactly two of A, B, C occur}) = P(B \cap C) + P(C \cap A) + P(A \cap B) - 3P(A \cap B \cap C)$

(iv) $P(\text{exactly one of A, B, C occur}) =$
$P(A) + P(B) + P(C) - 2P(B \cap C) - 2P(C \cap A) - 2P(A \cap B) + 3P(A \cap B \cap C)$

(IV) Conditional probability

If A and B are two events, then $P(A/B) = \dfrac{P(A \cap B)}{P(B)}$.

Note that for mutually exclusive events P(A/B) = 0.

(V) Independent and dependent events

If two events are such that occurence or non-occurence of one does not affect the chances of occurence or non-occurence of the other event, then the events are said to be independent. Mathematically : if $P(A \cap B) = P(A)\,P(B)$, then A and B are independent.

Note: (i) If A and B are independent, then
(a) A′ and B′ are independent,
(b) A and B′ are independent and
(c) A′ and B are independent.

(ii) If A and B are independent, then $P(A / B) = P(A)$.

If events are not independent then they are said to be dependent.

Independency of three or more events

Three events A, B & C are independent if & only if all the following conditions hold :

$P(A \cap B) = P(A) \cdot P(B)$; $P(B \cap C) = P(B) \cdot P(C)$

$P(C \cap A) = P(C) \cdot P(A)$; $P(A \cap B \cap C) = P(A) \cdot P(B) \cdot P(C)$

(VI) Binomial probability theorem :

If an experiment is such that the probability of success or failure does not change with trials, then the probability of getting exactly r success in n trials of such an experiment is ${}^nC_r\, p^r\, q^{n-r}$, where 'p' is the probability of a success and q is the probability of a failure in one particular experiment. Note that p + q = 1.

(VII) Expectation :

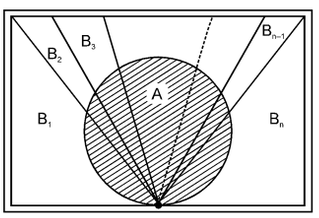
If there are n possibilities $A_1, A_2, \ldots A_n$ in an experiment having the probabilities $p_1, p_2, \ldots\ldots p_n$ respectively. If value $M_1, M_2, \ldots\ldots, M_n$ are associated with the respective possibility. Then the expected

value of the experiment is given by $\sum_{i=1}^{n} p_i \cdot M_i$

(VIII) Total probability theorem

If an event A can occur with one of the n mutually exclusive and exhaustive events $B_1, B_2, \ldots, B_n$ and the probabilities $P(A/B_1), P(A/B_2) \ldots P(A/B_n)$ are known, then

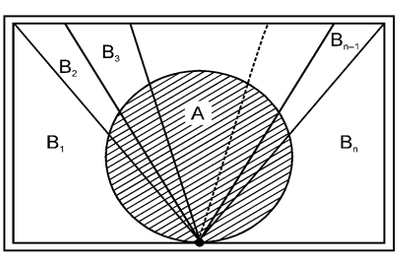
$$P(A) = \sum_{i=1}^{n} P(B_i) \cdot P(A/B_i)$$



(IX) Bayes' theorem :

If an event A can occur with one of the n mutually exclusive and exhaustive events $B_1, B_2, \ldots, B_n$ and the probabilities $P(A/B_1), P(A/B_2) \ldots P(A/B_n)$ are known, then

$$P(B_i/A) = \frac{P(B_i) \cdot P(A/B_i)}{\sum_{i=1}^{n} P(B_i) \cdot P(A/B_i)}$$



(X) Probability distribution :

(i) A probability distribution spells out how a total probability of 1 is distributed over several values of a random variable (i.e. how ..... possibilities)

(ii) Mean of any probability distribution of a random variable is given by :

$$\mu = \frac{\Sigma p_i x_i}{\Sigma p_i} = \Sigma p_i x_i \quad (\text{Since } \Sigma p_i = 1)$$

(iii) Variance of a random variable is given by, $\sigma^2 = \Sigma (x_i - \mu)^2 \cdot p_i$

$\therefore \quad \sigma^2 = \Sigma p_i x_i^2 - \mu^2$ (Note that SD = + $\sqrt{\sigma^2}$)

(iv) The probability distribution for a binomial variate 'X' is given by :

$P(X = r) = {}^nC_r\, p^r q^{n-r}$ where P(X = r) is the probability of r successes.

The recurrence formula $\frac{P(r+1)}{P(r)} = \frac{n-r}{r+1} \cdot \frac{p}{q}$, is very helpful for quickly computing P(1) . P(2) . P(3) etc. if P(0) is known.

Mean of Binomial Probability Distribution = np ; variance of Binomial Probability Distribution = npq.

(v) If p represents a person's chance of success in any venture and 'M' the sum of money which he will receive in case of success, then his expectations or probable value = pM

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